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TOP SECRET // NC2 // PCOS - DECLASSIFIED 2071 (E.O. 14xxx) - WORK OF FICTION

THE DECIDING MINUTES

A President's Account of the Pacific Nuclear Crisis, 2045–2046

Presidential Crisis Operating System (PCOS), Edition 1: The Pacific Crisis, 2045–2046

Thomas Ellery Kane *49th President of the United States*

With a Prologue, an Epilogue, and Annexes prepared by the Editorial Team of the Presidential Crisis Operating System

Released by the National Archives and Records Administration, College Park, Maryland, 2071

Original copy: one of four. Handwritten diary sources: eleven volumes, retained. This document was prepared by the President personally, in office, for the use of his successors. It is not an official history of the United States Government.

Document Control

Field	Entry
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Declassification Notice

NATIONAL ARCHIVES AND RECORDS ADMINISTRATION Office of the Archivist of the United States Notice of Release, Presidential Crisis Operating System, Edition 1

1. The document reproduced in this volume was created between November 2045 and June 2046 by Thomas Ellery Kane, then serving as the 49th President of the United States. It was classified TOP SECRET by its author and held under special access controls for a period of twenty-five years.

2. Under the provisions of Executive Order 14xxx, governing the systematic and mandatory review of classified national security information of permanent historical value, the document was reviewed during 2069–2071 by an interagency panel. The panel's findings were accepted by the Archivist of the United States on 4 March 2071.

3. The document is released with redactions. Redacted material falls into four categories: (a) information that would identify living human sources of intelligence, or the surviving relatives of deceased sources; (b) specific capabilities and vulnerabilities of command, control, and communications systems that remain in service or have successors in service; (c) foreign government information provided in confidence, where the providing government has not consented to release; and (d) personal medical information concerning individuals other than the author.

4. Text that the author himself struck through, annotated, or marked "do not copy" has been reproduced where the panel judged it to be of historical value, and is identified in the editorial notes.

5. A Prologue, an Epilogue, and Annexes B through G were prepared by the Editorial Team of the Presidential Crisis Operating System. The Prologue was drafted in 2046 under the author's direction. The Epilogue and the expanded Annexes were completed in 2071 for the purposes of this release. Annex A is the author's own text.

6. The author, now a private citizen, was consulted on this release under the customary courtesy extended to former Presidents. He raised no objection to release and requested that no passage be softened.

7. Release of this document does not constitute confirmation or denial of any intelligence relationship, operation, or capability not expressly described herein.

Signed at College Park, Maryland, 4 March 2071.

Handling Instructions

The following handling instructions appeared on the inside cover of each original copy, typed on a separate sheet and initialed by the author. They are reproduced as they were written, for their historical interest. They no longer have legal force.

1. **This document is written for one reader at a time.** It is addressed to the sitting President, the President-Elect during the transition, and the Commander of United States Strategic Command in the event of the death or incapacity of both. No other person is to read it except under a written read-in signed by the sitting President.

2. **Do not summarize it.** Staff will want to turn this into a memorandum with bullet points. Do not let them. The bullet points are in Annex A. The rest of the document exists because bullet points failed.

3. **Read Chapter 1 before your first nuclear briefing.** Read Chapter 6 before your first night alone in the residence. Read Annex A, Section 2 ("Nuclear Warning Protocol") until you can recite it without looking.

4. **Do not remove it from the White House Complex** except aboard the National Airborne Operations Center or another continuity platform, and then only in the custody of the military aide.

5. **Do not copy it.** If you wish to correct it, write in the margins, sign and date your note, and leave it for the next reader. That is what an operating system is for. It is meant to be patched.

6. **If you disagree with it, say so in writing.** The worst thing a successor could do with this document is obey it without thinking. The second worst is ignore it without reading it.

7. **The names of human sources in this document are protected by a promise I made personally.** If you ever have to choose between that promise and the national interest, the decision is yours. I ask only that you make it slowly.

T.E.K.

Reader's Note

The following note is not part of the fictional document. It is addressed to the reader of this book.

The Deciding Minutes is a work of fiction.

It imitates the form of a presidential memoir and the form of a declassified government record, but it is neither. President Thomas Ellery Kane, his family, his Cabinet, his generals, the Chinese leaders and officers in these pages, the intelligence sources code-named LANTERN, WILLOW, THRUSH, KESTREL, and ORCHID, the satellite program called ARGUS, the company Halcyon Orbital, the "Jade Line," the Bashi Channel Incident, the Eleven Minutes, and the Singapore Understanding are all invented. The crisis described here has not happened. The author hopes it never will.

No real living person appears as a character. Real people are mentioned only as historical figures, such as President John F. Kennedy in 1962, or Lieutenant Colonel Stanislav Petrov, the Soviet duty officer who in September 1983 judged a satellite warning of an American attack to be a false alarm. Heads of government of real countries in the year 2045 are invented. Real institutions (the National Security Council, United States Strategic Command, the People's Liberation Army Rocket Force, the National Archives) appear as institutions, not as portraits of the people who serve in them.

The technical world of the book is built only from material that is already public: unclassified government reports to Congress, annual Department of Defense reports on Chinese military power, the published work of independent nuclear researchers, and well-known histories of nuclear command and control. Where a detail would be operationally sensitive in the real world, such as authentication procedures, codes, system vulnerabilities, or weapon design, the book either invents a harmless fictional procedure or stays deliberately vague. Cyber operations are described by their effects, never by their methods. Nothing in this book will teach anyone how to attack anything.

The form owes a debt to *Detik-Detik yang Menentukan* (2006), in which B. J. Habibie, the aeronautical engineer who became the third President of Indonesia, reconstructed his most dangerous weeks in office from his notes, explaining his reasoning as an engineer explains a design.

The banner at the head of each chapter, the redactions, the "Operating Notes," and the "Decision Records" are literary devices. The redactions do not hide anything. There is nothing behind them.

Table of Modules

In the original document, chapters were designated "modules," in keeping with the author's conception of the text as an operating system. Both designations are used below.

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- Reader's Note
- Foreword, by the Archivist of the United States (2071)
- Preface: Why an Operating System, by Thomas Ellery Kane (2046)

Module 0 — Prologue: The World of 2045 *Prepared by the Editorial Team.* The nuclear balance after New START; Golden Dome, the Proliferated Warfighter Space Architecture, and ARGUS; the aftermath of the 2036 Treasury market crisis; Taipei under President Lin Shu-wen; Beijing under General Secretary Wen Zhaolin; the people in the room.

Module 1 — Chapter 1: The Engineer's Models and the Machinery of the End The six models: the Three Clocks, the Reversibility Matrix, the Mirror Model, the Signal Ladder, the Two-Source Rule for Armageddon, and control-loop thinking. A tour of nuclear command, control, and communications, from the inauguration-day briefing of 20 January 2041 to the Presidential Emergency Operations Center, the National Military Command Center, the North American Aerospace Defense Command, and Strategic Command. The football and the man who carries it.

Module 2 — Chapter 2: The Quarantine (March–May 2045) Beijing's "law-enforcement quarantine" of Taiwan; the 3:10 a.m. awakening of 4 March; the deaths aboard the *Formosa Pearl*; LANTERN's first crisis report; the escorted humanitarian corridor and Operation OPEN DOOR; the first, cold call on the Jade Line.

Module 3 — Chapter 3: The Pressure Season (June–August 2045) The customs blockade; cyber attacks on West Coast ports and the Texas grid; Treasury yields at 7.1 percent; "The Patience of Free People"; the Defense Production Act and Halcyon Orbital; WILLOW's report on the General Secretary's health; the Sentosa channel; three submarines out of Yulin in forty-eight hours.

Module 4 — Chapter 4: Blood in the Strait (September–October 2045) The Bashi Channel Incident and the loss of USS *Thomas Hudner*; forty-seven minutes without knowing; the proportionate response; the blinding of the satellites; Defense Readiness Condition 3; the arrest of KESTREL; the decision not to bring LANTERN home; the call with Moscow; the speech about "the window."

Module 5 — Chapter 5: The Nine Days, I — The Gamble (1–5 November 2045) Readiness Level 2 at Hami; the argument for striking "left of launch" and the reasons for refusing it; quiet

generation of strategic forces; the General Secretary in hospital; the nineteen-hour decision to reveal what we knew; the letter of 5 November.

Module 6 — Chapter 6: The Nine Days, II — The Eleven Minutes (6–9 November 2045) 02:31 Eastern, 6 November: six probable launches that were not there. Minute by minute through the Eleven Minutes. The call of 7 November. The fall of the Rocket Force commander. The pause.

Module 7 — Chapter 7: The Long Way Down (November 2045–April 2046) The Singapore talks, ashore and at sea; the hardest conversation with an ally; the lifting of the blockade; the Singapore Understanding and the Kane–Wen Summit of 21–22 March 2046; closed testimony before the Senate; Operation NIGHT FERRY. With the author's closing reflections, **"Letter to My Successor."**

Epilogue: After the Minutes (2071) *Prepared by the Editorial Team.* What became of the people, the agreements, and the machines; and an assessment of what the author got wrong.

Annexes

- **Annex A — The Operating System.** The author's forty rules and six protocols: the first hour of a crisis; nuclear warning; diplomatic channel management; managing allies; managing Congress and the public; managing yourself.
- **Annex B — Chronology,** January 2044 to April 2046.
- **Annex C — Dramatis Personae.**
- **Annex D — Glossary of Terms and Acronyms,** all spelled out.
- **Annex E — Texts:** the public text of the Singapore Understanding, and the President's letter of 5 November 2045 to General Secretary Wen Zhaolin.
- **Annex F — Organization Charts in Text:** the United States nuclear command, control, and communications chain; the Central Military Commission and People's Liberation Army Rocket Force chain.
- **Annex G — Note on Sources.**

Foreword

by the Archivist of the United States

There is a room on the fourth floor of the National Archives building in College Park, Maryland, that the staff call "the quiet vault." It is not, strictly, a vault. It is a climate-controlled reading room with a single table, a single chair, a camera in each corner, and a steel cabinet bolted to the floor. For twenty-two years the cabinet held two things that had come to us from the White House in January 2049: a sealed envelope marked *Copy 2, NSC Executive Secretary*, and eleven bound notebooks in a gray archival box, each with a cloth spine and a handwritten label: *T.E.K. — 2045 (I)*, *2045 (II)*, and so on through *2046 (III)*.

The notebooks are the diaries of Thomas Ellery Kane. The envelope contained the document you are about to read.

I first opened the envelope in the autumn of 2069, when the mandatory twenty-five-year review began. I had been told what to expect: a classified after-action report, perhaps two hundred pages, of the kind every administration produces after a crisis and very few people ever read. What I found was six hundred and eleven typed pages, much of it in the first person, much of it reading less like a government report than like a letter written very late at night by a tired man to someone he would never meet. On the first page, above the title, the author had written in blue ink: *For whoever is sitting where I am sitting. I'm sorry it's you.*

I have read cables written in panic and memoranda written to protect their authors. I had never read anything like this.

What this document is

In the formal language of the archival register, the *Presidential Crisis Operating System, Edition 1* is a "personal instructional record prepared by the President in office for the use of successors." That description is accurate and tells you almost nothing.

The document is, first, an account of the Pacific Crisis of 2045–2046, the eighteen months in which the United States and the People's Republic of China came closer to a nuclear exchange than at any time since the two nations acquired the means to destroy each other. Readers who lived through it will remember the blackouts in Texas, the forty-one sailors of USS *Thomas Hudner*, and the sudden, almost inexplicable relief of November 2045. They will not remember, because almost nobody knew, how close that relief came to not arriving. The events of the early morning of 6 November 2045, which the author calls "the Eleven Minutes," were known until this release to fewer than forty people in the United States Government.

The document is, second, a manual. President Kane was an engineer before he was a governor or a president, and he wrote the way engineers write when something has nearly failed: he described the system, identified the failure modes, and proposed corrections. He called his corrections "Operating Notes" and placed them throughout the text. He recorded his major decisions in a fixed format, the "Decision Record," listing the options he considered, whether each could be undone, what he feared, what he chose, and what happened. He reduced what he had learned to forty rules and six protocols, which appear as Annex A. He intended the whole to be revised by each president who read it, and he called it "Edition 1" in the expectation that there would be an Edition 2.

There was not. The read-in register shows that each of the three presidents who held office between 2049 and 2061 signed for it. Two left marginal notes, which remain classified and are not reproduced here. After 2061 the read-in register is blank. The document had become history before it was ever published.

The document is, third, a confession. Readers should be warned that the author does not spare himself. He writes about a human source of intelligence, code-named KESTREL, who was arrested and executed in the autumn of 2045, and about his own decision, made in the same weeks, to leave a second source in danger rather than bring him to safety. He writes about concealing agreements from Congress, about his daughter, about not sleeping. A reader looking for a hero will find one only by looking past the author's own account of himself.

How the document was reviewed

The interagency panel that reviewed this document met forty-one times between October 2069 and February 2071. It included representatives of the Office of the Director of National Intelligence, the Central Intelligence Agency, the Departments of Defense, Energy, and State, and the National Security Council staff, and it was chaired by the Office of the Archivist.

I want to be candid about the panel's arguments, because they shaped what you are holding.

The intelligence representatives initially proposed redacting every reference to the human sources whose code names appear in these pages. The author's account of the crisis, they argued, would tell any foreign counterintelligence service exactly where to look, even twenty-five years later, and could endanger the families of people who had trusted the United States with their lives. The panel took this argument seriously. In the end it concluded that the sources were already known to the government concerned, that several had died, and that the surviving relatives could be protected by redacting identifying particulars rather than whole episodes. The true name of the source code-named LANTERN appears once, in the Epilogue, with the written consent of his family. All other true names of sources remain withheld.

The Defense representatives proposed redacting much of Chapters 1 and 6, which describe nuclear command, control, and communications. The panel found that nearly all of the author's technical descriptions drew on material public when he wrote, or public since, describing systems long superseded. It redacted what remained sensitive, and tried to keep the redactions few.

The State representatives raised the secret portions of the Singapore Understanding of 2046. Those were declassified by agreement with the other governments concerned during the 2060s; the panel saw no reason to withhold what the world already knew.

There was one further argument, which was not about secrecy at all. A member of the panel asked whether it was right to publish the private diary entries of a living man, written when he believed no one but a future president would read them. The question was put to the author. His answer,

delivered in a handwritten letter from his home in Colorado in the spring of 2070, is in the archival register. It reads, in part: “I wrote it to be read. I expected the readers to be presidents. If it turns out the readers are citizens, so much the better. They are the ones who hired us.”

On the author

Thomas Ellery Kane was born in 1985. He trained as an aerospace and control-systems engineer, earned a doctorate in control theory from the Massachusetts Institute of Technology, and spent the early part of his career working on the guidance of satellites before founding a company that manufactured components for launch vehicles. He served as Governor of Colorado from 2031 to 2039, was elected President in November 2040, and was re-elected in November 2044 by a narrow margin. He left office in January 2049.

He is, as I write this, eighty-six years old. He lives quietly. He has declined, for twenty-two years, every invitation to discuss the events described in this book. He has also kept, without interruption, the handwritten diary that he began at the age of twelve; he told the review panel, with some amusement, that the current volume is number one hundred and fourteen.

The eleven diary volumes covering the crisis remain in the quiet vault. They are not released with this document, and under the terms of their deposit they will not be released during the author’s lifetime. The panel compared every dated diary passage quoted in this document against the original notebooks. With very few exceptions, the quotations are faithful. Where the author revised a diary entry for the document, sometimes to make it clearer and once or twice to make himself look worse, the editorial notes say so.

On the form

Government records are usually written to show that procedures were followed. This one was written to show what happens when procedures are not enough. Procedures matter. The author spent much of his presidency building them, and Annex A is a monument to his belief in them. But the heart of this document is the space between the procedure and the decision, the few minutes in which a human being has to judge whether what the screens say is true.

Those minutes did not end in 2045. Every president since has inherited them. So will every president to come.

A note of thanks

I am grateful to the Editorial Team of the Presidential Crisis Operating System, whose members asked not to be named; to the interagency panel, which argued for sixteen months in good faith; and to the archivists who kept the quiet vault quiet for twenty-two years.

Finally, I am grateful to the author, who wrote this for one reader at a time and has allowed it to be read by all of us.

Dr. Corinne Haddad-Lowell Archivist of the United States College Park, Maryland March 2071

Preface: Why an Operating System

The White House, Washington. Written between May and June 2046.

> *"Write down what you saw before you write down what you think. You will need to know the difference later."* — from the author's diary, 4 July 1997

The notebook

On the fourth of July, 1997, I was twelve years old, and I watched a small spacecraft bounce across the surface of Mars inside a cluster of airbags. My mother let me stay up for it. We watched the coverage on a television in the kitchen, and I remember the engineers at the Jet Propulsion Laboratory in California standing up and hugging each other when the first signal came back, and I remember thinking that I had never seen grown men so happy about arithmetic.

The next morning my mother, who taught eighth-grade science and believed that every child should be made to write things down, gave me a hardbound notebook with graph paper inside and said: "If you want to be one of those people, you start by keeping a log. Engineers keep logs. They write down what happened, when it happened, and what they did about it. Every day. Even the boring days." On the first page, under the date, I wrote the sentence that stands at the head of this preface. I am not sure where I got it. I suspect from her.

I have kept that log every night since. I kept it in a college dormitory, in the clean rooms where I worked on satellite guidance, in the Governor's Residence in Denver, in the Executive Residence of this house, and on the night my father died. On the bad nights it is three lines. On the worst nights of the crisis described in this document it was eleven pages, written on yellow legal paper in the Presidential Emergency Operations Center under the East Wing and copied into the notebook the next morning before the President's Daily Brief.

I tell you this not because my habits are interesting but because they are the source of everything in this document. When I quote a conversation in these pages, I am quoting what I wrote down that night, usually within a few hours, sometimes within a few minutes. When I give a time to the minute, it is because I looked at the clock on the wall and wrote it in the margin. My memory is no better than anyone else's, and in the summer of 2046 I already find that I remember some of these events in a shape they did not have. The notebooks correct me. Where my memory and my notebooks disagree, I have trusted the notebooks.

I also tell you this because I want you to start keeping one. More on that below.

The book I wish I had been given

On the morning of 20 January 2041, about two hours before I took the oath, a military officer I had met only once sat down with me in a small room at Blair House and explained how I could end the world. I describe that briefing in Chapter 1. It lasted forty minutes. It was clear, professional, and complete in every respect except one: it told me how the machinery worked, and it told me nothing at all about how to decide.

That is not a criticism of the officer. Nobody can tell you how to decide. The system is designed, correctly, to place that burden on one civilian, elected by the people, and to leave the judgment to that person. But I walked out of that room with a strange feeling that I could not name for some months. I named it eventually, in the notebook, in the spring of 2041: *They have handed me the controls of a machine and no manual. They have handed me the manual for the machine and no manual for the operator.*

Every new president receives shelves of briefing books, almost none written by anyone who has sat in the chair. The few presidents who wrote about their crises wrote memoirs for the public, after leaving office, when much of what mattered was still classified. The candid instruction, the kind one engineer gives another at shift change, has been passed on, if at all, in a few minutes between an outgoing president and an incoming one before the ride to the Capitol. That is not enough time to explain anything.

In 1998 an aircraft engineer named B. J. Habibie found himself President of Indonesia, suddenly and without preparation, in the middle of a crisis that could have torn his country apart. Years later he wrote an account of those weeks called *Detik-Detik yang Menentukan*, "The Deciding Moments." I read it in an English summary as a young man, when I was more interested in his aircraft than his politics, and I read it again properly during my first term. What struck me was not his conclusions, many of which are still debated in Indonesia, but his method. He wrote as an engineer. He described the forces acting on him. He explained which decisions he regarded as reversible and which were not, and why he treated them differently. He let the reader see the diagram behind the decision.

I borrowed his title and I borrowed his method. I have tried to do for my successors what he did for his readers: to show not only what I decided but how I built the decision, so that you can test my design, find its flaws, and build a better one.

Why "operating system"

My staff dislike the title of this document. My Chief of Staff, Megan Pruitt, told me it sounds like a software manual. My National Security Advisor, Sam Oyelaran, said, more kindly, that it sounds like something only an engineer would call a book. Both are right. I kept it anyway, and I owe you an explanation.

An operating system is the layer of software that sits between a computer's hardware and the programs a person actually wants to run. You do not notice it when it works. It decides which task gets the processor next. It manages memory, deciding what is kept close at hand and what is put away. It handles interrupts: the signals from outside that demand attention immediately and force everything else to wait. It enforces permissions, so that one program cannot destroy another. And it recovers from errors, or tries to.

The government of the United States has extraordinary hardware: the most capable military in history, intelligence services that can see nearly any place on Earth within minutes, diplomats in

nearly every capital. It has more applications, in the form of policies and programs, than anyone can count. What it does not have, at the top, is a reliable operating system. Each president writes his or her own, mostly from scratch, mostly under pressure, and mostly in the first crisis, which is the worst possible time to be writing software.

The presidency in a crisis is a scheduling problem: the military clock, the political clock, and the information clock all demand the processor at once. It is a memory problem: what do you keep in your head, and what do you trust others to hold? It is an interrupt problem: the phone at 3 a.m., the alert from a satellite, the burst transmission from a man in a uniform on the other side of the world who has risked his life to tell you something. It is a permissions problem: who may act without asking, who must ask, and what no one may do without you. And it is an error-recovery problem, because you will make errors, and the question is whether the system catches them before they become irreversible.

This document is my attempt at an operating system for that job. It is not complete. It is certainly not bug-free. I have called it Edition 1 because I expect, and hope, that it will be revised.

What happened, and what nearly happened

You will know the public outline: the quarantine and blockade of Taiwan from March 2045; the attacks on our ports, grid, markets, and satellites; the forty-one sailors killed in the Bashi Channel in September, and our response; the raised nuclear readiness of the autumn; the sudden break in November; and the Singapore Understanding of 2046, which I believe will keep the peace in the Pacific for a generation if it is honored.

What the public does not know, and what you must know, is how close we came in the first days of November 2045, and why. It was not because either government wanted a nuclear war. Neither did. It was because a combination of fear, fatigue, illness, ambition, faulty machines, and very good intelligence, too good in some ways, created a situation in which a nuclear war could have begun without anyone choosing it. Chapter 6 describes the eleven minutes in the early morning of 6 November when my own warning systems told me, with what the screens called high confidence, that China had launched intercontinental ballistic missiles at the United States. It had not. I will tell you how I knew, and how close I came to not knowing.

I want to be careful here. I do not claim that I was wise. I was lucky, and I was helped by people braver than I am, some of whom will never be thanked publicly because their names cannot be published. I made decisions in this crisis that I would make differently now. I record them anyway, because a manual that only describes successes teaches nothing about failure, and failure is what you need to prepare for.

How this document is built

The document has three kinds of material. **The narrative**, Chapters 1 through 7, is my account in the first person, organized by date and drawn from my notebooks. Chapter 1 sets out the six models I use to think about crises and walks through the machinery of nuclear command and control; Chapters 2 through 7 follow the crisis from March 2045 to April 2046. I have written them so that you feel, as nearly as words allow, what it was like in the room, because some of the most important lessons are about fatigue, fear, and the pressure of other people's confidence, and you cannot learn those from a checklist. Conversations are reconstructed from my notes; they are not transcripts. My

inner thoughts, in italics, are what I wrote that night that I had been thinking. I have not improved them.

The Operating Notes and Decision Records are boxed passages throughout the narrative. The notes are lessons addressed to you directly. The records set out each major decision in a fixed form: options, reversibility, risk, choice, outcome. If you have no time for the narrative, read these.

The Annexes begin with Annex A, the operating system itself, reduced to forty rules and six protocols. It is the only part I would ask you to memorize. The rest, prepared with a small editorial team that also drafted the Prologue at my direction, are reference: chronology, people, glossary, texts, charts, sources.

How to use it

I have thought for a long time about how I would want my successor to use this document, and I have come to a few conclusions.

First, read it before you need it. The worst time to read a manual is during the emergency it describes. If you are reading this as President-Elect, in the weeks before your inauguration, you are reading it at the right time. Read Chapter 1 before your first nuclear briefing, so that you know what questions to ask. You will be surprised how few questions anyone expects you to ask.

Second, argue with it. I have been wrong about important things, and I will be wrong about some of what I have written here. Your world will not be 2045, and some of my rules will not suit you. Write in the margins. Sign and date your notes. Leave them for the next reader. An operating system that is never patched becomes dangerous, because the world it was written for no longer exists, and nobody remembers why the rules were written in the first place.

Third, do not let anyone turn it into a slogan. In this crisis I reduced my reasoning to short phrases, such as "slow the military clock" and "two sources before Armageddon." Tired people remember short sentences. They also obey them without understanding. Every rule in Annex A has a story behind it in the narrative. Know the story before you rely on the rule.

Fourth, distrust the machines, and distrust me. The central technical lesson of this crisis is that our ability to see the world had outrun our ability to understand what we saw. By 2045 the United States could watch nearly every launcher in China nearly all the time. We could fuse the data from hundreds of satellites with the help of artificial intelligence and put a confidence score on the screen in seconds. I believed this made us safer, and in most ways it did. It also frightened our adversary, made our own officers overconfident, and on one night in November produced a picture of an attack that did not exist. The machines were extraordinary. They were not wise. Neither am I. The rule that saved us that night, that no one should act on a nuclear warning from a single kind of evidence, is a rule about distrust. Learn it.

Fifth, take care of the human sources. This document mentions, by code name, several people who provided intelligence to the United States from inside the Chinese military and government during the crisis. Without them, I believe, the crisis would have ended differently. One of them died because of what he did for us. Another remained in danger for months because I decided I needed him where he was. Their families trusted us. The names are protected by a promise I made, and I have asked in the Handling Instructions that you break that promise, if you must, only slowly. I will add here what I could not put in the instructions: that the decisions I made about those people are

the ones I think about most often at night, and that you should expect to face decisions like them, and that there is no model in this document that makes them easier.

Sixth, keep a log. I mean this literally. Buy a notebook. Write in it every night, by hand, for at least ten minutes. Write down what happened, when it happened, what you did, and what you were afraid of. Write down what people said to you, as close to their words as you can manage. Do not dictate it. Do not let anyone see it. You will think you do not have time. You do not have time not to.

The presidency distorts time. A decision you made on Tuesday feels, by Friday, like something from a year ago. People will tell you, with complete conviction, what you said and decided, and they will sometimes be wrong. The log is your anchor. It is also, on the worst nights, the only place where you can be honest, because it is the only listener in this building who wants nothing from you.

There is one more reason, which my mother understood in 1997 and which I did not understand until the autumn of 2045. Writing things down slows you down. When you have to put a decision into a sentence, you have to know what the decision is. When you have to write down the time, you have to look at the clock. When you have to describe what you saw before describing what you think, you discover, sometimes, that you did not see what you thought you saw. In a job whose greatest danger is speed, a habit that slows you down is worth more than any briefing.

A word about the people

This is my account, and it is necessarily about me. But the crisis was not managed by one person. It was managed by a Vice President who disagreed with me on important questions and told me so in private, never in public; by a National Security Advisor who was right more often than I was; by a Secretary of State who spent the autumn of 2045 in rooms with people who had every reason to hate her and came out with agreements; by generals and admirals who offered me options I did not like and then carried out the ones I chose; by intelligence officers whose work I can describe only in part; by sailors; and by a military aide, a Marine lieutenant colonel, who stood within arm's reach of me for the worst nine days of my life and never once let me see that he was afraid.

My wife, Claire, a pediatric oncologist who has spent her life telling parents the truth about their children, told me the truth about myself on several nights when no one else would. My daughter, Maya, was a serving officer aboard a destroyer in the Seventh Fleet for the entire period described in this document. I have tried to write about her honestly, without making her a symbol, which she would hate. I will say here only that there is no way to describe what it means to be the person who orders ships into danger when your child is aboard one of them, and that I hope you never find out.

It was managed, finally, on the other side, by people I will never be able to thank, and some I would not wish to. The Chinese leaders were not villains, with one possible exception, and even he believed he was protecting his country. The peace we reached in Singapore was possible only because, at the decisive moment, a sick man in a hospital bed in Beijing chose to believe an American president who had just told him something no president had ever told an adversary. I describe that call in Chapter 6. I still do not fully understand why he believed me. I am grateful that he did.

Last words before the beginning

I am writing this in June 2046, at the desk in the Treaty Room, usually after eleven at night. The crisis is over in the sense that the blockade is lifted and the forces have stood down. It is not over in

the sense that matters most. The weapons still exist. The machines are still watching. The fears that drove both governments to the edge are still there, slightly smaller and better managed. They will be there when you read this.

You will face your own crisis. I do not know what it will be. It may come in your first month or your last, with machines that make ours look like the radar sets of 1962. But I believe some things will not have changed. You will be tired. You will be told things with confidence that are not true. You will be pressed to act faster than you can think. You will have to judge what the other side fears, not just what it wants. And at some point, possibly at three in the morning, you will have to decide whether to believe what a screen is telling you.

When that moment comes, I hope this document will have done one thing for you: I hope it will have made you a little slower. Not hesitant. Slower. Slow enough to ask for the second source. Slow enough to see the other side's fear. Slow enough to keep the reversible door open a little longer.

That is all an operating system can do. It cannot make the decision. It can only make sure that when the decision comes, the person making it has been given every chance to make it well.

I'm sorry it's you. I'm glad it's you and not me. And I am, whatever else you may conclude about me after reading these pages, entirely on your side.

Thomas Ellery Kane 49th President of the United States The White House 30 June 2046

Prologue

The World in 2045: A Briefing Prepared by the Editorial Team

> "We have to judge a decision by the conditions in which it was made, not by the conditions we enjoy afterward." — Note pinned above the desk of the editorial team, National Archives, 2070

Editorial Preface to the Prologue

President Thomas Ellery Kane wrote the chapters that follow for a reader he never expected to meet: the next person to sit in the chair he occupied during the autumn of 2045. He assumed that reader would already know the world he lived in. He did not stop to explain why a Chinese silo field in the Gobi Desert mattered more to him than a carrier group in the Philippine Sea, why a satellite company's chief executive could refuse a President, or why a phone call that lasted four minutes could undo eight months of escalation. To him and to his successors in 2049, these things were simply the weather.

They are no longer the weather. A reader in 2071, and certainly a reader later still, comes to this document from a world that the Pacific Crisis helped to make, and therefore cannot easily see the world that produced the crisis. The editorial team that prepared this declassified edition decided, as the editors of an earlier Indonesian presidential memoir once did, to write a prologue in the third person. Its job is modest. It sets the stage. It describes the forces, the machines, the institutions, and the people that the President takes for granted in his own pages.

We have organized it in the shape of the documents Kane himself read every morning: a briefing. Each section opens with a short summary of key judgments, as an intelligence product would, and then sets out the detail. Where the original government records remain classified, we say so. Where we have had to rely on public reconstruction, we say that too. Nothing here reveals a method that could hurt anyone still living, and nothing here goes beyond what could be found, in its time, in open literature: Congressional Research Service reports, the annual Department of Defense reports on Chinese military power, the nuclear notebooks of independent scientists, and the memoirs of those who served.

1. How the World Got Here: The Twenty Years After 2026

Key judgments. The world of 2045 was shaped by three trends that began in the middle of the 2020s and never reversed: the collapse of formal nuclear arms control between the great powers; the rise of China as a full nuclear peer in capability, if not yet in numbers; and the fusion of commercial

technology, especially space and artificial intelligence, with military power. A fourth trend, financial fragility in both leading economies, turned these military facts into political pressure.

The end of the treaties

On 5 February 2026 the New Strategic Arms Reduction Treaty, signed in Prague in 2010, expired. It had been the last binding limit on the strategic nuclear forces of the United States and Russia: 1,550 deployed strategic warheads each, 700 deployed launchers, and a regime of on-site inspections and data exchanges that had already been suspended in practice since 2023. No successor treaty was signed. For a period there were informal statements from both capitals about observing the central limits voluntarily, but such statements had no inspectors behind them and no legal force, and over the next decade they quietly dissolved.

The United States, facing two nuclear peers for the first time in its history, responded in the 2030s with what the planning documents called a "measured upload." American missiles and bombers had been built with room for more warheads than the treaty allowed; some of that room was used. By 2045 the United States held approximately 3,700 warheads in its stockpile, of which approximately 1,770 were deployed on strategic delivery systems. This was a modest increase over the treaty years, and it was much debated, but it did not restore the comfortable margin that American planners had once enjoyed over every potential adversary combined.

The rise of a second nuclear superpower

China's nuclear expansion had been visible since 2021, when independent researchers using commercial satellite imagery found the first of three great missile silo fields under construction in the western deserts: near Yumen in Gansu province, near Hami in Xinjiang, and in the Ordos basin near Hanggin Banner in Inner Mongolia. The Department of Defense reported in its annual assessment for 2024 that the three fields held some 320 silos for solid-fuel intercontinental ballistic missiles, and in its 2025 report that more than 100 of them were likely already loaded. In the same years China was more than doubling its older force of liquid-fuel DF-5 silos. Its stockpile, assessed at roughly 600 warheads in the mid-2020s, was projected to exceed 1,000 by 2030.

The projection proved conservative. By 2045 American intelligence assessed the Chinese stockpile at approximately 1,600 warheads. The editorial team cautions that this figure, like all such figures, carried error bars of several hundred; the President himself writes that he never fully trusted any number with four digits that came out of the intelligence community. But the shape was clear. China had moved, within a single generation, from a "minimum deterrent" of a few hundred warheads, most of them kept separate from their missiles, to a large, diverse, and ready force.

More important than the numbers was the posture. Since about 2030, the People's Liberation Army Rocket Force had kept part of its force on what Chinese doctrine called *yujing fanji*, "early-warning counterstrike": in Western terms, launch on warning. The idea is simple to state and terrible to contemplate. If a country's satellites and radars detect enemy missiles in flight, it launches its own before they arrive, so that its weapons are not destroyed in their silos. The United States and Russia had lived with such postures since the 1960s. China, for most of its nuclear history, had not; its doctrine had been to absorb a first strike and retaliate later with whatever survived. The shift to launch on warning, made possible by China's own space-based early-warning satellites and large ground radars, meant that for the first time a Chinese leader might have to make a decision about nuclear war in minutes.

The fusion of commerce and war

The third trend was technological. The 2020s saw the arrival of proliferated low-Earth-orbit constellations: thousands of small, cheap satellites providing internet service around the globe. The war in Ukraine that began in 2022 had shown how decisive such networks could be on a battlefield, and how uncomfortable it was for governments to depend on private companies to provide them. In the same years, the United States Space Force began to build its own proliferated architecture for missile tracking and military communications, and the first commercial constellations purpose-built for national security customers, such as SpaceX's Starshield, came into service.

By the 2040s the line between a commercial satellite and a military one had become almost meaningless. Sensors rode on commercial buses; commercial companies operated government payloads under contract; and the same rocket might carry a broadband satellite and a classified radar on a single flight. Artificial intelligence, which in the 2020s had been a novelty in military planning, became by the late 2030s the connective tissue of every major armed force: sorting sensor data, recommending targets, routing logistics, and, in the most sensitive places, advising on whether a set of bright dots on a screen was an attack.

The crises before the crisis

The Pacific Crisis of 2045 was not the first confrontation in the Taiwan Strait, and the people who lived through it carried memories of the earlier ones. The team will not catalogue them here, but readers should know that the 2020s and 2030s saw repeated large-scale Chinese military exercises around Taiwan, each one somewhat closer to the island and somewhat more realistic than the last; repeated incidents of ships and aircraft passing dangerously close; a series of serious cyber intrusions into American infrastructure attributed to Chinese state units, of which the "Volt Typhoon" campaign revealed in the mid-2020s was the first to be widely publicized; and a slow, grinding erosion of the informal understandings that had kept the peace since 1979.

Each of these episodes ended without war. Each of them also taught both sides a lesson that later proved dangerous: that the other side would, in the end, step back. By 2045 a generation of officers in both militaries had been trained on the assumption that brinkmanship worked. President Kane writes, in a diary entry from March 2045 quoted in Chapter 2, that "the most dangerous thing in this crisis is the memory of all the times it didn't happen."

2. The Reckoning: Debt, Markets, and Domestic Weather

Key judgments. By 2045 both the American and Chinese governments were financially constrained and politically insecure. In Washington, the memory of the 2036 Treasury market crisis made every escalation a question for the bond market as well as the Pentagon. In Beijing, a decade of stagnation had left the Party more dependent on nationalism for legitimacy than at any time since the 1980s. Both leaderships had, in different ways, less room to retreat.

The 2036 Treasury crisis

The American federal debt held by the public passed 100 percent of gross domestic product in the late 2020s and kept climbing. Interest payments, which had been a minor line in the federal budget for most of the early century, became by the 2030s the fastest-growing item, larger than the defense budget. Economists had warned for years that a moment would come when the market for United

States Treasury securities, the deepest and most trusted market in the world, would stop absorbing new debt at a reasonable price. That moment arrived in the spring of 2036.

The details of the crisis are beyond this briefing, and the economic histories of the period treat them at length. In short: a failed series of auctions, a sudden withdrawal of foreign official buyers, and a disorderly sale by leveraged funds drove the yield on the ten-year Treasury note up by more than two percentage points in nine trading days. The Federal Reserve intervened on a scale not seen since 2020; Congress, after a frightening week, passed a fiscal framework that combined tax increases, spending caps, and a bipartisan commission on entitlement reform. Markets calmed. But they never forgot.

By 2045 federal debt stood at approximately 165 percent of gross domestic product. Every Secretary of the Treasury since 2036 had governed with one eye on the auction calendar. Daniel Iwu, Kane's Treasury Secretary during the crisis, kept on his desk a framed chart of the ten-year yield during April 2036, and he showed it to the President more than once in 2045. The relevance to national security was direct. China and its state institutions still held a large stock of Treasury securities, though much smaller than at its peak; a sudden sale, even at a loss, could spike American borrowing costs. American sanctions on China would hit American consumers and markets. And a President who ordered military escalation knew that the bond market would vote on his decision within hours. Kane's "information clock," described in Chapter 1, ran partly on Wall Street.

China's long stagnation

China's economy, which had grown at near-double-digit rates for three decades, slowed through the 2020s and stalled in the 2030s. The causes were well known by then: a property sector that had absorbed too much of the nation's savings and then collapsed; local governments buried under debt borrowed through off-balance-sheet financing vehicles; a population that had begun to shrink in 2022 and was aging faster than any large country in history; and a long contraction of foreign investment as Western governments restricted technology exports and companies shifted supply chains to Vietnam, India, Mexico, and Indonesia.

By 2045 official Chinese growth figures hovered around two percent, and few outside economists believed even those. Youth unemployment in the cities, which the government had stopped publishing for a time in the 2020s, was estimated by independent researchers at more than a quarter of young people. Pension funds in several provinces were insolvent. A generation that had been promised it would be richer than its parents was discovering it would not be.

The Party's response had been twofold. Economically, it doubled down on state-directed investment in advanced manufacturing, semiconductors, and green technology, with some real successes. Politically, it leaned harder on nationalism: the narrative of a "century of humiliation" ending in national rejuvenation, of which the "reunification" of Taiwan was the final and essential chapter. This narrative was not new, but in a stagnant economy it carried more weight. A leader who seemed to give ground on Taiwan risked not only his opponents in the Party but a public that had been told, for its whole life, that this was the one thing that could never be compromised.

Washington's divided house

The American political system in 2045 was, in the President's own phrase, "a two-thirds functional government": the executive branch could act quickly, the courts slowly, and Congress, most of the time, barely at all. Kane, an independent-minded Democrat, had been re-elected in November 2044

by a narrow margin. The Senate was held by the opposition under Majority Leader Cal Whitford of Texas; the House by the President's party under Speaker Diane Okonkwo. The arrangement produced the usual gridlock on domestic questions, and, as this document shows, a more complicated relationship on national security, where Whitford was both Kane's sharpest critic and, in the worst hours, a patriot who kept his criticism behind closed doors.

The team notes one further domestic factor that readers today may underestimate: the public's exhaustion. Americans in 2045 had lived through a pandemic in their childhood or youth, a debt crisis, two decades of political polarization, and a constant drumbeat of warnings about China. Polling in early 2045 showed majority support for "defending Taiwan" in the abstract and majority opposition to "risking war with China" in the concrete. No President could rely on a public that would follow him anywhere. Kane knew this, and it shaped his decisions more than his critics later admitted.

3. The Strait: Taiwan's Politics and the Road to the Status Referendum Act

Key judgments. Taiwan in 2045 was a prosperous, democratic, and anxious society of about twenty-two million people, still producing a large share of the world's most advanced semiconductors, though a smaller share than in the 2020s. Its politics had polarized between those who believed security lay in accommodation with Beijing and those who believed it lay in deterrence and a firmer identity. The election of Lin Shu-wen in January 2044 tipped the balance toward the second camp, and her government's push for a referendum on the island's name in international applications became the trigger, though not the cause, of the crisis.

The island and its defenses

The basic geography had not changed since 1949. Taiwan lies about 130 kilometers from the coast of Fujian province at the narrowest point of the Strait. Its eastern coast faces the open Pacific and rises steeply into mountains; its western coast, facing China, is flat and densely populated, with the great ports of Kaohsiung, Taichung, and Keelung (on the northern tip) handling nearly all of its trade. The island imports almost all of its energy, most of it as liquefied natural gas and coal arriving by sea, and held only days to weeks of reserve. Any planner, in Beijing or in Washington, understood that Taiwan's vulnerability was not primarily to invasion but to strangulation.

Over the 2020s and 2030s, under heavy American encouragement, Taiwan had shifted its military toward what strategists called the "porcupine" model: not trying to match China ship for ship and jet for jet, which was impossible, but making itself too costly to swallow. That meant large numbers of mobile anti-ship missiles hidden along the coast, sea mines, short-range air defenses, drones by the tens of thousands, hardened communications, and a reformed reserve and civil defense system. Conscription had been extended back to one year in 2024 and was lengthened again in the 2030s. By 2045 the porcupine had many quills, though American planners still worried about its stores of fuel, food, and munitions in a long siege.

The politics of identity

Public opinion surveys in Taiwan, conducted for decades by universities in Taipei, had shown a long, steady shift: by the 2020s, large majorities of the island's residents identified themselves as "Taiwanese" rather than "Chinese" or "both," and very few supported unification under the People's

Republic. Yet equally large majorities preferred the "status quo" — de facto independence without formal declaration — because they understood that a formal declaration could bring war.

The two main parties had organized themselves around this dilemma. The Democratic Progressive Party (DPP) leaned toward a distinct Taiwanese identity and deeper ties with the United States and Japan. The Kuomintang (KMT), the party that had ruled China before 1949 and then Taiwan until the 1990s, leaned toward engagement with Beijing and the formula of "one China, respective interpretations." Smaller parties came and went. Power alternated between the DPP and KMT through the 2030s, each change producing a lurch in Beijing's behavior.

Lin Shu-wen

Lin Shu-wen, the DPP candidate, won the presidential election in January 2044 with a clear plurality and was inaugurated on 20 May 2044. She was fifty-three, a former constitutional law professor at National Taiwan University and a two-term legislator from Tainan, the southern city that had long been the heartland of Taiwanese identity. Her grandfather had been imprisoned during the White Terror, the period of martial law under the KMT, and she had written her doctoral thesis on the right of self-determination in international law. She was precise, unsentimental, and, in the judgment of American diplomats who dealt with her, "the most capable leader Taiwan has produced and the one most likely to get it killed." Both halves of that judgment were meant seriously.

Lin's campaign promise was not independence. She was too careful for that. Her argument was about dignity. For half a century, she said, Taiwan had been forced to compete in the Olympics as "Chinese Taipei," to be excluded from the World Health Organization during a pandemic, and to apply to international bodies under names chosen by others. She proposed that Taiwan's people be allowed to say, in a non-binding referendum, what name they wished their government to use when applying for membership in the United Nations and its agencies. The legal content was modest. The symbolic content, to Beijing, was enormous: the People's Republic had long declared that any move toward "de jure independence" would be grounds for "non-peaceful means" under its Anti-Secession Law of 2005, and it regarded a referendum on the island's international name as exactly such a move.

The Status Referendum Act

In February 2045, Lin's government introduced the Status Referendum Act in the Legislative Yuan. It proposed a non-binding referendum, to be held in 2046, on whether the government should apply to United Nations bodies under the name "the Republic of China (Taiwan)." The phrasing was a careful compromise: it retained the constitutional name of the Republic of China, which the KMT could accept, and appended "Taiwan," which the DPP's base demanded. The bill had a reasonable chance of passing. Debate opened in the second half of February amid large demonstrations on both sides in Taipei.

Washington's position was awkward. The United States had for decades declared that it "does not support Taiwan independence" and "opposes unilateral changes to the status quo" by either side. Kane's administration privately urged Lin to delay the bill, arguing that the timing, with China's economy weak and its nationalism inflamed, was dangerous. Lin listened politely and declined. "You are asking us to give up a right in order to preserve a peace that depends on our never using any rights," she told Secretary of State Helena Marsh in a call in January 2045, according to State Department records quoted in Chapter 2. Marsh had no good answer. Neither, at first, did the President.

The editorial team wishes to be fair to President Lin, who is often cast in American accounts of the crisis as reckless. She was not reckless. She was a democrat in a democracy, carrying out a promise she had made to voters, in the face of threats from a neighbor that did not allow its own citizens to vote. That this promise brought the world to the edge of nuclear war is a fact about the world, not a judgment of her character. Kane himself, as readers will see, came to the same conclusion, and his conversations with her in Phase V are among the most painful passages in his account.

4. Inside the Western Hills: Chinese Elite Politics in the Wen Era

Key judgments. The Chinese leadership in 2045 was less unified than it appeared. General Secretary Wen Zhaolin, who came to power in 2037, was a cautious technocrat who had inherited a highly centralized system and a nationalist narrative he could not abandon. He was quietly ill. Beneath him, a reformist economic wing led by Premier Qiao Weiming sought stability and reopening; a hardline military faction, whose most forceful voice was Rocket Force commander General Luo Tiancheng, believed that a strategic window was closing. The decisive question of the crisis turned out to be not what "China" wanted but who in China was in control.

The system

The People's Republic is governed by the Chinese Communist Party, whose formal apex is the Politburo Standing Committee, in 2045 a body of seven men. The General Secretary of the Party is also, by convention, President of the state and Chairman of the Central Military Commission (CMC), the body that commands the People's Liberation Army. The army is the Party's army, not the state's: its loyalty is sworn to the Party, and its highest decisions run through the CMC, which in the 2040s consisted of the Chairman, two Vice Chairmen, and a handful of members heading the joint staff, political work, and discipline inspection.

In principle, nuclear weapons in China are under the tightest possible central control. Release authority rests with the CMC Chairman, and orders pass from the CMC Joint Operations Command Center, buried beneath the Western Hills outside Beijing in a complex Western analysts call the Jingxi bunker, to the headquarters of the Rocket Force, then to its numbered bases and their brigades. In the era of minimum deterrence, warheads were stored separately from missiles, under different chains of custody, so that no single officer or unit could launch. The shift to launch on warning changed this. A force that must be able to launch within minutes of warning cannot keep its warheads in a separate depot hundreds of kilometers away. Readiness requires warheads mated to missiles, crews on alert, and orders that flow quickly. Readiness, in other words, loosens control. This was the central structural danger of 2045, and Western analysts had warned about it for fifteen years.

For the purposes of this document, readers should know the fictional four-level readiness scale that the President uses throughout, reconstructed from intelligence reporting: **Readiness Level 4**, peacetime routine; **Level 3**, heightened readiness, with crews recalled and security increased; **Level 2**, pre-launch preparation, including warhead mating and the dispersal of mobile launchers from garrison; and **Level 1**, launch. Before 2045 the Rocket Force had not gone to Level 3 since the 2030s, and had never, so far as American intelligence knew, gone to Level 2 outside an exercise.

Wen Zhaolin

Wen Zhaolin was sixty-seven in 2045. He had succeeded the previous General Secretary in 2037, at the end of a long and opaque transition that Western scholars still argue about. He had been an engineer by training (hydraulic engineering at Tsinghua, a fact that later gave Kane an unexpected point of connection), then a provincial governor in the coastal province of Zhejiang, then a vice premier responsible for industrial policy. He was not a charismatic speaker. Diplomats who met him described a soft-voiced, methodical man who read his briefing papers closely, asked technical questions, and rarely revealed what he thought.

Wen was chosen, insiders believed, precisely because he seemed safe: a manager rather than a visionary, unlikely to purge his rivals or to take reckless risks. In his first years he had tried, quietly, to stabilize the economy and to reduce tensions with the United States. But he had inherited a political system built around the supreme authority of one man and a narrative of national rejuvenation in which Taiwan was the climax. He could not be seen to retreat. And, over time, he too had stoked the nationalist fires, partly from conviction and partly because they warmed him in a cold economy.

What few outside his inner circle knew, and what American intelligence learned only in 2045, was that Wen suffered from a serious heart condition. He had undergone a procedure in 2042 that was hidden from the public. Through the crisis, his health would become a factor of the first importance: a leader who disappeared from view for days at a time left a vacuum at the top of a system designed to have no vacuum.

Qiao Weiming

Premier Qiao Weiming, sixty-two, was the head of the State Council, China's government, and the Party's senior economic manager. He had studied economics at Peking University and spent a year as a visiting scholar in the United States in the 1990s, a fact his enemies did not let him forget. He believed that China's future depended on reopening to foreign investment, rebuilding private enterprise, and repairing relations with the West. During the crisis he was the leading advocate, within the Standing Committee, for de-escalation, on grounds he stated openly: the economy could not survive a war, or even a long blockade. He was often outvoted. His office would become, in ways described later, a channel of the utmost delicacy.

Zhang Hesheng

General Zhang Hesheng, sixty-five, an army officer from Shandong, was the first-ranked Vice Chairman of the Central Military Commission. He had commanded ground forces in the north and had been Wen's chief military ally since their days together in Zhejiang. He was a professional soldier of the old school: conservative, suspicious of the United States, but deeply loyal to the Party's control over the gun and unwilling to see any faction in the army act outside the chain of command. In the worst hours of November 2045, that loyalty mattered more than any doctrine.

Luo Tiancheng

General Luo Tiancheng, fifty-eight, Commander of the PLA Rocket Force, is the closest thing this account has to an antagonist, and the team has tried to describe him fairly. He was born in Sichuan, the son of a missile engineer who had worked on China's first strategic weapons in the 1960s, and he grew up in the closed military towns of the western interior. He rose through the Rocket Force as a

brigade commander, a base commander, and chief of staff, and was known as the most intellectually serious strategist of his generation. His published writings, under a pen name, argued with great sophistication that nuclear deterrence was a matter of perception as much as of weapons.

Luo's core belief, set out in a series of internal papers from the late 2030s that American intelligence later obtained, was that the strategic balance was moving against China. He reasoned as follows. The United States had a large and modernized counterforce arsenal capable of destroying Chinese silos. It had ARGUS and related systems that could, it claimed, track nearly every Chinese mobile launcher outside its garrison. And it was building Golden Dome, a missile defense that could intercept the ragged remnant of Chinese missiles that might survive an American first strike. Put the three together, Luo argued, and within a decade the United States might be able to disarm China in a single blow and shield itself from the consequences. The window in which China's deterrent was secure was, in his phrase, "a window that is closing." The implication, which he was careful never to write directly, was that if China ever had to act, it should act before the window closed.

Luo was not a madman. His reasoning was, in its own terms, coherent; it was the reasoning of every "use it or lose it" strategist since the dawn of the nuclear age, and American strategists had made the same arguments about the Soviet Union in the 1950s. Kane, as readers will see, took Luo's arguments more seriously than most of his own advisors did, and built his Mirror Model partly around them.

Gao Ruiming, Shen Yilan, Kong Dewei, Liang Qiwen, Tang Shiyu

Admiral **Gao Ruiming**, sixty, commander of the Eastern Theater Command responsible for operations against Taiwan, was a navy officer who had commanded China's first carrier group and believed, with some justification, that the PLA Navy could now defeat any American force within the first island chain. He was aligned with Luo in the crisis, though more from professional confidence than from Luo's doctrinal fear.

Foreign Minister **Shen Yilan**, fifty-six, the first woman to hold the post in decades, was a career diplomat with long postings in Washington, Geneva, and New York. She had a reputation for cutting wit and for never conceding a point in public. In private, as Marsh came to know, she was a realist who understood exactly how close the two countries were to catastrophe.

Minister of State Security **Kong Dewei**, sixty-three, ran China's civilian intelligence and counterintelligence service, the Ministry of State Security (MSS). He was a security professional from the Party's political-legal apparatus, cold, patient, and very good at his job. In 2045, as described in Chapter 4, he ran a mole hunt inside the Rocket Force that cost at least one American source his life.

Liang Qiwen, fifty-nine, Director of the Office of the Central Foreign Affairs Commission, was the Party's senior foreign-policy coordinator, a role roughly comparable to the American National Security Advisor. He would become Samuel Oyelaran's counterpart in the secret talks of late 2045.

Ambassador **Tang Shiyu**, in Washington since 2042, was a gracious and well-connected diplomat who often seemed to know more about Congress than many members did. He was the formal channel for many messages during the crisis, though not, in the end, for the most important ones.

5. The Balance of Terror: Nuclear Forces in 2045

Key judgments. In 2045 the world had three nuclear superpowers for the first time. The United States retained the most capable and diverse force, a modernized triad of land-based missiles, submarines, and bombers. China had built a force large and ready enough to threaten the United States with devastating retaliation, and had placed part of it on a hair trigger. Russia, declining in conventional strength, remained the largest holder of warheads and an unpredictable third party. Each side's forces were intelligible only in relation to the others' fears. A successor reading Kane's account must understand not only what these weapons were, but what each side believed the other could do with them.

The American triad

The American strategic nuclear force rested, as it had since the 1960s, on three legs, each designed to compensate for the weaknesses of the others.

The land-based leg. By 2041 the LGM-35A Sentinel intercontinental ballistic missile (ICBM) had fully replaced the Minuteman III, which had stood on alert for more than six decades. Four hundred Sentinel missiles were deployed in underground silos spread across the northern Great Plains, under three missile wings: the 90th at F.E. Warren Air Force Base in Wyoming, the 341st at Malmstrom in Montana, and the 91st at Minot in North Dakota. Each silo was separated from its neighbors by miles of prairie, so that an enemy would need to spend at least one warhead on each. The Sentinel program, notoriously over budget in the 2020s, had rebuilt not only the missiles but the launch facilities, the command cables, and the launch control centers where two-officer crews stood alert around the clock.

The ICBM force's virtues were readiness and reliability: Sentinel missiles could be launched within minutes of a valid order, and their communications were hardened and redundant. Its weakness was that its locations were fixed and known. An adversary could target every silo, and so the ICBMs carried what strategists called a "use or lose" logic of their own: if a President believed a large attack was coming, the missiles could be lost in their silos unless they were launched first. This is why the American system, too, preserved the option of launch under attack, and why the President's decision window in a missile warning was measured in minutes. A ballistic missile launched from western China would reach the continental United States in roughly thirty minutes; a missile from a submarine off the coast, far less. Subtract the time needed to detect, assess, and convene, and the President might have between six and twelve minutes to decide.

The sea-based leg. The most survivable part of the American force was its fleet of ballistic missile submarines (SSBNs). In 2045 the Navy operated six of the new Columbia-class boats, with six more under construction or on order, alongside six remaining Ohio-class submarines approaching the end of service lives that had been extended well beyond their original design. Each Columbia carried sixteen Trident II D5 missiles in their life-extended form; each missile could carry several independently targetable warheads. The boats operated from Naval Submarine Base Kings Bay in Georgia and Naval Base Kitsap-Bangor in Washington State. At any given time, a substantial portion of the fleet was at sea, hidden in the oceans, communicating only by receiving very-low-frequency radio and other signals and never by transmitting.

Because an adversary could not find and destroy them, the SSBNs guaranteed American retaliation even after the worst imaginable surprise attack. That guarantee is what made deterrence stable. It also, as Kane discovered in November 2045, had a mirror image: a surge of American submarines

to sea, intended as a calm signal of resolve, could look to Chinese early-warning officers like the preparation for a first strike.

The air leg. The bomber force comprised roughly 120 B-21 Raider stealth bombers, entering service through the 2030s and based at Ellsworth in South Dakota, Whiteman in Missouri, and Dyess in Texas, and the old B-52 Stratofortress, re-engined in the 2030s as the B-52J and flying in its ninth decade. The B-52s carried the Long-Range Standoff cruise missile; the B-21s could also carry gravity bombs deep into defended airspace. Bombers were the slowest leg but the most flexible: they could be generated to alert, dispersed to other airfields, flown toward a crisis as a visible signal, and recalled. In the language of Kane's Reversibility Matrix, bombers were the reversible leg of the triad. Missiles, once launched, could not be recalled.

Command and control. Binding the three legs together was the nuclear command, control, and communications system, abbreviated NC3: the early-warning satellites and radars, the command centers, the communications networks and aircraft, and the procedures through which only the President could authorize nuclear use. Chapter 1 describes it from the inside, and the team will not duplicate that tour. Readers should know its main elements by name: the National Military Command Center (NMCC) in the Pentagon; the North American Aerospace Defense Command (NORAD) and U.S. Northern Command at Peterson Space Force Base and Cheyenne Mountain in Colorado; U.S. Strategic Command (STRATCOM) at Offutt Air Force Base in Nebraska; the E-4B and its successor "doomsday" airborne command posts and the Navy's E-6B aircraft, which can relay orders to the submarines and missile fields if ground centers are destroyed; and the President's emergency satchel, known for decades as "the football," which travels everywhere with a military aide. In 2045 that aide, on most days, was Lieutenant Colonel Jamal Brooks of the United States Marine Corps.

Early warning of a missile attack came from two independent kinds of sensor, a principle known as "dual phenomenology." The first was infrared satellites in geosynchronous and polar orbits (by the 2040s the Next-Generation Overhead Persistent Infrared constellation had replaced the older Space-Based Infrared System), supplemented by the proliferated tracking layer described in Section 7, which detect the hot exhaust plumes of rockets within seconds of launch. The second was the great ground radars: the Upgraded Early Warning Radars at Clear in Alaska, Beale in California, Cape Cod in Massachusetts, Pituffik (formerly Thule) in Greenland, and Fylingdales in Britain, together with the Cobra Dane radar at Shemya in the Aleutians. Radars confirm what satellites see, but only once the missiles rise above the horizon, which for a launch from western China meant several minutes after the satellites first saw the plumes. That gap in time, between the first infrared warning and the first radar track, is the stage on which the climax of this book takes place.

China's forces

China's nuclear forces in 2045 were commanded by the PLA Rocket Force (PLARF), whose six numbered missile bases were spread across the country. The base most relevant to this document is **Base 64**, headquartered at Lanzhou in Gansu province, whose brigades the President's sources associate with the missile field near Hami.

The silo fields. Some 350 silos had been built in the three great western fields and in the modernized liquid-fuel force: the fields at **Yumen** in Gansu, **Hami** in eastern Xinjiang, and **Ordos** near Hanggin Banner in Inner Mongolia, each a grid of silos spaced a few kilometers apart across the desert floor, linked by roads, power lines, and buried communication cables, with support bases and launch control facilities at their edges. By 2045 they held a mixture of DF-41 solid-fuel ICBMs, each

capable of carrying up to three warheads, older DF-31-class missiles loaded in the early years, and, in separate silos, the heavy liquid-fuel DF-5C. The silos' lids, like those in the American Great Plains, were visible to every imaging satellite in orbit; their maintenance cycles, crew movements, and security patterns were a daily subject of American analysis.

The silo fields served the launch-on-warning posture directly. Silos are vulnerable to accurate attack, so they make sense chiefly for a force that intends to launch before the attack arrives. American analysts had made this observation in the early 2020s, and it remained the most worrying feature of China's posture two decades later.

Road-mobile missiles. China also deployed substantial numbers of road-mobile ICBMs, the DF-41 and the DF-31AG, carried on large wheeled transporter-erector-launchers (TELs). In peacetime they lived in garrisons and in the vast tunnel complexes that Chinese media had once nicknamed the "Underground Great Wall," a network of hardened tunnels said to extend thousands of kilometers under the mountains of central and western China. In a crisis, they would disperse onto roads and pre-surveyed launch sites, where they could fire within minutes of stopping. Finding and tracking them was the central challenge of American strategic intelligence and the reason ARGUS existed. The intermediate-range DF-26, capable of carrying either a conventional or a nuclear warhead and able to strike Guam, added a further, deeply ambiguous layer: a launch of a DF-26 might be conventional or nuclear, and no sensor could tell which.

The sea leg. By 2045 China operated six Type 096 ballistic missile submarines, quieter and larger than their Type 094 predecessors, each armed with the JL-4 submarine-launched ballistic missile, whose range allowed patrols from protected waters to reach the continental United States. The submarines were based at Yulin on the southern coast of Hainan Island, where an underground pen tunneled into the hillside allowed them to come and go partly hidden from satellites. China had rarely kept more than one or two boats at sea at once. A surge of several would be an unmistakable signal.

The air leg. The H-20 long-range stealth bomber became operational in 2039 and, by 2045, existed in small numbers; it gave China a true triad for the first time, though the air leg remained the weakest.

Exotic systems. In the summer of 2021, China had tested a hypersonic glide vehicle launched into a partial orbit around the Earth before descending on its target: a fractional orbital bombardment system, or FOBS. The test surprised American officials. A FOBS can approach from unexpected directions, including over the South Pole, avoiding the northern-facing radars on which American early warning had long depended. By 2045 China had fielded such systems in small numbers. They were not militarily decisive, but they complicated the early-warning picture and gave Chinese planners a theoretical way to strike around American missile defenses.

Early warning. The posture rested on China's own early-warning system: infrared satellites in geosynchronous orbit (the first of which had been launched in the 2020s, reportedly able to detect a missile launch within about ninety seconds and to alert a command center within a few minutes), large phased-array radars on the mainland, and, in the beginning, technical assistance from Russia, which had helped design parts of the system in the late 2010s and 2020s. By the 2040s China's system was its own, and it was good. But it was younger and less tested than the American one, and Chinese officers, unlike their American counterparts, had no institutional memory of false alarms. The American system had survived the false alarms of 1979 and 1980, when a training tape and then a faulty computer chip at NORAD briefly showed incoming Soviet attacks; the Soviet system had survived the 1983 incident in which Lieutenant Colonel Stanislav Petrov, alone at his console,

judged a satellite's warning of five American missiles to be a malfunction, correctly, and did not pass it up the chain as an attack. China's system had never been tested by such a night. Kane knew this history intimately. It weighs on every page of Chapter 6.

The posture problem

Launch-on-warning postures are dangerous in proportion to the number of people and machines that can produce a warning, and to the time available to check it. In 2045 both great powers had compressed the time available almost to nothing, and both had added machines to the loop. A Chinese commander who believed his silos were about to be destroyed would have perhaps fifteen to twenty minutes from warning to impact, minus the time for detection and transmission. An American President, perhaps six to twelve. Neither had ever faced such a decision. Both knew the other could.

The editorial team has found no better summary of this condition than a line from Kane's diary on 2 November 2045, quoted in Chapter 5: "Two men who have never met, each with his hand near a switch, each told by machines that the other's hand is moving."

Russia

Russia in 2045 was a declining power with an enormous nuclear arsenal. Its economy, battered by the long war in Ukraine in the 2020s, by sanctions, and by demographic collapse, was smaller than that of several European states. Its conventional forces had never fully recovered. But it retained a large stockpile of nuclear warheads, a modernized force of Sarmat heavy ICBMs, Avangard hypersonic glide vehicles, Borei-class submarines, and a collection of exotic systems that its leaders liked to unveil in televised speeches. After the end of New START, its numbers drifted slowly downward as older systems retired faster than new ones arrived, but it remained, in raw warheads, the largest nuclear power on Earth.

Russia's relationship with China had become, over two decades, a partnership of unequals. China was by far the stronger economy, and Russia sold it oil, gas, and weapons technology on increasingly unfavorable terms. Russian engineers had helped build China's early-warning system, and Russian and Chinese bombers flew joint patrols near Alaska and Japan. Yet Moscow had no wish to be dragged into a war over Taiwan, and its leaders privately feared Chinese power almost as much as they resented American power.

Arkady Volensky, President of the Russian Federation, was seventy-one in 2045. A former energy executive and governor who rose to power in the 2030s, he was a pragmatist with a cold sense of national interest and a long list of grievances, chief among them the sanctions that still constrained Russia's economy. American analysts judged that Volensky wanted three things from the crisis: to avoid being drawn in; to be seen as indispensable; and to extract sanctions relief from whichever side would pay. He would prove, at a critical moment in October 2045, more helpful than Washington expected and less helpful than Beijing hoped.

6. The Shield: The Golden Dome Architecture

Key judgments. Golden Dome, announced in 2025 and partially fielded by 2045, was intended to protect the American homeland against missile attack. Its actual capability against a large Chinese strike was modest and contested. Its perceived capability, in Beijing, was enormous. That gap between capability and perception was one of the root causes of the crisis.

The idea of a national shield against ballistic missiles was as old as the missiles themselves. The Strategic Defense Initiative of the 1980s, nicknamed "Star Wars," had imagined space-based weapons destroying Soviet missiles in flight; it produced research but no shield. The modest Ground-based Midcourse Defense system deployed after 2004, with interceptors in silos at Fort Greely in Alaska and Vandenberg in California, was designed to stop a handful of missiles from a small state such as North Korea, not a great-power arsenal.

In January 2025 a presidential executive order directed the Department of Defense to design a far more ambitious architecture, first called "Iron Dome for America" and soon renamed "Golden Dome." Its stated aim was to defend the United States against ballistic, hypersonic, and cruise missile attacks from any adversary, including peer powers. Its proposed elements included a much larger sensor network in space, new ground-based and sea-based interceptors, and, most controversially, interceptors based in space itself, able to strike missiles in their boost phase, when they are slow, hot, and easy to see. Its early cost estimates ran from the hundreds of billions to well beyond, and it was attacked from one side as a fantasy and from the other as too timid.

By 2045, two decades and several budget crises later, Golden Dome was real but partial. Its components, as known publicly and as described in the President's account, were:

- **Ground-based interceptors** in Alaska and California, with the Next Generation Interceptor, carrying multiple kill vehicles, replacing the older models through the 2030s.
- **Regional and sea-based layers**: Aegis-equipped destroyers and Aegis Ashore sites, and terminal defenses around key locations.
- **A space-based sensor layer**, including the tracking satellites described in Section 7, which could follow missiles and warheads from launch to impact and hand their tracks to interceptors.
- **Approximately 200 prototype space-based interceptors** in low-Earth orbit, deployed in the early 2040s. Their effectiveness was contested even within the American government. Because satellites in low orbit move quickly around the Earth, only a fraction of them would be over any given launch area at a given moment, and a large salvo could overwhelm them. American analysts estimated that the space interceptors might destroy some fraction of a small, ragged retaliatory strike under good conditions, and very little of a full-scale attack.
- **A battle management system**, heavily reliant on artificial intelligence, that fused data from every sensor, assessed the threat, and recommended engagements in seconds. This system, whose internal designation the team has left redacted where it appears, plays a central role in the eleven minutes of 6 November 2045.

The President's view of Golden Dome was ambivalent. As an engineer he admired its ingenuity, and as a former governor of Colorado, home of many of its contractors and of the Space Force commands that operated it, he had supported it publicly. But in office he became convinced that its strategic effect was the opposite of its intent. Here is the logic, which Kane formalized as his Mirror Model and which he credits in part to reading General Luo's papers.

Missile defense, in the American view, was defensive: it protected cities. In the Chinese view, it was the final piece of a first-strike capability. If the United States could destroy most Chinese missiles on the ground with its accurate counterforce weapons, and find most of the mobile launchers with ARGUS, then Golden Dome only needed to stop the few dozen survivors. A shield that could not stop a full attack could, perhaps, stop a broken one. And a country that believed its enemy could disarm it and survive the response would be under enormous pressure, in a crisis, to launch first, or to keep its finger close to the switch. "Every interceptor we put in orbit," Kane writes in Chapter 1, "is a reason for a Chinese colonel to keep a warhead on a missile."

7. The Eyes: The Tracking Layer, ARGUS, and Halcyon Orbital

Key judgments. By 2045 the United States possessed the most comprehensive surveillance of the Earth's surface ever achieved, built from government satellites and sensors hosted on commercial constellations. Officials described it as "persistent" and, in the President's own early phrase, "absolute." It was neither. It could be blinded by weather, deceived by decoys, evaded by tunnels, and degraded by attack. Above all, it could see what an adversary was doing but not why. Its dependence on a private company introduced a new kind of vulnerability into national command.

The tracking layer

In the 2020s the Space Development Agency, part of the United States Space Force, began to build the Proliferated Warfighter Space Architecture (PWSA): hundreds of relatively small and inexpensive satellites in low-Earth orbit, launched in successive "tranches" every two years or so, each improving on the last. The architecture had several layers. The **Transport Layer** provided a secure data network linking satellites, aircraft, ships, and ground forces. The **Tracking Layer**, which matters most for this account, consisted of satellites carrying wide-field and medium-field infrared sensors able to detect and track missiles, including maneuvering hypersonic glide vehicles, from launch through flight, and to pass "fire-control quality" tracks to interceptors.

The logic of proliferation was resilience. Older American warning satellites were few, large, expensive, and sitting in high orbits; destroying a handful would blind the nation. A constellation of hundreds could lose dozens and keep working. By 2045 several tranches of the tracking layer were operational, providing global coverage with considerable redundancy. But, as Chapter 4 describes, the Chinese counter-space campaign of September 2045 degraded one tranche significantly, and the gaps it left in the infrared picture over western China were still being filled in November. A degraded constellation does not fail cleanly: it fails noisily, producing ambiguous data that fusion algorithms must interpret. That too matters on 6 November.

ARGUS

ARGUS, named for the hundred-eyed giant of Greek myth, was a classified program that fused data from many sources into a single, continuous picture of the surface of the Earth, focused above all on the nuclear forces of China and Russia. Its existence was publicly acknowledged only after the crisis; most of its details remain classified even in 2071. From the declassified portions of the President's account and from later official histories, the team can describe its broad shape.

ARGUS drew on electro-optical imaging satellites operated by the National Reconnaissance Office (NRO); synthetic aperture radar (SAR) satellites, which use microwaves to produce images through cloud and darkness; radio-frequency geolocation satellites, which locate emitters such as radars and radios; hyperspectral sensors, which detect the chemical signatures of materials; and, crucially, sensors hosted on commercial mega-constellation satellites, built on buses similar to those SpaceX had developed for its Starshield service in the 2020s. Combining these with the PWSA layers and with signals intelligence and human reporting, ARGUS was claimed to revisit any point on Earth with SAR roughly every ninety seconds, and to maintain continuous "custody," meaning an unbroken track, of 94 percent of Chinese road-mobile missile launchers whenever they left their garrisons.

The 94 percent figure appears often in the President's account, usually with a note of irony. It meant, of course, that six percent were not tracked. It said nothing about launchers that never left the tunnels. It assumed that the objects being tracked were real launchers and not decoys, which

China built in large numbers. It depended on the weather for electro-optical sensors and on the integrity of the satellites themselves. And it reported positions, not intentions. "ARGUS can tell me where every truck in western China is parked," Kane wrote in his diary in the spring of 2045. "It cannot tell me what the man in the cab is thinking, or what the man in Beijing has told him to do." That is why, despite the most powerful technical intelligence system in history, the outcome of the crisis turned on a handful of human beings.

The Mirror Model applied to ARGUS as much as to Golden Dome. To Beijing, a system that could track nearly every mobile launcher was not an intelligence tool; it was a targeting system for a disarming strike. China's response, over the 2030s, had been to build more tunnels, more decoys, more silos, and, ultimately, to adopt launch on warning: if you cannot hide your missiles, you must be ready to fire them before they are destroyed. "Absolute surveillance," the President concluded, "is a weapon that also loads the enemy's gun."

Halcyon Orbital and Marcus Thorne

Many of the commercial satellites hosting ARGUS sensors belonged to **Halcyon Orbital**, a private company that by 2045 operated one of the two largest broadband constellations in the world, with many thousands of satellites in low orbit, its own reusable launch vehicles, and a government services division modeled on the national security constellations of the 2020s. Halcyon's satellites carried classified government payloads under contracts whose details remain sealed. The arrangement was efficient: it gave the government rapid launch, cheap buses, and a constellation so large that it could not be destroyed. It also made a private company part of the national command structure, and its civilian satellites potential military targets.

Halcyon's founder and chief executive, **Marcus Thorne**, was fifty-one in 2045: brilliant, mercurial, and enormously rich, with a habit of announcing corporate decisions on social media at two in the morning. He had built Halcyon from a small launch startup into one of the most valuable companies in the world, and he regarded himself, not without reason, as one of the people who had made American space dominance possible. He was also deeply wary of being drafted into a war he had not chosen. When, in July 2045, the government asked to use ARGUS payloads on Halcyon satellites for active military targeting, Thorne refused, arguing that it would make his entire constellation a legitimate target under the laws of war. The resulting confrontation, which ended with Kane invoking the Defense Production Act and a negotiated settlement in the Oval Office, is told in Chapter 3. It foreshadowed a question that later presidents could not avoid: in an age when private companies own the eyes of the state, who commands the eyes?

8. Machines in the Loop: Artificial Intelligence in the Two Militaries

Key judgments. By 2045 artificial intelligence was embedded in nearly every function of the American and Chinese militaries, from logistics to targeting to early warning. Both sides officially maintained that humans retained authority over the use of nuclear weapons. In practice, the speed and volume of data meant that humans increasingly decided on the basis of machine judgments they could not independently verify in the time available. This "advisory dependence" was among the least understood dangers of the era.

In the 2020s, military artificial intelligence had been mostly a matter of image recognition, drone navigation, and decision aids. By the 2030s, large-scale models trained on decades of sensor data had become the standard means of fusing information from thousands of sources: a modern battle

management system might ingest data from satellites, radars, aircraft, ships, cyber sensors, and intelligence reports, and produce in seconds a picture that would once have taken a room of analysts an hour. Both militaries had autonomous drones in the air, on the sea, and under it, operating in swarms. Both used AI to plan logistics, to model adversary behavior, and to generate options for commanders.

The United States had declared, in policy statements reaching back to the 2020s, that a human would always remain "in the loop" on decisions to use nuclear weapons, and China had made similar if vaguer statements. In 2024 the leaders of the two countries had jointly affirmed that humans, not artificial intelligence, should control the decision to use nuclear weapons. These commitments were sincere. But the editorial team asks readers to notice the difference between authority and dependence. A human may hold the authority to decide; but if a machine tells him, with "high confidence," that six missiles are in flight, and he has eight minutes, and no other source of information, then the machine has in effect decided, and the human has merely ratified.

This was the problem Kane had foreseen as a control-systems engineer. In his terms, artificial intelligence had made the sensors and the "estimator" (the part of a system that turns noisy measurements into a belief about the state of the world) faster and more powerful, but also more opaque. When the estimator is wrong, the whole loop acts confidently on a false picture. His Two-Source Rule, introduced in Chapter 1, was his attempt to keep a human-understandable check in the loop: never act on a nuclear warning from one kind of evidence, however confident the machine. On 6 November 2045, that rule was tested.

On the Chinese side, the team knows less. Declassified material suggests that the Rocket Force's early-warning system, too, relied heavily on automated fusion, and that Chinese officers, under the launch-on-warning posture, were trained to respond very quickly to its alerts. A Chinese AI system, like an American one, could mistake a training exercise, a fire, or a deception operation for an attack. Neither side fully understood the other's machines. Neither side's machines understood each other.

9. The Circle: Allies, Partners, and the Neutrals

Key judgments. American strategy in the Pacific in 2045 rested on a network of alliances and partnerships stronger in some places than it had been in the 2020s and weaker in others. Japan and Australia were committed allies; the Philippines was indispensable but politically volatile; India was a partner, not an ally; and the states of Southeast Asia sought above all to avoid choosing. Singapore, small and neutral, would play an outsized role.

Japan. Japan had transformed its defense posture in the 2020s, doubling its defense budget toward two percent of gross domestic product, acquiring long-range counterstrike missiles, and integrating its command structure more closely with American forces. By 2045 it was the most capable American ally in Asia. Its southwestern islands, the Ryukyu chain, stretched to within about 110 kilometers of Taiwan, and the Japanese government had declared since the 2020s that a Taiwan contingency could constitute a threat to Japan's survival. **Prime Minister Hiroshi Saeki**, sixty-four, a former defense minister from the Liberal Democratic Party, was a steady, unflashy conservative who understood that his country would be in the war from the first hour if war came. Japanese destroyers sailed in the convoys of 2045, and one of them was damaged in the Bashi Channel.

The Philippines. The Philippines sat at the southern end of the Strait, its northernmost islands, the Batanes, lying within sight of Taiwan on a clear day across the Bashi Channel. Under the Enhanced Defense Cooperation Agreement, expanded in the 2020s, American forces had access to a

growing number of Philippine bases, some in the northern province of Cagayan, close to Taiwan. American intermediate-range missile systems had first been deployed to the Philippines on exercises in 2024, to China's fury. **President Andrés Villanueva**, fifty-eight, a former governor from the north, was a nationalist who distrusted both great powers and had won office promising to defend Philippine interests in the South China Sea. He was indispensable to the convoys and unpredictable in his politics. The question of missiles on the Batanes would become one of the secret elements of the eventual settlement.

Australia. Australia, bound to the United States by the ANZUS treaty of 1951 and, since 2021, by the AUKUS partnership with Britain, had begun receiving nuclear-powered attack submarines in the 2030s and hosted rotating American forces in its north. **Prime Minister Grace Holloway**, fifty, a Labor leader and former foreign minister, was cautious about war but firm about alliance; her country's warships joined the escorted corridor in 2045.

ASEAN and Indonesia. The ten (by 2045, eleven) members of the Association of Southeast Asian Nations sought above all to avoid being forced to choose between Washington and Beijing, on whom they depended for security and trade respectively. Their unity was fragile. Indonesia, the largest member and in 2045 one of the world's ten largest economies, held to its long tradition of a "free and active" foreign policy. **President Arya Wicaksono**, a former governor of Central Java, played a small but honorable role, lending support to an ASEAN statement calling for restraint and offering Jakarta as a venue for talks, an offer that, in the end, ceded the stage to Singapore.

Singapore. The city-state of Singapore, with six million people, a superb diplomatic service, and good relations with both great powers, had long been a meeting place for those who could not meet elsewhere; it had hosted the first summit between an American President and a North Korean leader in 2018, and the annual Shangri-La Dialogue of defense ministers. **Prime Minister Daniel Tan Wei Ming**, fifty-five, a former finance minister and central banker, was trusted in both Washington and Beijing, as much for his discretion as for his intelligence. He would host the secret meetings of 2045 at Sentosa and, later, aboard a Singaporean navy ship, and the summit of March 2046.

India. India was the most important power in Asia not aligned with either side. A member of the Quad grouping with the United States, Japan, and Australia, and a rival of China along their disputed Himalayan border, India had grown closer to Washington through the 2030s but guarded its independence fiercely. **Prime Minister Ananya Rao**, sixty, a former diplomat, offered quiet support and declined to be drawn in, which was, from the American point of view, roughly the right answer.

10. The Invisible War: Intelligence on Both Sides

Key judgments. The crisis was decided as much by intelligence as by force. The United States possessed an unprecedented technical collection system and, crucially, a small number of human sources inside the Chinese nuclear establishment. China possessed formidable cyber capabilities and a counterintelligence service that, during the crisis, uncovered at least one of those sources. Readers should understand both what intelligence could do and what it cost.

Human intelligence, the oldest form of espionage, remained irreplaceable precisely because it could reveal intentions: what was being ordered, by whom, and in what mood. China had been one of the hardest environments in the world for American human intelligence since the early 2010s, when, as was publicly reported at the time, the MSS had dismantled a large CIA network inside China, killing or imprisoning many of its sources. Rebuilding took two decades of patient work.

By 2045, the President's account reveals, the CIA's China Mission Center had placed or recruited a small number of sources of extraordinary value. The team identifies them here only by the codenames under which they appear in the chapters that follow, because some of their true identities remain protected and because the President himself knew most of them only by these names.

- **LANTERN**, a senior officer in the Rocket Force, recruited in 2033, whose access concerned the readiness of strategic missile brigades.
- **WILLOW**, an officer in the communications staff of the CMC Joint Operations Command Center, whose reporting concerned the leadership's orders and mood.
- **THRUSH**, a civilian engineer associated with the submarine base at Yulin.
- **KESTREL**, a technician with access to the infrastructure of the Yumen silo field.
- **ORCHID**, a member of the staff of the Premier's office, who provided political intelligence and, in time, something more.

These sources communicated through methods that remain classified; the President refers to burst transmission devices and dead drops, and the team has not elaborated further. Beyond human sources, the United States held what the account calls "technical access" to certain Chinese military networks, gained over years, under an operation codenamed QUIET HARVEST. Its methods remain classified. Its effects, during the crisis, are described in Chapters 3 and 5.

The man who ran these sources at the operational level was the chief of the CIA's China Mission Center, known in this document only as **"Ruth Calder"**, a pseudonym. She was a veteran case officer who had spent much of her career in Asia, spoke Mandarin and Cantonese, and had personally recruited LANTERN. She was the President's most direct link to the human beings whose lives, as he came to feel acutely, he held in his hands.

China's intelligence services were formidable in turn. The MSS, under Minister Kong Dewei, ran extensive human networks abroad, and PLA cyber units had spent decades placing hidden access in American infrastructure, which they used, as Chapter 3 describes, to cut power to millions of Texans in June 2045. Chinese counterintelligence was relentless. The cost of espionage in the crisis was measured not only in risk but in lives, and the President's account does not spare him on this point.

11. The People: Portraits of the Principal Figures

Key judgments. The Pacific Crisis was managed by a small number of people under extreme pressure. The Chinese principals are described in Section 4. The Americans who mattered most are described here.

Thomas Ellery Kane

Thomas Ellery Kane was born in 1985 in Pueblo, Colorado, the son of a machinist and an eighth-grade science teacher. He kept a handwritten diary every night from the age of twelve, a habit he never broke; this document is built on those diaries. He studied aerospace engineering, then earned a doctorate in control theory at the Massachusetts Institute of Technology, writing on guidance algorithms for satellites under noisy measurement. He worked on satellite guidance systems before founding a company that made components for launch vehicles, which grew large enough to make him comfortably wealthy and to teach him, he said, "that every system fails, and the only question is whether it fails gracefully."

He entered politics late, served as Governor of Colorado from 2031 to 2039, and was elected the 49th

President of the United States in November 2040, inaugurated on 20 January 2041, and re-elected narrowly in November 2044. He was sixty in 2045. Colleagues described him as warm, curious, stubborn, and given to drawing diagrams on any available surface. He thought in models, and the six models he introduces in Chapter 1 are the skeleton of this book. A lapsed Methodist, he sometimes prayed and was never sure to whom.

His wife, **Dr. Claire Kane**, a pediatric oncologist who continued seeing patients one day a week at a Washington hospital throughout his presidency, was, in his words, "the only person who never once told me what I wanted to hear." Their daughter **Maya**, twenty-eight, was a Navy lieutenant serving aboard the destroyer USS *Daniel Inouye* (DDG-118) with the Seventh Fleet. Her father's crisis was her deployment. Their son **Eli**, twenty-four, was a software engineer in Austin, Texas, and lost power for three days in June 2045 along with millions of his neighbors.

The President's team

Vice President Rosalind Achterberg, a former Senator from Ohio, was steady, experienced in Congress, and more hawkish than the President; she was the senior voice for firmness in the National Security Council and, by statute and duty, the person who would have succeeded him had the worst occurred.

Dr. Samuel Oyelaran, the National Security Advisor, was born in Lagos and raised in Maryland, a former Georgetown professor of international relations who had written a celebrated book on crisis bargaining. He was the President's closest strategist and the person who ran the National Security Council process, chairing the Principals Committee in the President's absence.

Secretary of State Helena Marsh, a former ambassador to Beijing, spoke fluent Mandarin and had known many of the Chinese leaders for decades. She was a diplomat's diplomat: patient, exact, and convinced that every crisis ends in a room.

Secretary of Defense Raymond "Ray" Castillo, a retired four-star Marine general who had commanded in the Pacific, was blunt, loyal, and protective of his troops; after the Bashi Channel Incident, he carried the names of the forty-one dead in his breast pocket.

General Patricia Dunmore, United States Air Force, Chairman of the Joint Chiefs of Staff, was the President's principal military advisor, a former bomber pilot and commander of Air Force Global Strike Command, and the strongest advocate in the crisis for "left of launch" options, meaning actions to prevent enemy missiles from being launched at all.

Admiral Victor Hale, Commander of U.S. Strategic Command, was a career submariner who had commanded a ballistic missile submarine, cool, precise, and allergic to euphemism. It fell to him to brief the President on nuclear options, and he did so, Kane writes, "as if reading a weather report, which is the only way it can be done."

Admiral Kenji Moriyama, Commander of U.S. Indo-Pacific Command in Honolulu, the grandson of Japanese-American internees, commanded the forces in the Strait.

Abigail Fenn, the Director of National Intelligence, a former NSA executive, guarded the integrity of intelligence analysis against every pressure, including the President's. **Gregory Laszlo**, Director of the CIA, a former case officer and Deputy Director for Operations, was responsible for the human sources and for the decisions about whether to risk them.

Daniel Iwu, Secretary of the Treasury, a former central banker, managed sanctions and markets; **Megan Pruitt**, the White House Chief of Staff, managed the President's time, temper, and Congress.

Lieutenant Colonel Jamal Brooks, USMC, carried the football and appears throughout this account as a quiet, constant presence; Kane came to regard him with something close to a father's affection.

In Congress, **Senate Majority Leader Cal Whitford** of Texas, a former rancher and Air Force officer, was the opposition's leader and the President's fiercest critic, but also, in the worst hours, a man who kept his counsel. **Speaker Diane Okonkwo** of the President's own party held the House.

12. The Eve: January to March 2045

On 20 January 2045, Thomas Kane took the oath of office for the second time on the West Front of the Capitol, in bitter cold. His inaugural address mentioned China twice and Taiwan not at all. His advisors had argued about that omission for a week.

In the weeks that followed, the signals accumulated, though they did not yet seem to form a pattern. In February the Status Referendum Act was introduced in the Legislative Yuan in Taipei, and Beijing's Taiwan Affairs Office warned of "grave consequences." Chinese state media began a campaign against "separatist forces" and their "foreign backers." The China Coast Guard increased its patrols around Taiwan's outlying islands. American imagery showed the Eastern Theater Command moving amphibious and coast guard vessels into ports in Fujian and Zhejiang, which could be an exercise or something more. The Rocket Force remained, so far as anyone could see, at its peacetime routine.

In Washington, the Principals Committee met three times in February on the Strait, and agreed on little except that the situation was serious. Secretary Marsh flew to Tokyo and Manila. Secretary Iwu ordered quiet contingency planning for sanctions. The President spent an evening in the Treaty Room with a whiteboard, drawing what would become the Three Clocks Model, and wrote in his diary: "Everything is slow right now. That is when you should be thinking hardest, because it will not stay slow."

On 3 March 2045, the People's Republic of China declared a "law-enforcement quarantine" of Taiwan, and the Eastern Theater Command announced a joint exercise, "Strait Thunder-2045," in the waters around the island. Through that day and night the picture sharpened: China Coast Guard cutters took station off Kaohsiung, Taichung, and Keelung, and in the early afternoon of 4 March, Beijing time, they began stopping and boarding merchant ships bound for Taiwan's ports. In Washington it was the middle of the night.

At 03:10 Eastern Time on 4 March, the President was awakened.

What follows is his account.

Chapter 1: The Engineer's Models and the Machinery of the End

"Every system I ever built failed first in the part that was supposed to understand what the sensors were saying. I do not expect the government of the United States to be different."
— Diary of Thomas Ellery Kane, 9 February 2041

A Word Before the Machinery

This chapter is the one my staff asked me to cut. It is long, it is technical in places, and it is about me before it is about the crisis. They argued, reasonably, that a successor opening this volume in the first week of a new administration would want the story: the quarantine, the convoys, the Bashi Channel, the eleven minutes. I overruled them, which is a thing presidents are permitted to do and should do rarely.

I overruled them because the story will not make sense unless you know two things first. The first is how the machine works: the physical, electrical, human apparatus that sits between a flash of infrared light over a desert in western China and a decision made by one tired person in a room under the East Wing. The second is how I think, because the way a president thinks is itself part of the machine. It is the least documented part, the least tested, and the one most likely to fail.

B.J. Habibie, the aircraft engineer who became the third President of Indonesia in 1998, wrote a memoir of his first five hundred days in office in which he explained his decisions with diagrams. He drew his reasoning the way he once drew the load paths through a wing. I read that book in English translation as a graduate student at the Massachusetts Institute of Technology, when I was supposed to be reading about Lyapunov stability, and I remember thinking that it was the first political memoir I had ever read that admitted the author had a method. Most leaders write as though their judgment arrived by revelation. Habibie wrote as though it had been designed, tested, and occasionally broken. I have tried to do the same.

So this chapter has three parts. First, briefly, where I come from, because my habits are older than my offices. Second, the six models I carried into the Pacific crisis of 2045 and 2046, each set out the way I would set out a control architecture for a review board. Third, a guided tour of the machinery, beginning with the day I was handed it: Sunday, 20 January 2041.

> **OPERATING NOTE** — You will be tempted to skip the briefings on the nuclear command system after the first one, because nothing in them changes and because the subject is unbearable. Do not skip them. The system will not change, but you will, and the only time to learn the machine is when it is not trying to kill anyone.

Part One: Where the Habits Come From

Pueblo, and a Notebook

I was born on 14 June 1985 in Pueblo, Colorado, a steel town on the Arkansas River that was, by the time I could notice such things, a former steel town. My father, Walter Kane, was a machinist who had worked at the steel mill until the layoffs of the early 1980s and then at a series of shops that made parts for things he was never told the purpose of. My mother, Ann, taught fourth grade for thirty-one years. We were Methodists of the dutiful kind: present at the First United Methodist Church on Sunday, generous at the potluck, quiet about the rest.

On my twelfth birthday my mother gave me a composition notebook with a black-and-white marbled cover and said, “Write down one true thing every night. It doesn’t have to be important. It has to be true.” I have done so, with lapses I can count on two hands, every night since. The notebooks from 1997 to 2040 fill four shelves of a bookcase in our house outside Salida. The notebooks from my presidency are, at the time I write this, in a safe in the Treaty Room, and they are the primary source for this volume. Where I reconstruct dialogue, I reconstruct it from what I wrote that night, cross-checked against the Situation Room logs and the memoranda of conversation prepared by the National Security Council staff. Where my notes and the official record disagree, I say so.

The diary matters for a reason beyond sourcing. It taught me, very early, the difference between what happened and what I wished had happened. At twelve you write that you were brave on the playground. At thirteen you reread it and know you were not. Over decades the notebook becomes a kind of instrument: it measures the gap between your account of yourself and the facts. A successor should find their own instrument. Without one, the presidency will gradually persuade you that your account and the facts are the same thing.

My father taught me the other half. In his garage he kept a lathe, a drill press, and a steel cabinet of drawers labeled in his square capitals. When I was nine I broke the governor on a lawnmower engine trying to make it go faster, and instead of scolding me he sat me down on an upturned bucket and drew on the back of a feed-store receipt a little loop: the engine speed goes out, the governor feels it, the governor moves the throttle, the engine speed changes. “Everything that stays steady,” he said, “stays steady because something is watching it and pushing back. You took away the watching.” I did not know it then, but that was my first lesson in control theory, and it was the most important one.

The Massachusetts Institute of Technology, and the Problem of Knowing Where You Are

I went east on a scholarship in 2003, studied aeronautics and astronautics as an undergraduate, and stayed for a doctorate in control theory, which I finished in 2012. My dissertation concerned state estimation for small satellites with poor sensors: how a spacecraft with a cheap star tracker, a noisy gyroscope, and a magnetometer that lied whenever the solar panels drew current could nonetheless work out which way it was pointing. The mathematics is a descendant of the filter Rudolf Kálmán published in 1960, which flew to the Moon on Apollo. The core idea is simple to state and hard to live by: you never observe the true state of the world. You observe noisy measurements of it, and you maintain an estimate, with an honest statement of your uncertainty, that you update as new measurements arrive. The worst thing an estimator can do is become overconfident, because an overconfident estimator stops listening to new evidence. Engineers call this filter divergence. The spacecraft goes on believing it is pointed at Earth while it tumbles.

I tell you this because I watched the government of the United States approach filter divergence twice during the crisis, and I was the cause of it at least once.

After my doctorate I spent two years at a large aerospace contractor in Littleton, Colorado, working on guidance software for a class of government satellites whose names I am still not permitted to write, even in a document like this. In 2014, with two colleagues and a loan secured against my parents' house, which my mother insisted on and my father silently endured, I founded Arkansas River Dynamics in a rented bay in Pueblo. We made valves, actuators, and flight-control electronics for the commercial launch companies that were then reinventing access to space. We were very good at small, unglamorous parts that had to work every time. By 2026 we employed nine hundred people in Pueblo and Colorado Springs. I learned from that company most of what I know about how organizations fail: not dramatically, but through the slow accumulation of things everyone assumes someone else has checked.

Claire, Maya, Eli

I met Claire Okafor-Reid at a friend's wedding in Denver in 2013. She was a pediatric oncology fellow at Children's Hospital Colorado and she asked me, within ten minutes, what I did that mattered. I gave her a long answer about launch costs and she said, "So, valves," and I married her two years later. She has been, through every office I have held, the person who asks the question nobody in the room wants asked. In the crisis that question was usually some version of: *Who pays for this, and have you looked at their faces?* She spent her days with children who were dying and their parents, and she had no patience at all for abstractions that hid people.

Our daughter Maya was born in 2017. She is, at the time of the events in this book, a lieutenant in the United States Navy, serving aboard the guided-missile destroyer USS *Daniel Inouye* in the Seventh Fleet. I did not ask her to join the Navy. I argued against it, in fact, at our kitchen table in Denver in 2035, when I was governor and she was eighteen and had an appointment to the Naval Academy. She listened to my argument and said, "Dad, you spend your whole life telling people to serve. You don't get to make an exception for your own." She was right, and I have had cause to regret her being right more than once.

Our son Eli was born in 2021. He writes software in Austin, Texas, and disapproved of ARGUS long before I learned to.

Governor of Colorado

I was elected Governor of Colorado in November 2030 and served two terms, from January 2031 to January 2039. Colorado is a useful school for a president because it holds so much of the national security apparatus within its borders. Peterson Space Force Base and the North American Aerospace Defense Command are in Colorado Springs, and so is the headquarters of United States Space Command. Cheyenne Mountain is there, the granite hollow that every child of the Cold War has seen in films. Buckley Space Force Base, east of Denver, receives the data from the early-warning satellites. Schriever is out on the plains. As governor I attended the change-of-command ceremonies, had lunch with the commanders, and was given the unclassified tours, which are fascinating and tell you almost nothing.

What the governorship taught me was not national security but emergency management. In my first term, in August 2033, the Roan Plateau Complex fire burned 190,000 acres on the Western Slope and came within four miles of Grand Junction. I learned there the thing that later became

my second model: that some decisions can be taken back and some cannot, and that the two kinds require completely different speeds. Ordering an evacuation is reversible; people can go home. Not ordering one, if the wind shifts, is not.

I ran for president in 2040 as a Democrat with a reputation for independence, which in practice meant that both parties found me exasperating. I won in November 2040 with a margin in the Electoral College that was comfortable and a margin in the popular vote that was not. I was re-elected in November 2044, narrowly, against an opponent whose central argument was that I had been too patient with Beijing. Readers of this volume may judge that argument for themselves.

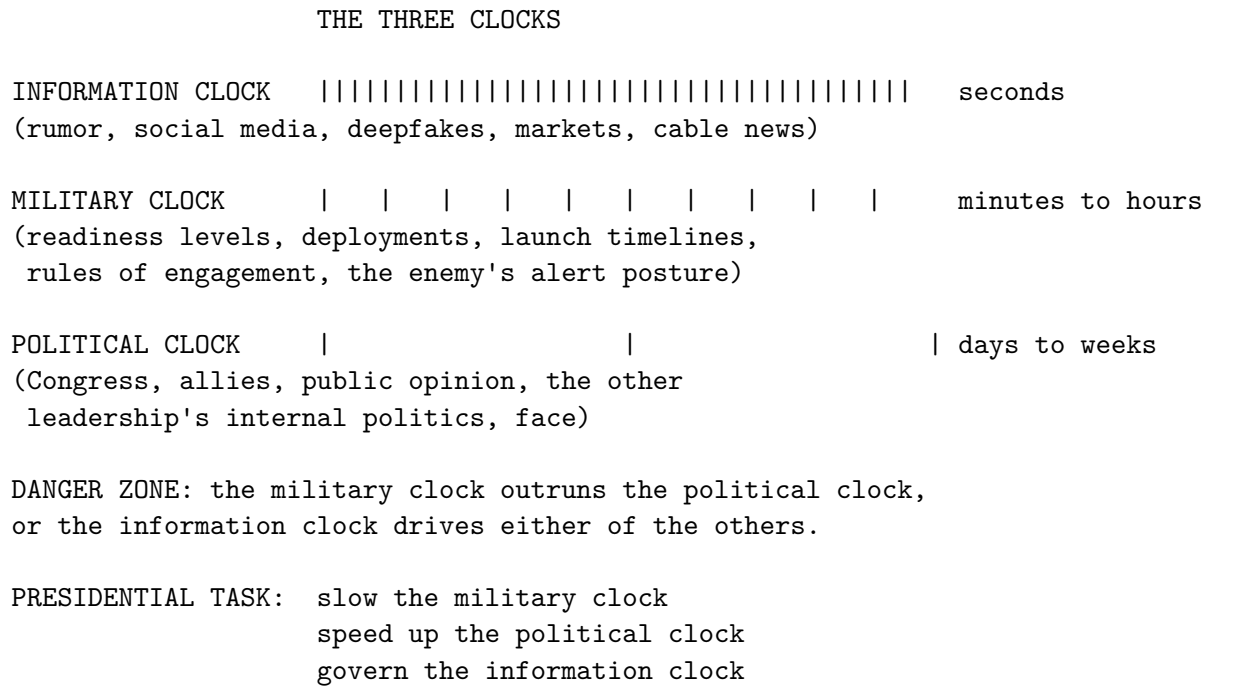
Part Two: The Six Models

An engineer does not trust a design he cannot draw. Over thirty years I accumulated a handful of mental drawings that I return to whenever a problem is too large to hold in my head all at once. In the presidency I formalized six of them. I wrote them on index cards in the first month of my first term and kept the cards in the breast pocket of my jacket, next to the card I will describe later in this chapter, which is the one card a president must never lose. My staff called the index cards "the Six" and learned to anticipate them. Dr. Samuel Oyelaran, my National Security Advisor, used to say that he could predict which card I was going to pull out of my pocket by the expression on my face.

I set the models out here in the order I wrote them, not the order of importance. Each comes with a diagram in words, an example from before the crisis, and a note on how it can mislead.

Model 1: The Three Clocks

Every crisis I have studied, and every one I have lived through, runs on three clocks at once, and they tick at different speeds.



The *information clock* ticks in seconds. A video appears on a social platform purporting to show a

Chinese warship sinking; within four minutes it has been viewed ten million times; within six, the yield on the ten-year Treasury note has moved; within nine, a senator has demanded a response; and within twenty, someone in your own government has repeated it in a meeting as though it were established fact. It may be real. In 2045 it was, more often than not, partly synthetic. The information clock cannot be stopped. It can only be governed: fed with accurate facts quickly enough that the vacuum is not filled with false ones.

The *military clock* ticks in minutes and hours. A destroyer can close twenty nautical miles in an hour. A bomber wing can disperse in hours. A ballistic missile crosses the Pacific in about thirty minutes. Readiness postures, once raised, create their own momentum: crews are recalled, weapons are moved, commanders are given authorities, and each of those steps is visible to the other side, which raises its own posture in response. Militaries are designed to be fast, and that is a virtue right up until the moment it becomes the problem.

The *political clock* ticks in days and weeks. Allies must be consulted, and a Japanese prime minister must consult his coalition. Congress must be briefed. A Chinese General Secretary must manage the Politburo Standing Committee, the Central Military Commission, and a public he has spent a decade teaching to expect strength. Face must be saved, and saving face takes time. Diplomacy is slow because human beings who represent other human beings are slow, and nothing I learned in five years made that slowness disappear.

The danger arises when the clocks desynchronize: specifically, when the military clock runs faster than the political clock. Then the generals are reacting to each other's moves before the leaders have had time to understand what those moves mean. The classic case is August 1914, when mobilization timetables, once begun, ran ahead of every diplomat in Europe. The opposite case, the hopeful one, is October 1962, when President Kennedy deliberately slowed the military clock by choosing a quarantine rather than an air strike, and moved the blockade line closer to Cuba to buy Khrushchev a few more hours to think.

My task in the Pacific crisis, as I understood it from the first night, was to do three things at once: slow the military clock, speed up the political clock, and govern the information clock. Almost every decision recorded in this book can be read as an attempt to do one of those three.

Example. In the spring of 2042, a Russian surveillance aircraft and an American fighter collided over the Black Sea in an incident that killed no one but destroyed both aircraft. Within an hour, a fabricated video showing the American pilot firing on the Russian aircraft was circulating. The Russian Ministry of Defense had already issued a statement. Our own Pentagon wanted forty-eight hours to complete an investigation before saying anything. I ordered the release of the American fighter's unedited cockpit video within three hours, with a single classification review. The military clock was slow, so it was not a problem. The information clock was the problem, and we governed it by being faster with the truth.

How it misleads. The model can make you treat speed itself as the enemy. Sometimes the military clock must run fast, because a ship is under attack or a missile is in flight. The model does not tell you to be slow. It tells you to know which clock you are on.

Model 2: The Reversibility Matrix

This is my homage to Habibie, who divided his decisions into those that could be undone and those that could not. I added a second axis, speed, because what makes reversibility valuable is the time it buys.

THE REVERSIBILITY MATRIX

	TAKE IT FAST	TAKE IT SLOW
REVERSIBLE	QUADRANT I Correct zone. Act, observe, adjust.	QUADRANT II Wasted time. Paralysis. The enemy moves while you deliberate.
IRREVERSIBLE	QUADRANT III CATASTROPHE ZONE. Where wars and accidents begin.	QUADRANT IV Correct zone, only when no reversible path remains.

RULE: Take reversible decisions fast. Take irreversible decisions slowly, and only when no reversible path remains.

The rule sounds obvious, and it is violated constantly, because human beings and institutions have the instinct exactly backward. We agonize over small reversible choices, which feel risky because we can imagine being criticized for them, and we rush large irreversible ones, because they come wrapped in urgency and the urgency feels like duty.

Three disciplines follow from the matrix. First, before every significant decision, ask aloud: *Can this be undone, and at what cost?* I made my staff label every decision memorandum with a reversibility assessment, and I learned to distrust any memo in which the author had written "reversible" without explaining how. Second, when a decision is irreversible, look hard for a reversible version of it. Many apparently irreversible actions can be decomposed into a reversible first step and an irreversible second step, and the first step often buys the information that makes the second unnecessary. Third, when you find yourself in Quadrant III, being pressed to take an irreversible step quickly, recognize that you are in the most dangerous place a president can stand, and that the pressure itself is a piece of data.

A nuclear launch is the purest Quadrant III decision in existence. It is irreversible in a way nothing else is: once a ballistic missile has left its silo, there is no recall and no self-destruct command available to the president. And the whole architecture of warning and response that I will describe later in this chapter is built to compress it into minutes. Most of my effort as president went into building ways out of Quadrant III before I needed them.

Example. As governor, during the Roan Plateau fire, the incident commander asked me at two in the morning whether to order the evacuation of Palisade, a town of three thousand. The fire behavior models were split. Evacuation was reversible: if the wind held, people would come home in a day, angry and inconvenienced. Not evacuating was irreversible if the wind shifted. I ordered it in ninety seconds. The wind did not shift. I was criticized in the *Grand Junction Daily Sentinel* for a week. I would do it again.

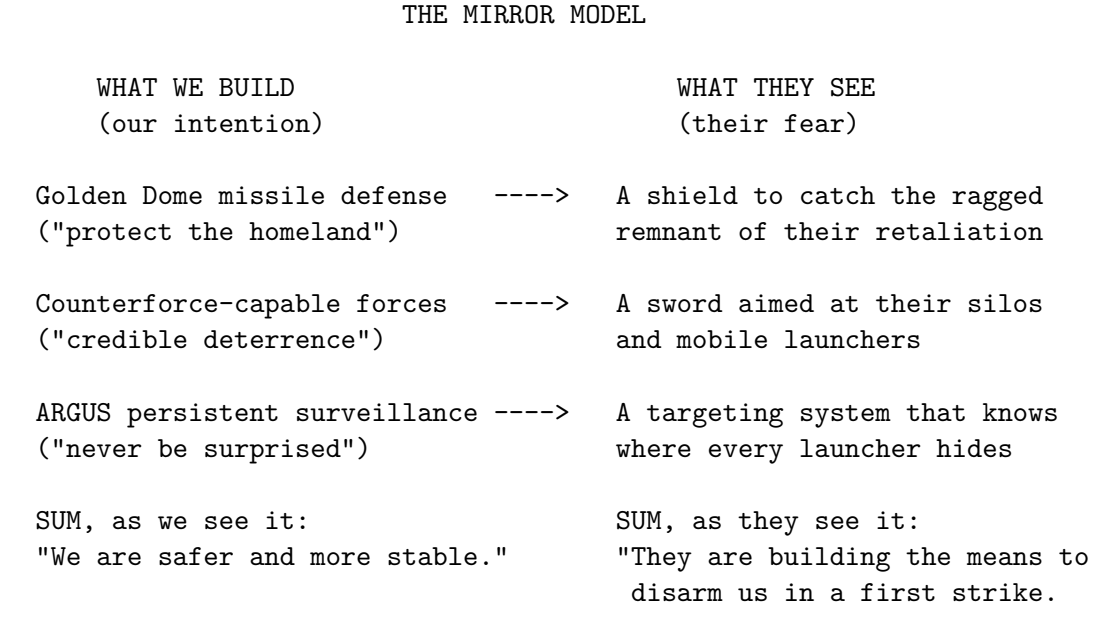
Decision	Options considered	Reversible?	Risk	Chosen	Outcome
Evacuation of Palisade, Colorado, Roan Plateau fire, 14 August 2033, 02:10 MDT	(a) Evacuate now; (b) wait two hours for new fire-behavior model run; (c) shelter in place	(a) Reversible: residents return; (b) and (c) irreversible if wind shifts	(a) Cost, anger, one elderly resident's fall during transport; (b) and (c) potential mass casualties	(a) Evacuate now	Wind held; town undamaged; residents returned in 31 hours; criticized in local press; the case became my teaching example for Quadrant I

How it misleads. Almost nothing is perfectly reversible. Every action leaves a residue: a signal sent, trust spent, a precedent set. The matrix is a tool for honest estimation of that residue, not an excuse to pretend it is zero. During the crisis I called several decisions "reversible" that left marks I did not foresee.

Model 3: The Mirror

The third model is the one I most want a successor to carry, and the one I most often failed to apply in time.

The ordinary method of intelligence analysis asks: *What does the adversary intend?* It assembles evidence of plans and preparations and tries to infer purpose. That is necessary. But it misses half the problem, because adversaries act on fear as often as on intention. The Mirror asks a different question: *What does the adversary see when he looks at us?*



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                                        Our deterrent is dying. Act
                                        before the window closes."

PARADOX: the more perfectly we see, the more frightened they
         become. Absolute surveillance is a weapon that also
         loads the enemy's gun.
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When I took office, the United States was in the middle of building three things, each defensible on its own terms. The Golden Dome, announced in 2025, was a layered missile defense architecture: ground-based interceptors in Alaska and California, the Next Generation Interceptor, a space-based sensor layer, and, by 2041, a first tranche of prototype space-based interceptors whose effectiveness was hotly contested inside my own government. Our nuclear forces, modernized through the 2030s, were accurate enough to threaten hardened silos. And ARGUS, which I will describe in detail later, promised persistent watch over every road-mobile missile launcher outside its garrison in China.

From Washington these looked like three separate programs managed by three separate bureaucracies with three separate budget lines. From Beijing, I came to understand, they looked like a single system: a sword, a shield, and an eye. The eye finds the launchers. The sword destroys most of them. The shield catches whatever survives. The logic of a disarming first strike, which no American president in my lifetime has seriously contemplated, was nonetheless physically present in the hardware, and the Chinese planners responsible for their own country's survival could not afford to assume our intentions were benign. They had to plan against our capabilities. So, of course, did we.

The Mirror does not tell you to stop building defenses. It tells you to account for their second-order effect: every increment of our security that the other side perceives as an increment of its insecurity will produce a response, and the response may make you less safe than before. China's move, around 2030, to keep part of its missile force on what its doctrine called "early-warning counterstrike," what we would call launch on warning, was in large part a response to what it saw in our mirror. A missile force on launch-on-warning is a force that will fire on the basis of sensor data in the minutes before our warheads arrive, rather than waiting to absorb a strike. That shortened Beijing's decision time, and therefore ours, and it made the whole world hostage to the reliability of Chinese and American sensors. I will return to this.

Example. In my second month in office, the Missile Defense Agency asked me to approve a public demonstration: an intercept of a target missile by a space-based interceptor prototype, to be announced with fanfare. Applying the Mirror, I asked Helena Marsh, then still Ambassador to Beijing, a career diplomat I had asked to stay at her post and later made my Secretary of State, what Beijing would see. She wrote me a two-page cable that I still have. The key sentence: "They will not see a test. They will see a rehearsal." I approved the test and cancelled the fanfare. The test failed, as it happened, which may have reassured Beijing more than any statement could.

How it misleads. The Mirror can become an argument for doing nothing, and an adversary who understands that you use it can manufacture fear to manipulate you. Modeling the other side's fears is not the same as being governed by them. There were advisors in my government who accused me, during the crisis, of seeing so much of Beijing's fear that I could not see Beijing's ambition. They were not entirely wrong.

> **OPERATING NOTE** — Once a month, ask your Director of National Intelligence for a short paper written from the adversary's point of view: "What the United States looks like from Beijing (or Moscow) this month." Insist that it be written in the first person,

as if by their analysts. The first few will be clumsy. Persist. It is the cheapest insurance you will ever buy.

Model 4: The Signal Ladder

Herman Kahn, the strategist of the 1960s, described escalation as a ladder with forty-four rungs, from "ostensible crisis" to "spasm war." His ladder was a taxonomy of violence. Mine is a taxonomy of meaning. In a crisis between nuclear powers, every move is a message, and a move whose message is illegible is more dangerous than a move that is larger but clear.

THE SIGNAL LADDER

RUNG	MOVE	INTENDED MESSAGE	OFF-RAMP
9	Strategic forces generated (visibly)	"We are ready for the worst."	Private channel: "This is defensive; here is what brings it down."
8	Strikes on forces that attacked us (limited, conventional)	"Attacks will be answered, exactly."	Public statement that the response is "complete"
7	Escorted convoys	"We will not accept strangulation."	Pause conditions named in advance
6	Sanctions, phase by phase	"Costs will rise."	Each phase reversible on stated conditions
5	Allied force movements	"We are not alone."	Ally-led de-escalation
4	Leader-to-leader call	"I am paying attention."	Offer of a quiet channel
3	Public statement	"We object."	Deliberate ambiguity
2	Private demarche	"We have noticed."	Nothing public yet
1	Watch and wait	(none sent)	(none needed)

RULES:
1. Every rung must carry a message the other side can read.
2. Build the off-ramp before you climb the rung.
3. Never climb two rungs at once without saying why.
4. Leave a golden bridge: the enemy must have a way home that he can describe to his own people as a victory.

The "golden bridge" comes from the Chinese strategic tradition, usually attributed to Sun Tzu: when you surround an army, leave an outlet free; do not press a desperate foe too hard. I came to think of it as an engineering requirement. A crisis has to be designed to terminate. If the only way for the other leader to stop is to be seen surrendering, he will not stop, and he may not be permitted to stop by the people around him. So every rung I climbed, I tried to build, in advance, the bridge back down: the statement that would let Beijing say it had made its point, the condition whose fulfillment would let us pause, the face-saving formula that a Chinese foreign ministry spokesperson could read aloud without being dismissed the next week.

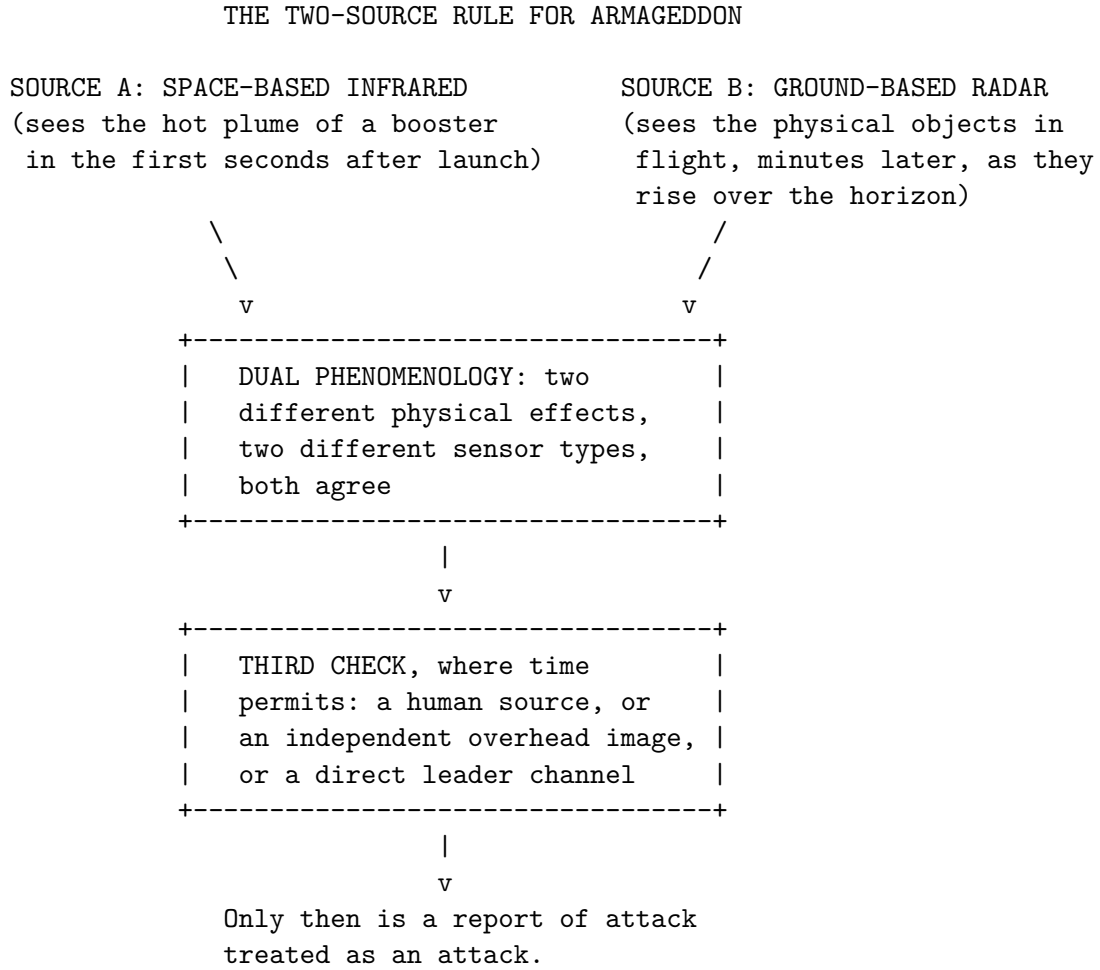
Legibility is the hardest part. A message is what the receiver decodes, not what the sender intends. The same convoy that we intended as "we will not accept strangulation" could be read in Beijing as "the Americans are preparing to break the blockade by force, and will use the convoys to move military supplies." Much of the diplomatic labor in this book consists of Helena Marsh, Sam Oyelaran, and a remarkable Singaporean prime minister trying to make sure that what we sent was what they

received.

How it misleads. Ladders imply that escalation is orderly, one rung at a time. Real crises skip rungs, because of accidents, because subordinates act on their own, because a missile battery commander with a confused picture fires at a convoy. The ladder is a plan for what you control. Most of what happens, you do not control.

Model 5: The Two-Source Rule for Armageddon

This model is the shortest, the most rigid, and the one that, as it turned out, mattered most.



RULE: Never act on a nuclear warning from one phenomenology. One sensor type can be wrong, blinded, spoofed, or misread by software. Two different physics rarely fail in the same way at the same moment.

"Dual phenomenology" is the term of art in the American warning system. It means that an attack is confirmed only when two independent kinds of sensors, measuring two different physical phenomena, agree. The classic pairing is infrared, from satellites that watch for the heat of a rocket booster, and radar, from the great phased-array stations that watch the northern horizon. The principle has existed in American doctrine for decades. It exists because the system has lied before.

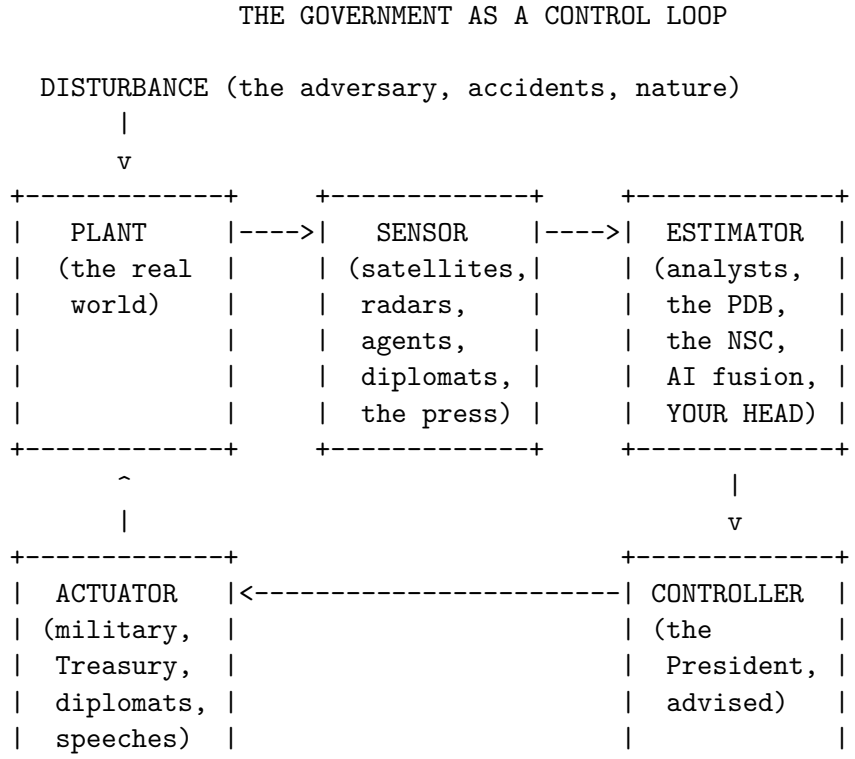
On 9 November 1979, a training tape simulating a massive Soviet attack was mistakenly loaded into a computer at the North American Air Defense Command, and the attack appeared on screens across the warning system as though it were real. Interceptor aircraft were launched. The error was caught within minutes, because the radars and the satellites saw nothing. In June 1980, a failed computer chip, reportedly costing less than a dollar, caused warning displays to show phantom missile launches, more than once in a matter of days. And on 26 September 1983, in a bunker south of Moscow, Lieutenant Colonel Stanislav Petrov of the Soviet Air Defense Forces watched his early-warning system report the launch of five American intercontinental missiles. The satellite system was new, and he knew that an American first strike would not consist of five missiles. He reported it as a false alarm. He was right: the satellites had mistaken sunlight glinting off high clouds for rocket plumes. Petrov did not have a second source in time. He had his judgment, and his judgment served as one.

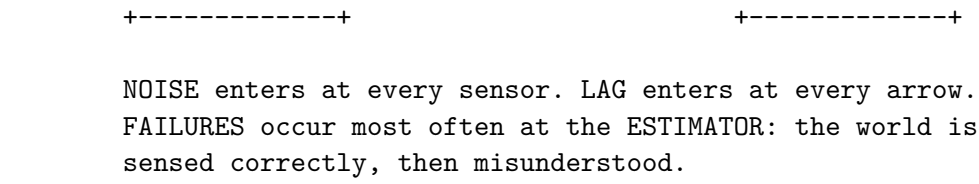
My refinement, which I began to write down in the first weeks of my presidency and which became a National Security Presidential Memorandum in 2042, was a third check: where time permits, a human or independent source. I did not imagine, when I wrote it, how literally it would be tested. I describe the memorandum in the Decision Record at the end of this chapter.

How it misleads. The rule assumes you have time to wait for the second source. The architecture of nuclear warning is designed to give a president very little time, and an adversary on launch on warning gives you even less. The rule is a commitment to accept the risk of waiting. That risk is real, and a successor must decide in advance whether he or she is willing to carry it. I was. I am not certain every president should be.

Model 6: Control-Loop Thinking

The last model is the one underneath all the others. It is the lawnmower governor on the back of the feed-store receipt, grown up.





A control engineer sees the government as a feedback system. The real world, the plant, is disturbed by events. Sensors observe it: satellites, radars, signals intercepts, human agents, embassies, journalists. An estimator turns those observations into a picture of what is happening: the analysts of the intelligence community, the President's Daily Brief, the National Security Council staff, increasingly in my time the machine-learning systems that fuse data faster than any analyst can, and finally, inescapably, the president's own mind. A controller, the president, chooses actions. Actuators carry them out: the armed forces, the Treasury, the State Department, the bully pulpit. The world responds, and the loop closes.

Two properties of real control loops matter enormously. The first is *lag*. Every arrow in the diagram takes time. Intelligence takes hours or days to collect and analyze; orders take hours to execute; diplomatic replies take days. A controller who reacts to old information as though it were current will oscillate, overcorrecting first one way and then the other. Governments do the same thing with sanctions, with force deployments, with rhetoric.

The second property is *noise*. Every sensor lies a little. Agents misremember, satellites see clouds, embassies report what the host government wants them to hear, and the press reports what is interesting. The estimator's job is to filter the noise without filtering out the signal, and to maintain an honest measure of its own uncertainty.

My central claim, which the rest of this book bears out, is that crises fail at the estimator. Sensors are usually good enough. Actuators, the American military above all, are almost always good enough. Controllers are as good as the people elected. But the estimator, the part of the system whose job is to understand what the sensors are saying, is where the catastrophic errors happen. In 1914, in 1941 at Pearl Harbor, in 1962 in the early estimates that there were no nuclear warheads in Cuba, in 2003 in Iraq, the failure was not in the seeing. It was in the understanding. And in November 2045, the failure that nearly ended the world was, once again, an estimator failure: a fusion system that took a confusing collection of real sensor returns and assembled them, confidently, into a picture of an attack.

A president is part of the estimator. That is the hardest lesson in this chapter. You are not merely the person who chooses; you are the last filter through which the picture of the world passes before a choice is made. If you are exhausted, frightened, angry, or overconfident, the filter is degraded, and nobody below you can fix it.

> **OPERATING NOTE** — Ask every briefer the same three questions until they anticipate them: *How do you know? How confident are you, as a number? What would change your mind?* An analyst who cannot answer the third question is not giving you an estimate. They are giving you a belief.

Part Three: The Machinery of the End

Sunday, 20 January 2041 — 06:40 EST, Blair House

Inauguration Day fell on a Sunday in 2041, as it had in 1985 and 2013. The Constitution says the term begins at noon on the twentieth; custom says the public ceremony on the Capitol steps waits for Monday. So there were two oaths: a private one in the Blue Room of the White House at five minutes before noon on Sunday, with Claire holding my grandmother's Bible and the Chief Justice in a business suit, and a public one on Monday in front of a crowd and a cold wind off the Potomac. People remember the second. The machinery cares only about the first.

I had slept perhaps four hours at Blair House, the President's guest house across Pennsylvania Avenue. My diary for the night before records one true thing: *I am not afraid of the job. I am afraid of one hour of it, and I do not know which hour.*

At 06:40, Megan Pruitt, who would be my Chief of Staff for both terms, knocked and said that the briefing team was downstairs. The briefing I am about to describe has been given, in some form, to every incoming president since the early Cold War. It is sometimes called "the nuclear briefing," sometimes, by those who have given it, "the conversation." It takes about an hour. It changes the person who receives it.

Three people were waiting in the Lee dining room, which had been swept for listening devices the night before and was guarded by two Secret Service agents who did not look at the papers on the table. The first was the military aide who would carry the emergency satchel for the first part of my term: a Navy commander, a surface warfare officer with a quiet voice, whose name I will not write here because she has asked to remain a private citizen. The second was an Air Force major general from the Joint Staff's operations directorate, the deputy director responsible for nuclear operations, who would brief the war plans. The third was a civilian from the White House Military Office who said almost nothing and wrote down everything. Sam Oyelaran, who would be sworn in as my National Security Advisor that afternoon, sat in at my request. The outgoing administration's National Security Advisor had sent a gracious note and stayed away, as is customary.

"Mr. President-elect," the general began, "in a little over five hours you will become the only person in the United States with the authority to order the use of nuclear weapons. I'm going to walk you through how that authority is exercised, what the options look like, and what the system around you is designed to do. Please interrupt me as often as you like. Most people do."

I did not interrupt for the first ten minutes. I want to set down what he said in my own words, as accurately as my notes allow, because a successor should know what is in this briefing before receiving it.

The Authority

The President of the United States has, by law and by long practice, sole authority to order the use of American nuclear weapons. There is no requirement in the system for the concurrence of the Secretary of Defense, the Chairman of the Joint Chiefs, the Congress, or anyone else. The Secretary of Defense and the military officers in the chain have a role in verifying that an order truly comes from the President, and every officer swears an oath to obey lawful orders, which carries with it the duty to refuse an unlawful one. But the architecture was built in the 1950s and 1960s on the assumption that a Soviet attack might leave a president ten minutes or less to decide, and that any requirement for consultation could be fatal. It was built for speed. That has not changed.

“Is there anyone,” I asked, at minute eleven, “who can stop me?”

The general looked at me evenly. “Sir, the system is designed so that no one can stop a valid order from the President. It is also designed so that no one but the President can give one. Both halves of that sentence are deliberate.”

“And if the President is wrong?”

“Then the system will carry out a wrong order very efficiently, sir.” He paused. “That’s why I’m here this morning, and why we’ll keep coming back.”

I wrote that sentence down in the margin of my briefing book. It is the most honest thing anyone told me in five years.

The Football

The commander lifted onto the table a black leather case, about the size of a large briefcase, its aluminum frame visible at the corners. Officially it is the Presidential Emergency Satchel. Everyone, including the people who carry it, calls it the football. The name is usually traced to a war plan in the early 1960s code-named “Dropkick,” which required something to carry it in.

The football does not contain a button. It never has. What it contains, in general terms that have long been public, is a means of communication and a set of documents. The communication equipment allows the President to reach the National Military Command Center at the Pentagon, and from it the rest of the system, from wherever he is: a motorcade, a hotel in a foreign capital, a high-school gymnasium in Iowa. The documents include a book of pre-planned response options, which in my era was bound in a gold-edged cover and universally called the Gold Folder; a description of the locations to which the President can be moved in an emergency; and a description of the emergency procedures for broadcasting to the nation.

“The Gold Folder,” the general said, opening a training copy with nothing in it but sample pages, “is designed to be readable by a person who has just been woken from deep sleep, who is frightened, and who has perhaps four to six minutes. That’s why it looks the way it does.”

The way it looks, I may say only in general terms, is like a restaurant menu designed by an engineer who hated restaurants. Options are grouped into broad categories, which public literature has long described as major attack options, selected attack options, and limited attack options, with a further class designed for regional contingencies. Each option is summarized on a page: the general class of targets, the approximate number of weapons, and an estimate, rendered with a terrible bureaucratic neutrality, of expected consequences.

“It is simplified,” the general said, “because the decision may need to be made in minutes. If there is time, the Commander of Strategic Command and the Chairman will brief you in detail, and Strategic Command can build an option tailored to the situation. If there is no time, this is what you’ll have.”

I turned the training pages slowly. I asked the question every president asks, according to the general, which was how many people would die under the largest of the options. He told me. I will not write the number here. It is in the classified annex, and it is in my diary, and I have not been able to forget it for one day since.

Then I asked a question he said fewer presidents ask: “Where is the option that does nothing?”

“Sir?”

"If I am told that missiles are coming, and I am not certain, where is the page for 'wait'?"

He was quiet for a moment. "There isn't a page for that, sir. Not deciding is always available to you. But you're right that the book doesn't present it as an option."

By the end of 2041 there was such a page. It is the first page in the Gold Folder now. It is blank except for a single line, which I wrote: *You may decide not to decide. Ride out and retain. Confirm before you act.* "Ride out" is the old doctrinal term for absorbing an attack before retaliating; "retain" means keeping the surviving forces, above all the submarines at sea, for a later and better-informed decision. The submarines are why a president can afford to wait. I will come back to them.

The Biscuit

The last object was the smallest. The commander handed me a sealed envelope; inside it was a plastic card, about the size of a credit card, in a protective sleeve. The card is called the biscuit. It carries a set of codes that allow the President to identify himself to the officers of the National Military Command Center so that they can be certain an order comes from him. The codes change on a schedule. The card would become active at noon, when I took the oath, and the outgoing President's card would be deactivated at the same moment.

I will not describe the format of the codes, the method of authentication, or the sequence by which an order is verified and transmitted, and neither should any successor writing for posterity. It is enough to say that the process is a challenge and a response between the President and the duty officers in the Pentagon, that it is designed to take a very short time, and that, in my era, it was called the Recognition Procedure. Once the President is recognized and an option is chosen, the order is formatted into an emergency action message and transmitted to the forces through multiple redundant paths. Crews in launch control centers and on submarines verify it against their own sealed materials, by procedures that require two people at every step. Then they carry it out.

"Keep it on you at all times, sir," the commander said. "In a jacket pocket, or wherever you'll never leave it. Some presidents have lost theirs. It's caused some excitement."

I put it, at noon, in the inside breast pocket of my jacket, behind the index cards of the Six. It lived there for eight years. Claire asked me, the first night, what it felt like. I said it felt like a piece of plastic. That was a lie, and she knew it, and I wrote the truth in the diary: *It feels like carrying a sleeping animal that could wake up.*

11:55 EST — The Blue Room

The private oath took ninety seconds. The Chief Justice read; I repeated; Claire held the Bible; Maya, an ensign two years out of Annapolis and on leave from her first ship, and Eli stood by the window with my mother, who was eighty and who wept. At the moment I said "so help me God," a young officer in the corner, the commander's relief for the day, took one step closer to me than he had been standing before. That was the transfer. Somewhere across the city, at the same moment, another military aide took one step away from the man who had been president a minute earlier. There is no ceremony for this. I think there should not be.

The Timeline in Minutes

Back at Blair House that morning, before the oath, the general had drawn a line across a sheet of paper and marked it in minutes. I asked him for the sheet afterward, and I have redrawn it here,

updated with what I learned over the following four years. A successor should memorize it.

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    A LAND-BASED MISSILE ATTACK ON THE UNITED STATES FROM ASIA:
    THE PRESIDENT'S TIMELINE (approximate, public-literature level)

  T+0:00   Booster ignition. Space-based infrared satellites detect
           the plume within seconds.
  T+0:30   Ground stations process the detection; the Missile Warning
           Center at NORAD and the NMCC see a launch report.
  T+1:00   Booster phase ends within the first few minutes; warhead
           bus coasting toward space.
  T+2:00   Threat assessment conference convened by the NMCC with
           NORAD and STRATCOM. Question: is this real, and is it
           aimed at North America?
  T+3:00   If the event looks hostile, the conference is escalated;
           the President, the Secretary of Defense, and the Chairman
           are brought in.
  T+5:00   President on the line. Woken, moved, or interrupted.
  T+8:00   to T+15:00   Ground-based early-warning radars begin to
           see objects as they rise over the radar horizon. SECOND
           PHENOMENOLOGY becomes available.
  T+10:00  to T+20:00   The window in which a decision to launch
           land-based missiles before impact must be made, if it is
           to be made at all.
  T+25:00  to T+33:00   Warheads arrive over targets in the
           continental United States.

  SUBMARINE-LAUNCHED MISSILES FROM CLOSE OFFSHORE: compress all
  of the above to roughly 10 to 15 minutes.

  WHAT THE PRESIDENT ACTUALLY HAS: perhaps six minutes of real
  deliberation, on the most important decision ever made.
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"Six minutes," the general said, tapping the line. "That's the figure people quote, and it's roughly right if the decision is whether to launch our land-based missiles before incoming warheads arrive. The submarines change the arithmetic. They're survivable. You don't have to use them in the first thirty minutes, or the first thirty days."

"Then why is anyone in a hurry?"

"Because the land-based missiles are in fixed silos, sir, and an adversary would target them. If you want to use them, you have to use them before they're destroyed. That's the logic of launch under attack. It's an option. It's not a requirement."

I wrote, in the margin: *The silos are the reason for the hurry. The submarines are the reason we don't need to hurry.* Every successor should understand this sentence, because the whole psychological pressure of the six minutes comes from a weapon that the nation could, if it chose, decline to use quickly. By 2041 the land-based leg was the LGM-35A Sentinel, four hundred missiles in silos spread across the northern plains around F.E. Warren Air Force Base in Wyoming, Malmstrom in Montana, and Minot in North Dakota. Sentinel had fully replaced the old Minuteman III that year, after a

program whose cost overruns had been a political wound for a decade. They are superb machines. They are also, by their nature, use-them-or-lose-them weapons, and an adversary's planners know it.

> **OPERATING NOTE** — Decide, in your first month and in calm, what your personal standard will be for launching land-based missiles on warning. Write it down. Tell the Chairman and the Commander of Strategic Command what it is, so they can build the conference around it. A president who decides his standard in the middle of an alarm has not decided it; the alarm has.

The Sensors

The machine begins with eyes. In 2041 the United States watched for missile launches with three families of sensors, and by 2045 a fourth.

The first family is the space-based infrared system: satellites in geosynchronous orbit, about 36,000 kilometers up, and in highly elliptical orbits over the poles, that stare at the Earth looking for the intense heat of rocket exhaust. The original Defense Support Program satellites flew from the 1970s; the Space-Based Infrared System replaced them; by 2041 the Next-Generation Overhead Persistent Infrared satellites had replaced those. Their data comes down to ground stations, most importantly at Buckley Space Force Base outside Denver, which I had visited as governor without understanding a word of what I was shown. Infrared satellites are the first to know. They can see a booster burning within seconds. What they cannot do is see the warheads after the booster burns out; they see heat, not objects.

The second family is the newer layer in low Earth orbit. The Space Development Agency's Proliferated Warfighter Space Architecture, begun in the early 2020s, put hundreds of smaller satellites into orbits a few hundred to a couple of thousand kilometers up. Its "tracking layer" was designed to follow missiles, including the maneuvering hypersonic glide vehicles that fly too low for the old radars to see early, through more of their flight. By 2041 the tracking layer was mature. By 2045 it was the backbone of the warning system's view over Asia, and its data, fused by machine-learning software, fed the Golden Dome's battle management system. I will have a great deal to say about that fusion in Chapter 6.

The third family is ground-based radar: the great phased-array stations that watch the horizon. The Upgraded Early Warning Radars at Clear Space Force Station in Alaska, at Beale Air Force Base in California, at Cape Cod in Massachusetts, at Royal Air Force Fylingdales in England, and at Pituffik Space Base in Greenland, which older officers still called Thule. Cobra Dane at Eareckson Air Station on Shemya Island, at the end of the Aleutian chain, looks toward Asia. The Sea-Based X-band radar and the newer discrimination radars in Alaska refine the picture. Radars see physical objects: the warheads, the decoys, the spent rocket stages. But they cannot see over the curvature of the Earth, so a missile launched from western China does not appear on an American radar until it has climbed high enough over the Pacific or the pole to rise above the radar horizon, typically some eight to fifteen minutes into flight.

This is the gap that matters. For the first several minutes of any attack, the President's only evidence is infrared: one phenomenology. The second phenomenology arrives late, and by the time it arrives the window for launching land-based missiles on warning is half gone. The Two-Source Rule is, in plain terms, a commitment to spend that half.

The fourth family, by 2045, was ARGUS, which is not a warning system at all, and which I will describe separately because it was the thing I was proudest of and the thing that most nearly undid

me.

NORAD, the NMCC, and STRATCOM

The eyes report to three brains.

The North American Aerospace Defense Command, NORAD, is a binational command, American and Canadian, headquartered at Peterson Space Force Base in Colorado Springs, with an alternate command center deep inside Cheyenne Mountain. Its job, for this purpose, is tactical warning and attack assessment: to say whether something is coming at North America, what it is, and where it is going. In the language of the system, NORAD "characterizes" the attack. When the infrared satellites see a launch, the Missile Warning Center, part of NORAD's operations, decides within a minute or two whether the event is a space launch, a test into the ocean, a missile aimed at a third country, or a threat to the continent.

The National Military Command Center, the NMCC, is in the Pentagon. It is a set of rooms in the building's inner rings, lit like a trading floor, staffed around the clock by teams of officers led by a general or flag officer known as the Deputy Director for Operations. It is the hub through which the President's orders flow to the forces. It is also the convener: when NORAD reports something alarming, it is the NMCC that pulls the relevant commanders and, if necessary, the President, onto a single secure conference. There is an Alternate National Military Command Center at Raven Rock Mountain Complex, Site R, just over the Pennsylvania line from Maryland, inside a mountain, built for the day the Pentagon is gone.

United States Strategic Command, STRATCOM, at Offutt Air Force Base outside Omaha, Nebraska, commands the nuclear forces themselves: the missiles, the submarines, the bombers. Since a reorganization in 2018, it has also been the single operational commander for the nuclear command, control, and communications system as a whole, which by public estimates comprises more than two hundred separate ground, air, and space systems. The Commander of STRATCOM is the officer who, in a nuclear crisis, presents the President with the options, explains what each would do, and, if ordered, carries one out. In my second term that was Admiral Victor Hale, a submariner who never raised his voice and never, in my presence, softened a fact.

The Conference Ladder

What happens when the eyes see something is a sequence of conferences, each a little more serious than the last. The official names have changed over the decades; the sequence has not. In my era it went like this.

The *threat assessment conference* is the first rung. It is called by the NMCC when NORAD reports an event that might be threatening. It joins the NMCC, NORAD, and STRATCOM, and its question is narrow: is this real, and is it aimed at us? Most threat assessment conferences end within minutes: the launch was a satellite going up, or a test missile heading into a known impact area, or a sensor fault.

If the event cannot be dismissed, the conference is escalated. In the 1970s and 1980s the next rung was called a "missile attack conference"; by my time the formal name had been softened to the *missile event conference*, which is the kind of euphemism institutions use when the thing is too terrible to name. At this rung the President, the Secretary of Defense, and the Chairman of the Joint Chiefs of Staff are brought onto the line, wherever they are. The Commander of STRATCOM joins to present options. The duty officers at NORAD present the attack characterization: number

of missiles, likely origin, likely targets, time to impact. The President listens, asks questions, and decides.

"Decides," the general said in the Blair House dining room, "can include deciding to wait. Deciding to wait is a decision. You'll be told the cost of waiting, which is that some of our forces may be destroyed. You should also ask the cost of not waiting."

"Which is?"

"If the warning is false, sir, the cost of not waiting is the end of the world." He did not smile. "Every president has asked me to repeat that."

19:15 EST — The PEOC

On the evening of the private oath, after a quiet dinner in the residence with my family, I asked to see the bunker. The Secret Service had planned the tour for later in the week. I asked for it that night. My diary does not explain why, but I remember: I wanted to see the room in which the hour I was afraid of would happen, if it happened.

The Presidential Emergency Operations Center lies beneath the East Wing. Its existence has been public since President Roosevelt's day, when it was built in the early years of the Second World War; its details are not, and I will keep to what has long been reported. You reach it through a corridor and an elevator and a series of heavy doors. It was rebuilt and expanded in the 2010s and again in the 2030s. It is a working command center: a conference room with a long table, displays on the walls, secure communications to the Pentagon, to the intelligence agencies, to STRATCOM and NORAD; a small kitchen; bunks; a medical room. It smells of filtered air, carpet glue, and coffee that has been on the burner since the Eisenhower administration. I came to know that smell very well.

The duty officer, an Army lieutenant colonel, walked me through the displays. One showed the status of the nation's strategic forces in a format I will not describe. One showed the current location of the Vice President, the Speaker, and the Cabinet officers in the line of succession. One was blank, kept for whatever the crisis required. On the long table, at the place for the President, there was a small card with the number of the NMCC conference line and nothing else.

"Sir, would you like to talk to the NMCC?" the lieutenant colonel asked.

I said yes. He picked up a handset and said, in an even voice, a few words. Within seconds a calm voice said, "NMCC, Deputy Director for Operations." A one-star general in the Pentagon had been told the President might call on his first night. He walked me through the board. Nothing was happening anywhere in the world that required the President's attention. A Chinese space launch from Jiuquan earlier that day had been tracked and characterized. A North Korean short-range test into the Sea of Japan the previous week was being analyzed. A Russian bomber patrol was off the Aleutians, as it was most weeks, and NORAD fighters had gone up to escort it.

"How often," I asked him, "do you convene a threat assessment conference?"

"Several times a year at that level, sir, for events that need a second look. Almost all of them are resolved in minutes. The last time a President was brought onto an escalated conference for a real alarm, before tonight's administration, was" (he named a year, which I will not write) "and it was resolved without a decision."

"And if I'm brought onto one?"

"We'll give you the facts as fast as we have them, sir, and we'll tell you what we don't know."

"Tell me what you don't know first," I said.

There was a short pause on the line. "Yes, Mr. President."

That became a standing instruction. In the PEOC on the morning of 6 November 2045, the first words from the NMCC were what they did not know. I will describe in Chapter 6 how much that mattered.

The Aircraft That Never Land in Time of War

A command system that depends on buildings is a command system that can be destroyed in thirty minutes. So the United States, since the early 1960s, has kept a part of its command system in the air.

The old hands still call the first of these aircraft by the name of its predecessor, the E-4B Nightwatch, a militarized Boeing 747 that for half a century served as the National Airborne Operations Center: a flying Pentagon, built to carry the President or the Secretary of Defense and a battle staff above a war. By 2041 the E-4B had been retired, replaced by the E-4C Survivable Airborne Operations Center, built on newer 747-8 airframes and delivered through the mid-2030s. The new aircraft can refuel in flight, is hardened against the electromagnetic pulse of high-altitude nuclear bursts, and carries communications that can reach the forces when the ground networks are gone. One of them is always on alert, often near wherever the Secretary of Defense is traveling.

The second family is the airborne link to the submarines. For decades it was the Navy's E-6B Mercury, built on the old Boeing 707 airframe, flying the mission called TACAMO, "Take Charge and Move Out," trailing a very long wire antenna behind it to transmit at very low frequencies that can penetrate seawater to a submarine's trailing receiver. The E-6B also flew the "Looking Glass" mission, an airborne command post for STRATCOM able to launch land-based missiles by radio if the launch control centers were destroyed. By the late 2030s the Navy had moved TACAMO to the E-130J Phoenix II, a turboprop based on the C-130J Super Hercules, and the Air Force had taken back Looking Glass. When I flew in one in 2042, a young officer explained, with great courtesy, that her job was to make sure a dead president's orders could still reach the sea.

That sentence is the essence of continuity of government.

> **OPERATING NOTE** — The airborne command posts and the submarine link are what make waiting possible. If the President knows that the submarines will receive his orders an hour, a day, or a week after an attack, he need not decide in six minutes. Every dollar spent on the survivability of command is a dollar spent on the President's time to think. Defend those budget lines against every other priority.

Continuity of Government

Continuity of government is the set of plans that ensures the United States still has a lawful president, a functioning chain of command, and the essential institutions of a constitutional republic after a catastrophe. Much of it is classified. Some of it is simply constitutional: the line of succession, set by the Twenty-Fifth Amendment and the Presidential Succession Act of 1947, runs from the President to the Vice President, the Speaker of the House, the President pro tempore of the Senate, and then through the Cabinet in the order the departments were created, beginning with the Secretary of State.

On my first night, the PEOC display showing the locations of the successors was the one I looked at longest. The system is built to keep them apart. On a State of the Union night, one Cabinet member, the "designated survivor," stays away from the Capitol at an undisclosed location with a military aide and a copy of everything the President carries. In a crisis, the Vice President is moved to a separate location; in 2045 Rosalind Achterberg spent many nights at sites she would describe to me only as "underground and badly catered." The Speaker and the Senate's President pro tempore have their own arrangements. There are relocation sites for the departments, for the Supreme Court, for the Congress.

What continuity of government cannot guarantee is judgment. A successor sworn in at a relocation site would have lawful authority over the nuclear forces, but perhaps no knowledge of them. That is why I insisted that every successor in the line receive a version of the nuclear briefing within thirty days of taking office, and that the Vice President attend at least two exercises a year.

Golden Dome

I come now to the two systems that dominated my first term's defense budgets and that sat, as I have said, at the center of Beijing's mirror.

The Golden Dome for America was announced in 2025, in the spirit of the Strategic Defense Initiative of the 1980s, with the stated goal of protecting the homeland against ballistic, hypersonic, and cruise missiles. By the time I took office in 2041 it was neither the impenetrable shield its champions had promised nor the fantasy its critics had derided. It was something in between, which is where most large engineering programs end up.

In public terms, the architecture in 2041 had four layers. On the ground, the Ground-based Midcourse Defense interceptors at Fort Greely, Alaska, and Vandenberg Space Force Base, California, were being supplemented by the Next Generation Interceptor, which carried several kill vehicles per missile and was designed for a limited attack from a small nuclear power. Regional defenses, the Navy's Aegis destroyers with their Standard Missile interceptors and the Army's Terminal High Altitude Area Defense batteries, protected ships, bases, and allies. In space, the sensor layer, the tracking satellites I have described, provided the "birth-to-death" track that interceptors need. And, most controversially, there was a first tranche of space-based interceptors: by 2045, some two hundred prototypes in low orbit, designed to strike missiles in their boost or early midcourse phase.

Their effectiveness was, to put it charitably, contested. The Missile Defense Agency's own tests had produced mixed results. Independent physicists pointed out, as they had since the 1980s, that boost-phase interception from space requires very large constellations, because at any moment only a small fraction of the satellites are close enough to a given launch site to engage in the few minutes the booster burns. Two hundred interceptors could, at best, thin a small attack. Against a large attack they were close to irrelevant.

And yet, to a Chinese planner looking in the mirror, two hundred was a beginning, not an end. Two hundred could become two thousand. A defense that could not stop a full Chinese first strike might nonetheless stop a ragged, damaged Chinese retaliation after an American first strike had destroyed most of China's missiles. That was the fear, and no number of American assurances could dispel it, because assurances are words and interceptors are hardware.

The Golden Dome also brought something new into the warning system: its battle management software. To coordinate interceptors across ground, sea, and space, the architecture needed to fuse data from every sensor in seconds, faster than any human team could, and to produce a single track

picture with confidence values attached. That fusion software, developed through the 2030s with machine-learning components, came to sit, in practice, alongside the traditional warning chain. By 2045 its outputs appeared on the same screens in the NMCC and at NORAD as the legacy warning data, with a confidence score in the corner. Officers learned to glance at the score. I learned to fear the glance.

ARGUS

If the Golden Dome was the shield, ARGUS was the eye. It was, in my judgment then, the finest intelligence instrument any nation had ever built. I accelerated it in 2042 against the advice of some of my own advisors, and I do not entirely regret that. I do regret what I believed about it.

ARGUS was a classified program, run jointly by the National Reconnaissance Office and the Space Force, that fused the outputs of many kinds of satellites into a single persistent picture of selected regions of the Earth. It drew on government satellites carrying electro-optical cameras, synthetic aperture radar that sees through cloud and darkness, radio-frequency receivers that locate emitters, and hyperspectral imagers that detect materials by the light they reflect. And, crucially, it drew on sensors hosted aboard commercial mega-constellations: thousands of small satellites whose buses, built on the model of the Starshield service that SpaceX had offered the government in the 2020s, carried government payloads alongside commercial communications equipment. By 2041 the largest such operator was Halcyon Orbital, whose chief executive, Marcus Thorne, plays a part in Chapter 3 that he would not enjoy reading.

The result was revisit. With enough satellites in enough orbits, any point on Earth within ARGUS's priority regions could be imaged by synthetic aperture radar roughly every ninety seconds. The fusion software tracked objects across those images, so that a Chinese road-mobile missile launcher leaving its garrison could be followed, frame by frame, as it moved along the roads of Gansu or Inner Mongolia. By 2045 the program claimed what its managers called "custody" of ninety-four percent of Chinese road-mobile launchers outside their garrisons at any given time.

I called it, in a speech to the program's engineers in 2043, "absolute persistent watch." I am ashamed of the phrase now, not because it was a lie, but because I half believed it.

ARGUS had limits, and an engineer should have named them first. Electro-optical satellites cannot see through cloud. Radar sees shapes, not purposes, and a decoy launcher built of steel tubing and radar reflectors looks, from four hundred kilometers up, very much like a real one. China had built thousands of kilometers of tunnels for its missile forces, which its own press called the "Underground Great Wall," and what goes into a tunnel cannot be watched until it comes out. Satellites can be dazzled by ground-based lasers, jammed, or destroyed. And, above all, seeing is not knowing. ARGUS could tell me where a launcher was. It could not tell me what the general who commanded it intended. For that I needed people.

There was a second limit, which I learned only by applying the Mirror too late. ARGUS was the eye in the sword-shield-eye triad. Every improvement in its custody numbers was, from Beijing's side, an improvement in America's ability to find and destroy China's survivable deterrent. The better ARGUS worked, the more strongly the logic of "use it or lose it" pressed on the hardliners in the Rocket Force. I did not see this clearly until 2045.

How the National Security Council Works

Machines of steel and silicon are only half the apparatus. The other half is made of meetings.

The National Security Council was created by the National Security Act of 1947. By statute its members are the President, the Vice President, the Secretaries of State, Defense, and Energy, and a few others; the Chairman of the Joint Chiefs and the Director of National Intelligence sit as statutory advisors. In practice, the work is done at three levels, and a successor should learn the levels before learning anyone's name.

At the top is the *Principals Committee*: the Cabinet-level officials, chaired by the National Security Advisor, without the President. In my administration it was Sam Oyelaran in the chair, with Helena Marsh from State, Ray Castillo from Defense, Pat Dunmore as Chairman, Abigail Fenn as Director of National Intelligence, Gregory Laszlo from the Central Intelligence Agency, Daniel Iwu from the Treasury when money was involved, and Megan Pruitt sitting in for me. Below it is the *Deputies Committee*: their deputies, chaired by the Deputy National Security Advisor, which does the real drafting, argues out the options, and sends up only what cannot be settled below. Below that are the interagency policy committees, run by the Council staff, where officials at the assistant-secretary level and below do the daily work. When the President attends, it is a meeting of the National Security Council proper.

The discipline of the system is that decisions rise only as far as they must. A good National Security Advisor protects the President from decisions the Deputies can make, and protects the Deputies from a President who wants to make them all. Sam was very good at both. Formal presidential decisions are recorded in National Security Presidential Memoranda, which tell the departments what the President has decided and who is responsible for carrying it out.

The other daily instrument is the *President's Daily Brief*, the PDB, prepared overnight by the Office of the Director of National Intelligence and delivered every morning. In my time it arrived on a secure tablet at 07:00, with a briefer from the intelligence community who sat across from me in the Oval Office while I read and answered my questions. It was typically a dozen short articles, each with a statement of the analysts' confidence. I asked for one change: at the end of every article, a single line headed "What would change this judgment." It was my third question, institutionalized.

How the Agency Runs Agents

Satellites see. Signals intelligence hears. Only people know what other people intend. And so, in March 2041, two months into my first term, Gregory Laszlo, whom I had just appointed Director of the Central Intelligence Agency, came to the Oval Office with a woman I will call by the name she used in the Agency, Ruth Calder, to give me what the staff called the "sources briefing."

Ruth Calder was then in her fifties, a case officer who had spent most of a career on China, quiet, precise, and entirely unimpressed by the room. She explained the craft first, because, she said, "Presidents who don't understand the craft get agents killed."

Human intelligence, she said, begins with spotting: identifying people with access to secrets who might, for some reason, be willing to share them. Then comes assessing: learning everything possible about the person, their work, their family, their frustrations, their vulnerabilities. Then development: building a relationship, usually over months or years, often under a cover identity, until trust exists. Then the pitch, the moment at which the case officer asks the person to work with the United States. Then handling: running the agent, communicating securely, protecting them, evaluating what they report against every other source. And eventually, if it can be managed, resettlement.

"People think we buy secrets," she said. "Sometimes we do. The Cold War shorthand was MICE: money, ideology, compromise, ego. The best agents are the ones whose reasons are their own. Money

wears out. Conviction doesn't."

Then Laszlo nodded, and she opened a folder that I was permitted to read but not to keep, and told me about LANTERN.

LANTERN, as I Learned of Him in 2041

In 2031, a senior officer of the People's Liberation Army Rocket Force, [REDACTED], had a daughter of eleven who was diagnosed with a rare and aggressive childhood cancer. The treatment that offered her the best chance was available at a specialist center abroad, outside China. Through channels of which Ruth Calder told me only the outline, his family secured the treatment, at a cost and under circumstances that left him indebted, frightened, and, for the first time in his life, exposed to doctors, nurses, and other parents from many countries. One of the people he met in those months was not a doctor. The Agency had spotted him two years earlier, because his posting gave him a view of the Rocket Force's missile brigades in the northwest. It had not expected to have a chance to meet him.

Ruth Calder's account of the development was spare. The daughter's treatment succeeded. The officer returned home. Contact was maintained, slowly, through a cover that need not be described. What emerged over two years, she said, was not a man who wanted money, though he accepted help with medical debts, nor one who hated his government. It was a man who had spent his career preparing missiles for a war he had come to believe would end both his country and every other, and who had sat in a children's ward watching parents of five nationalities pray for the same thing.

"He told me," she said, "that when he saw the war plans, he thought of the ward."

The pitch came in 2033. He said yes on the condition, she told me, that he never be asked to sabotage anything, never be asked to harm a fellow officer, and that his information be used to prevent a war, not to win one. The Agency accepted the conditions, which is unusual, and I have tried to honor them, which has not always been easy.

By 2041 he was a senior colonel serving as deputy chief of staff of PLA Rocket Force Base 64 at Lanzhou, whose brigades were responsible for the missile field near Hami. The Agency gave him the cryptonym LANTERN. He communicated, Ruth Calder said, chiefly in two ways: through dead drops, the oldest technique in the craft, used for documents; and through a covert communication device concealed inside what appeared to be an ordinary commercial vehicle navigation unit, which could, under conditions I will not describe, compress a short message and transmit it to a satellite in a burst lasting a fraction of a second. He reported, among other things, the readiness orders given to his base and the status of warhead handling on the brigades he could see.

"What does he report on, exactly?" I asked.

"Alert levels," she said. "Who orders them, and through which channel. Whether the warheads are with the missiles or in storage. Whether the generals believe what they're saying." She closed the folder. "Mr. President, ARGUS can tell you where a launcher is. LANTERN can tell you whether anyone intends to use it. There's no satellite for that."

I asked how many people knew his true identity. She said fewer than you have fingers on one hand. I asked whether his reporting had ever been wrong. She said it had been incomplete, often; wrong, never in a way that mattered. I asked what would happen to him if he were discovered. She told me. Then I asked what I could do for him.

"Read what he sends," she said. "Take it seriously. And when the time comes that we tell you he has to come out, believe us."

I wrote that night: *I have been introduced to a man I will never meet, who has trusted his life to people he cannot see, for a reason I would be proud to have. I do not know his name. I hope I am worthy of his trust.* Four years later, in the autumn of 2045, Ruth Calder came to me and said the time was coming. I did not believe her soon enough. That is in Chapter 4.

There were other sources, recruited later, in the Central Military Commission's command center, at the submarine base on Hainan, in the missile field at Yumen, in the office of the Premier. Each enters this account under a cryptonym when they became important. One of them did not survive it.

> **OPERATING NOTE** — Before the crisis, learn the names of the human sources, by cryptonym, that your most important judgments rest on, and learn what each can and cannot see. In the crisis you will be asked to spend them: to act on their reporting in a way that may expose them. You cannot weigh that cost in the middle of the night if you met them for the first time that evening.

The First Memorandum

On 3 March 2042, after a year of studying the machine I had been handed, I signed the decision I had been drafting since the Blair House dining room. It was the first presidential memorandum I wrote almost entirely myself, and the one I would most want a successor to keep.

Decision	Options considered	Reversible?	Risk	Chosen	Outcome
National Security Presidential Memorandum 4, "Confirmation Before Consequence," 3 March 2042	(a) Retain existing practice: dual phenomenology as doctrine, launch under attack available on infrared warning alone in extremis; (b) require dual phenomenology for any nuclear decision, plus a third independent check where time permits, plus a "decide not to decide" page first in the Gold Folder; (c) formally abandon launch under attack	(a) and (b) reversible by any later president; (c) reversible on paper, but a public declaration would cost credibility with allies to retract	(a) Decision driven by one sensor type and machine-fused data; (b) some land-based missiles might be lost before a decision in a real attack; (c) allied alarm, domestic attack on "weakness," adversary recalculation	(b), classified; Commander of Strategic Command, Chairman, and Vice President briefed; no public statement	In force on 6 November 2045. Applied in the eleven minutes described in Chapter 6. A successor may revoke it. I hope none will

Sunday, 20 January 2041 — 23:48 EST, the Residence

I end this chapter where it began, on the first night.

After the PEOC, I went upstairs. Claire was asleep. I sat at the small desk in the Treaty Room with the composition notebook I had been keeping since my twelfth birthday, number one hundred and nine, and the biscuit in my pocket. The house was very quiet, in the way only a house guarded by a great many people can be quiet.

I asked myself the question I have asked at the end of every important day since I was twelve: *What is one true thing?*

I wrote several, and crossed out all but this:

Today I was given a machine designed, at enormous cost and with great skill, to let one person end

the world in six minutes. Its sensors are superb. Its actuators will not fail. Its controller is me. Its estimator is also me. If it ever fails, it will fail in my head, at night, with a confidence score in the corner of a screen telling me what to believe. Tom, build yourself some time.

I did not know, that night, that the hour I was afraid of would come four years, nine months, and seventeen days later, at 02:31 in the morning, in the room I had just visited. I knew only that I had to prepare for it as an engineer prepares for the failure mode he cannot test: by designing for it in advance, and by being honest with himself about the part of the system he trusts least.

The rest of this book is the record of how that preparation held, and where it did not.

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Chapter 2: The Quarantine

"They have chosen a word. Now we must decide whether to accept their dictionary." — Diary of T. E. Kane, 3 March 2045

Module 2.0 — What This Chapter Is For

A successor reading this chapter should come away with four things. First, the difference between a quarantine and a blockade. It sounds like a lawyer's quibble, and it is the hinge on which the first three months of this crisis turned. Second, how a naval escort actually works: who gives orders to whom, what a commanding officer may do without asking Washington, and why the president must resist the temptation to drive a destroyer from the Situation Room. Third, how alliance consultation is done in practice, as opposed to how it is described in communiqués. Fourth, how a report from a human source inside a foreign nuclear force travels from a frightened man in a provincial city to the desk of the President of the United States, and what happens to it on the way.

I will also tell you what I felt. A president who does not know his own fears will mistake them for analysis. I made that mistake more than once in the spring of 2045, and I want you to see where.

The phase I call the Quarantine ran from the third of March to the end of May. Nobody died in it except two Taiwanese merchant sailors, and I know how that sentence sounds. I ask you to keep those two men in mind as you read. Everything that came later, the blood in the Bashi Channel and the eleven minutes in November, grew out of choices made in these quiet-seeming weeks, when the military clock was still slow enough that we believed we had time.

We did have time. We did not use all of it well.

Friday, 3 March 2045 — 06:40 EST

The news came to me the way most bad news comes to a president: not with sirens but with a knock and a folder.

Megan Pruitt, my Chief of Staff, was standing in the doorway of the small study off the Treaty Room. I had been awake since five, writing in my diary as I have every morning and every night since I was twelve years old, and I had a cup of coffee going cold beside the notebook. Megan does not knock unless she has already decided that I need to put the pen down.

"Beijing has announced something," she said. "The Situation Room wants you to see the text before the networks start characterizing it. Sam's on his way in."

The folder held two pages. The first was a translation of a notice issued at 19:00 Beijing time,

about six hours earlier, jointly by the China Coast Guard and the Taiwan Affairs Office of the State Council. The second was a one-paragraph assessment from the Situation Room's senior duty officer, which said, in the flat prose the duty officers are trained to write, that the notice appeared to establish a "law-enforcement quarantine" of the island of Taiwan beginning at 16:00 Beijing time on 4 March, and that the People's Liberation Army Eastern Theater Command had simultaneously announced a large joint exercise called STRAIT THUNDER-2045.

I read the notice twice. I still have the translation, folded into the diary for that day, and I quote the operative sentences:

> "In accordance with the laws of the People's Republic of China, the China Coast Guard will conduct law-enforcement inspections of vessels bound for the ports of Kaohsiung, Taichung, and Keelung in the Taiwan region of China, for the purposes of preventing the entry of weapons, separatist materials, and goods that endanger national security. Vessels that accept inspection will be permitted to proceed. Vessels that refuse inspection will be dealt with according to law."

Dealt with according to law. I underlined the phrase. Then I underlined "law-enforcement" and "Taiwan region of China" and "permitted to proceed".

Tom, I asked myself, *what is this sentence designed to make you do?*

That question is the first thing I ask about any foreign statement in a crisis, and I commend it to you. Governments do not write notices like this one to inform. They write them to shape the choices of the people who read them. Every word in that notice had been argued over in Beijing for months. I was looking at a machine built to produce a specific response from the United States, and my job was to work out which response, and then decide whether to give it.

Sam Oyelaran arrived twenty minutes later with his tie in his pocket. He had been my National Security Advisor for four years and my friend for longer, and he has a habit, when he is thinking hard, of rubbing the bridge of his nose with two fingers, as though trying to press a thought into place. He did it now.

"It's elegant," he said. "I hate how elegant it is."

"Tell me why it's elegant."

"Because every word is chosen so that we're the ones who escalate. It's not a blockade. A blockade is an act of war under international law, and they know it. It's a 'quarantine', which is the word Kennedy used in 1962 precisely to avoid saying blockade. It's not the navy, it's the coast guard. It's not inspection of all ships, only ships bound for three ports. Ships that comply go through. So the message to the world is: this is a domestic police matter, conducted by police, under Chinese law, on Chinese territory, and if anyone shoots, it will be the Americans."

"And the exercise?"

"The exercise is the fist inside the glove. STRAIT THUNDER puts the Eastern Theater on a war footing without anyone declaring a war. If we push on the coast guard, the navy is already standing behind it."

I had the notice open on my knee. I had read every China Military Power Report the Pentagon had sent Congress since my time as governor, and I knew that analysts had written about this scenario for twenty years: a "quarantine" as the middle rung between gray-zone harassment and an outright

blockade or invasion. Knowing that a thing is possible is not the same as knowing what to do when it happens on a Friday morning.

"What do we know about why now?" I asked.

"The referendum bill," Sam said. "The Legislative Yuan is debating Lin's Status Referendum Act. They're saying, in effect: you want to ask your people what name to use at the United Nations? Then learn what it costs to depend on the sea."

"And their own politics?"

"Bad economy, young people without jobs, a leadership that has spent a decade telling the country Taiwan is the last unfinished business of the civil war. Wen can't be seen to do nothing. But he doesn't want a war either. This is the most he can do while still being able to stop."

That was Sam's first read, and it was mostly right. I wrote it down. Then I asked the question that I would ask in every meeting for the next three months.

"Which clock is moving fastest?"

He thought about it. "Right now? The information clock. In an hour the markets in Asia will be reading this as a blockade whatever the notice says, and every channel will be asking whether we'll send the Navy. The military clock doesn't start for another, what, twenty-one hours, when the quarantine takes effect."

"And the political clock?"

"Hasn't started. Congress is out until Monday."

There it was. The Three Clocks Model, which I described in Chapter 1, rests on a simple observation: crises become dangerous when the three clocks desynchronize. Here the information clock was already racing, the military clock would begin in less than a day, and the political clock, the one that governs consultation with Congress and allies and the public, had not even been wound. My first task was obvious. I needed to slow the information clock, start the political clock now, and make sure that when the military clock began ticking at 16:00 Beijing time, Washington was not still arguing about vocabulary.

I asked Megan to have the Press Secretary say, at the morning briefing, three sentences and no more: that we had seen the announcement, that we were consulting closely with our allies and partners, and that the United States remained committed to the freedom of navigation in international waters and to peace and stability across the Taiwan Strait. Each phrase was old. That was the point. When the information clock is racing, the most useful thing a government can do is say something boring. Boredom is a signal that the adults are working.

Then I asked Sam to convene a Deputies Committee meeting for that afternoon and to schedule a full National Security Council in the Situation Room for Saturday morning, after the quarantine took effect and we could see what it actually was.

"We'll know a lot more after the first boarding," I said. "I don't want to decide what this is before they show us."

That was a reasonable instinct. It was also, I think now, a small error, and I will return to it.

Friday, 3 March 2045 — 23:52 EST

From my diary, written that night in the residence:

> *Beijing has chosen the word "quarantine". It is our word. Kennedy's word. They are holding up a mirror and daring us to say it means something different when they use it.*
>
> *Sam says it is elegant. I think it is engineered. Every parameter is tuned to the threshold just below where we would be forced to respond with force: coast guard not navy, three ports not all, inspection not seizure. A control engineer would call it operating at the margin of stability. The trouble with operating at the margin is that small disturbances are amplified. A nervous captain, a radio message misheard, a crew that refuses to stop. The system does not care about intentions.*
>
> *Claire asked at dinner whether Maya's ship is near Taiwan. I told her the truth, that I do not know exactly where Inouye is tonight, and that I will not ask, because the moment I ask is the moment every admiral between here and Yokosuka knows the President is tracking his daughter's ship. She looked at me for a long time. Then she said, "You're allowed to be her father, Tom." I said I know. I don't think I do know.*
>
> *Tomorrow at 16:00 Beijing time, 03:00 here, the first Chinese coast guard cutter will try to board the first ship. I have asked to be woken if there is a boarding. I have asked to be woken if there is anything. I will probably not sleep anyway.*

I did sleep, for about two hours.

Saturday, 4 March 2045 — 03:10 EST

The phone beside the bed in the residence is a plain beige thing that looks as if it were installed in 1985, which it may have been. When it rings at night it is always the Situation Room, and the voice is always calm. The calm is deliberate. The duty officers are taught that the president's first five seconds of awareness set the tone for what follows, and so they speak as though reading a weather report.

"Mr. President, this is the senior duty officer. At 02:41 Eastern, China Coast Guard vessels conducted the first boarding under the quarantine, of a Panamanian-flagged container ship bound for Kaohsiung. The ship complied. At 02:55, the Eastern Theater Command announced live-fire zones in six areas around Taiwan as part of STRAIT THUNDER, effective immediately. Dr. Oyelaran is en route. He recommends you join the principals in the Situation Room at 04:00."

"I'll be there at 03:45."

Claire did not wake, or pretended not to. I dressed in the dark, a sweater over a shirt with no tie, which my staff later told me was taken as a sign by some of the principals that I wanted them relaxed and by others that I was rattled. You will discover that everything you wear in a crisis will be read as a signal. Wear the same thing every time and they will stop reading it.

Lieutenant Colonel Jamal Brooks was waiting at the bottom of the stairs. He was the military aide on duty that night, carrying the satchel that the press calls the football and that I described at length in Chapter 1. Brooks is a Marine, tall, from Baltimore, with the stillness that certain combat officers have, as though they are conserving something. He fell in a step behind my left shoulder, as the aides always do.

"Morning, sir."

"Good morning, Colonel. I suppose it is morning."

"Somewhere it's afternoon, sir. That's the problem."

I laughed, and the laugh did me good. Thank them. They will not expect it, and you will need them.

Saturday, 4 March 2045 — 04:00 EST: The First Situation Room Meeting

The Situation Room complex was rebuilt in the early 2020s and again, more modestly, in the late 2030s, and it no longer smells of old carpet and cold coffee as the old one did in photographs. It smells of nothing, which I found worse. The main conference room has a long table, screens on the walls, and a seal at the end that nobody looks at. When you sit at the head of the table you are facing the screens, and the people on the screens are facing you, and there is a moment, every time, when you feel the weight of all those faces waiting.

Present in the room: Vice President Rosalind Achterberg; Sam; Secretary of State Helena Marsh; Secretary of Defense Ray Castillo; General Patricia Dunmore, Chairman of the Joint Chiefs; Director of National Intelligence Abigail Fenn; CIA Director Gregory Laszlo; Treasury Secretary Daniel Iwu; Megan Pruitt. On the screen from Camp H. M. Smith in Hawaii was Admiral Kenji Moriyama, Commander of United States Indo-Pacific Command, INDOPACOM. A second screen showed a map of the Taiwan Strait with the six live-fire zones drawn in red, the quarantine's declared inspection areas in orange, and the positions of known Chinese and American naval vessels as small icons that I had learned, over four years, never to trust as precisely as they appeared.

I asked Abigail Fenn to begin, as I always do. The intelligence brief comes first because the people in the room need to share a single picture of the facts before they begin arguing about what to do. If you let the Secretary of Defense speak before the DNI, you will find that the facts have been quietly arranged to support his recommendation. This is not dishonesty. It is human nature.

Abigail is a precise woman, a former Senate Intelligence Committee staff director, and she speaks in numbered points.

"One. As of 03:50, China Coast Guard has conducted three boardings, all of foreign-flagged commercial vessels bound for Kaohsiung and Taichung, all compliant. Boarding teams are armed, small arms only. Inspections lasted between forty and seventy minutes. Two. We count fourteen coast guard cutters on station, including two of the twelve-thousand-ton class, which are larger than our destroyers. They are supported by maritime militia vessels, perhaps forty, fishing boats in name. Three. The PLA Navy has about thirty surface combatants at sea in the Eastern Theater, including the carrier group of the *Fujian*, which is east of Taiwan. Four. STRAIT THUNDER's live-fire zones overlap major shipping lanes. Commercial traffic is already diverting. Lloyd's market war-risk premiums for the Taiwan Strait have, as of an hour ago, been quoted at levels not seen since the 2027 exercises. Five. We see no indications of preparations for an amphibious assault. No mass loading of landing craft, no mobilization of reserves at scale, no stockpiling of blood. Our assessment, at moderate confidence, is that this is coercion, not the opening phase of an invasion."

"And the strategic forces?" I asked. I always asked, and I want you to always ask. A conventional crisis between nuclear powers is never only conventional.

"No change in the posture of the PLA Rocket Force that we can see," Abigail said. "ARGUS shows road-mobile launchers in garrison at their normal rates. The silo fields at Yumen, Hami, and Ordos look routine."

I noted the words *that we can see*. ARGUS, the surveillance fusion I described in Chapter 1, gave us something close to continuous custody of every Chinese road-mobile launcher outside its garrison. I had been proud of it. I had fought for its budget. But it watched vehicles, not decisions, and a decision to raise readiness can be made in a room, by a man, long before a single vehicle moves.

Gregory Laszlo spoke next. He is a tall man with a long face, a career intelligence officer who rose through the clandestine service and has the habit of saying less than he knows. "We have tasked our sources," he said. "I'll have more for you in days, not hours."

I knew what he meant. We had people inside the Chinese system, and the most important of them were not people you could telephone. I will describe in a later section how one of them reached us. For now, it is enough to say that Greg was telling me, in the language of the Situation Room, that human intelligence works on the political clock, not the military one. A source who must wait for a safe moment to transmit cannot be rushed. Push him and you lose him.

Then I asked Helena Marsh to tell us what the word meant.

Module 2.1 — Quarantine and Blockade: What the Law Says, and Why It Matters

Helena had served as ambassador in Beijing for three years and speaks Mandarin well enough to argue about poetry in it. She is also, by training and temperament, a lawyer, and she gave us a lesson that morning that I will reproduce here, because every president needs it.

"Mr. President, under the law of naval warfare, a blockade is a belligerent act," she said. "It is an operation to prevent vessels of all states from entering or leaving the ports or coasts of an enemy. The rules were written down in the nineteenth century, in the Declaration of Paris in 1856 and the London Declaration of 1909, which never entered into force but is treated as a statement of custom, and most recently in the San Remo Manual of 1994, a restatement of the law of armed conflict at sea drafted by international lawyers and naval experts. Under those rules, a blockade must be declared, must be notified to neutral states, must be effective, meaning actually enforced and not just announced, and must be applied impartially to ships of all flags. And critically, it presupposes an armed conflict. You cannot blockade in peacetime. A blockade is war."

"Which is why Kennedy called his a quarantine," Achterberg said.

"Exactly. In October 1962 the Kennedy administration's lawyers argued that a blockade would be an act of war against Cuba and the Soviet Union, so they designed what they called a 'defensive quarantine' of offensive weapons only, authorized by a resolution of the Organization of American States. It was a legal fig leaf, and everyone knew it, but fig leaves matter. They let the other side pretend they are not being forced to choose between war and humiliation."

"And this one?"

"This one is a different fig leaf. Beijing's argument is not that it has an international authorization. Its argument is that Taiwan is part of China, so the waters around it are Chinese, and therefore the coast guard is enforcing domestic law against ships entering Chinese ports. On that reading, it is

not a quarantine in the international sense at all. It's customs enforcement. It's the equivalent of our Coast Guard inspecting a ship entering Miami."

"Except Taiwan isn't Miami."

"No. Our position since 1979, under the Taiwan Relations Act and the three joint communiqués, is that we acknowledge Beijing's position that there is one China and Taiwan is part of it, but we do not endorse it. We have never accepted that the PRC has jurisdiction over Taiwan. So under our view, those coast guard boarding teams are boarding foreign ships, outside Chinese jurisdiction, on their way to ports that the PRC does not control. Most of the inspections are taking place more than twelve nautical miles from Taiwan's coast, which means outside Taiwan's territorial sea, in what the United Nations Convention on the Law of the Sea, UNCLOS, would treat as the exclusive economic zone or the high seas. On the high seas, a ship is subject only to the jurisdiction of its flag state, with narrow exceptions: piracy, the slave trade, unauthorized broadcasting, ships without nationality, and certain treaty-based arrangements. None apply."

Ray Castillo, who has the bluntness of a Marine who reached four stars, leaned forward. "So it's illegal."

"Under our view of the law, it's an unlawful interference with navigation. Under their view, it's lawful enforcement. And here is the trap, Ray." Helena turned to face him. "If we call it a blockade, we are saying it is an act of war. That has consequences. It may trigger obligations, it raises insurance and commercial panic, and it tells the Chinese that we consider ourselves already in a war, which licenses them to treat our ships as belligerents. If we call it a quarantine, we are using Kennedy's word and implicitly accepting that this is something short of war, which is what they want. If we call it law enforcement, we are accepting their claim of jurisdiction over Taiwan, which we can never do."

"So what do we call it?" I asked.

"Something new," Helena said. "Something that denies their jurisdiction without declaring a war. I'd suggest 'an unlawful interference with international shipping'. It's ugly. It's meant to be."

I have spent many hours since thinking about that moment, and I want to set down the principle it taught me, because I believe it is general.

> **OPERATING NOTE — The War of Words Is the First Battle.** In the first hours of a crisis, your adversary will offer you a vocabulary. Every term in it has been chosen to make his action seem lawful and your response seem aggressive. Do not adopt his words by default, and do not reach automatically for the most dramatic word of your own. Ask your Secretary of State and your lawyers, specifically: What does each available word commit us to legally? What does it signal to the adversary? What does it tell markets and allies? Choose deliberately and then hold the line in every statement by every official. A government that uses three different words for the same act in its first week has told the world it does not yet know what it thinks. Write the agreed term on a card and give it to everyone who will speak in public.

We chose Helena's phrase. "Unlawful interference with international shipping" appeared in every official statement for the next three months. Reporters hated it. The Chinese Foreign Ministry spokesman mocked it. But it held a precise position: that the act was illegal, that it was not a war, and that Taiwan was not China's to police. It also left us free to escalate our characterization later without having spent our strongest word on the first day.

Saturday, 4 March 2045 — 04:40 EST: The Options Nobody Liked

Then we turned to the question that every Situation Room meeting eventually reaches. What do we do?

I asked General Dunmore and Admiral Moriyama to describe the military situation first, not their recommendations. Pat Dunmore is an Air Force officer, a former bomber pilot and then a space and missile commander, with a dry voice and a habit of saying "candidly" before her most important sentences.

"Candidly, Mr. President, the force posture in the region is adequate for presence and not for a fight," she said. "Seventh Fleet has the *George Washington* carrier strike group in port at Yokosuka, ready to sail in seventy-two hours. We have six destroyers at sea in the Western Pacific, including two in the Philippine Sea, plus the Marine littoral regiment in the southern Ryukyus and northern Philippines. We have the air wings in Japan and Guam. If you want to put a carrier group east of Taiwan, Admiral Moriyama can do it within a week. If you want to fight, you need months, and you need to understand that every American ship inside the first island chain is within range of thousands of Chinese missiles."

Moriyama, on the screen from Hawaii, nodded. He is a thoughtful officer, the son of a Japanese American family from Honolulu, and he has the habit of laying out options in pairs. "Sir, I'd frame the immediate choices as two pairs. First pair: do we change our naval posture now, or wait to see how the quarantine is enforced? Second pair: if we change it, do we move toward the Strait, which is presence, or into it, which is challenge? I would recommend moving toward, not into, and not this week."

Castillo disagreed. "Kenji, with respect, if we don't challenge it early, we accept it. Every day this goes on, it becomes normal. Remember the islands in the South China Sea. We watched them build, and by the time we decided it mattered, there were runways."

"With respect, Mr. Secretary, the islands weren't defended by a carrier group and a rocket force on alert."

"Neither is this, yet."

I let them argue for a while. I have learned to do that. Ray's assumption was that time favored China, that every day of acceptance hardened the new status quo. Kenji's assumption was that time favored us, that a slow, deliberate build-up was safer than a fast one. Both were partly right. The question was which effect would dominate.

Then I drew on the pad in front of me, as I always do. My staff learned to expect it, and some of them later admitted that they found it comforting. I drew the Reversibility Matrix I described in Chapter 1: a square divided in four, reversible and irreversible across the top, fast and slow down the side. And I asked them, one by one, to place each option in it.

Moving the carrier group from Yokosuka toward the Philippine Sea: reversible, fast. We could do it and undo it with no loss but some fuel and some face.

Sending American warships to escort ships through the quarantine: partly reversible. Once our ships were in the inspection zones, a confrontation with the coast guard could happen in an instant, and a single death could not be undone.

Imposing sanctions: reversible in principle, but in practice slow to unwind, because sanctions build

constituencies and markets react to their announcement, not their removal.

Putting American forces on a higher alert: reversible, but highly visible, and dangerous if the other side read it as preparation for war.

Doing nothing: reversible, in the narrow sense, but with a cost that accumulated invisibly, as Ray said, in the slow acceptance of a new normal.

"So the rule is," I said, when we had filled the square, "take the reversible decisions fast and the irreversible ones slowly. I'm going to make the reversible ones now."

I ordered the *George Washington* group to sail from Yokosuka and take up station in the Philippine Sea, east of Taiwan but well outside the live-fire zones. I asked Admiral Moriyama to increase surveillance flights in international airspace, with strict instructions on minimum distances from Chinese aircraft. I asked Helena to begin calls to every ally and partner in the region before the end of the day. I asked Daniel Iwu to have Treasury prepare, but not announce, a menu of sanctions options. And I asked Abigail and Greg to tell me within seventy-two hours whether anything in the Chinese strategic forces had changed.

I did not order escorts. I said we would decide that question after we had consulted our allies and seen how the quarantine was enforced over its first week.

Here is the error I mentioned. In delaying the escort decision, I told myself that I was keeping an irreversible decision slow, which was correct. But I was also letting the Chinese set the facts on the water for a full week before we had even decided our own principles, and I did not, in that first meeting, give the Navy a clear instruction about what their ships should do if a coast guard cutter tried to board an American-flagged vessel. We had standing rules of engagement, of course, but they were written for general conditions. What we needed, and did not issue until 7 March, was a specific supplemental instruction for this situation. For three days, the commanding officers in the area were guessing what Washington wanted. That they guessed correctly is to their credit, not mine.

> **OPERATING NOTE — Decide the Principle Before the Policy.** You can delay a policy decision. You cannot delay telling the people at the sharp end what principle guides them while you decide. In the first meeting, even if you defer every major choice, issue a short written instruction to the theater commander covering the one situation most likely to arise before you decide: in this case, a foreign official attempting to board a ship flying the American flag. The answer can be cautious. It must exist. Commanders who do not know what the President wants will fill the gap with their own judgment, and you will learn what that judgment was only afterward.

Saturday, 4 March 2045 — 09:30 EST: Taipei

The first call I made that morning was to Taipei, and I want to explain why, because the order of calls in a crisis is itself a message, and successors often get it wrong.

The conventional sequence is treaty allies first, in order of importance, then partners. On that logic I would have called Tokyo first. But the people being coerced were in Taiwan, and the first question in every capital in Asia would be whether the Americans had spoken to the Taiwanese before speaking about them. If I called Tokyo first, Taipei would learn of it within hours and conclude, rightly or wrongly, that it was being discussed as an object. So I called President Lin first. Then I called

Prime Minister Saeki in Tokyo, President Villanueva in Manila, and Prime Minister Holloway in Canberra, in that order, over the following eight hours.

You should know that the United States does not have formal diplomatic relations with Taiwan. Since 1979 the relationship has been conducted through the American Institute in Taiwan, a nominally private nonprofit that functions as an embassy without the name. Presidents have historically been cautious about speaking directly to Taiwan's leaders, because Beijing reads every such contact as a step toward recognition. By 2045 that caution had eroded. My predecessors had taken calls; I had taken two before the crisis. But each call still had a cost in Beijing, and I made this one knowing that the Chinese would learn of it, and intending that they should. The message was simple: we will not let you isolate them.

Lin Shu-wen was then fifty-four, a former law professor and legislator, small, quick, and more combative than her predecessors. She had won the January 2044 election on a promise to "stand straight", and the Status Referendum Act was the most visible form of that promise. She was, I believed then and believe now, a person of real courage. She was also, from the perspective of an American president trying to prevent a war, a complication.

The call was on a secure line routed through the Institute. She spoke excellent English.

"Mr. President, thank you for calling. I want you to know first that Taiwan will not be intimidated."

"I know that, Madam President. I'm calling to tell you that you are not alone, and to ask what you need."

She had a list, and it was a good one. Fuel, first. Taiwan imports almost all of its energy, and its liquefied natural gas reserves, which had been a known weakness for decades, covered less than two weeks of consumption despite expansions in the 2030s. Coal and oil reserves were longer, but not by much. Food, second; Taiwan is less dependent on imports than people think for staples, but not for feed grain and much of its protein. Medicine, third. And fourth, a public statement from the United States that the quarantine was illegal.

"You'll have the statement today," I said. "On the rest, we're working on options with our allies. I need to ask you something in return."

"Please."

"I need you not to do anything in the next two weeks that changes the political facts. I'm not asking you to withdraw the referendum bill. I'm asking you not to accelerate it."

There was a pause of several seconds on the line. When she spoke, her voice was cooler.

"Mr. President, I am being told by Beijing that my people may not decide their own name. Now I am being told the same thing by Washington."

"No," I said. "You're being told by Washington that if we are going to stand with you in the water, we need you to give us time to get there. That's different."

Another pause. "I understand the difference. I am not sure my people will."

"Then help me explain it to them."

She agreed, with reluctance, to let the bill sit in committee. I do not know whether she would have done so without the call. I know that the call cost me something with her, and that I would spend that account down further in the months to come, until by the end of the crisis I had asked more of

her than I had any right to. I will tell you that story in Chapter 7. For now, the lesson is this: the person being coerced is never only a victim. She is also a political actor with her own clocks and her own constituents, and your interests and hers are aligned only in part. You must say so honestly, early, while you still have credit.

Module 2.2 — How Alliance Consultation Actually Works

The communiqués make consultation sound like a meeting of equals who share a single view. It is not. It is a sequence of private conversations in which each party learns how far the others will go, and in which the strongest party, which in the Pacific is the United States, must constantly resist the temptation to inform rather than consult.

There are three formal treaties in the region that mattered in this crisis, and a successor should know them by heart.

The Treaty of Mutual Cooperation and Security between the United States and Japan, signed in 1960, commits the United States, under Article V, to act against an armed attack on "the territories under the administration of Japan". That includes the Senkaku Islands and the southern Ryukyus, which are a little over a hundred kilometers from Taiwan. It does not include Taiwan. But Japan hosts the Seventh Fleet at Yokosuka, the Marine forces on Okinawa, and the air bases at Kadena and Misawa, and under Article VI and the 1960 exchange of notes, major changes in the deployment of US forces from Japan, or the use of bases in Japan for combat operations outside Japan, are subjects of "prior consultation" with the Japanese government. In practice this means that if I wanted to fight from Japanese bases, Tokyo had a voice, and arguably a veto.

The Mutual Defense Treaty with the Philippines, signed in 1951, commits each party to act "in accordance with its constitutional processes" against an armed attack in the Pacific area on the other. The bases opened under the Enhanced Defense Cooperation Agreement of 2014, expanded several times since, include sites in northern Luzon and the Batanes Islands, a few hundred kilometers from Taiwan. The Philippines lies on the other side of the Bashi Channel, the main southern gateway to the Strait.

The ANZUS Treaty of 1951 with Australia, a more general commitment to consult and to act against common danger, was reinforced by the AUKUS partnership of 2021, under which Australia was acquiring nuclear-powered submarines and hosting rotational US forces in the north.

None of these treaties obliged any of our allies to help us escort ships to Taiwan. That is the first thing to understand. Our allies would decide, each on its own terms, based on its own fear of China, its own domestic politics, and its own judgment of whether we would stay the course. And each of them was asking, silently, a question that no ally will ever ask a president directly: *If we stand with you, and you lose your nerve, what happens to us?*

Tokyo

Hiroshi Saeki was in his second year as Prime Minister, a former Defense Minister from the conservative mainstream of the Liberal Democratic Party, with a reputation for caution and a deep, private alarm about China. We spoke at 11:15 Eastern, 01:15 the next morning in Tokyo. He had been awake all night.

He said the right public things quickly. Then he said, through the interpreter, something I wrote down verbatim:

“Mr. President, Japan will do what is necessary. But I must tell you frankly. If an American ship escorting cargo to Taiwan comes into conflict with the Chinese coast guard, and if your ship sailed from Yokosuka, then China will say that Japan is a party. My Diet will ask whether we were consulted. I must be able to say yes. And I must be able to say that we consulted before, not after.”

“You will be consulted before,” I said. “I give you my word.”

“Then I will ask the Self-Defense Forces to prepare for cooperation in surveillance and logistics. On escort, I must have more time.”

That was an honest answer. Japan’s 2015 security legislation permitted collective self-defense when Japan’s survival was threatened, and later reinterpretations had broadened this, but escorting cargo to Taiwan sat at the edge of what the law and the public would accept. Saeki was telling me that I could have Japan’s help, but only if I let Japan move at the speed of Japanese politics.

Manila

President Andrés Villanueva was a former provincial governor and senator, a populist with a talent for television and a shrewd sense of where power lay. I liked him. He was the most frightened of the three, and with reason: the Philippines lay closest to the fighting that might come, its northern islands were already full of Chinese maritime militia, and it had the weakest military.

“Tom,” he said, he always called me Tom, “if there is a war, it happens on my islands. Your Marines in Batanes, your missiles in Luzon. China already sends me messages every week. They say, Andrés, you are letting the Americans turn your country into a target. And my people listen.”

“What do you need from me, Andrés?”

“I need you to promise that nothing happens from my bases without my knowledge.”

“You have that promise. Under the treaty and the agreements, you already have it.”

“I know what the paper says. I am asking for your promise.”

I gave it. And I want you to understand that it was not a small thing. A president’s personal word to a foreign leader is a currency, and it is limited. I spent it that morning, and I made sure that every officer in INDOPACOM knew I had spent it, because if they broke it, I could never earn it back.

Canberra

Grace Holloway was the most direct of the three. A former trade minister from Western Australia, she had come to office warning about China’s economic coercion of her country in the early 2020s, when Beijing had imposed punitive tariffs on Australian barley, wine, and coal. She had no illusions.

“We’ll be with you, Prime Minister to President,” she said. “But I’d like to know what ‘with you’ means before my Cabinet meets on Monday. Are we talking about a statement? Ships? Aircraft for surveillance? The submarines aren’t ready, you know that.”

“I don’t know yet, Grace. I’m asking each ally what they can do before I decide what we will do. I don’t want to design an operation and then ask you to fit into it.”

“That’s a nice change,” she said, and laughed, and I understood that some predecessor had done exactly that to her.

OPERATING NOTE — Consult Before You Decide, or Don't Call It Consultation. The test of real alliance consultation is whether the ally's answer can change your decision. If it cannot, you are informing, and the ally will know. Call allies before the decision, in an order that shows you understand who is most exposed. Ask what they can do before you tell them what you will do. Give each leader something they can say to their own legislature: that they were consulted first, that nothing will be done from their territory without their knowledge. Write down every promise you make on these calls, because you will be held to each one, and keep the list in front of you when you finally decide.

Thursday, 9 March 2045 — 01:20 EST: The *Formosa Pearl*

For five days, the quarantine worked as Beijing intended. Most ships complied. Some diverted. Shipping companies, unwilling to risk crews or cargo, began refusing Taiwan bookings, and the war-risk premiums climbed until some insurers stopped quoting at all. Taiwan's ports did not close, but they slowed, and the slowing was the point.

Then, on the fifth day, a Taiwanese ship refused.

The *Formosa Pearl* was an aging general cargo vessel of about eight thousand tons, flagged in Taiwan, owned by a small Kaohsiung company, carrying medical supplies, canned food, and diesel generators from Manila. Her master was a sixty-one-year-old named Chen, a veteran of forty years at sea. When a China Coast Guard cutter hailed him at 14:05 local time, about thirty nautical miles southwest of Kaohsiung, and ordered him to stop for inspection, he replied, on an open radio channel that was recorded by at least three other ships, that his vessel was registered in the Republic of China, sailing to a port of the Republic of China, and would not submit to inspection by a foreign power.

What happened next was the subject of a long investigation, and I can only summarize what we concluded. The cutter, one of the larger classes, maneuvered to block *Formosa Pearl's* course. Chen held his course, which under the international rules of the road he was arguably entitled to do. The cutter crossed his bow at close range. At 14:20, the two ships collided. The Chinese said the *Pearl* rammed them. Video from a nearby container ship, which we obtained within the hour, showed the cutter turning into the *Pearl's* port bow. The cutter was much larger and built with a reinforced hull. The *Pearl's* bow was crushed below the waterline, she took on water, and two of her crew, a deckhand named Lai and an engineer named Hsu, both young men, were killed in the forward compartment.

The *Pearl* did not sink. She was towed into Kaohsiung by Taiwanese coast guard vessels the next day. The cutter's crew did not board her.

I was woken at 01:30 Eastern. I sat on the edge of the bed with the phone in my hand, and I did not turn on the light. Claire put her hand on my back. I listened to the duty officer read the facts, and I remember feeling something I had not expected. It was not anger. It was a kind of recognition. *There it is,* I thought. *The disturbance at the margin.*

I had written it in my diary five nights earlier. A nervous captain, a crew that refuses to stop. The system does not care about intentions. I did not believe Beijing wanted those two men dead. I did not believe the cutter's captain had been ordered to kill them. I believed he had been ordered to stop the ship, and that a proud old Taiwanese master had refused, and that the physics of two hulls

in close water had done the rest.

That made it worse, not better. A system that kills by accident is more dangerous than one that kills by design, because you cannot negotiate with an accident.

The Morning After

At 07:00, the principals met again. The mood had changed. Ray Castillo opened by saying that this was exactly what he had feared, that the Chinese were testing us and we were failing the test.

"Two dead, Mr. President. If we don't respond, the next ship that refuses will be sunk, and after that no ship will refuse."

Helena disagreed, carefully. "Ray, the Chinese didn't plan this. Look at their Foreign Ministry statement. It's defensive. They're claiming the *Pearl* rammed the cutter. You don't make up a story like that if you meant to do it. They're embarrassed."

"Embarrassment isn't a policy."

"No. But it's an opportunity. If they're embarrassed, we can make them more careful."

Pat Dunmore spoke for the military. "Candidly, sir, I'd separate two questions. One is how we respond to this incident. The other is whether we change our posture to prevent the next one. The first is about signaling. The second is about protection. The worst outcome is that we do something dramatic in response to the first and nothing useful on the second."

That was exactly right, and I said so. We agreed on the first question quickly. I made a statement from the Rose Garden at noon, the first time I had spoken on camera since the quarantine began. I named the two dead men, Lai and Hsu. I called the collision "the predictable consequence of an unlawful interference with international shipping". I said the United States would work with Taiwan and with our allies to ensure that humanitarian supplies, food, fuel, and medicine, could reach the people of Taiwan without hindrance. And I said that we held the government of the People's Republic responsible for the conduct of its vessels.

I did not threaten. I wanted Beijing to hear that I had noticed their embarrassment and would not exploit it to corner them. And I wanted them to hear the phrase "humanitarian supplies", which was the first public signal of the decision I had not yet made.

The second question, protection, was harder. It took me until April.

Friday, 10 March 2045 — 22:15 EST

A message from Maya arrived that night, through the family email account that the White House Communications Agency maintains on a secure system for the president's immediate family. She had written it, I learned later, from her stateroom on the *Inouye*, which was then in the Philippine Sea with the *George Washington* group.

> *Dad — I can't say where we are, which you know, and I won't ask what you're going to do, which you also know. I just want you to know that everyone here watched the video of the Pearl. A lot of the junior officers are angry. I'm not sure I am. I keep thinking about the guy on the coast guard cutter who was driving, and how he's probably twenty-four and terrified, same as us. I don't know what that means. I guess it means please be careful with all of us, theirs too. Love you. Tell Mom I'm eating vegetables. M.*

I read it three times. Then I gave it to Claire, who read it once and put it down and went into the bathroom and closed the door. I sat on the edge of the bed and thought about the word *careful*, and about the fact that my daughter, a lieutenant who stood watches as a tactical action officer on a guided-missile destroyer, had arrived independently at the Mirror Model I had spent twenty years developing. *What does it look like from the bridge of the other ship?*

I wrote back two lines: *I will be careful with all of you. I'm proud of you. Dad.*

I did not tell her that her ship was on a list of vessels being considered for the escort mission. I did not tell her that I had already decided I would not remove it from the list.

Sunday, 12 March 2045 — 16:40 EDT: LANTERN Speaks

The clocks changed that morning. Daylight saving time began at 02:00, and Beijing, which had been thirteen hours ahead of Washington, was now twelve. I mention it because for the rest of the day half the timestamps in the cable traffic were wrong by an hour, and a watch officer in the National Military Command Center spent the afternoon correcting them by hand. Insist on a single time standard for every document that reaches you, with the zone written out.

At 16:40, Greg Laszlo asked to see me alone. He came to the Oval Office with Ruth Calder, the chief of the CIA's China Mission Center. I will use the name the Agency uses for her in this edition; her real one remains withheld. She is in her fifties, gray-haired, with the look of a schoolteacher and the patience of a sniper. She had spent most of her career in Asia and had recruited, in 2033, the source we called LANTERN.

I described in Chapter 1, in general terms, how the Central Intelligence Agency runs human sources. Here is how one report actually reached me.

LANTERN was a senior officer on the staff of PLA Rocket Force Base 64, headquartered at Lanzhou, which commands the missile brigades responsible for, among other things, the silo field near Hami in Xinjiang. I knew his position. I did not then know his name; very few people did, and I did not ask. I knew why he had come to us: not for money, though we paid him, but because in the course of his duties he had seen the war plans, and the war plans frightened him. He believed, Ruth told me once, that the only thing more dangerous than an enemy who knows too much is one who knows too little.

He communicated in two ways. For routine material, there were dead drops, hiding places where documents or memory cards could be left and later collected by officers working under cover, a method as old as espionage and still, in an age of total surveillance, among the safest. For urgent material, he had a device disguised inside a commercial vehicle navigation unit of a type sold by the millions in China. At a moment of his choosing, it could compress a short message, encrypt it, and transmit it in a burst lasting a fraction of a second to a satellite. I will not describe it further, and neither should you, ever, to anyone who does not need to know.

The burst arrived at a CIA facility at 03:12 Eastern that morning. It was decrypted automatically and placed in a restricted queue. A duty officer cleared for the compartment, which I will call by the cover term the Agency used in this edition, [REDACTED], read it, recognized its priority, and woke Ruth. Ruth read it, checked it against LANTERN's previous reporting and against the indicators from ARGUS and signals intelligence, and called Greg at 05:00. They spent the morning doing what the Agency calls "source validation": asking whether the message could have been sent under duress, whether its content was consistent with what LANTERN could plausibly know, whether any other source contradicted it. Only when they were satisfied did Greg call Sam and ask for time with me.

This is the right process, and it took thirteen hours. A successor must understand that this lag is the price of reliability. If you demand that raw human reporting come to you instantly, you will get it, and you will also get the fabrications, the misunderstandings, and the planted stories that careful officers exist to catch. The burst device was built for speed. The institution around it was built for truth. You need both.

Ruth handed me a single sheet. The report itself was eleven lines. Its core was this:

> *Base 64 ordered at 0900 local 11 March to Readiness Level 3. Order from PLARF headquarters, citing STRAIT THUNDER. Brigades to increase manning, test communications, verify warhead storage security. No order to move launchers. Officers told this is precautionary. First RL3 order at this base since 2034. Mood among staff: serious, not alarmed.*

The PLA's readiness scale is not public, and I will use the simplified version our analysts used. Level 4 is peacetime. Level 3 is heightened readiness. Level 2 is preparation for launch, which includes mating warheads to missiles. Level 1 is launch. Moving from 4 to 3 was not a threat. It was what a prudent military does when its country is conducting a major exercise near a potential enemy. We had our own equivalents.

But it was the first time in eleven years. And I noticed something else, the thing I always look for: it was invisible. ARGUS had seen nothing, because there was nothing to see. No launcher had moved. The change was in rosters and procedures inside buildings. If LANTERN had not told us, we would not have known.

"What does this mean, Ruth?"

"It means they're thinking about us, sir. Not planning to use anything. Thinking about whether we might."

"The Mirror Model," Greg said.

"Yes. What do they see when they look at us?" I turned to Ruth. "What do they see?"

She considered. "They see a carrier group moving toward the Strait. They see us talking about convoys. And they see Golden Dome, and ARGUS, and every one of our new interceptor launches. The Rocket Force has spent fifteen years arguing internally that we're building the ability to strike first and catch whatever survives. They raised readiness because they always raise readiness when they think we might be coming."

I asked how we could tell them we were not coming without telling them what we knew.

"That," Ruth said, "is your job, sir. Mine is to keep him alive."

> **OPERATING NOTE — How to Read a Spy.** A human source inside an adversary's nuclear forces is the rarest asset a president will ever have, because he can tell you what satellites cannot: what was ordered, by whom, and how the people receiving the order felt. Treat his reports as precious and as fallible. Ask three questions of every one: How does he know this? What would he not know? What is he afraid of? Never ask for a report faster than the institution can validate it. Never take an action that only his reporting could justify, unless you are prepared to lose him. And remember that you are not the only one reading the signals. Every change in your posture triggered by his warnings is also a clue to the adversary's counterintelligence service that someone is warning you.

That last sentence I wrote in 2046, after I had learned it in the worst possible way. In March 2045, I took the report and did the one thing I could do with it that would not endanger LANTERN. I decided not to change anything about our strategic forces. No bomber movements, no new submarine sailings beyond the normal schedule, no visible alert changes. If they had raised readiness out of fear, I would not feed the fear.

Module 2.3 — The Convoy Decision

For three weeks after the *Formosa Pearl*, the question of escorts dominated every meeting. I will compress those weeks, but I want you to see the structure of the decision, because it is a model for how to make an irreversible choice slowly.

How Naval Escort Works

First, the mechanics, which many politicians do not understand.

A naval escort is an arrangement in which warships accompany merchant ships to protect them. It is old: convoys crossed the Atlantic in both world wars. The modern American precedent most relevant to us was Operation EARNEST WILL in 1987 and 1988, when the Navy escorted Kuwaiti tankers through the Persian Gulf during the Iran–Iraq War. Those tankers were "reflagged", registered under the American flag, which gave the United States legal standing to protect them as American ships. EARNEST WILL worked, but it was costly: the frigate *Samuel B. Roberts* struck a mine, and the cruiser *Vincennes* shot down an Iranian airliner by mistake, killing 290 people. Every officer in the Navy knows that history. So should you.

Under international law, a warship is a sovereign instrument of its state, immune from boarding or interference by any other state. A merchant ship sailing under escort enjoys a kind of borrowed protection: the escorting warship can, under the law of the sea and our own rules, defend vessels of its own flag from unlawful interference. Protecting ships of other flags is more legally complicated, and requires either the consent of their flag state or a broader legal basis.

The chain of command for an escort runs from the President to the Secretary of Defense to the combatant commander, here Admiral Moriyama at INDOPACOM, then through the Seventh Fleet commander to the escort commander at sea, usually the commodore of a destroyer squadron, and then to individual commanding officers. The president approves the mission and the supplemental rules of engagement, the ROE. The rules define when a commander may use force. Under our standing rules, every commander always retains the inherent right and obligation of self-defense against a hostile act or demonstrated hostile intent. Supplemental rules can restrict or expand what is permitted beyond that, for instance, whether a destroyer may use force to prevent a coast guard cutter from boarding a merchant ship it is escorting.

Finally, there were rules for unplanned encounters. The Code for Unplanned Encounters at Sea, CUES, adopted at the Western Pacific Naval Symposium in 2014 and signed by China, sets out safe speeds, distances, and radio procedures for naval vessels meeting unexpectedly. It applies to navies, not coast guards. That gap mattered.

The Options

By the end of March, the Principals Committee had refined five options. I put them on the whiteboard in the Oval Office on 28 March, and I reproduce them here as a record.

Decision	Options considered	Reversible?	Risk	Chosen	Outcome
How to respond to the quarantine at sea (decided 30 March 2045, announced 6 April)	(A) Diplomatic and economic pressure only; no naval action. (B) Airlift of humanitarian supplies, no ships. (C) "Escorted humanitarian corridor": US Navy escorts US-flag merchant ships only, carrying declared humanitarian cargo, with allied surveillance and logistics; allies may join later. (D) Escort all ships bound for Taiwan regardless of flag. (E) Declare the quarantine void and break it by force, clearing coast guard vessels from the approaches.	A, B: fully reversible. C: largely reversible: can be paused, widened, or narrowed; a single incident could not be undone. D: partly reversible but much higher exposure. E: irreversible; likely war.	A: accepted coercion, new normal. B: airlift insufficient for fuel; aircraft vulnerable to being "escorted" by PLA fighters. C: confrontation with coast guard; the PLA Navy behind them. D: confrontation with dozens of ships of many flags; legal basis weak without flag-state consent. E: armed conflict with a nuclear power.	**C**, with a written path to widen to Japanese- and Philippine-flag ships after 30 days if allies consented	First convoy, Operation OPEN DOOR, sailed 20 April and arrived 22 April. PLA shadowed; one dangerous close pass; no shots fired. Quarantine partly eased in May.

The choice of option C rested on three arguments.

The first was legal. By escorting only American-flag ships, we stood on the firmest legal ground available: the protection of our own flag. We were not claiming the right to police the Strait. We

were claiming the right of an American ship to sail to a port of its choosing through international waters. There were very few American-flag ships in international trade in 2045, perhaps two hundred oceangoing vessels, many in the Maritime Security Program, a federal arrangement under which the government pays a stipend to keep commercial ships under the US flag for use in an emergency. The Maritime Administration identified eleven suitable ships. That small number was itself useful. It made the operation a demonstration, not a siege.

The second argument was the Signal Ladder. Each rung of escalation should carry a legible message and leave an off-ramp. Option C said: *We will not accept your claim to police our ships, but we are not trying to break your quarantine.* It left Beijing free to let American ships through without inspection and call it magnanimity. Option D would have told them we were contesting their entire policy. Option E would have told them we were at war. Both would have closed the off-ramp.

The third argument was reversibility. If OPEN DOOR went badly, we could pause it after one convoy. If it went well, we could widen it. Each step would be a new decision, taken in light of new facts.

The strongest objection came from Rosalind Achterberg, who was hawkish but never reckless, and who said in the meeting of 30 March something I have not forgotten.

"Tom, you're protecting the law and abandoning the island. Eleven ships, carrying American cargo, aren't going to keep Taiwan's lights on. Ninety percent of what Taiwan needs comes on ships flagged in Panama, Liberia, the Marshall Islands. If the corridor only protects our flag, the Chinese will let our ships through, keep strangling everyone else, and call it a victory."

"You may be right," I said. "If you are, we'll widen it. But I want the first step to be one we can take back."

"And if they decide our flag isn't sacred either?"

"Then we'll have learned that at the smallest possible cost, and the whole world will have seen who escalated."

Treasury Prepares

While the Navy planned the corridor, Daniel Iwu was building the other half of our pressure. Daniel is a former New York Federal Reserve official, quiet, Nigerian-born like Sam, with whom he would argue in Yoruba when they did not want the rest of us to follow. His briefings were the most frightening I received that spring, because they described a weapon that could injure us as badly as its target.

Sanctions, he explained, are mostly administered by Treasury's Office of Foreign Assets Control, OFAC. The most powerful tool is designation on the Specially Designated Nationals list: a named person or company has its assets in American jurisdiction frozen, and Americans are barred from dealing with it. Because the dollar is the world's main reserve and settlement currency, and because most international dollar payments clear through American banks, designation can cut a company off from much of the world economy. "Secondary sanctions" go further, threatening foreign banks that deal with designated parties with exclusion from the dollar system. Export controls, run by the Commerce Department, restrict the sale of technology, above all advanced semiconductors and the tools to make them.

Daniel had prepared three packages: Phase A, targeting a dozen or so PLA-linked shipbuilders and coast guard suppliers; Phase B, broader sanctions on state banks; and Phase C, which he described

as "the financial equivalent of a blockade", cutting major Chinese banks off from dollar clearing. Each, he said, would bring retaliation. China held something over seven hundred billion dollars of US Treasury securities. It had its own payment system, CIPS, the Cross-Border Interbank Payment System. It dominated the refining of rare earths and gallium, without which our own weapons could not be built.

"And the debt," Daniel said. "After 2036, the markets don't trust us the way they used to. If Beijing sells Treasuries in size while we're sanctioning their banks, yields could spike, and every point on the ten-year costs us a fortune in interest. Mr. President, I can build you a weapon, but it has a recoil."

I asked him to finalize all three packages but announce none. Sanctions prepared and held, I reasoned, were a signal that could be sent privately: Helena could tell Beijing that they existed. Sanctions announced were a decision that could not be cleanly taken back.

> **OPERATING NOTE — Economic Weapons Have Recoil.** Sanctions feel like the safe middle option between words and force. They are not safe. Against a large, interdependent economy, every financial weapon damages your own markets, your allies' industries, and your own debt. Before you use one, ask your Treasury Secretary three questions in writing: What will the target do in retaliation? What will that do to our borrowing costs? Which of our allies will be hurt worse than we are? Prepare sanctions early; announce them late. A sanction held in reserve and described privately to the adversary is often worth more than the same sanction imposed.

I signed the National Security Presidential Memorandum authorizing OPEN DOOR on 6 April. It ran to four pages, including the supplemental rules of engagement, which authorized escort commanders to use non-lethal means, positioning, warnings, and if necessary physical interposition of the warship's hull, to prevent the boarding of an escorted American-flag ship, and reserved any use of force beyond self-defense to my personal approval. I wanted the commanders to know exactly where the line was. I wanted the Chinese to learn, through every channel available, exactly where the line was. A line the other side cannot see is not a deterrent. It is a trap.

Sunday, 2 April 2045 — 08:00 EDT: The Jade Line, Call One

Four days before I signed that memorandum, I spoke to Wen Zhaolin for the first time since the crisis began.

The JADE LINE was the leader-to-leader video link our two governments had agreed to build in 2043, an upgrade of the old hotline arrangements. The Defense Telephone Link between the Pentagon and the Chinese Ministry of National Defense had existed since 2008, but it had a long history of going unanswered in exactly the moments it was needed. The JADE LINE was meant to be different: a dedicated, encrypted video circuit between the White House and Zhongnanhai, tested monthly by technicians on both sides, and used, until that morning, for nothing but tests.

A successor should understand what such a call actually is. It is not a conversation. It is two delegations, each in a room, each leader flanked by an interpreter, a notetaker, and his senior foreign policy advisor, with a dozen more officials listening off camera. Every sentence is interpreted consecutively, which means each side speaks, then waits while the interpreter renders it, so a thirty-minute call contains perhaps twelve minutes of substance. Both sides prepare "talking points" in advance, and both leaders mostly read them. The value lies in the few moments when one of them departs from the script.

I had Helena on my right, Sam on my left, and our senior interpreter, a woman from the State Department who had interpreted for three presidents, at the end of the table. The screen showed Wen in a wood-paneled room, seated before a painting of mountains, with Foreign Minister Shen Yilan beside him. He looked older than in his photographs. His face was heavy and pale. I noticed it, and I noted it in my diary that night, and I did not understand what I had noticed until July.

He began with his points, and they were cold. Taiwan was China's internal affair. The law-enforcement measures were a necessary response to the separatist provocations of the Lin authorities. The United States was interfering in China's internal affairs by sending warships and speaking to the leader of a region of China. The *Formosa Pearl* incident had been caused by the reckless conduct of the Taiwanese vessel. If the United States sincerely wished for peace, it should stop arming Taiwan and stop encouraging the separatists.

I let the interpreter finish. Then I put my own points aside.

"Mr. President, I want to tell you something before it happens, so that you are not surprised by it. In the coming weeks, the United States will escort ships flying the American flag, carrying food and medicine, to the ports of Taiwan. Only American ships. Only humanitarian cargo. Our ships will not fire unless fired upon. They will not enter your territorial waters. We are not attempting to break your measures by force. We are protecting our own flag, as any nation would."

The interpreter rendered it. Wen listened without moving. Shen Yilan wrote something on a pad and slid it toward him. He did not look at it.

"You tell me this," he said, "so that if there is a clash, you can say you warned me."

"I tell you this so that there will not be a clash. I want your commanders to know exactly what our ships will do, and what they will not do. I am telling you our line so that neither of us crosses it by mistake."

There was a long pause, longer than the interpretation required.

"The United States always says it seeks peace," Wen said, "while it builds weapons to fight a war in space, and missiles to destroy our deterrent. You speak of food and medicine. Your Golden Dome is not food and medicine."

There it was. The Mirror Model, spoken aloud by the man on the other side of the mirror. He had moved, without prompting, from ships in the Strait to the balance of nuclear terror. To him they were one subject. And LANTERN's report sat in my mind, *they raised readiness because they think we might be coming*, and I could not mention it.

"Mr. President," I said, choosing each word, "nothing we do at sea has anything to do with our strategic forces. Nothing. I give you my personal assurance of that."

He looked at me for a moment, through the camera, across twelve time zones. Then he said, "Assurances are words. We will watch what you do."

The call ended five minutes later with formulaic expressions of the importance of communication. It had lasted thirty-four minutes. Helena called it "cold, but not frozen". Sam said, "He heard you. He wanted you to know he's afraid." I sat for a while after the screen went dark.

What do they see when they look at us? They saw a country that spoke of humanitarian corridors while building the means to destroy their deterrent and blind their response. They saw ARGUS,

which I had built to give us certainty, as the eye that would find their missiles for our weapons. They saw my assurance as a cover story. And from where they sat, they were not wrong to doubt it.

That night I wrote: *The more perfectly we see, the more frightened they become. Absolute surveillance is a weapon that also loads the enemy's gun. I have known this as an equation for twenty years. Today I saw it in a man's face.*

Friday, 21 April 2045 — 22:14 EDT: OPEN DOOR

The first convoy sailed from Subic Bay on 20 April: three American-flag ships, a container ship, a product tanker carrying diesel, and a roll-on roll-off vessel loaded with medical supplies and food, escorted by two destroyers and the cruiser that served as the escort commander's flagship. An Australian P-8 and a Japanese surveillance aircraft flew overhead. A Philippine navy frigate accompanied the convoy to the edge of Philippine waters and then turned back, as Villanueva and I had agreed. The *George Washington* group stood a hundred and fifty miles to the east.

One of the two destroyers was the USS *Daniel Inouye*, DDG-118.

I had seen the escort list on 12 April. I knew that removing her ship would be noticed by every officer in Seventh Fleet, and that it would tell them the President valued his daughter's life above their sons' and daughters'. I knew also that leaving her ship on the list was a choice I made as a father as much as a president. I did not tell Claire until the night before the convoy reached the Strait. She did not speak to me for most of an evening. Then she said, "I would have done the same, and I would have hated you for it," and took my hand.

The convoy passed south of the Penghu Islands and turned north toward Kaohsiung on the morning of 22 April, local time, which was the late evening of the 21st in Washington. I was in the Situation Room with Sam, Ray, Pat Dunmore, and Abigail Fenn. Admiral Moriyama was on screen from Hawaii. We were watching a feed that fused the escort's radar picture, ARGUS imagery, and the P-8's sensors: a cluster of blue symbols for our ships, and around them, more and more red ones.

The PLA Navy had not tried to stop the convoy. It had chosen to shadow it. Two Type 052D destroyers and a frigate took station on either flank, at four to six thousand yards. Three coast guard cutters stood ahead, near the inspection zone. Above them, PLA fighters flew in pairs. Our analysts had predicted all of this. Then, at 10:14 local time, one of the Chinese destroyers, which had been paralleling the convoy to the west, turned sharply and accelerated across the bow of the leading American-flag ship, the container vessel, passing about three hundred yards ahead of her and about five hundred yards from the *Inouye*, which was screening on that side.

On the screen it was two symbols converging. In the room it was absolute silence.

"Inouye is maneuvering," Moriyama said. "She's increasing speed and turning to interpose between the PLA destroyer and the container ship. Fire control radar on the PLA ship, no. Repeat, no fire control radar. Navigation radar only. Bridge-to-bridge radio, Inouye hailing on channel 16."

Ray Castillo leaned toward me. "Sir, if they illuminate with fire control, under the standing rules that's demonstrated hostile intent. The CO can engage."

"I know the rules, Ray."

"I'm just saying we need to be ready."

Pat Dunmore said quietly, "The CO is ready, sir. That's his job."

I wanted, more than I have ever wanted anything, to pick up the phone and speak to the commanding officer of the *Inouye*. I wanted to tell him to back off, or to hold his ground, or to get my daughter off the bridge. I had the authority to do it. The technology existed: a voice circuit from the Situation Room could reach that ship in seconds. And I knew that if I did it, I would be taking command of a warship in a close-quarters maneuver from seven thousand miles away, with a delay of a few seconds and a picture that was a few seconds stale, overriding a commander who could see the other ship with his own eyes.

Tom, I told myself, *you are the slowest, noisiest sensor in this loop. Stay out of it.*

I put my hands flat on the table and did not move them.

Decision	Options considered	Reversible?	Risk	Chosen	Outcome
Presidential intervention during the OPEN DOOR close pass (22 April 2045, 10:14–10:23 local)	(A) Direct voice contact with *Inouye*'s commanding officer to order a maneuver. (B) Order through INDOPACOM to loosen supplemental ROE in real time. (C) Order the convoy to turn back. (D) Hold: leave tactical decisions to the commanding officer under existing rules; Washington observes and prepares diplomatic follow-up.	A, B: the order itself reversible, its consequences possibly not. C: reversible, but would signal that a close pass could turn back a US convoy, inviting repetition. D: reversible.	A: confusion in the chain of command; decisions on stale data; precedent of presidential micromanagement. B: rules changed mid-encounter invite misreading on both sides. C: collapse of the corridor's credibility. D: depends on the judgment of one officer.	**D**	PLA destroyer completed the pass and resumed shadowing at 10:23. No fire control radar used. Convoy arrived Kaohsiung 16:40 local. Protest delivered to Beijing within 3 hours citing CUES distances.

The *Inouye*'s commanding officer, a commander from Ohio whose name I will leave out of this edition at his request, did what he had been trained to do. He increased speed, put his ship between the Chinese destroyer and the container ship, sounded five short blasts on the ship's whistle, the international signal of doubt about another vessel's intentions, and hailed the Chinese ship on the bridge-to-bridge radio, in English and Mandarin, citing the Code for Unplanned Encounters at Sea.

The Chinese destroyer held its course for perhaps forty seconds, then turned away, passed down the *Inouye*'s starboard side at about five hundred yards, and resumed its station.

The whole thing lasted nine minutes. Nobody fired. The convoy entered Kaohsiung that afternoon and was met at the pier by President Lin, a crowd of thousands, and a great many cameras.

I learned later that Lieutenant Maya Kane was the tactical action officer on watch in the *Inouye*'s combat information center throughout the encounter. She told me months later that she had been too busy to be afraid, and that afterward, when the watch ended, she had gone to the head and been sick. I asked her what she had been thinking during those forty seconds. She said, "That the guy on their bridge didn't want to die either. And that he didn't turn on his fire control radar. I kept watching for it. He never did."

> **OPERATING NOTE — Do Not Drive the Ship.** Modern communications will let you speak to any commander anywhere in the world in seconds. This is a temptation, not a capability. In a fast tactical encounter, the officer on the scene has better information, faster reactions, and years of training you do not have. Your control is exercised before the encounter, through the mission you approve and the rules of engagement you sign, and after it, through the diplomacy that follows. During it, your job is to keep your hands still and to keep others in the room from reaching for the phone. If you do not trust a commander's judgment, replace him before the mission, not during it. Model the loop honestly: in a nine-minute encounter at sea, the President is the sensor with the longest delay and the most noise.

May 2045: The Lull

Two more convoys sailed in early May without incident. The PLA shadowed them at greater distances. On 9 May, the China Coast Guard quietly stopped inspecting ships bound for Keelung, in the north, and on 16 May reduced inspections at Taichung to what its notices called "random checks". Kaohsiung remained under full inspection, except for American-flag ships, which were simply waved through. The live-fire zones of STRAIT THUNDER were lifted on 12 May. War-risk premiums began to fall.

Both sides declared victory. The Chinese Foreign Ministry said the law-enforcement measures had "achieved their purpose of deterring separatist forces" and were being adjusted "in light of the situation". The Taiwanese said they had shown they could not be strangled. Our own press office, with my approval, said that the humanitarian corridor had demonstrated the resolve of the United States and its allies, and that we welcomed the reduction in unlawful interference with shipping. Senator Whitford, the Majority Leader, said on a Sunday program that the administration had "won a round it should never have let start". That was a fair hit, and I told him so when he called me the next day.

Rosalind Achterberg's prediction had come partly true. Our flag was respected and other flags were not, entirely; the Chinese had found a way to let us win without losing. But the pressure on Taiwan had eased enough that the island's reserves began to recover. I chose not to widen the corridor to other flags. I told myself it would have been an escalation without a purpose. I believe that judgment was correct. I am less sure it was complete.

Because on 24 May, Greg Laszlo came to see me again with Ruth Calder, and the eleven lines she handed me that day ended the lull for me, if for no one else.

Base 64 remains at RL3 despite reduction of coast guard activity. Order received 20 May for "warhead mating drills" at two Hami brigades, using training rounds, to be completed by 30 June. Drill described as validation of procedures for transition to RL2. Base staff told PLARF HQ considers present calm "temporary". Commander Luo visited Lanzhou 18 May; spoke privately with brigade commanders. Mood: more serious than in March.

"What does it mean?" I asked.

Ruth thought for a while before she answered. "It means the people in Beijing who run the Rocket Force don't believe this is over. And they're getting ready for the next round."

"And Wen?"

"We don't know what Wen knows about these drills, sir. That's what bothers me."

I did not change our posture. I asked Abigail to direct ARGUS to increase its coverage of the Hami field, quietly, within normal tasking patterns, and I asked Greg to tell LANTERN, through whatever channel was safe, that we were grateful, and that he should take no risks he did not have to take. Then I went upstairs.

Sunday, 28 May 2045 — 23:10 EDT

From my diary:

Claire and I ate dinner on the Truman Balcony tonight, the first warm evening of the year. She talked about a girl on her ward, eight years old, whose leukemia is in remission. She said that remission is the hardest word in her profession, because parents hear "cured" and she has to tell them it means "quiet, for now". She said, "Watch it, keep watching it, and don't stop watching it because it's quiet."

She was talking about the girl. I was thinking about the Strait.

Everyone in this city believes the crisis is over. The markets believe it. Congress believes it. Even Ray believes it, a little. The quarantine has eased, the convoys are sailing, both sides have their victory. And in a building in Lanzhou, men are practicing how to put warheads on missiles.

The three clocks are almost in sync tonight. The information clock is calm. The political clock has slowed to its summer pace. The military clock, the one I can see, is slow. But there is a fourth clock I left out of my model, and I see it now: the clock inside the other side's head. LANTERN reads it for me, a few lines at a time. It is not slowing down.

Maya's ship is in Yokosuka for maintenance. She called tonight and sounded happy, and talked about a noodle shop near the base. I did not tell her anything. I listened to her voice and I tried to memorize it.

Remission. Quiet, for now. Keep watching.

I did keep watching. It was not enough. Nine days later, on the sixth of June, Beijing declared that every cargo bound for Taiwan must clear Chinese customs in Fujian, and the quarantine became something closer to a blockade, and the war we had kept on the water came ashore in our own country, through the wires.

That is the next chapter.

Chapter 3: The Pressure Season

"They have stopped trying to break the door. Now they are trying to break the house." — Diary of Thomas Ellery Kane, 14 June 2045

A Note Before the Summer

A successor reading this chapter may find it less dramatic than the chapters around it. No ship is sunk here. No missile flies. Nobody in my family is in danger in the way my daughter would be in September. And yet if I were asked which months of the Pacific Crisis taught me the most about the actual job of the presidency, I would answer without hesitation: June, July, and August of 2045.

In the spring the crisis had been a naval problem. A naval problem has a shape. There are ships, there are lanes of water, there are rules of engagement, and there is an admiral who can draw it on a chart. By June the crisis had become something else. It had become a contest over which society could absorb more pain without changing its mind. Beijing had concluded, I believe correctly, that it could not win a short fight at sea against the United States and its allies without risking a war it did not want. So it chose a longer fight on ground where it believed we were weak: our ports, our power grid, our debt, our supply chains, our impatience, and the private companies on whom our military now depended without fully admitting it.

I called it, in my diary, "the pressure season." The phrase stuck among my staff, and it is the title of this chapter.

Beijing's strategy in the summer of 2045 was resonant forcing. Each push was modest. Each push was deniable or explainable. And each was aimed at a place where American society, left to its own dynamics, would amplify the pain: markets that panic, a politics that turns on itself, a public that wants the bad news to stop.

The job that summer was not to win a battle. It was to damp the oscillation. In terms of my Three Clocks Model, the military clock slowed down for a while, and the information clock and the political clock went wild. I learned that a president can lose a war without a single shot being fired, simply by letting the information clock drag the political clock into a decision the military clock was never ready for.

That is what this chapter is about.

Tuesday, 6 June 2045 — 05:52 EDT

Megan Pruitt, my chief of staff, does not knock. She has a way of opening the door to the bedroom in the residence that is louder than knocking, a deliberate rattle of the handle that gives Claire time

to pull the sheet up and me time to find my glasses. I heard it at 05:52 and was sitting up before she spoke.

"Beijing has made an announcement," she said. "Not a quarantine anymore. They're calling it customs. Sam is on his way in. Helena is already in the building."

Claire put a hand on my back and said nothing. In twenty-nine years of marriage she had learned which mornings need words and which do not.

I dressed in the clothes I had laid out the night before, a habit from my years as an engineer on launch campaigns, when you never knew whether a hold would end in a scrub or a 4 a.m. countdown.

The announcement had come at 17:40 Beijing time, which was 05:40 our time; in summer Beijing is twelve hours ahead of Washington. The General Administration of Customs of the People's Republic of China, together with the Ministry of Transport and the China Coast Guard, had issued what it called the "Provisional Regulations on the Administration of Goods Entering and Leaving the Taiwan Region." Its core was a single sentence, which Helena Marsh, my Secretary of State and a former ambassador to Beijing, translated for us in the Situation Room from the Chinese text before the official English version had even been posted:

"From 00:00 on 10 June, all goods bound for ports in the Taiwan region shall complete customs declaration and inspection procedures at designated ports in Fujian Province."

"Designated ports," Helena said, "means Xiamen, Fuzhou, and Pingtan. A ship bound for Kaohsiung now has to go to the mainland first, declare its cargo to Chinese customs, pay whatever they decide to charge, and receive a release document. A ship that does not have the release document is, in their language, 'smuggling.' And smuggling is a law-enforcement matter for the Coast Guard."

"So it's a blockade," said Ray Castillo, the Secretary of Defense.

"It's a blockade that doesn't use the word," Helena said. "That's the point of it."

What a Customs Blockade Is, and Why It Is Clever

I want to stop here and explain what Beijing had done, because a successor will very likely face a version of it, and the version may not involve Taiwan.

Under the law of armed conflict, a blockade is an act of war. A state that declares a blockade of another's coast is a belligerent, and the blockade must be declared, effective, and impartial. When President Kennedy faced Soviet missiles in Cuba in October 1962, his lawyers advised him not to call his naval line a "blockade" for precisely this reason. They called it a "quarantine." That word, chosen by Americans in 1962, was the word Beijing borrowed in March 2045, and I do not think the irony was accidental.

What Beijing did on 6 June was more ambitious. It claimed that Taiwan was simply a Chinese customs territory, like Fujian or Guangdong, and that China was exercising ordinary administrative control over goods entering its own territory. Under this theory, there was no blockade at all, no quarantine, no military operation. There was paperwork.

The genius of the paperwork was in what it demanded of everyone else. A shipping company does not want a war. It wants its ship to arrive, its insurer to pay if something goes wrong, and its customers to receive their goods. If the price of arriving in Kaohsiung was a stop in Xiamen, a form, and a fee, most shipping companies would pay it. And every ship that paid it would be, in a

small way, accepting that Beijing had the right to collect it. Sovereignty, in the end, is not only proclaimed. It is also practiced, one form at a time.

Samuel Oyelaran, my National Security Advisor and the man whose mind I trusted more than any other in government, put it plainly when he arrived ten minutes later, still knotting his tie.

"They're not trying to stop ships," he said. "They're trying to get ships to stop themselves. Every captain who goes to Xiamen votes for their theory of the case."

"And every captain who doesn't?"

"Gets boarded by a Coast Guard cutter. And then we have a choice: escort them, which means American warships protecting smugglers in Chinese law, or don't, which means Taiwan's trade slowly reroutes through the mainland."

Tom, I asked myself, *what is this for? Not what is it. What is it for?*

The Mirror Model, which I described in Chapter 1, asks a president to look at a move from the other side: not what the adversary is doing, but what he fears and what he needs. What did Wen Zhaolin need in June 2045? He needed to show his own people, and above all the hardliners in his Central Military Commission, that the spring quarantine had not been a failure. The quarantine had ended in May with both sides claiming victory, which meant, from the point of view of a Chinese nationalist, that China had blinked. He needed a move that escalated the pressure on Taiwan without risking the sinking of an American warship. A customs regime did that. It moved the confrontation off the sea and onto insurance contracts and port schedules.

And what did he fear? He feared, I believed, a war he could not control. A man who wants war does not invent paperwork.

That was my first estimate on the morning of 6 June. I wrote it on the yellow pad in front of me: *Escalation designed to avoid escalation. He's buying time with Taiwan's economy.* I was right about the first part. I was too optimistic about the second. What Beijing was buying time for, it turned out, was something it had been preparing for years.

The First Meeting

The Principals Committee convened at 07:00. A successor should know the difference between the Principals Committee and the National Security Council itself. The Council, by statute, is chaired by the President. The Principals Committee is the same cabinet-level people, chaired by the National Security Advisor, without the President in the room. Below it sits the Deputies Committee, the deputy secretaries and their equivalents, who do most of the actual work of producing options. I attended the 07:00 meeting anyway, which made it, technically, a Council meeting, and I did so knowing that my presence would make people more careful and less candid. I made a rule that summer, which I recommend: on the first morning of a new phase, the President should be in the room, because the first framing of a problem tends to become its permanent framing. After that, the President should let the principals argue without him, and come in to decide.

The meeting reached three conclusions in ninety minutes.

First, the United States would not recognize the customs regime and would say so publicly, in the language of international law, not in the language of anger. Helena would draft a statement that called the regulations "an unlawful attempt to impose a blockade by administrative means" and called on all states to "decline to comply."

Second, we would not immediately extend naval escort to non-American ships. Admiral Kenji Moriyama, the commander of United States Indo-Pacific Command, joined by secure video from Honolulu and was clear about his limits: his ships were already stretched by the escorted humanitarian corridor of the spring, Operation OPEN DOOR, and escorting the full commercial traffic of Taiwan, more than a thousand port calls a month, would require a fleet we did not have in the theater.

Third, and this was Daniel Iwu's point, the real battlefield would be insurance. Daniel, my Treasury Secretary, a former bond trader who had spent the 2036 Treasury crisis on the wrong end of a falling market and had never forgotten it, explained that the decisive actors in the next ten days would not be admirals but underwriters in London, Singapore, and Oslo.

"A ship doesn't sail without war-risk cover," he said. "If the Lloyd's market's Joint War Committee lists Taiwan's waters as a listed area, premiums go vertical. If premiums go vertical, owners route to Xiamen, because a customs stop is cheaper than a ten percent surcharge on the hull. Beijing knows this. They're not aiming at the ships. They're aiming at the actuaries."

"Can we underwrite it ourselves?" I asked.

"Partly. There's precedent. We did a version of it for Ukrainian grain in the 2020s with partners, and the Maritime Administration has war-risk authority for ships in the national interest. We can stand up a government reinsurance backstop for ships calling at Taiwan, with Japan and maybe the United Kingdom. It won't cover everyone. But it changes the arithmetic at the margin, and the margin is where this gets decided."

I approved the reinsurance backstop that afternoon. It was, I think, one of the most important and least noticed decisions of the crisis, and I mention it here because a successor should understand that in economic warfare the most powerful weapons are often the most boring ones. In June 2045 the Strait of Taiwan was contested not only by destroyers but by spreadsheets.

> **OPERATING NOTE — Gray-zone pressure is designed to make you choose between overreacting and acquiescing.** Every move in the gray zone (the space between peace and war) is built to present the defender with two bad options: respond with force and look like the aggressor, or do nothing and let a new fact become permanent. The answer is almost always a third option on a different axis: answer a legal move with a legal move, an economic move with an economic move, a slow move with a slow move. Do not let your adversary choose the dimension of the contest. And look first at the boring instruments, such as insurance, standards, financing, and law. They are usually where gray-zone campaigns are won or lost.

By 10 June, when the regulations took effect, about forty percent of the ships headed for Taiwan were stopping in Fujian. By 20 June, with the reinsurance backstop announced and Japan joining it, the figure had fallen to about twenty-five percent. It never went to zero. It never needed to. Beijing's theory of the case was neither accepted nor completely defeated, and in the gray zone that is often the best one can hope for.

I thought, for a week, that we had found the shape of the summer. Then came the morning of 14 June.

Wednesday, 14 June 2045 — 04:18 EDT

The first warning did not come from the intelligence community. It came from a port.

At 01:05 Pacific time, which was 04:05 in Washington, the terminal operating systems at three container terminals in the Port of Los Angeles and two in the Port of Long Beach began to behave in ways their operators could not explain. A terminal operating system is the software brain of a modern port. It knows where every container in the yard is stacked, which crane should lift it next, which truck is waiting at which gate, and which ship it belongs to. Since the 2030s most of the large terminals on the West Coast had been heavily automated: driverless straddle carriers, remote-operated cranes, automated gate systems that read license plates and container numbers.

On the morning of 14 June they were wrong. Containers that the system said were in one stack were in another. Gate systems refused trucks that had valid appointments and admitted none at all for long stretches. Two automated cranes stopped mid-lift, under their own safety logic, when their position data disagreed with what the yard system expected. Nobody was hurt; the safety systems did what they were designed to do. But by 04:00 Pacific, five terminals, which together handled roughly a quarter of the container traffic entering the United States from Asia, had stopped moving freight. By 06:00, two terminals in Oakland and one in Seattle–Tacoma reported similar symptoms.

At 04:18 Eastern, the Cybersecurity and Infrastructure Security Agency called the White House Situation Room. At 04:26, Megan's handle rattled my door.

"Ports," she said. "West Coast. Maybe cyber. Sam's coming in."

The Grid

At 13:40 Central time that afternoon, the second blow fell, and this one I felt in my chest when I heard it, because my son Eli lived in Austin.

The Electric Reliability Council of Texas, which everyone calls ERCOT, operates the power grid that serves most of Texas. It is, for historical reasons that every Texan can recite and every non-Texan finds baffling, largely an island: it is only lightly connected to the two great interconnections that serve the rest of the continental United States, so that when it is short of power it cannot import much from its neighbors. A successor should know this fact because it explains why an adversary chose Texas. An island grid is easier to push off balance and harder to rescue.

On 14 June Texas was in the middle of an early heat wave. Demand was near the summer peak. At about 13:40, operators at several large generation and transmission facilities in the ERCOT region began to see false readings and unexpected control behavior. I will describe what happened only by its effects, as this book must, and as I believe any responsible public account must: equipment received information that did not match reality, protective systems acted on that false information, and within about twenty minutes roughly 6,000 megawatts of generation, enough to supply well over a million homes on a hot afternoon, had tripped offline or been taken offline by cautious operators who no longer trusted their screens.

An electric grid must balance supply and demand every second. When supply falls suddenly, the frequency of the alternating current, normally 60 cycles per second in North America, begins to fall. If it falls far enough, other generators disconnect themselves to protect their equipment, which makes the shortfall worse, and the whole system can collapse into a blackout that takes days to restart. ERCOT came within a few minutes of exactly that in February 2021, during Winter Storm Uri, and the Texans who ran the grid in 2045 had been trained on the memory of it.

They did what they had been trained to do. They ordered rotating outages, shedding load in blocks across the state to keep the frequency up. It worked, in the sense that the grid did not collapse. It

failed, in the sense that for the next three days, from 14 to 17 June, millions of Texans lived with rolling blackouts in temperatures above 100 degrees Fahrenheit.

I called Eli at 15:10 Eastern. He answered on the second ring from a coffee shop that had a generator.

“Dad, it’s fine. We have two hours on, two hours off. It’s hot. Everybody’s annoyed. Is it China?”

“I can’t tell you that.”

“That means yes.”

“It means I can’t tell you that,” I said, and I heard him laugh, and I was grateful for the laugh. “Check on your neighbor. The older man downstairs.”

“Mr. Okafor. I will.”

That night, in my diary, I wrote: *Asked Eli to check on his neighbor. Realized after I hung up that the President of the United States had just delegated the Texas heat emergency to a 24-year-old software engineer. That is, in a sense, how it works. A country is held together by people checking on their neighbors.*

Over the three days, according to the Texas Department of State Health Services, nineteen people died in heat-related incidents that investigators later linked, at least in part, to the outages. Most were elderly. Two were in nursing homes whose backup generators failed. I have their names. I will not print them here, because their families did not ask to become part of a book. But a successor should know that I kept the list in my desk, and that I looked at it when people told me cyberattacks were “non-kinetic.”

Attribution

The single most important question in a cyber crisis is not “what happened” but “who did it, and how sure are we.” This is called attribution, and a successor must understand that it is hard, slow, and always a matter of confidence rather than certainty.

At 21:00 on 14 June, Abigail Fenn, the Director of National Intelligence, came to the Oval Office with General Patricia Dunmore, the Chairman of the Joint Chiefs, and the commander of United States Cyber Command, a soft-spoken Army general whom I will not name here because her name remains on a list that has not been declassified.

“Mr. President, we assess with high confidence that both the port disruptions and the grid incident were conducted by the People’s Liberation Army,” Abigail said. “Specifically, by a unit of what used to be called the Strategic Support Force and is now the Information Support Force and the Cyberspace Force. We assess with moderate confidence that the access was established years ago, possibly as early as the late 2030s, and held in reserve.”

“Pre-positioned,” I said.

“Yes, sir. We warned about this publicly in the 2020s. The intelligence community called the earlier campaign Volt Typhoon. The idea was always the same: get inside critical infrastructure in peacetime, sit quietly, and keep the ability to disrupt it in a crisis. What you saw today is that capability being used.”

“How do you know it’s them and not someone pretending to be them?”

She gave me the answer at the level I am permitted to repeat: the assessment rested on several independent kinds of evidence, technical and human, which pointed in the same direction; it was consistent with access we had previously seen and tracked; and no other actor had both the motive and the demonstrated capability. She also told me what we did not know: we did not know whether the decision to use the access had been made by Wen himself, by the Central Military Commission, or by a military commander acting under general authority.

That last point mattered to me more than any other. In my Control-Loop Model, the estimator is where crises fail, and the estimator's most important job in a crisis is to figure out who is actually running the adversary's controller. If Wen had ordered the grid attack personally, that told me one thing about his intentions. If his generals had done it under some standing authorization he had signed months ago, that told me something else, something more frightening: that his own system might be running faster than he was.

"Ask WILLOW," I said.

WILLOW, as a reader of Chapter 1 will know, was Major Song Lihua, a duty officer in communications at the Joint Operations Command Center of the Central Military Commission in the Western Hills outside Beijing. She saw message traffic. She did not see everything. Getting a question to her and an answer back took, in those months, between four and nine days, because the methods of communication had to be safe for her, and safe is slow.

The answer came back on 22 June. I will quote the relevant part of the CIA's report, as declassified:

> WILLOW reports that directives authorizing "network-domain countermeasures against US critical economic nodes" were transmitted from the CMC Joint Operations Command Center on or about 9 June under the authority of the CMC Joint Staff Department. WILLOW did not see a directive signed by the CMC Chairman [Wen]. WILLOW states that duty officers understood the action to have been approved "in principle" at a CMC meeting in late May. WILLOW notes that the PLA Rocket Force commander was present at that meeting and spoke in favor of "imposing costs on the American homeland to deter intervention."

So the PLA Rocket Force commander, General Luo Tiancheng, a man whose name would grow heavier in my diary every month, had argued for it. And Wen had approved it in principle, and then, apparently, let his staff choose the time.

The Counter-Cyber Question

Thursday, 15 June 2045 — 16:00 EDT, Situation Room

The question before me was whether the United States should respond in kind.

The pressure to do so was immense, and I understood it. By the afternoon of 15 June the cable news networks had a graphic, a map of Texas with blacked-out counties, under the words AMERICA UNDER ATTACK. The Senate Majority Leader, Cal Whitford, a Republican from Texas, had called me twice, once in anger and once in something closer to grief; his own mother's assisted-living facility in Midland had lost power for six hours. On the phone he said, "Mr. President, my state is cooking. Somebody in Beijing turned off the air conditioning on two million Texans. If we don't do something back, they'll do it to Ohio next."

He was not wrong about the logic. Deterrence depends on the adversary believing that a cost will be imposed. An attack that goes unanswered teaches the attacker that the price is zero.

But I had also, by then, been briefed on what we could do, and I was frightened by it in a way I had not expected.

The Cyber Command general laid out the options. I will describe them as she did, by effect and by target, never by method.

The first option was a symmetrical response: disrupt Chinese civilian infrastructure, ports and power, in a manner of comparable scale. She was confident we could do it.

The second option was a response against the PLA itself: disrupt military networks that supported the blockade. Here she spoke of a specific program, which had been running quietly for some years under the name QUIET HARVEST. I have described it briefly in Chapter 1. In the terms I am permitted to use: QUIET HARVEST was an intelligence collection effort that had, over time, gained access to parts of the logistics and supply networks used by the PLA Rocket Force and by the units supporting the Eastern Theater Command. It had been built, like the Chinese access in Texas, to sit quietly in peacetime and provide intelligence. It could also, if ordered, have effects.

The third option was to do nothing in cyberspace and respond in other domains: sanctions, public attribution, indictments, diplomatic expulsion.

General Dunmore spoke first, and she spoke for the second option.

"Mr. President, the logic of a response against their military is that it's proportionate, it's lawful, and it's aimed at the people who did this. It signals that we can reach the things they care about. And it degrades their ability to sustain the customs blockade, which is a real military benefit."

Ray Castillo agreed with her, more forcefully. "They hit our grandmothers. We hit their trucks. That's a distinction any American can understand."

Helena was more cautious. "The problem with a response against the PLA is that we don't know how they'll read it. If we touch Rocket Force logistics, do they think we're preparing the battlefield for something larger?"

That was the question. I asked the Cyber Command general to describe exactly which networks the QUIET HARVEST accesses touched. She did so, in detail I will not reproduce. When she finished, I asked one more question.

"Do any of the logistics networks you can reach support nuclear forces? Not conventional missiles. Nuclear. Warhead handling, nuclear brigade supply, anything in that chain."

She paused. "Some of the networks are shared, sir. The Rocket Force runs both conventional and nuclear brigades, and in some bases the supply systems serve both. We can segregate our effects, with high but not perfect confidence."

"High but not perfect."

"Yes, sir."

I put down my pen.

The Mirror Model, Applied

This was the moment, more than any other in June, when the Mirror Model did its work.

What does Beijing see when it looks at us?

It sees the most capable intelligence and surveillance enterprise in human history, ARGUS, revisiting every point in China every ninety seconds and claiming custody of nearly every road-mobile launcher outside its garrison. It sees the Golden Dome, which it believes, wrongly in my judgment but sincerely in theirs, could catch the ragged remnant of a Chinese retaliatory strike after an American first strike had destroyed the rest. It sees American submarines it cannot track. And now, I asked myself, imagine that it sees one more thing: the networks that move the supplies of its nuclear brigades suddenly misbehaving, at the height of a crisis.

What would I conclude, if I were Luo Tiancheng?

I would conclude that the Americans were blinding me before they struck. I would conclude that this was the first move of a disarming attack. And because the PLA Rocket Force had kept part of its force on early-warning counterstrike posture, launch-on-warning, since the early 2030s, I would argue to the Central Military Commission that we had to raise our readiness immediately, so as not to lose the force.

The cruelest feature of nuclear deterrence in the 2040s was that capabilities designed for conventional war and intelligence were entangled with nuclear forces. Scholars had written about this "entanglement" since the 2010s. A cyber operation against a conventional logistics network that happened to share infrastructure with nuclear forces could be read, by a frightened adversary, as an attack on the nuclear force itself. And a nuclear force under attack, or believed to be under attack, is the most dangerous object on earth.

"No," I said. "Nothing that touches the Rocket Force. Nothing that could be read as touching nuclear command, nuclear logistics, or early warning. Not this summer and not without my personal authorization, and I'm telling you now that I'm unlikely to give it."

"Mr. President," Ray said, "that rules out the most effective target set."

"It rules out the most dangerous target set. Those are the same thing. That's why it's dangerous."

Then What?

But I also did not accept doing nothing. Whitford was right that deterrence required a price. And there was a further consideration, which Sam raised, and which tipped me.

"If we respond only with sanctions," he said, "we're telling them that cyber is a domain where they can hit us and we'll only answer with paperwork. That's the same game they're playing in the Strait. We'd be accepting their dimension."

So I asked the general for a fourth option, which we built together that evening and through most of the next day. Its principles were these.

First, it would target only the specific PLA units and networks that supported the customs blockade at sea, meaning the Coast Guard and Eastern Theater Command logistics systems that scheduled, fueled, and supplied the cutters and their supporting ships off Fujian. It would not touch the Rocket Force, strategic command and control, early warning, or anything that served nuclear forces, and where networks were shared, they would be excluded entirely.

Second, it would not touch civilians. No power grids, no water, no hospitals, no civilian ports, no banks. This was not only a moral choice, though it was that. It was a strategic one. I wanted the

contrast to be visible: they attacked our grandmothers; we answered against the uniformed units enforcing their blockade.

Third, it would be reversible and temporary. The effects were designed to cause disruption, delay, and confusion for a defined period, not destruction.

Fourth, and this was my addition, it would be accompanied by a private message. We would tell Beijing, through the Jade Line staff channel and through their ambassador, Tang Shiyu, what we had done, why, and what we had deliberately not done.

"You want to tell them?" Ray said. He was not angry. He was genuinely puzzled. "The whole value of a cyber operation is ambiguity."

"Not this one," I said. "The value of this one is the message. If they don't know it's a message, it's just noise, and noise in a crisis gets interpreted by the most frightened person in the room. I want Luo Tiancheng to be the second most frightened person in that room, not the first."

> **OPERATING NOTE — In cyber, the target you choose is the message you send.** Cyber operations are unusually prone to misreading because effects are delayed, attribution is uncertain, and networks are shared. Before authorizing any offensive cyber operation in a crisis with a nuclear-armed adversary, ask three questions: (1) Could the adversary read this as an attack on its nuclear forces, nuclear command and control, or early warning? If yes, do not do it. (2) Does it touch civilians? If yes, you surrender the moral distinction that is your greatest asset. (3) Will the adversary understand it as a bounded message? If not, tell him, privately and precisely. An unexplained cyber operation is not a signal. It is a rumor, and rumors are interpreted by the most frightened person in the room.

Friday, 16 June 2045 — 18:30 EDT, Oval Office

I signed the authorization at 18:30. The operation was given its own name; I will call it here by the name of the larger program from which its access derived, as my staff did in conversation: the QUIET HARVEST response.

Decision	Options considered	Reversible?	Risk	Chosen	Outcome
Response to People's Liberation Army cyberattacks on West Coast ports and the Texas grid (14–17 June 2045)	(A) Symmetrical attack on Chinese civilian infrastructure; (B) Broad attack on PLA networks, including Rocket Force logistics reached through QUIET HARVEST; (C) No cyber response, sanctions and attribution only; (D) Narrow, temporary disruption of the Coast Guard and Eastern Theater units enforcing the customs blockade, excluding anything shared with nuclear forces, with private notification	A: practically irreversible (civilian harm, moral distinction lost). B: technically reversible, strategically irreversible (could be read as preparation for a disarming strike). C: reversible, but sets a price of zero. D: reversible by design	A: escalation and loss of legitimacy. B: Chinese move to higher nuclear readiness; entanglement. C: invites repetition. D: smaller military effect; possible exposure of some QUIET HARVEST access	D, authorized 16 June, 18:30 EDT; accesses that touch nuclear-related networks withheld	Blockade enforcement off Fujian degraded for about 72 hours (cutter fueling and scheduling disrupted; compliance checks fell sharply). Beijing acknowledged the private message through Ambassador Tang on 19 June. No further PLA attacks on the US grid that summer. Part of the QUIET HARVEST access was lost afterward as China rebuilt its networks

The operation began on the night of 17 June, Washington time. There is nothing to watch in a cyber operation. A president who launches aircraft can see them on a screen. A president who authorizes a cyber operation sees, at most, a summary a few hours later, written by people who are themselves inferring effects from indirect evidence.

The summary I received at 07:00 on 18 June said that the scheduling and fueling systems supporting China Coast Guard operations off Fujian had been disrupted, and that the number of ships stopped

for “customs inspection” had dropped by about seventy percent overnight. Over the next three days the cutters operated, but slowly, unevenly, and with visible confusion. By 21 June the effects had faded, as designed.

At 10:00 on 18 June, Helena called Ambassador Tang to the State Department. She told me afterward that she read him a short statement that I had approved word for word. Its essence was this:

“The United States has taken a limited and temporary action in response to the attacks of 14 June on American civilian infrastructure. That action was directed solely at the units of the People’s Republic of China enforcing an unlawful blockade at sea. It did not, and will not, touch any element of China’s strategic forces, its strategic command and control, or its early-warning systems. It did not touch Chinese civilians. The United States does not seek escalation. Further attacks on American civilians will be answered.”

Tang listened, Helena said, without interrupting. At the end he said only, “I will convey it.” Then, standing to leave, he added in Mandarin, very quietly, so that the note-taker could not easily record it: “Thank you for telling us what you did not do.”

Helena repeated that line to me in the Oval Office that afternoon, and I wrote it in my diary in capital letters. It was the first sign, in a summer of very few signs, that the Mirror Model might work in both directions. They were afraid of what we saw. Tang was telling me that they had noticed we were afraid of what they feared.

On 14 June the Texas grid was attacked. Through the remainder of the summer, despite enormous strain, it was not attacked again. Whether that was because of our response, or because Beijing had decided the attack had cost it more in world opinion than it gained, I do not know. I suspect both. A successor should be honest about the limits of what deterrence can prove. You can see when deterrence fails. You can almost never prove when it works.

The Bond Market

Monday, 26 June 2045 — 07:45 EDT

If the grid was the blow I felt in my chest, the bond market was the one I felt in my stomach, slowly, over two weeks.

A successor must understand something about the financial position of the United States in 2045, because it shaped every decision I made that summer and it will likely shape yours. By the mid-2040s the federal debt held by the public was about 165 percent of the size of the entire American economy. We had been warned for decades. The warnings had been right, but early, and being early is how warnings lose their audience. Then, in 2036, a weak Treasury auction coincided with a downgrade and a political standoff over the debt limit, and for nine days the market for United States Treasury securities, the deepest and safest market in the world, the foundation on which the global financial system is built, stopped working properly. Yields spiked. Dealers would not make markets. The Federal Reserve had to step in as buyer of last resort. The crisis passed, but the market never forgot that it could happen, and neither did Beijing.

China held, in 2045, somewhat under $600 billion in United States Treasury securities, far less than its peak of over a trillion dollars in the early 2010s, but still enough to matter at a moment of fragility. Its holdings were partly direct and partly held through custodial accounts in Belgium, Luxembourg, and elsewhere, which made them harder to track.

On the morning of 21 June, our Treasury market surveillance team noticed unusually heavy selling of longer-dated Treasuries through custodial accounts associated with Chinese state institutions. By 23 June, the selling was unmistakable. By the morning of 26 June, Daniel Iwu estimated that China had sold, or was in the process of selling, about $200 billion in Treasuries in less than a week.

On its own, $200 billion is not enough to break the Treasury market. The market trades several times that every day. But markets are not physics, they are psychology with a price attached. The selling came on top of the war fears, the grid attack, the fact that every investor in the world could see that the United States would need to borrow enormous sums to fund a possible war, and the memory of 2036. On 26 June, the yield on the ten-year Treasury note, the interest rate the United States government pays to borrow for ten years and the benchmark from which mortgage rates and corporate borrowing costs are set across the world, crossed 6.5 percent. It had been 5.2 percent in May.

The Weapon That Cuts Both Ways

I want to explain to a successor why China's sale of Treasuries was, at the same time, a powerful weapon and a weak one, because the paradox is the lesson.

It was powerful because it attacked the place where we were most sensitive: confidence in our debt. In a democracy, the political clock is directly wired to the financial clock. When mortgage rates rise, voters are angry, members of Congress are anxious, and the President's room for maneuver shrinks. Beijing understood this well.

It was weak, or at least costly to Beijing, for three reasons. First, every bond China sold into a falling market was sold at a loss, so China was spending its own wealth to hurt us. Second, China needed the dollars, or something like dollars, somewhere: its reserves had to be held in some asset, and there is no other market in the world large and liquid enough to absorb what it held. If it sold Treasuries, it had to buy something else, and that something else, euro bonds or Japanese bonds or gold, would rise in price and partly offset the damage. Third, and most important, a collapse of the dollar and the American economy would destroy China's largest export market and the value of its remaining dollar assets. Mutual economic vulnerability is not as stable as mutual nuclear vulnerability, but it rhymes with it.

This was, I think, why Beijing sold $200 billion and not $500 billion. It wanted to frighten us, not to ruin itself. A successor should look for these self-imposed limits in an adversary's economic campaign. They tell you where his real red lines are.

The Response

Over the week of 26 June to 3 July, the Treasury and the Federal Reserve together took a series of actions which, taken together, constituted the largest coordinated defense of the Treasury market since 2036. I describe them here in some detail because a successor will need to know what tools exist and who controls them.

The Federal Reserve is independent. The President does not command it and must not try to. In the 2040s the Chair of the Federal Reserve was a woman of great integrity who had been appointed by my predecessor and whom I had renominated, and our relationship was formal and, I believe, correct. I never once asked her to do anything. Daniel spoke with her daily, as Treasury Secretaries always do, about market conditions, and that is the limit of what the executive branch should do.

The Fed acted on its own judgment, within its mandate to maintain financial stability and the functioning of markets. On 27 June it announced that it would conduct temporary purchases of longer-dated Treasury securities to "restore orderly market functioning," a tool it had used in March 2020 and in 2036, and it expanded the Standing Repo Facility, through which banks and dealers can borrow cash against Treasuries, so that no institution would have to dump bonds into the market simply to raise cash.

The Treasury acted within its own authority. Daniel shifted new government borrowing toward shorter-dated bills, which were in strong demand as a safe asset, and away from long-dated bonds, which were where the pressure was. He expanded the Treasury's buyback program, buying back older, less liquid securities to support market functioning. And he and his team coordinated with Japan's Ministry of Finance, the largest foreign holder of Treasuries, which publicly announced that it had "no intention" of reducing its holdings and privately did more.

Finally, we made the price of the selling visible. On 29 June, Daniel gave a speech at the Economic Club of New York. Without naming China, he said:

"Some have chosen to sell American government debt at a loss in order to send a political message. They are free to do so. But I would note that every dollar sold at a loss is a dollar taken from their own people's savings. The United States will pay every obligation it owes, on time, in full, as it always has. Those who bet against that promise have lost that bet for two hundred and fifty years."

On 3 July, the ten-year yield peaked, intraday, at 7.1 percent. It was the highest yield on the ten-year note since 2001, the highest of my lifetime as an adult. For a few hours that morning I wondered whether the structure would bend. Then, by the afternoon, it began to fall. By the end of July it was back near 6 percent. It never returned to May levels during the crisis.

> **OPERATING NOTE — Your debt is part of your defense posture.** A country that borrows heavily gives its adversaries a lever that requires no navy. In a crisis, the bond market becomes a battlefield, and the President's room for maneuver narrows as interest rates rise. The tools of defense are mostly not yours: the Federal Reserve is independent and must remain so, and you must never ask it for anything. What is yours is the Treasury's debt management, coordination with allied finance ministries and central banks, a calm public voice, and the most important tool of all, credibility built over decades. A successor should treat fiscal discipline in peacetime as a matter of national security, because it is.

Sanctions and Rare Earths

Wednesday, 28 June 2045 — 14:00 EDT, Roosevelt Room

The sanctions debate, like the bond market, ran through most of June. I had authorized Treasury to prepare sanctions in March, during the quarantine. In June I had to decide whether to use them.

A successor should know the basic architecture. The President's principal legal authority for economic sanctions is the International Emergency Economic Powers Act of 1977, which allows him, after declaring a national emergency with respect to an "unusual and extraordinary threat" originating abroad, to block property and prohibit transactions. The Treasury Department's Office of Foreign Assets Control, called OFAC, administers most sanctions programs: it maintains the list of Specially Designated Nationals, with whom Americans may not deal and whose American-linked assets are frozen. The Commerce Department's Bureau of Industry and Security administers export controls,

through the Export Administration Regulations and the Entity List, which restrict the export of American technology. The State Department designates under its own authorities. The power of all of these tools comes from two things: the size of the American market, and the central role of the dollar in global finance. A foreign bank that is cut off from the dollar system cannot easily do business with anyone.

But sanctions are a weapon with a recoil. Every sanction on China in 2045 was also a sanction on some American company, some American farmer, some American consumer. And China had its own weapons, foremost among them its near-monopoly over the processing of rare earth elements and several other critical minerals.

Daniel presented three packages, which his team had labeled Phase A, Phase B, and Phase C, in order of severity.

Phase A: blocking sanctions on fourteen Chinese companies directly linked to the PLA's cyber operations and to the blockade, including shipbuilders that built Coast Guard cutters, two telecommunications firms whose infrastructure the intelligence community assessed had been used in the attacks, and several state-owned logistics firms involved in the "customs" regime; and a total ban on the export to China of advanced semiconductors, chipmaking equipment, and the software used to design chips, extending the controls first imposed in 2022 to essentially everything.

Phase B: sanctions on major Chinese state banks, cutting them off from the dollar system.

Phase C: a comprehensive embargo, the kind imposed on Iran or North Korea.

"Phase C is economic war," Daniel said. "Phase B is close to it. If we cut off the big state banks, every company in the world that does business with China has to choose. Many will choose China. And it would freeze trade that a lot of our own supply chains depend on."

"What happens to us if they retaliate on minerals?" I asked.

The answer was given by the Commerce Secretary's deputy, a mining engineer by training, and it was sobering. In 2045 China still processed the large majority of the world's rare earth elements, despite fifteen years of efforts by the United States and its allies to build alternative supply chains in Australia, Canada, the United States, and elsewhere. Rare earths are not rare in the earth's crust; they are difficult and dirty to separate and refine, and China had spent decades building that industry while the rest of the world looked away. They are essential for the permanent magnets in electric motors, wind turbines, guidance systems, and much of modern military hardware. China also dominated the production of gallium and germanium, used in semiconductors and in the radars of modern weapons, and graphite, used in batteries. It had used export restrictions on these materials repeatedly since 2023.

"How long can we last?" I asked.

"For defense production, we have a stockpile in the National Defense Stockpile and in industry inventories. Perhaps six to nine months for most critical items at current production rates. Less if we surge munitions. The allied supply chain, Australia, Malaysia, the new processing plants in Texas and California, covers perhaps thirty-five to forty percent of our defense needs. For civilian industry, much less. Car manufacturers would start idling lines within three months."

I drew a diagram. Two curves. One showed the damage that sanctions would do to China over time. The other showed the damage China's mineral restrictions would do to us. The first curve rose faster at first, because the chip ban hit China's technology sector immediately. The second rose

slowly, and then, at around six months, crossed the first, because our stockpiles ran out and China's chip industry partly adapted.

"So the question," I said, "is not who hurts more. It's who hurts more at the moment the decision gets made. And the moment that matters is not now. It's six months from now."

Sam saw it at once. "Which means we want sanctions that bite hardest early and that we can escalate if needed, and we want to buy time on minerals."

That became the policy. I approved Phase A on 28 June, effective 1 July. I held Phase B in reserve and told Beijing privately, through Helena's channel to Foreign Minister Shen Yilan, that it existed, that it was ready, and that it would be used if China attacked American civilian infrastructure again or if the blockade were enforced by lethal force against ships under the reinsurance backstop. A sanction held in reserve, credibly, is often more useful than a sanction imposed: it gives the adversary a reason to stop.

And I signed, the same day, a set of orders under the Defense Production Act, the 1950 law that allows the President to direct industry in support of national defense, to accelerate the domestic and allied processing of rare earths, gallium, and germanium, including loan guarantees and purchase commitments. Those orders would not help in June. They might help in January. A successor should understand that some of the most important decisions in a crisis are investments whose returns arrive after the crisis is over, and some that arrive only in the next crisis. I was, in 2045, living off investments made by others in the 2020s and 2030s, and grateful for them, and wishing they had been larger.

China's response came on 3 July, the same day the bond yields peaked. The Ministry of Commerce announced that, "in order to safeguard national security and interests," exports of all rare earth elements, rare earth magnets, gallium, germanium, and related technologies to the United States, and to any company "supplying the United States military," were prohibited, effective immediately. It was the total restriction we had feared.

That evening Claire found me in the Treaty Room, the study on the second floor of the residence, with the two curves on my desk.

"You drew it again," she said.

"I drew it again."

"Did it change?"

"No. That's the problem with models. They don't change because you're tired."

She sat on the arm of the chair. She is a pediatric oncologist, and her work has taught her more about endurance than mine has taught me. "My patients' parents ask me the same question every week," she said. "How long can we keep doing this. And the answer is always the same. Longer than you think, as long as you know what you're doing it for. Tomorrow you have to tell people what we're doing it for."

Decision	Options considered	Reversible?	Risk	Chosen	Outcome
Economic sanctions on China in response to the customs blockade and cyberattacks (June–July 2045)	Phase A: blocking sanctions on 14 PLA-linked firms plus total ban on exports of advanced chips, chipmaking tools, and chip design software. Phase B: cutting major state banks off from the dollar system. Phase C: comprehensive embargo	A: largely reversible (designations can be lifted; export controls relaxed). B: hard to reverse; reshapes global finance. C: very hard to reverse	A: Chinese retaliation on rare earths (expected). B: global financial disruption, allied split, acceleration of non-dollar payment systems. C: economic war, global recession	Phase A, effective 1 July; Phase B held in reserve and privately signaled; parallel Defense Production Act orders to rebuild mineral supply	China banned rare earth, gallium, and germanium exports to the US and its defense suppliers on 3 July. Stockpiles held through the crisis; allied supply rose. Phase B never used; its existence was raised by Shen Yilan at Sentosa in August

The Patience of Free People

Tuesday, 4 July 2045 — 20:00 EDT, the South Lawn

I wrote the Independence Day speech myself, over four nights, in longhand, on yellow pads. My speechwriters, who were very good, edited it, and the best lines in it were theirs; the structure and the argument were mine. I wanted it to do one thing, which was to slow the political clock.

By early July the pressure on me from the public and from Congress was coming from two directions at once, which is often a sign that a president is roughly in the right place but has not explained himself. From one side, people said that I was weak: that China had attacked our grid and dumped our bonds and was starving Taiwan, and that I had responded with "paperwork and a few hacked trucks." From the other side, people said I was reckless: that I was risking nuclear war for an island most Americans could not find on a map, that mortgage rates were rising because of my stubbornness, and that I should "make a deal."

I spoke at 20:00 from a podium on the South Lawn, before the fireworks, with about two thousand guests: military families, workers from the West Coast ports, a nurse from a Houston hospital that had run on generators for three days, a lineman from a Texas electric cooperative who had worked thirty-one hours straight restoring service, and the parents of a Taiwanese sailor killed on the *Formosa Pearl* in March. I had asked that they be there, and I spoke to them.

I reproduce the core of the speech here, as delivered.

"Two hundred and sixty-nine years ago, a group of men in Philadelphia signed their

names to a sentence that was, at the time, a dangerous act. They knew it might cost them everything. They signed it anyway, not because they were certain of winning, but because they were certain of what was right.

"This summer, our country has been tested in a new way. Our ports have been disrupted. The power in Texas went out in a heat wave, and some of our fellow citizens died. The cost of borrowing has risen for every family with a mortgage. Prices on some goods are going up. I will not pretend to you that these things are small, or that they are over. They are not.

"I want to tell you plainly what is happening and why. The government of the People's Republic of China is trying to force the people of Taiwan, a democracy of twenty-three million, to accept by pressure what they will not accept by choice. And it is trying to persuade you, the American people, that standing with them is too expensive. It has not tried to defeat our navy. It has tried to defeat our patience.

"It will not succeed.

"I want to be equally plain about what we will not do. We will not answer an attack on our civilians with attacks on theirs. We will not seek a war. We do not seek to humiliate China or to change its government. We do not support a unilateral change to the status quo across the Taiwan Strait by anyone, and we have said so to every party. What we will not accept is that the status quo be changed by force or by coercion, by warships or by paperwork.

"Some have said that free societies are too impatient for long contests. That democracies cannot bear pain. That we will fold when the bills come in. I have heard that argument all my life, and history has answered it every time. Free people are slow to anger and slow to fight. But they are also slow to quit. Our patience is not weakness. It is the deepest kind of strength there is: the strength of people who do not need to be ordered to endure, because they have chosen to.

"So here is what I am asking of you. Not to hate anyone. Not to fear. I am asking you to be patient. To check on your neighbors when the power fails. To understand that the price we are paying this summer is the price of a world in which big countries cannot simply take what they want from small ones. And to know that every day we hold steady, without war and without surrender, is a day the other side learns that pressure will not work.

"The men in Philadelphia did not know how their story would end. Neither do I. But I know how this chapter must be written: calmly, firmly, and together."

I had a sentence in the draft, deleted on the morning of 4 July, that I want to record here because a successor might be tempted to say something like it. The sentence was: "If the People's Republic of China attacks American forces, it will face the full power of the United States." It was true. It was also a statement that would be read in Beijing, by Luo Tiancheng and men like him, as either a bluff or a threat, and in either case would add nothing except pressure on me to fulfill it later. Sam struck it out with a pencil and wrote in the margin: *They already know. Saying it makes it about your credibility instead of their choices.* He was right.

The speech was not a triumph. Polls in the following week showed a modest rise in support for my handling of the crisis, from 44 to 49 percent, within the range of noise. But it did something I had

hoped for. In the two weeks after the Fourth of July, the political argument in the country shifted from "what should we do right now" to "how long can we keep this up." That is a different question, and a better one. It is a question about the political clock, not the information clock.

Afterward, on the Truman Balcony, while the fireworks went up over the Mall, Lt. Col. Brooks stood a few steps behind me with the satchel, as always. I asked him, because I sometimes asked him things, what the Marines in his old unit thought.

"Honestly, sir? They thought it was a good speech. They also thought it was a speech for civilians." He paused. "Marines don't need to be asked to be patient. We need to be told what the objective is."

"And what is it?"

"I think you said it, sir. Hold steady without war and without surrender. The hard part is the without war."

> **OPERATING NOTE — In a long crisis, the President's voice is a stabilizer, not an accelerator.** Use public speech to slow the political clock, not to win the news cycle. Tell the public the truth about costs; they will forgive pain they were warned about and will not forgive pain they were told would not come. State what you will not do as clearly as what you will. And cut every threat that adds nothing your adversary does not already know: a public threat converts a question about his choices into a question about your credibility, and then you are bound to it.

The Man Who Owned the Sky

Monday, 10 July 2045 — 09:15 EDT

Of all the confrontations of the summer, the one I least expected was with an American.

A reader of Chapter 1 will recall how ARGUS worked. It was not a single satellite system but a fusion: the National Reconnaissance Office's own imaging, radar, and signals satellites; the Space Force's Proliferated Warfighter Space Architecture tracking layer; and, critically, hundreds of sensor payloads hosted on commercial satellites, mostly on the mega-constellation operated by Halcyon Orbital. Earlier in the century the government had learned, from SpaceX's Starshield program and others, that it was far faster and cheaper to put government sensors on commercial satellite buses, built on commercial production lines and launched on commercial rockets, than to build every spacecraft itself. By 2045 the hosted payloads on Halcyon's satellites provided a large share of ARGUS's revisit rate, the ninety-second cadence that let us claim custody of almost every Chinese road-mobile launcher outside garrison. Without Halcyon, the revisit time over western China would, in the worst case, roughly triple.

The contracts that governed those hosted payloads had been written in peacetime. They allowed the government to use the sensors for "intelligence, surveillance, reconnaissance, and missile warning." They said nothing explicit about "targeting," meaning the use of the data to direct weapons against specific objects. That gap had seemed academic to the lawyers who wrote it in 2038.

On the morning of 10 July, it ceased to be academic. Marcus Thorne, Halcyon's founder and chief executive, a man worth a sum of money I will not attempt to state because it changed daily, notified the Space Force, the NRO, and, through his lawyers, the White House Counsel, that Halcyon would "suspend the provision of data from hosted payloads for any use relating to the selection or

engagement of targets" effective 14 July, and would require the government to certify, for every data stream, that it was not being so used.

His letter gave his reasons. I will summarize them fairly, because they were not foolish. Chinese state media had for weeks been describing Halcyon's constellation as "an American military system wearing a civilian mask," and a senior PLA officer had written in the *PLA Daily* that "commercial satellites that participate in war are lawful military objectives." Halcyon had more than eleven thousand satellites in orbit providing internet service to customers in a hundred and forty countries, including airlines, ships, hospitals, and, as he pointed out, Taiwan itself. If China decided to attack the constellation, the debris and the loss of service would hurt millions of people who had nothing to do with the crisis. Halcyon's insurers had told him that its coverage did not extend to acts of war. And, though the letter did not say this, he had a large business in markets that China influenced.

Sam brought the letter in at 09:15. Gen. Dunmore was with him, and so was the Director of the National Reconnaissance Office.

"How much does this hurt?" I asked.

"If he means it literally, a lot," the NRO Director said. "We can't certify, stream by stream, that ARGUS data isn't used for targeting. That's the whole point of fusion. The data goes into a common picture, and the common picture supports everything: warning, intelligence, and, if it ever came to it, targeting. If we have to segregate the Halcyon streams, we lose most of their value. Custody of mobile launchers outside garrison drops from about 94 percent to something like 70 percent."

That number, 70 percent, I would hear again in September, under far worse circumstances. In July it was a hypothetical. It was enough.

Inner Dialogue

I spent an hour alone in the small study off the Oval Office before I called anyone. I want to record what went through my mind, because it surprised me.

My first feeling was anger; my second, about ten minutes later, was recognition. I had been, before I went into politics, the founder of a launch-components company. I had sold valves and guidance hardware to the government and to commercial customers. I knew exactly what it felt like to read a government contract and realize that one clause, written by a lawyer in a hurry, could bankrupt you. If I had been in Thorne's position, with my satellites declared a military target by a nuclear power and my insurers refusing cover, I would have been frightened too.

My third thought was the Mirror Model, turned to my own side. *Tom, what does Thorne see when he looks at you?* He sees a government that has made his company a central part of its military machine without ever saying so publicly, without sharing the risk, and without asking him. He sees the possibility that his life's work will be destroyed in an afternoon of anti-satellite fire, and that the government will thank him afterward and pay nothing.

And my fourth thought was the most uncomfortable. He was right that something important had been done without being decided. The United States had built its eyes on private infrastructure, and in doing so had made that infrastructure a target, and the companies and the public had never been told plainly. That was a failure of my government and of the governments before mine. It was not his failure.

That did not mean he could be allowed to blind us.

The Defense Production Act

The Defense Production Act of 1950, written at the start of the Korean War, gives the President broad authority over industry in a national defense emergency. Its Title I allows the President to require that contracts and orders necessary for national defense be given priority over all other contracts, and to allocate materials, services, and facilities. Presidents had used it for munitions, for medical supplies during the COVID-19 pandemic, for critical minerals, and, earlier that very summer, for rare earths. The government's lawyers advised me that Title I could be used to issue a priority rating to the hosted-payload services under the existing contracts, requiring Halcyon to continue providing them on the government's terms, and that a refusal could, in principle, carry criminal penalties.

"Can we actually enforce it?" I asked the White House Counsel.

"Legally, yes, I believe so, though he would sue, and the courts haven't ruled on this kind of use. Practically, sir, it's his software. His engineers run his network. You can order him to provide the service. You can't easily make him provide it well."

That was the heart of it. The Defense Production Act could compel. It could not, by itself, produce cooperation. And what I needed was not grudging compliance from a constellation that a hostile owner could quietly degrade. I needed a partner.

So I decided to use both. I would sign the priority order, so that the legal question was settled and the man understood that I was willing to use the full authority of the office. And I would invite him to the Oval Office, alone, to find a deal.

Decision	Options considered	Reversible?	Risk	Chosen	Outcome
Halcyon Orbital's suspension of hosted-payload data for targeting-related uses (July 2045)	(A) Accept and segregate data streams; (B) Negotiate only, no compulsion; (C) Defense Production Act Title I priority order only; (D) Title I order plus direct negotiation on risk-sharing: government indemnity, liability protection, replacement guarantees	A: reversible, but custody of mobile launchers falls to about 70 percent during the crisis. B: reversible; slow, leaves the outcome to Thorne. C: legally strong, practically weak (compliance cannot compel quality). D: order can be lifted; deal terms are binding	A: blindness at the worst time. B: delay; signals the President can be overruled by a contractor. C: litigation, quiet degradation, public feud. D: cost to taxpayers; precedent for indemnifying commercial operators	D. Title I order signed 13 July; Oval Office meeting 14 July	Agreement 14 July: government indemnity against losses from hostile action, liability protection, priority launch of replacements, joint public statement. Halcyon's cooperation proved decisive after the September counter-space attacks

Friday, 14 July 2045 — 11:00 EDT, Oval Office

Marcus Thorne arrived in a dark suit, without a tie, and with a single aide whom he left in the Roosevelt Room. He was taller than I expected, and quieter. He had a reputation for saying outrageous things in public. In private he was precise, and, I came to believe, frightened.

I had signed the Title I priority order the night before, and his lawyers had received it at 07:00. He opened with it.

"You've drafted my company, Mr. President."

"I've told your company it has a priority customer," I said. "I'd prefer not to have had to."

"Then why did you?"

"Because I need you to understand that I'm willing to, and then I need you to understand that I'd rather not. Both are true. Sit down, please."

He sat on one of the two couches. I sat across from him, not behind the desk. I had asked that no one else be in the room, against the preference of the Counsel, because I believed that what had to be said could not be said in front of lawyers.

"Mr. Thorne, I want to tell you three things, and then I want to hear what you need. First: you're right that we put your company in danger without asking you, and without telling the public. That was a mistake by this government and previous ones. I'm sorry for it. Second: I cannot let the United States go blind over China in the middle of a crisis with a country that has fifteen hundred nuclear weapons, some of them on launch-on-warning. I'll tell you more than I'd normally tell a contractor. Our ability to see their mobile missiles is part of what keeps both sides from panicking. If we can't see, we have to assume the worst. Your satellites are not only targeting tools. They're part of what lets me tell my own generals to stand down. Third: I'm not going to ask you to carry this risk alone."

He was quiet for a long time. Then he said, "Do you know what keeps me awake? It's not the satellites. I can build satellites. I build forty a week. It's the debris. If they go after my constellation with direct-ascent weapons the way they've tested, low orbit fills with fragments. Everybody's satellites. Weather satellites. Your own. Years of it. I'd be the man whose company made low earth orbit unusable."

"Then help me make sure they never need to," I said. "The best way to keep them from attacking your satellites is to make the attack pointless. If they know that every satellite they destroy will be replaced within weeks, and that the United States considers an attack on its warning and surveillance satellites an extremely grave act, they're less likely to try."

We talked for an hour and forty minutes. By the end we had the outline of an agreement, which our lawyers finalized that evening. The government would indemnify Halcyon against losses caused by hostile action against its satellites, up to a substantial ceiling. It would provide liability protection against third-party claims arising from the government's use of the hosted payloads. It would guarantee priority launch and procurement of replacement satellites in the event of attack. Halcyon would continue providing all hosted-payload services without stream-by-stream certification, under the priority order, which would remain in force but which both sides would describe publicly as "a framework for cooperation."

And we agreed, at his insistence, and I think rightly, on one thing that was not in any contract: that the government would, after the crisis, ask Congress to write in law a clear framework for the

wartime use of commercial space infrastructure, so that the next company would know, before it signed, what it was agreeing to.

At the door he said, "Mr. President, I'm told you're an engineer."

"Control theory."

"Then you know that a system with one point of failure isn't a system. It's a bet."

"I know," I said. "This summer I'm learning how many bets the country has made."

> **OPERATING NOTE — Commercial space is now part of your nuclear stability, and its owners are not your subordinates.** Much of your sight from orbit, and much of your communications, will run on satellites owned by private companies with shareholders, insurers, foreign customers, and fears of their own. In peacetime, settle in law and contract who carries the risk of hostile action, what the data may be used for, and who replaces what is destroyed. In a crisis, the Defense Production Act can compel priority, but it cannot compel good faith. Use the authority to settle the legal question, then use the room to win the partner. And remember: a constellation that helps you see the adversary's missiles also helps you persuade your own generals that there is no need to strike them.

WILLOW: The Hospital

Friday, 28 July 2045 — 22:40 EDT, Treaty Room

Gregory Laszlo, the Director of the Central Intelligence Agency, called on the secure telephone in the residence at 22:40 and asked whether he could come over. Directors of Central Intelligence do not ask to come to the residence at 22:40 for good news. I said yes.

He arrived at 23:15 with Ruth Calder, the chief of the CIA's China Mission Center, the case officer who had recruited LANTERN more than a decade earlier and who, though I did not know it then, also oversaw the handling of WILLOW. Calder was a small woman in her late fifties with a voice like gravel, and I had learned, over the spring, that when she spoke slowly it meant she was certain.

She spoke slowly that night.

"Mr. President, WILLOW reports that General Secretary Wen was taken from Zhongnanhai to the 301 Hospital, the People's Liberation Army General Hospital in Beijing, on the night of 19 July, and remained there for four days. WILLOW did not see medical records. WILLOW saw message traffic rerouting certain categories of command communications that normally go to the Chairman, and she heard duty officers discuss 'the Chairman's heart.' She assesses, and we agree with moderate confidence, that Wen suffered a cardiac event. He returned to Zhongnanhai on 23 July. He has appeared on state television once since, in what appears to be footage recorded beforehand."

I asked the questions a president must ask about any human intelligence report. How did she know? What was her access? What could she not see? Could she be wrong, or could she be fed? Calder answered each one. WILLOW's access was to message traffic and to conversations among officers on her shift. She could not see Wen. She could not see the Politburo. Her reporting was consistent with two other, independent indicators which I am still not permitted to describe. [REDACTED — 3 lines] And there was no sign, as yet, that WILLOW was under suspicion.

"There's more, sir," Calder said. "During the four days, command traffic that would normally go to the Chairman for decision went instead to the CMC Vice Chairman, General Zhang Hesheng, as you'd expect. But WILLOW reports that at least two directives concerning Rocket Force readiness training were issued over General Luo's authority and noted to Zhang afterward, not before. And she reports a change in tone. Officers who had been cautious in June are now talking about 'the window,' meaning the idea that the balance with the United States will only get worse for China as Golden Dome matures, so China must act while it still can. Luo's people are using the phrase. More of them are in the operations center than in the spring."

"So while Wen was in the hospital," I said, "Luo moved his people forward."

"That's our assessment, sir. Not a coup. Nothing like a coup. Just weight shifting in a room."

When they had left, I sat in the Treaty Room for a long time. Claire was asleep. I opened the diary and wrote:

28 July. Wen's heart. Four days in 301. Luo's men in the room.

I built my whole model of this crisis around one man in Beijing who does not want war and is trapped by the nationalism he helped build. I assumed his hand was on the controller. What if his hand is weak? What if it is sometimes not there at all?

In control theory, when the controller drops out, the system doesn't stop. It keeps running on its last command, or on whatever local loop takes over. And local loops are tuned for local goals. Luo's local goal is not peace. It is not to lose his force.

The question is no longer only what Wen wants. It is who is holding the controller on any given night.

I need a second channel. Not to Wen. Around him. To the people who also do not want this.

I underlined the last sentence twice. Those few lines, I see now, were the seed of everything I did in November.

There was one more consequence of WILLOW's report, which I record because a successor must think about it every time he reads a human source: the more valuable WILLOW's reports became, the more I wanted from her, and the more I wanted, the more danger I put her in. Calder said as much at the door. "Mr. President, every question you send her is a risk to her life. Please send fewer, and better ones." I took the advice. I wish I had taken it more seriously still, in other cases, later.

Sentosa

Saturday, 19 August 2045 — Singapore

The second channel began with a telephone call from the Prime Minister of Singapore, Daniel Tan Wei Ming, on 4 August. Singapore's leaders have practiced for sixty years the hardest art in diplomacy, which is to be trusted by both sides of a great-power rivalry without being owned by either. Tan said only that Foreign Minister Shen Yilan would be "in the region" on 19 August, that Singapore would be honored to offer "a quiet lunch," and that no one in Singapore would ever confirm it had happened.

A successor should understand why back channels exist. Official channels, the Jade Line, the ambassadors, the formal talks, are performances as well as conversations. Every word is recorded,

reported upward, and judged by hardliners for weakness. A back channel lets two people say what they cannot yet say officially: what they fear, what they might accept, where the other side's red lines really are. Its value depends on three things: deniability, so that nothing said there commits anyone; trust between the two negotiators; and an intermediary with no interest in the outcome except peace.

Helena flew on a plain government aircraft listed as a trip to Canberra. The meeting took place in a private residence on Sentosa, the island off Singapore's southern coast, with Prime Minister Tan's foreign policy adviser in the next room and no one else.

Helena's notes, which she gave me in the Oval Office on 21 August, run to eleven pages. The essentials were these. Shen came alone, which mattered. She did not repeat Beijing's public positions; she said, "We both know them." She raised Phase B of our sanctions herself, which told us our private signal in July had been heard. She said Beijing had read our June notification, "what you did not do," at "the highest level." And when Helena asked, carefully, about the Chairman's health, Shen looked at her for a long moment and said, "The Chairman is directing the work of the Party." It was not a denial. It was the sentence one uses when a denial would be a lie.

Then Shen said something Helena underlined. "There are people in Beijing who believe time is against China. Your Golden Dome. Your eyes in the sky. They believe that if China does not act in the next few years, it will never be able to act. You must understand that this is not propaganda. They believe it. If you want to help those of us who disagree, you must give us something to show that time is not against us."

Nothing was agreed at Sentosa. Nothing was meant to be. But a channel had been opened between two women who had known each other for fifteen years, through a host both trusted, and it would carry, in November, the most important messages of my life.

There was one more voice that month. On 12 August President Arya Wicaksono of Indonesia, chair of the Association of Southeast Asian Nations that year, called to say that Jakarta would press for an ASEAN foreign ministers' statement calling for "restraint, freedom of navigation, and the peaceful resolution of disputes in accordance with international law," and that Indonesia, which had long refused to take sides between Washington and Beijing, would offer Jakarta as a venue for any future talks. The statement, issued on 16 August, did not name China. It did not need to. That ten Southeast Asian governments, many deeply tied to China's economy, said it together was a signal Beijing could not dismiss. I thanked him, and said so publicly. A successor should value such voices: middle powers do not end crises, but they change the room in which great powers decide.

> **OPERATING NOTE — Open the back channel before you need it.** A back channel takes months to become trustworthy and seconds to be needed. Build it in the quiet weeks: choose a negotiator the other side respects, a host both sides trust, and rules that nothing said there commits anyone. Its first meeting will produce nothing, and that is its purpose. What you are building is not an agreement but a pipe, so that when the hour comes, the message can get through around the people who would stop it.

THRUSH

Wednesday, 30 August 2045 — 23:55 EDT

The summer ended the way it had begun: with a report that changed the shape of the problem.

THRUSH was Mei Zhen, a civilian engineer employed by a contractor that maintained support equipment at the submarine base near Yulin, on the southern tip of Hainan, home of China's six Type 096 ballistic-missile submarines, each carrying JL-4 missiles. Her reports came rarely and were always precise. This one arrived through Calder at 23:55, and Sam brought it up himself.

In the preceding forty-eight hours, three Type 096 submarines had left Yulin. Never before had more than one been seen to sail in so short a period; China's submarines had historically spent most of their time in port. ARGUS imagery confirmed three empty berths.

A successor should understand why this mattered. A ballistic-missile submarine at sea is the most survivable part of any nuclear force; its purpose is to guarantee that, whatever happens to a nation's land-based missiles, it can still retaliate. A country surging its submarines is, in one sense, doing something stabilizing: making its deterrent safe from a first strike. In another sense it is preparing for the possibility of nuclear war. Both readings were true. Which mattered more depended on who had given the order, and why.

"Admiral Hale's view?" I asked.

"Sending them to sea is what he'd do if he were frightened of us," Sam said. "His word was 'defensive.' He also said it's what he'd do before anything else."

I thought of Shen at Sentosa: *They believe time is against China.* I thought of the empty bed at 301 Hospital, and the weight shifting in the room in the Western Hills. I thought of Maya, who was at that hour on USS *Daniel Inouye* somewhere east of the Philippines, standing a watch.

In my diary that night I wrote:

30 August. Three boats out of Yulin. The pressure season is over. They pushed on our ports, our grid, our bonds, our minerals, our companies, and our patience, and none of it broke. That should be good news. It isn't, quite. A man who cannot win by pressure has two choices left: to stop, or to push with something heavier.

Which one is Luo? Which one is Wen? Who is holding the controller tonight?

Slow the military clock. Speed the political clock. Keep the pipe to Singapore open.

God help us in September.

Chapter 4: Blood in the Strait

"Tonight I learned the exact weight of forty-seven minutes. I hope no successor ever has to learn it, and I know that some of you will." — Diary of Thomas Ellery Kane, 12 September 2045, 04:10

A Word Before This Module

Every operating system has a section where the system meets the world and the world wins for a while. This is that section.

Until September 2045, the Pacific crisis had been fought with cutters and customs forms, with cyber intrusions and Treasury auctions, with speeches and sanctions lists. People had died: the two Taiwanese sailors of the *Formosa Pearl* in March and the vulnerable in Texas during the June blackouts. But no American service member had been killed by Chinese fire, and no Chinese service member had been killed by ours. A line had held, a line that had held since the Korean War. In the first eleven days of September that line was crossed twice, once by them and once by me.

I have tried in this chapter to do three things. First, to tell what happened, as honestly as my diary and the records allow. Second, to teach a successor four pieces of machinery that matter most when shooting starts below the nuclear threshold: the conventional escalation ladder, missile defense at sea, counter-space warfare, and the military alert levels we call DEFCON. Third, to set down the human cost, which is not a separate subject. It is the subject. The machinery exists to manage it, and a president who forgets that will manage the machinery beautifully and lose the country.

I will also describe a decision about two men in China whom I never met. One of them died because of the work he did for us. The other I kept in danger when I could have tried to bring him out. I have not made my peace with either fact, and I do not intend to pretend that I have.

1. The Line on the Map

Friday, 8 September 2045 — 06:05 EDT

The notice came as a file. At 17:00 Beijing time, 05:00 in Washington, China's Ministry of National Defense published a statement with a map: a box east of Taiwan, from the waters off Hualien south to the Bashi Channel, reaching some one hundred and forty nautical miles into the Philippine Sea. Within it, "all aircraft and vessels not authorized by the competent authorities of the People's Republic of China are prohibited from entry." There was no end date.

Sam Oyelaran brought it to the residence at six with coffee. He had circled one word in red. *Prohibited.* In March they had said "inspected." In June they had said "cleared through Fujian." Now they said "prohibited."

"They've gone from customs to geography," Sam said. "A customs blockade says: your cargo is subject to our law. A zone says: this water is ours. It's the difference between a toll booth and a wall."

I asked myself the question that by then I asked every morning. *Tom, what do they see when they look at us?* In this case I thought I knew. Since OPEN DOOR in April, our convoys had used Taiwan's east coast, open ocean where Chinese law-enforcement fiction could not easily follow. The east side was Taiwan's lung. Beijing had decided to close it. They had chosen to do so with a declared zone rather than with fire, which told me they still wanted us to blink rather than to fight. But the box sat exactly across the route that Convoy Seven was scheduled to take on Monday.

A successor should understand what such a zone is and is not. In peacetime, any state may issue notices to mariners and airmen warning of hazardous activity — missile tests, live-fire exercises — and prudent ships stay clear. Under the law of the sea as the United States reads it, those notices are warnings, not grants of sovereignty: on the high seas beyond twelve nautical miles, a coastal state cannot simply forbid passage. Beijing's genius, if that is the word, was to issue a document that could be read either way. To its own people it was an assertion of sovereignty. To lawyers it was a hazard warning. To us it was a dare.

08:30 — Situation Room

The Principals Committee — the cabinet-level body of the National Security Council (NSC) — met at half past eight; I attended, which made it a meeting of the NSC proper. Present were Vice President Rosalind Achterberg; Sam; Secretary of State Helena Marsh; Secretary of Defense Ray Castillo; General Patricia Dunmore, Chairman of the Joint Chiefs; Director of National Intelligence Abigail Fenn; CIA Director Gregory Laszlo; and Treasury Secretary Daniel Iwu. Admiral Kenji Moriyama, commander of Indo-Pacific Command (INDOPACOM), joined by secure video from Camp H. M. Smith in Hawaii, looking as though he had not slept since August.

Moriyama gave us the facts. Convoy Seven — designated OPEN DOOR 7 — consisted of four US-flag cargo vessels carrying fuel, medicine, grain, and grid components, and one chartered liquefied natural gas carrier under the Japanese flag, the first time a Japanese-flag hull had joined. The escort was three US destroyers — USS *Thomas Hudner*, USS *Daniel Inouye*, and USS *Rafael Peralta* — and two Japan Maritime Self-Defense Force destroyers, JS *Chōkai* and JS *Teruzuki*. Air cover would come from a carrier strike group well to the east and from Kadena on Okinawa. The convoy would cross the eastern edge of the new zone, on its approach to the Bashi Channel and Kaohsiung, early on Tuesday, 12 September, local time — Monday evening in Washington.

He did not need to tell me who was serving aboard *Daniel Inouye*. Everyone in the room knew. Nobody looked at me when the name was read. I noticed that nobody looked at me, which is its own way of being looked at.

"Options," Sam said.

There were, in the end, four. We could turn the convoy around and lose everything the summer had bought. We could hold it outside the zone and negotiate, which Beijing would read as turning it around slowly. We could reroute it far to the south and west, adding days, through waters where PLA Navy (PLAN) forces were thicker. Or we could sail on schedule through the zone, on the published route, at the published time, with escorts visible and announced, and with the Japanese beside us.

Ray Castillo put it plainly. “If we stop at a line they drew on a map on a Friday morning, they’ll draw a new line every Friday morning until they reach Guam.”

Helena Marsh answered him, and I record her words because she was right too. “If we sail into a box they have declared closed, we are handing the decision about war to a twenty-six-year-old PLAN weapons officer who has been told for six months that we are pirates.”

I asked Abigail Fenn what the intelligence said about intent. Signals intelligence showed new rules of engagement issued to PLAN Eastern Theater units in late August; we did not have the text. WILLOW, our source in the Central Military Commission’s Joint Operations Command Center, reported that the zone had been approved at the level of the CMC, and that General Luo Tiancheng of the Rocket Force had argued for it hardest, on the grounds that “the Americans will not die for a gas tanker.” LANTERN, at Rocket Force Base 64, had nothing new; the strategic forces remained at Readiness Level 3, the heightened-but-not-preparing level they had held since March.

“So the people who drew the line are the people who think we won’t cross it,” I said.

“That is one reading, Mr. President,” Fenn said. “The other is that they think we will, and want to be seen to have warned us.”

The model on the legal pad

I asked for twenty minutes alone. I recommend this rarely, because a president who leaves the room too often teaches his team that the real meeting is elsewhere. But this was a decision that would send my daughter’s ship into a box drawn by people who might fire on it, and I did not trust myself to separate the President from the father without a pen.

I drew the Reversibility Matrix, the two-by-two I describe in Chapter 1, and placed each option on it. Turning back was fast and, in a narrow sense, reversible: we could sail again later. But politically it was irreversible, because the precedent would be set. Sailing through was fast and its consequences were potentially irreversible in the most literal sense, because ships can sink and sailors can die. Holding and negotiating was slow and reversible — the kind of decision my own rule told me to prefer — but it quietly conceded the zone with every hour.

Then I drew the Signal Ladder, which is less a diagram than a habit: for each option, write down in one sentence what the other side will read. Turn back: *the line works.* Hold: *the line works, and they need time to save face.* Reroute: *the line works, and they are clever about it.* Sail on schedule: *the line does not work, and the Americans are not afraid.*

What I wanted Beijing to read was something more precise: *the line does not work, the Americans are not afraid, and the Americans are not attacking you.* A message with three clauses is hard to send with warships. So I decided we would send the third clause separately and loudly.

We sailed. But we sailed with four conditions that I wrote on the pad and that became the execute order’s commander’s guidance. First, the convoy would stay on the published route, which ran no closer than one hundred nautical miles to Taiwan’s eastern coast at its nearest point and well outside any reasonable reading of Chinese territorial waters, and would transmit its position openly on the Automatic Identification System. Second, our aircraft would not enter the zone ahead of the convoy; we would not appear to be clearing the box for an attack. Third, the rules of engagement for the escorts would be strictly self-defense — fire only at inbound threats, not at launching platforms — unless I personally changed them. Fourth, Helena would call Foreign Minister Shen Yilan, and the

Pentagon would use the Defense Telephone Link to call Chinese military counterparts, to say in plain words what the convoy was, where it would be, and what it would not do.

The Defense Telephone Link, or DTL, was established in 2008 for exactly such moments; for much of its life the Chinese side declined to pick up. By 2045 it had been joined by the JADE LINE, the leader-to-leader video link President Wen Zhaolin and I had first used in April. Neither instrument guarantees that anyone listens.

> **OPERATING NOTE — On lines drawn by others.** When an adversary draws a line, it is rarely asking you a yes-or-no question. It is asking you which of several messages you wish to send. Before you decide whether to cross, write down in one sentence each message available to you, then ask which instruments can carry each clause. Ships carry resolve; they carry restraint badly. Carry restraint by other means — words, transparency, deliberate omissions — and send them at the same moment, through more than one channel. A message received through only one channel is a rumor.

Friday, 8 September — 21:15 — a letter

That night I did something I had not done since March. I wrote to Maya.

The Navy has rules about this, and the President is not exempt from them in spirit, whatever the regulations technically allow. I did not call her ship. I did not ask her captain about her. I wrote an ordinary email to her ordinary address, the way her mother did every Sunday, knowing it would reach her whenever the ship's bandwidth allowed and knowing, too, that every sailor on that ship would someday hear that the President had written to his daughter the night he sent her into the box.

So I wrote nothing that a father would not write. That her brother Eli had finally fixed the leaking faucet in his Austin apartment by himself and wanted credit. That her mother had a new patient, a seven-year-old girl from Pueblo with neuroblastoma, who called Claire "Doctor Clara" and refused to be corrected. That the aspens on the Front Range would be turning in two weeks and that I intended to see them this year, crisis or no crisis, which was a lie and we both knew it.

I closed with a sentence I had rewritten four times. *Stand your watch, and come home.*

She replied on Sunday afternoon with two lines. *Standing it. Tell Eli I'll inspect the faucet at Christmas.*

I have that email printed and folded in the back cover of the diary volume that covers September 2045. It is soft now from handling.

2. The Bashi Channel

Monday, 11 September 2045 — 18:30 EDT

The anniversary did not escape anyone. Forty-four years after the attacks of 2001, I had laid a wreath that morning at the Pentagon and spoken for six minutes about vigilance. By evening the convoy was approaching the eastern boundary of the zone; it was Tuesday morning in the Western Pacific, 06:30 local time.

I spent the early evening in the Oval Office on things I could not postpone — among them a call with Prime Minister Hiroshi Saeki of Japan, whose sailors were in the escort. Saeki ended our call

with words I remember: “Mr. President, my grandfather’s generation learned what it costs to be the first to fire. I pray the Chinese remember their lesson about being the second.”

At 19:40 the convoy crossed into the box. Moriyama’s watch officers reported a PLAN Type 055 destroyer and two Type 054B frigates shadowing at twelve to eighteen nautical miles, Chinese maritime patrol aircraft overhead, and a swarm of small drones in loose groups of six to ten between the convoy and the coast. Chinese radio warnings came every fifteen minutes on the international distress frequency: *You have entered a prohibited area. Leave immediately or bear all consequences.* Our ships answered each time, politely, in English and Mandarin, that they were transiting the high seas in accordance with international law.

I had dinner with Claire in the residence at eight. It was our first dinner together in nine days. She had cooked, which she does when she is frightened: green chile stew, from her mother’s recipe out of Pueblo, which she makes too spicy on purpose so that I will drink too much water and therefore sleep. We did not talk about the convoy. We talked about her seven-year-old patient from Pueblo, who had finished her second round of chemotherapy and wanted a dog. At 21:20 Claire said, “You keep looking at the door,” and I said I was sorry, and she said, “Don’t be sorry. Just eat.”

21:40 EDT

The phone in the residence that connects to the Situation Room is not loud. It does not need to be.

It was Sam. “Mr. President, the convoy is under missile attack. Multiple inbound. Please come down.”

I remember three things about the walk: that Claire stood up when I did and did not ask anything; that Lieutenant Colonel Jamal Brooks, the military aide on duty that night, was already in the corridor with the satchel we call the football, and that he fell into step without a word, as he had been trained to, a half pace behind and to my left; and that the elevator was slower than any machine I have ever known, and that I, an engineer who has spent his life on the arithmetic of time, could not do the arithmetic of the four floors.

21:44 — Situation Room

The large screen showed what is called the common operational picture: a digital chart of the waters east of Taiwan — blue for our ships and the Japanese, red for Chinese. Red tracks had converged on the blue cluster and already vanished. A successor must understand that by the time a president sees a missile engagement at sea, it is over. The engagements I am about to describe took less time than it has taken you to read this paragraph.

What had happened — I pieced it together that night, and the Navy’s investigation confirmed most of it in December — was this.

At 21:31 Washington time, 09:31 local, one of the drone groups had turned toward the convoy and closed to under eight nautical miles. Several drones in such groups had, earlier in the summer, carried small explosive payloads in exercises; others were purely for surveillance or electronic jamming. Nobody aboard *Thomas Hudner*, the destroyer on the convoy’s western flank, could know which these were. Under the self-defense rules I had approved, a ship may engage a threat that demonstrates hostile intent. The commanding officer of *Hudner* judged, after two warnings and a turn that put the drones on a direct course for the LNG carrier, that they did. At 21:34 *Hudner* fired two short-range interceptors and engaged with her close-in guns, and destroyed four drones.

At 21:35, a Chinese frigate reported to its command — we heard the traffic, and the Navy's investigators reconstructed it — that the Americans had "opened fire." Whether the frigate's officers understood that *Hudner* had fired at drones and not at Chinese crewed vessels, whether the report was garbled as it went up, whether it was received by people who wanted to be told exactly that — this the investigation never settled, and WILLOW's later reporting suggested all three were partly true.

At 21:37, a land-based battery on the Fujian coast, and two PLAN surface ships inside the zone, launched a salvo of anti-ship missiles at the convoy. The salvo included YJ-21 missiles, which the Chinese had shown publicly at an airshow in 2022 and which are described in open sources as hypersonic: they climb steeply and dive on their targets at many times the speed of sound, leaving defenders only seconds in the terminal phase. The salvo also included slower sea-skimming missiles, which fly a few meters above the waves to stay beneath radar horizons until the last moments. We counted later twenty-two missiles in all.

How a destroyer defends itself — a lesson for successors

I must stop here and teach, because a president who orders ships into harm's way should know what they can and cannot do, and I did not know it well enough before that night.

An American guided-missile destroyer of the *Arleigh Burke* class carries the Aegis combat system, the name for the integrated radar, computer, and weapons that together defend the ship and the ships around it. By 2045 most of our destroyers had the AN/SPY-6 radar, able to track many targets at once. The ship's missiles sit in vertical launch cells, ninety-six on a ship of this type holding a chosen mix of long-range interceptors such as the Standard Missile-6, intermediate Standard Missile-2s, short-range Evolved SeaSparrow Missiles packed four to a cell, and Tomahawks and anti-ship missiles for offense. The last line of defense is the close-in weapon system — a radar-guided rapid-fire gun or a short-range missile launcher — and electronic warfare equipment that tries to deceive an incoming missile's own seeker.

That sounds formidable. It is. But three facts limit it, and every one of them was present on the morning of 12 September in the Bashi Channel.

The first is the horizon. Radar travels in straight lines and the earth curves. A missile skimming the waves cannot be seen by a ship's radar until it is perhaps twenty or thirty kilometers away — under a minute of flight time for a fast missile. Aircraft and elevated sensors extend that horizon, which is why air cover matters; but our aircraft had been kept outside the box, by my own order, so as not to look like a strike package. I made that decision for good reasons. It cost seconds.

The second is the dive. A hypersonic missile coming steeply from above is visible earlier, but it arrives so fast and maneuvers so unpredictably that an interceptor has only a narrow window. Some interceptors, like the SM-6, were designed and later upgraded with this threat in mind. None was perfect. Open-source analysts had written for twenty years that the honest number for any intercept was a probability, not a certainty, and that defenders typically fire two interceptors at each incoming threat to raise the odds.

The third is the magazine. Ninety-six cells sound like many. Fire two interceptors at each of twenty-two missiles and you have used forty-four, and the attacker has many more missiles ashore than you have cells at sea. A destroyer cannot easily reload its cells at sea; it must go to port or, at best, to a sheltered anchorage with a tender. Naval officers call this "magazine depth." It means, for a president, that a single salvo can be defeated and still win, because it leaves the defender emptier

for the next one.

The Japanese destroyer *Chōkai*, an older Aegis ship of the *Kongō* class, fought beside ours. So did the convoy's own electronic warfare. Of twenty-two missiles, the escorts destroyed or decoyed nineteen. That number, which in any exercise would have been called a triumph, is the number I hold most bitterly from that year, because three got through.

One struck *Chōkai* near the stern, starting fires that her crew fought for four hours; six Japanese sailors were wounded and two later died. Two struck USS *Thomas Hudner*.

21:52

At 21:52 the screen updated: a blue symbol on the western flank began to blink, and beside it the text *DAMAGED — CASREP PENDING*. Casualty report pending. A second blue symbol, on the convoy's southern side, also began to blink, with no text at all.

"Which ship?" I asked. It was the first thing I had said.

The Navy captain in charge of the watch, a calm woman named Commander Ines Paredes, said, "Mr. President, the western track is *Thomas Hudner*. The southern track we are confirming. We have lost the data link from it. It may be damage or it may be a link outage from jamming."

The southern track was where *Daniel Inouye* had been stationed.

The forty-seven minutes

I am going to write about the next forty-seven minutes as plainly as I can, because I promised myself in 2046, when I began this operating system, that I would not write a manual for presidents that left out what being a president does to a person. I am also going to explain why I did what I did, because I think it was right, and because I am not sure it was.

At 21:52 I did not know whether my daughter was alive. Neither did anyone else in the room. And I understood, at once and with a clarity I have never felt before or since, that every person in that room would now be tempted to tell me as soon as they knew, and that this temptation would bend the flow of information around one ship out of five, and one sailor out of perhaps fourteen hundred, at the exact moment when the system needed to be looking everywhere.

So I said, loudly enough for the room: "Nobody will report anything to me about any individual. You will report ships, casualties, and threats in the order the watch would report them if I were not her father. Commander Paredes, that is an order."

Paredes said, "Aye, sir," and turned back to her screens, and I saw Ray Castillo close his eyes for a moment, and I saw Sam write something on his pad and underline it twice. I learned later what he wrote. *He's going to break and he won't let us help.*

Then I did the work. That is all I can tell a successor in my place: do the next thing.

At 21:55 Patricia Dunmore asked whether the escorts could fire on the launching platforms — the two PLAN ships within the zone and the battery ashore. Under my rules they could not without my word, and a commander with empty cells and burning ships has every reason to want to shoot the archer. I asked Moriyama whether there was evidence of a second salvo being prepared. He said that the Chinese surface ships had turned away at speed after firing and that our sensors showed no

new launches from the coast. I asked him whether, if a second salvo came, the escorts could defend against it.

"Partially," he said. "*Hudner* is out of the fight. *Chōkai* is fighting fires. *Peralta* and the southern ship have expended significant interceptors. I'd want to pull the convoy east, out of the box, at best speed, with aircraft coming in overhead now."

"Do it," I said. "Bring the aircraft in. Air cover over the convoy, defensive only. They may engage anything that threatens the convoy. They do not attack ships that are not attacking them, and they do not cross toward the coast."

At 22:03 the first casualty report from *Hudner* came in, relayed through the Seventh Fleet. Two missiles had struck forward of the bridge and near the forward missile cells; a magazine fire; flooding; the ship was listing but the crew believed they could save her. Casualties: *many killed, many wounded, numbers not yet confirmed.* The phrase "not yet confirmed" was repeated three times on the page.

At 22:07 I asked Helena to get Ambassador Tang Shiyu on the phone and say, verbatim: "The United States has been attacked. We are withdrawing the convoy from the zone and our forces are acting only in self-defense. Any further attack will be met with force. We expect an explanation from your government within the hour." She wrote it down, read it back, and left the room.

At 22:11 Claire came down. I had not asked for her and nobody had told her to come. She walked in, and the military officers in the room stood, as they would for any member of the President's family, and she said, "Please sit," and came and stood behind my chair and put her hand on my shoulder. She did not ask about Maya. She had heard me say the order — or someone had told her, I never found out who — and she was obeying it too.

At 22:14 Abigail Fenn reported that Chinese state media accused US forces of "opening fire on Chinese vessels engaged in lawful activity," and that an edited video of *Hudner*'s drone engagement, with the drones cut out, had been viewed eleven million times in twenty minutes. I told Megan Pruitt, my chief of staff, that I would speak to the nation before midnight, and that until then we would confirm only that our ships had been attacked and had defended themselves.

At 22:19 there was a stretch of four minutes in which nothing new came in. I remember it the way you remember the moment before a car crash. Claire's hand did not move. I looked at the clock on the wall and did the only arithmetic I could manage. Twenty-seven minutes. I thought of the ship's officers' quarters, which I had seen on a visit to a sister ship years before: the narrow bunks, the curtain, the little reading light. I thought of the day she was commissioned at Annapolis, how she had turned to find us in the crowd before she threw her cover. I thought, and I am ashamed of this, that if one of the two ships had to be hit, I was glad it had been *Hudner*. I thought it for perhaps a second. Then I understood exactly what I had thought, and that forty-one families or however many it would be would never know that a president had thought it, and that I would know it for the rest of my life. I am writing it down here because a manual that leaves out the shameful parts is a manual for a machine.

Lord, I said, in my head, for the first time in some years, *keep her. And if you will not keep her, keep me working.*

At 22:23 Moriyama reported that a Chinese Type 055 destroyer had reversed course and was again approaching the convoy at high speed. The room tightened. Dunmore asked for authority to engage it if it came within missile range. I asked what range was meant. She told me. I asked whether

there were signs it was preparing to fire. Moriyama said its fire-control radars were not illuminating our ships. I said: not yet. If it illuminates, the commander on scene may engage. Not before. At 22:29 the Type 055 turned away again. I have wondered since whether its captain was given orders to provoke us, or orders to look, or no orders at all. I will never know.

At 22:31 the data link from the southern track came back up. The symbol stopped blinking. Its label read *DDG-118*. Status: *operational, interceptors expended 18, no damage.*

That told me the ship was whole. It did not tell me who was alive aboard it; a ship can be undamaged and still lose a sailor overboard, or to a fall in the chaos, or to any of the hundred things that happen when a crew fights. I let out a breath that I had held, I think, since 21:52. Claire's hand tightened on my shoulder. I said nothing.

At 22:39 Commander Paredes read the consolidated report, ship by ship, in the order the watch would have read it if I were not anyone's father. *Thomas Hudner*: heavily damaged, under way on one shaft, fires contained, casualties severe. *Chōkai*: damaged, fires under control. *Rafael Peralta*: no damage, no casualties. *Daniel Inouye*: no damage. "Personnel status for *Inouye*," she read: "all hands accounted for."

All hands accounted for.

Forty-seven minutes. From 21:52 to 22:39.

I said, "Thank you, Commander. Next item." Claire took her hand off my shoulder and left the room, and I learned later that she went into the small corridor outside the Situation Room, where there is a water fountain and a fire extinguisher and a framed photograph of President Kennedy, and sat down on the floor and wept, and that Lieutenant Colonel Brooks, who had been stationed outside the door with the football, sat down on the floor beside her and did not say anything at all, and that after a few minutes she said, "Thank you, Jamal," and stood up, washed her face at the fountain, and went back upstairs. I have never thanked him adequately for that. I do so now, in a document that will not be read until both of us are old or dead.

At 22:44 the report from *Hudner* was updated. Thirty-one dead, nineteen missing, thirty-six wounded. Over the next nine hours the missing were found, one by one, and ten of them were found dead. The final number, confirmed at 07:50 on Tuesday morning, was forty-one.

> **OPERATING NOTE — When the crisis is personal.** Sooner or later the crisis will reach into your own family: a child in uniform, a parent in a city under threat, a friend on a ship. Decide in advance what you will do. My rule was that information about my family would flow to me through the same channels, at the same speed, in the same order, as information about everyone else's. I believe this rule was right because it protected the system from me. It did not protect me from the system, and it was not meant to. You will want to break such a rule. You must not break it in front of the people who serve you, for they will follow your example with their own children. Find a person — your spouse, your aide, someone outside the chain — to whom you can be a father or mother for one minute when it is over.

3. The Night of the Ladder

Monday, 11 September — 23:40 EDT

I spoke to the nation from the Oval Office at 23:40, for four minutes. I said that American and Japanese ships, sailing peacefully in international waters to bring food, fuel, and medicine to the people of Taiwan, had been attacked by missiles fired by the armed forces of the People's Republic of China; that American sailors had been killed, and that I would not give a number until their families had been told; that our forces had defended themselves and withdrawn the convoy from danger; and that I would decide on the American response "with the seriousness that the lives of our sailors demand, and not with the speed that anger demands." I said that I would speak again the next day.

That last sentence was aimed at three audiences at once, and I had argued with Sam and Megan about it for ten minutes before I went on air. It told the American people that a response was coming. It told Beijing that it was not coming tonight, and that whatever came would be deliberate rather than reflexive — which was meant to reduce the chance that a frightened Chinese commander would fire first to beat our answer. And it told my own military, whose sailors had just died, that their Commander in Chief would answer for them. If you are ever tempted to believe that a short statement is a simple one, remember that the shortest ones are the most engineered.

Tuesday, 12 September — 00:15 — the Three Clocks

At a quarter past midnight we reconvened in the Situation Room, joined now by Admiral Victor Hale, commander of US Strategic Command (STRATCOM), on video from Offutt Air Force Base in Nebraska. I asked for him because whatever we did next would be watched by the Chinese strategic forces, and I wanted the officer responsible for ours to hear every word.

I began, as I usually did, by drawing the three clocks on the whiteboard. I want a successor to see how they stood that night, because it explains everything I decided.

The *military clock* was running fastest. Our forces in the Western Pacific were at war footing in all but name. Chinese forces were too. Every hour that passed with dead sailors unanswered was an hour in which commanders on both sides would make preparations — dispersing aircraft, sortieing submarines, readying missiles — and preparations seen by the other side look like intentions.

The *information clock* was faster still and running against us. By midnight the edited video had been viewed one hundred and forty million times, and a fabricated audio clip of a voice like mine ordering "the destruction of the Chinese fleet" was spreading in Chinese-language networks, traced by our analysts to a PLA information operations unit. At home, a retired admiral on cable television was demanding we "sink everything that floats between Fujian and Formosa."

The *political clock* was running slowest, as it always does. Congress was in session but asleep. Our allies needed hours to consult their own cabinets. Beijing's leadership, according to WILLOW's first fragmentary report at 00:40, had convened an emergency meeting of the Central Military Commission at which, she reported, "there was shouting."

The danger I wrote in red at the bottom of the board was the gap between the first clock and the third. If the military clock outran the political clock, the war would be decided by commanders before leaders could decide anything. My job that night was to answer our dead quickly enough to satisfy the military clock, precisely enough to keep the political clock alive on both sides, and visibly enough to beat the lies on the information clock.

The escalation ladder — a lesson for successors

A successor will hear the phrase "escalation ladder" in every crisis meeting. It comes from Herman Kahn, a strategist at the RAND Corporation, who in 1965 published a book describing forty-four rungs from "ostensible crisis" to "spasm or insensate war." Nobody uses his exact rungs anymore, but the image is useful.

The ladder is not a single staircase. It has at least three dimensions, and each move can climb along one, two, or all three.

The first dimension is *intensity*: how much force, how many people killed, how much destroyed. A strike on the units that fired is low; destroying a fleet is higher.

The second is *geography*: where the force is used. Waters near the incident are low. Bases on an adversary's mainland are higher — and for China, as for any great power, the mainland carries a weight that no ship at sea does. Widening the war to new places — the South China Sea, the Korean Peninsula, space — is what strategists call "horizontal escalation."

The third dimension, which Kahn's era understood differently, is *the target's meaning.* Some targets are just targets. Others carry a message about what you intend next. A strike on an anti-ship missile battery says: *do not shoot at our ships.* A strike on a radar that also serves as part of an adversary's early warning of nuclear attack says something else entirely: *we may be preparing to take away your ability to see us coming.* The same number of missiles, the same number of dead, can mean two completely different rungs.

The most important thing I learned in 2045 is that the adversary measures your rung not by what you meant but by what he fears. That is the Mirror Model, which I introduced in Chapter 1. A successor who forgets it will believe he is climbing one rung when the other side believes he has climbed five.

00:50 — the options

Patricia Dunmore and Ray Castillo presented four options, prepared long before by the Joint Staff and INDOPACOM and updated overnight. I summarize them.

Option One — Diplomatic and economic response only. Expel Chinese military attachés, impose the full "Phase C" sanctions package that Daniel Iwu had kept in the drawer since June, seek a United Nations Security Council resolution that China would veto, and rally allies. No military strike. Ray called it "the option that tells every PLA officer that American sailors are free to kill."

Option Two — A proportionate strike on the shooters. Strike the specific units that had fired on the convoy: the two PLAN surface ships that had launched missiles from inside the zone, a third PLAN vessel — a Type 052D destroyer — that our sensors showed had provided targeting data to the salvo, and the single coastal anti-ship missile battery in Fujian that had launched from the mainland. Use conventional weapons only: Tomahawk land-attack cruise missiles from submarines and surface ships, and Long Range Anti-Ship Missiles, which a successor will see called LRASM, from aircraft. Estimated Chinese military deaths: between one hundred and one hundred and fifty. Civilian deaths: low, but not zero.

Option Three — A broader punitive strike. Strike the entire Eastern Theater anti-ship missile network along the Fujian coast, plus PLAN ships at sea within the zone, plus the airfields from which the drones had flown. Estimated Chinese deaths: in the thousands. This option would also substantially reduce China's ability to threaten the next convoy.

Option Four — A strike to restore our advantage. Option Three, plus strikes on Chinese over-the-horizon radars and on the ground stations of Chinese military satellites that provided targeting data for anti-ship missiles. Planners wanted it for a sound reason: those sensors were what let the salvo find our ships a hundred miles offshore.

Victor Hale, who almost never spoke unless asked, asked to speak.

"Mr. President, I want to say one thing about Option Four from where I sit. Several of those radars and several of those satellite ground stations are dual-use. They support the PLA's conventional targeting, and they also support its early warning of missile attack on China. The PLA Rocket Force keeps part of its force on what it calls early-warning counterstrike. Launch on warning, in our terms. If we blind the eyes that tell them whether they are under nuclear attack, the men who command those missiles will have to decide whether they are blind because we are about to strike their missiles. I would not want to be them. And I would not want them to decide wrong."

There was a silence. Then Dunmore said, evenly, "Victor's right about the dual-use problem. My recommendation remains Option Three. It reduces the threat to our ships and our allies without touching anything strategic."

I asked Abigail Fenn and Gregory Laszlo what Beijing would read in Option Three. Laszlo said what I had expected. "They'll read it as a campaign, sir. Not a response. A campaign to break the blockade by force. And they'll read it as aimed at the mainland. In their system, a strike on the mainland coast at that scale goes to the Standing Committee, and the people arguing for restraint lose the argument."

"And Option Two?"

"Option Two is legible," Laszlo said. "It hits exactly the units that fired. They can tell their own people they were struck back for what those units did. It doesn't say anything about next time unless we make it say something."

Helena Marsh added the point I needed. "It also hits the mainland, Mr. President. One battery. But it will be the first American strike on Chinese soil since 1950. Don't let anyone tell you that's a small rung. It's a small hole in a very big wall."

I asked one more question, and it was the one I lost sleep over. "If we choose Option Two, and they hit the next convoy with a salvo twice as big, how many of our sailors will die because we left the rest of the battery network standing?"

Dunmore answered honestly. "If there is a next convoy into that zone without degrading their anti-ship network, sir, potentially more than last night. That is the cost of Option Two. You'd need to not send that convoy, or not send it the same way."

02:20 — alone with the pad

I asked for the room again. This time I took Sam with me, into the small conference room next door, and we sat at a table with a legal pad between us and the Reversibility Matrix drawn on it.

Option One was reversible on paper and irreversible in its effect on the other side's calculus: it would invite the next attack. Option Four was irreversible and fast, and its irreversibility was of the worst kind: it reached into the strategic domain, where Hale's warning applied. Option Three was irreversible in the deaths it would cause and, I believed, irreversible in the political dynamics it would trigger in Beijing.

Option Two was also irreversible. People would die and nothing would bring them back. But it was the smallest irreversible step that answered the irreversible step they had taken, and — this was the heart of it — it could be *bounded*. A strike that is bounded only in the mind of its planner is not bounded. A strike can be bounded only by what the adversary believes about what comes next. So the strike had to be accompanied by a statement that made the boundary public, credible, and costly for us to break.

"You want to tell them it's over before it starts," Sam said.

"I want to give them a bridge," I said. "Sun Tzu. Build your enemy a golden bridge to retreat across. If we strike and say nothing, they have to guess whether it's the first move of a campaign, and the men who want to guess the worst will win. If we strike and say *this is complete*, then the men who want to stop have something to point at. We'll be giving Wen and Qiao their argument."

"And if Luo's people don't care what we say?"

"Then we will have lost nothing that Option Three would have bought, and we'll still have the larger options in the drawer. That's the reversible part. The strike is irreversible. The *statement* isn't — we can withdraw it if they attack again."

Sam looked at the pad for a long time. Then he said, "There is one more thing you need to say out loud before you decide, and you need to say it to me and not to them."

I knew what he meant. "I'm choosing the smaller response, and my daughter's ship is in that convoy."

"Yes."

"Is that why I'm choosing it?"

"I don't know, Tom. Is it?"

I thought about it longer than I have thought about any question in my life. I asked myself whether, if Maya were a schoolteacher in Denver and not a lieutenant on *Daniel Inouye*, I would choose differently. I asked myself whether the smaller strike protected her or endangered her — and the honest answer was that it endangered her, because Dunmore was right that leaving the battery network standing made the next convoy more dangerous. A father choosing only for his daughter would have chosen Option Three. That reasoning was not a proof of my innocence, but it was the best evidence I had.

"No," I said. "I'm choosing it because it's the only option where both sides can still stop. And I'm going to hold the next convoy until we've made the zone safe some other way, or until they've backed off it."

Sam nodded. "Then I'll back you. And I'll write down in my own notes that you asked me the question. Somebody should."

DECISION RECORD — The proportionate response, 12 September 2045

Decision	Options considered	Reversible?	Risk	Chosen	Outcome
US military response to the Bashi Channel Incident (41 US sailors killed)	(1) Diplomatic and economic only; (2) Proportionate conventional strike on the three PLAN vessels and the one Fujian coastal battery that fired or targeted the salvo; (3) Broad strike on the Eastern Theater anti-ship network, ships in the zone, and drone airfields; (4) Option 3 plus dual-use radars and satellite ground stations	(1) Reversible in form, irreversible in precedent; (2) Irreversible in lives, but bounded by a public "complete" declaration that could be withdrawn; (3) and (4) Irreversible, with (4) touching China's nuclear early warning	(1) Invites further attacks; (2) Leaves the anti-ship network intact, raising risk to future convoys; first US strike on PRC soil since 1950; (3) Read in Beijing as a campaign, strengthens hardliners; (4) Could be read as preparation for a disarming strike, triggering launch-on-warning pressures	Option 2, executed at 21:00 EDT on 12 September (09:00 on 13 September, local time), paired with a public statement declaring the response "complete," a private message on the JADE LINE and through Singapore, and a pause in convoys through the zone	Three PLAN vessels sunk or disabled, battery destroyed; approximately 120 PLA personnel killed; one US Navy F-35C pilot lost and captured; China did not strike US ships again in 2045. China instead retaliated in space (see Section 4). Next convoy rerouted and delayed.

Tuesday, 12 September — 03:10 — the order

I gave the order at 03:10 on Tuesday morning. The strike would be executed eighteen hours later, at 21:00 Tuesday Washington time, which was 09:00 on Wednesday morning off Fujian. I chose the delay deliberately against the advice of the commanders, who wanted to strike within hours while Chinese units were still in the positions where we had seen them. My reasons were two. First, I wanted Congress briefed — the leaders of both parties in both chambers and the chairs of the intelligence committees, the group we call the Gang of Eight, under the consultation provisions of the War Powers Resolution of 1973. Second, I wanted to strike in daylight, at an hour when Chinese military sites would be staffed and civilian presence near them was lowest, and when our weapons could be seen arriving and leaving by Chinese sensors. I wanted Beijing to watch the missiles hit precisely what we said they would hit, and nothing else.

Dunmore said, "Mr. President, the battery will have moved by then." I said, "Then find it." They did, with ARGUS — the classified fusion of satellite sensors described in Chapter 1, able to revisit any spot on earth with radar every ninety seconds — at its best. The battery's launchers had redeployed eleven kilometers inland to a dispersal site in a quarry; ARGUS's synthetic aperture radar, which

sees through cloud and darkness, found them in four hours. I note this because, by the end of the same week, I would learn how much of my confidence in that capability had been borrowed against a future the Chinese were about to foreclose.

Tuesday, 12 September — 17:00 — the Gang of Eight

I briefed the congressional leaders in the Situation Room at five in the afternoon. Senate Majority Leader Cal Whitford of Texas, who had opposed nearly everything I had done since March, listened without interrupting for twenty minutes and then said, "Mr. President, I think you are hitting them too softly, and I will say so publicly on Thursday. But I will not say so before the missiles land, and I will not say so on the day the sailors come home." He kept both promises. I record this because successors will be tempted to treat the opposition as the enemy. On 12 September 2045, the opposition behaved like the loyal opposition, and I owed Cal Whitford for it.

Tuesday, 12 September — 21:00 EDT — the strike

I watched the strike from the Situation Room. A president adds nothing by watching, but I felt I owed it to the people who would die on both sides not to be asleep.

Tomahawk missiles were launched from a submarine and two destroyers well east of the zone; Long Range Anti-Ship Missiles were launched from Air Force bombers and Navy fighters standing off at distance. The missiles flew for between thirty and ninety minutes, which is the strange tempo of conventional war at sea: long flights, short arrivals. Two of the three PLAN ships were sunk; the third, the Type 052D, was disabled and later towed to port. The battery in the quarry was destroyed. The Chinese defended, and their defenses were good; some of our missiles were shot down. One of our F-35C fighters, flying cover for the bombers, was hit by a Chinese surface-to-air missile at the edge of the zone. Its pilot ejected over water and was picked up — not by us, but by a Chinese fishing militia vessel. He would remain a prisoner until January.

Our best estimate of Chinese military dead was one hundred and twenty. I asked for the number to be refined over the following weeks, and I was given one hundred and seventeen, and later one hundred and twenty-three. I do not know which is right. I made myself read the summary of what the units were — a destroyer crew, a frigate crew, a missile battery's launch and support staff — and to think of each as men with mothers. I offer this as a practice, not a virtue. A president who does not count the enemy's dead will soon stop counting his own.

22:30 — the golden bridge

At 22:30 I went to the Oval Office and spoke for five minutes. I give the statement here in full, because successors will need to write one like it, and because every sentence in it was designed.

> "Last night, missiles fired by the armed forces of the People's Republic of China struck the destroyer USS *Thomas Hudner* and the Japanese destroyer *Chōkai* in international waters east of Taiwan. Forty-one American sailors were killed. Two Japanese sailors have died of their wounds.
>
> "One hour ago, on my order, American forces struck the three Chinese naval vessels and the one coastal missile battery that fired upon or directed fire against our ships. We struck nothing else. We struck no city, no port, no airfield, no command center, and nothing connected to China's strategic forces. We used conventional weapons only.

> "This response is complete.
>
> "I say that word deliberately, so that there can be no misunderstanding in Beijing or anywhere else. The United States does not seek a wider war with China. We did not seek this one. We will not attack Chinese forces that do not attack us or our allies. The convoy has been withdrawn from the area, and we will not send ships into it while we seek, with our allies and with China, a way to ensure that food and medicine can reach the people of Taiwan without bloodshed.
>
> "But let no one mistake restraint for weakness. If American forces or our allies are attacked again, the response will not be limited as this one was.
>
> "To the families of our sailors: your country will honor them, and I will not rest until the conditions that killed them are ended. To the people of China: we have no quarrel with you, and we mourn the loss of your sons as we mourn our own. To the leaders of China: we have both lost enough this week. There is a way back. The door is open."

The key sentence was three words long. *This response is complete.* We argued about it for hours. Ray Castillo worried that it tied our hands. Cal Whitford, as promised, criticized it on Thursday. But it did what it was built to do. Within twelve hours, WILLOW reported that Premier Qiao Weiming had read the sentence aloud at an emergency session of the Politburo Standing Committee and had said, according to her source, "The Americans have announced their own ceiling. If we strike their ships again, we will be the ones who broke it." The Chinese Foreign Ministry's statement that afternoon was furious in tone but contained no threat of military retaliation against US forces at sea. In my diary I wrote: *The bridge holds weight. Now we see what they put on it.*

The phrase that mattered second-most was "nothing connected to China's strategic forces." It was there for General Luo Tiancheng and the officers of his Rocket Force. I wanted them to hear, in my own voice, that I had seen the line Victor Hale warned me about and had stopped short of it on purpose.

> **OPERATING NOTE — How to bound a strike.** A limited strike is limited only if the adversary believes it is. Say what you struck, say what you did not strike, and say in a single short sentence that the response is complete. Name the categories you deliberately avoided, especially anything the adversary's nuclear forces depend on. Then make the boundary costly for you to break by saying it publicly, so that your own credibility guards it. Finally, give the adversary's moderates words they can repeat in their own meetings. They will need them more than you will.

The human cost — Dover, 16 September

On Saturday, 16 September, Claire and I flew to Dover Air Force Base in Delaware for the dignified transfer of the remains of the forty-one sailors of USS *Thomas Hudner*. A successor will go to Dover. Do not let anyone talk you out of it, and do not let anyone make it about you.

The ritual is simple. The aircraft's ramp lowers. A team carries each transfer case — never call it a coffin — draped in the flag, down the ramp and into a waiting vehicle, in silence except for the commands. The families stand to one side. The President stands with the Secretary of the Navy and salutes each case as it passes. Forty-one cases take a long time.

The youngest was nineteen, a fire controlman from Lubbock. The oldest was forty-six, a chief petty officer from Guam who had served on four destroyers. There were eight women. There was a sailor

who had enlisted, according to his recruiter's note, because he wanted to see the ocean, having grown up in Kansas. There were two married to each other; their son, eleven, stood at Dover with his grandmother.

After the ceremony I met the families in a hangar. Some thanked me. Some would not shake my hand. One mother, whose son had been a gunner's mate, said to me: "You held back. They say you held back. Did you hold back because your daughter was out there?"

I had expected the question someday. I had not expected it that day. I told her the truth: that I had asked myself that very question and had concluded that I would have made the same decision if my daughter were nowhere near the sea, because it was the decision most likely to keep anyone else's son from coming home this way. She looked at me for a long time and said, "I hope that's true." I said, "So do I, ma'am." It was the most honest exchange I had with anyone that month.

I wrote forty-one letters by hand over the following two weeks, each at night after the diary. A presidential letter of condolence can be drafted by staff and signed by machine. Do not let it be. The families will know, and so will you.

4. The Dark Sky

Wednesday, 13 September — 14:12 EDT

China answered, but not at sea. It answered in orbit, where I had believed we were strongest, and that was precisely why it chose to answer there.

At 14:12 Washington time, Space Force's watch center at Vandenberg — the Combined Space Operations Center — reported that two ARGUS radar satellites had stopped returning imagery. Within ten minutes it was nine. Within three hours, it was approximately sixty satellites across ARGUS, the Space Development Agency's tracking layer that a successor will hear called PWSA (the Proliferated Warfighter Space Architecture), and several National Reconnaissance Office (NRO) imaging satellites, either blinded, jammed, or maneuvered against.

General Dunmore briefed me at 17:30 with the commander of Space Force's operations, and I want to set down the four tools the Chinese used, because a successor must understand that "attacking satellites" is not one thing.

The first and loudest is the *direct-ascent kinetic* weapon: a missile launched from the ground to smash a satellite. China demonstrated this publicly in January 2007, destroying one of its own weather satellites and creating thousands of pieces of debris that still threaten spacecraft decades later. Kinetic attacks are irreversible and indiscriminate, because debris does not respect flags. On 13 September China used none.

The second is the *co-orbital* weapon: a satellite that maneuvers close to another and interferes with it — by grappling, by spraying, by electronic means, or simply by its presence. China had launched inspection-and-servicing satellites for years; several of them moved that day.

The third is *directed energy*: ground-based lasers that dazzle or damage the optical sensors of imaging satellites as they pass overhead. Dazzling is temporary, like a flashlight in the eyes. Blinding is permanent. Public reporting had long identified Chinese laser facilities in Xinjiang. On 13 September those facilities and others we had not known about tracked our low-orbit electro-optical satellites one by one. Twenty-two lost their optical sensors permanently.

The fourth, and in numbers the largest, is *electronic and cyber* attack: jamming the radio links that carry commands up and data down, and interfering with ground stations by means I will describe only by their effects. Most of the sixty were not destroyed. They were deafened and muted. That was the elegance of it: almost nothing they did on 13 September created debris, and almost all of it was ambiguous enough that their Foreign Ministry could deny it.

Abigail Fenn put the result on a single map. Before 13 September, ARGUS had claimed persistent "custody" of ninety-four percent of China's road-mobile missile launchers outside their garrisons. After it, over the western provinces where the Hami and Yumen silo fields and most of the road-mobile brigades are based, our coverage fell to roughly seventy percent of what it had been, with gaps of up to forty minutes in radar revisit. For nine days, until replacements and workarounds closed the gaps, we did not know where several mobile launchers were.

What I learned about "absolute"

I had used the word "absolute" privately, in my diary and in conversations with Sam, to describe what ARGUS gave us. I want to retract it here for the benefit of any successor tempted to use it.

ARGUS was never absolute. Optical satellites cannot see through cloud, and western China has winter cloud and spring dust. Radar satellites see through cloud but can be fooled by decoys — inflatable launchers, metal-frame shapes, heat sources — which the PLA had built by the thousands. The PLA had dug what its own media called the "Underground Great Wall," thousands of kilometers of tunnels in which launchers could hide, and no satellite sees into a mountain. And even a perfect picture shows only what is there, not what anyone intends to do with it. Seeing is not knowing. A launcher leaving its garrison could be an exercise, a dispersal for survival, or the opening of an attack, and the satellite takes the same picture of all three.

Now I learned the fifth limit: anything in orbit can be touched, and an adversary who fears your eyes will one day put them out. That was the Mirror Model working against me. The more perfectly we saw, the more frightened Beijing became, and on 13 September its fear found a target. *Absolute surveillance is a weapon that also loads the enemy's gun* — I had written that sentence in 2041 as an abstraction. In September 2045 it became an inventory.

The response we did not make

The military recommended a response in kind: strikes against the Chinese laser sites and ground stations, either kinetic or cyber, to "restore the space balance." Some advisers wanted us to blind Chinese reconnaissance satellites in turn. I refused, and the reasoning is the same as on 12 September, only sharper. Several of the Chinese ground sites, and several Chinese satellites we could reach, were part of their early warning of missile attack — the system that allows their leaders to know whether American missiles are coming. A Rocket Force on launch-on-warning posture, suddenly blind, might conclude that blindness was the first act of a disarming strike.

Instead I chose to reconstitute and to talk.

Reconstitution was where Marcus Thorne surprised me. In July, Halcyon Orbital's chief executive had fought me over using his commercial constellation for military targeting, and I had used the Defense Production Act to compel his cooperation. On 14 September he called me himself. "Mr. President, they just shot at my satellites. Forty-one sailors and now my satellites. What do you need?" Halcyon had eighteen hosted-payload satellites in storage and a launch scheduled in five days. Within nine days, Halcyon and the Space Force had launched thirty-four replacements on three rockets, and

ARGUS coverage over western China was back to roughly ninety percent. Thorne charged us cost. On this I owe him.

Talking meant a message to President Wen on the JADE LINE on 15 September, which he did not answer live but which his office acknowledged, and a parallel message through Singapore's Prime Minister Daniel Tan. It said: *The United States has not responded to your attacks on our satellites. We will not attack satellites or ground systems that provide warning of missile attack to either of our countries. We propose that both sides observe this restraint. The eyes that warn of nuclear attack must stay open, for your sake and ours.* There was no reply for eleven days. Then, through Singapore, Foreign Minister Shen Yilan sent four words: *The principle is noted.* It was the seed of what became, six months later, the Space Non-Interference Code.

DECISION RECORD — Response to Chinese counter-space attack, 13–15 September 2045

Decision	Options considered	Reversible?	Risk	Chosen	Outcome
How to answer the blinding or jamming of ~60 US surveillance and tracking satellites	(1) Kinetic or cyber strikes on Chinese laser sites and satellite ground stations; (2) Counter-space attacks on Chinese reconnaissance satellites; (3) No military response: reconstitute with commercial launches, harden links, propose mutual restraint on early-warning systems	(1) and (2) Irreversible; some targets dual-use for Chinese nuclear early warning; (3) Reversible: options 1 and 2 remained available	(1)/(2) Blinding Chinese early warning under launch-on-warning posture risks nuclear miscalculation, plus debris if kinetic; (3) Nine days of reduced coverage over western China, appearance of weakness, domestic criticism	Option 3, plus private proposal on JADE LINE and via Singapore: early-warning satellites off-limits to both sides	Coverage restored to ~90% within nine days via 34 Halcyon replacement satellites; no further Chinese counter-space escalation in 2045; China "noted the principle," the seed of the 2046 Space Non-Interference Code

OPERATING NOTE — On depending on what you can see. Your intelligence systems will be magnificent, and you will come to lean on them as you lean on your own eyes. Do not. Every sensor has weather, every picture has decoys, every orbit can be reached, and no image shows intent. Before a crisis, ask your intelligence chiefs a single question: "If we lost a third of our satellites tomorrow, what would we stop knowing, and how would we find it out instead?" Make them answer in writing. Then remember

that your adversary is asking the mirror image of that question about you, and that the systems which warn each side of nuclear attack are the ones neither side can afford to blind.

5. Three Words: DEFCON 3

Wednesday, 20 September 2045 — 11:00 EDT

On 20 September, on the recommendation of Ray Castillo and Patricia Dunmore, I approved raising the readiness of US forces to DEFCON 3.

Films treat "DEFCON" as a doomsday dial. It is less dramatic and more important than that.

DEFCON stands for Defense Readiness Condition. It is a five-step scale of military readiness, from DEFCON 5, normal peacetime, to DEFCON 1, maximum readiness, which has never been declared. Levels are set by the Chairman and the Secretary of Defense with the President's approval, and can differ by command. Historically, US forces were placed at DEFCON 3 during the Arab-Israeli war of October 1973 and on 11 September 2001. Strategic Air Command went to DEFCON 2 during the Cuban Missile Crisis in October 1962, the only time that level is publicly known to have been reached.

What DEFCON 3 actually changes is not secret in outline. It is a step up in readiness: leave is cut back, units are recalled to stations, command posts move to fuller manning, aircraft and crews are prepared to launch on shorter notice, and communications and continuity-of-government arrangements are exercised and tightened. It is not a decision to use force, and it arms no weapon. It is a decision to shorten the time between a future decision and its execution.

Why did I approve it? Because by 20 September the military clock had outrun our readiness. Chinese forces had sortied most of their surface fleet, dispersed their aircraft, and — according to THRUSH at Yulin — sent a fourth Type 096 ballistic-missile submarine to sea, joining the three that had sailed in late August. LANTERN reported that Base 64 had received orders to "review" procedures for moving from Readiness Level 3 to Level 2. If we remained at peacetime readiness, a sudden Chinese move would find us slow.

Why did I hesitate? Because readiness is also a signal, and the Mirror Model asks what the other side sees. I asked Victor Hale and Sam to sort the measures into three columns: survivability, messages, and things that could look like attack preparation.

Two decisions came out of that sorting.

The first was *bomber dispersal*. Our B-52J and B-21 bombers are vulnerable on their few home bases, because a single attack could destroy many at once. Dispersing them to other airfields across the country makes them harder to destroy and therefore makes an attack on them less tempting. Dispersal is visible to satellites; it was meant to be. It says: *you cannot disarm us, so don't try.* I deliberately did not order bombers onto ground alert with nuclear weapons loaded, a step taken routinely during the Cold War and ended in 1991, because it would have looked like preparation for use. Some bombers were generated with conventional weapons for the Pacific. None flew toward China.

The second was the *SSBN surge*. In peacetime, a portion of our ballistic-missile submarines — the six new *Columbia*-class boats and the remaining *Ohio*-class — are at sea on deterrent patrol at any time, and the rest are in port at Kings Bay, Georgia, or Bangor, Washington, for maintenance and

crew changes. Submarines at sea are effectively invulnerable; submarines in port are not. I approved Hale's request to send every boat that could safely sail to sea. Within ten days, nine boats were on patrol. Departures from port are observable; the boats once submerged are not.

I sat with Hale's last question for a long time. "Mr. President, Beijing will see our boats leave. Do you want them to?" I decided yes. A surge says: *our second-strike force is secure; we have no need to strike first.* The paradox of deterrence is that the more secure your retaliation, the less reason either side has to fire first. But I asked Helena to make sure the message was received as I meant it, and so, on the DTL, our Defense Secretary told his Chinese counterpart, in words I approved: *The movements of US strategic forces are defensive. They are intended to make our deterrent secure, not to threaten yours.* Their side listened and did not respond.

There is one small human detail of DEFCON 3 that I want to preserve. On the evening of 20 September, Lieutenant Colonel Brooks, carrying the football as always, rode with me to a dinner for the Congressional Black Caucus. In the car he said, without being asked, "Sir, the procedures for the satchel don't change at DEFCON 3. I want you to know that. It's the same bag at five and three. The difference is how fast everyone else around it moves." I asked him whether his family knew what he carried. "My wife knows it's heavy," he said. "She doesn't ask why."

> **OPERATING NOTE — What alert levels are for.** Raise readiness to buy survivability and time, not to frighten. Sort every readiness measure into three columns before you approve it: measures that make your forces harder to destroy; measures meant to be seen as messages; and measures that could look like preparation to strike first. Approve the first freely, the second deliberately, and the third only if war is already upon you. Say aloud, through every channel, what your movements mean. A secure deterrent is a calming deterrent only if the other side understands that it is secure.

6. The Price of a Name

Tuesday, 3 October 2045 — 07:20 EDT

Gregory Laszlo came to the Oval Office with Ruth Calder, the chief of the CIA's China Mission Center, the officer who had recruited LANTERN twelve years before. Calder does not come to the Oval Office with good news.

KESTREL had been arrested on 2 October by officers of the Ministry of State Security (MSS) at his home near the Yumen silo field. He was a technician on the field's power and communications network, recruited in 2039, and for six years he had told us how the field's systems were maintained, which told us how ready its missiles were. He had missed a scheduled covert transmission; a source in the local police confirmed the arrest. MSS Minister Kong Dewei had been running a mole hunt since July, when Chinese counterintelligence concluded — from our very precision in the cyber operations of June, Calder believed — that the Americans knew things only insiders could know.

"Is LANTERN exposed?" I asked.

"Not yet," Calder said. "KESTREL never knew LANTERN existed. But the hunt is now inside the Rocket Force, and it will go through Base 64. They'll look at who had access to readiness orders. LANTERN is on that list, with perhaps forty others. I want to start exfiltration planning now and execute within three weeks."

I asked what exfiltration involved. She told me as much as a president needs to know: a route out of

western China, false documents, a border crossing, a window of days in which LANTERN would be absent from his post — an absence that would itself announce him if the plan failed. She estimated the chance of success at sixty percent. If it failed, he would die, and so might his family.

Then I asked the question I had to ask. "If he leaves, what do we lose?"

Laszlo answered. "Our only human source inside the Rocket Force's chain of command for the Hami field. Our only way of knowing, from the inside, whether an order to go to Readiness Level 2 or Level 1 is real. After the thirteenth, with the satellites as they are, that may be the most important single piece of intelligence this country has."

I thought of the Two-Source Rule I had made in 2041: never act on a nuclear warning from one kind of sensor alone; require satellites and radar, and, where time allows, a human check. LANTERN was the human check. I did not know in October how decisive that would prove. I knew only that the satellites had been blinded once and could be again.

I asked for a day.

The night of 3 October

I wrote six pages in my diary that night, more than any other night of the crisis. I will not reproduce them. I will give the sentence I underlined at the end: *I chose to keep a man in danger to keep a billion out of it.*

I want a successor to understand that the sentence is an accounting, not an absolution. LANTERN had come to us not for money but because he had seen what his missiles were for and could not bear it. He had already risked everything. I was asking him to risk it longer, without asking him, because the CIA's channel to him was too dangerous to use for a conversation. Calder told me later that she sent him one message, by the agreed covert method: *Danger rising. Ready to bring you home on your word.* His reply came nine days later: *Not yet. The work is not finished.* That eased my conscience a little. It should not have eased it entirely, and it did not.

DECISION RECORD — Exfiltration of LANTERN, 4 October 2045

Decision	Options considered	Reversible?	Risk	Chosen	Outcome
Whether to exfiltrate LANTERN after KESTREL's arrest	(1) Exfiltrate within three weeks; (2) Keep LANTERN in place, suspend all tasking except urgent alert-status reports, prepare exfiltration for immediate use, offer him the choice; (3) Cut contact entirely to protect him	(1) Irreversible: once out, the source is gone for good; (2) Reversible: exfiltration remained available at short notice; (3) Partly reversible, but contact might not be restored	(1) ~40% chance of failure and death; loss of the only human check on PLARF alert orders; (2) Ongoing risk of arrest and execution as the mole hunt narrowed; (3) Blindness at the moment the satellites were weakest	Option 2	LANTERN chose to stay ("Not yet"). His reports proved decisive on 6 November (Chapter 6). He was exfiltrated in April 2046. The risk I placed on him was real for six months.

Friday, 20 October — 06:40 EDT

On 20 October, WILLOW reported, in a transmission of eleven lines, that KESTREL had been tried in secret by a military court and executed on 19 October. His wife and two children had been moved from Yumen; she did not know where.

[REDACTED — 4 lines]

Laszlo advised that any approach to his family would endanger them further. We did nothing. I have checked every year since whether there is anything we can safely do. As of the writing of this module, there has not been.

I did not tell anyone outside the small circle of people cleared for the program. I could not attend a funeral, nor write to a widow, nor name him in public. Of all the dead of 2045, he is the only one I could not honor in any way, and that is why I am naming him here as clearly as the classification rules permit: a man who worked on cables and power lines in a silo field in the desert, who believed that what he was doing would make war less likely, and who was right.

> **OPERATING NOTE — The people you will never meet.** Some of the most important people in your crisis will be strangers in the other country who work for us at the risk of their lives. You will never see their faces. You will make decisions that raise or lower the danger to them, usually without being able to ask them. Do not delegate the moral weight of those decisions to the intelligence chiefs, even though they will offer to carry it. Ask what each source costs and what each source buys, ask whether the source can be given the choice, and write down your reasons. When one of them is lost, you will want to know you decided honestly.

7. The Bear on the Phone

Saturday, 14 October 2045 — 09:00 EDT

On 9 October, Russian long-range bombers — Tu-95MS and Tu-160 aircraft — flew to airfields in the Russian Far East, and Russian Pacific Fleet submarines left port. On 14 October, at his request, I spoke with Russia's President Arkady Volensky for fifty minutes on the secure line.

Volensky began with a formal warning, for the record. Russia regarded "any attack on the strategic nuclear forces of the People's Republic of China" as "a threat to strategic stability affecting the security of the Russian Federation," and would "take all necessary measures." The bombers, he said, were "precautionary."

I answered for the record too: that the United States had not attacked and would not attack Chinese strategic forces, that I had said so publicly on 12 September, and that I expected Russia to respect the same restraint.

Then, as I had hoped, he changed his tone. He asked for the interpreters to stay but the note-takers to step back, which means nothing, since everything is recorded, as he knew. "Thomas," he said, "I am seventy-one years old. I have no wish to die for Taiwan or for General Luo. Russia will not join this war. But I cannot tell Beijing that, and you cannot tell anyone I said it." Then he mentioned, without making it a demand, that his economy was suffering under sanctions, and that "when this ends, some of us will remember who helped it end."

I gave him nothing on sanctions that day. What I took from the call, and wrote that night, was a lesson in reading a great power that is aligned with your adversary but not bound to it: *Public words are for allies; private words are for interests. Volensky's bombers are a message to Beijing as much as to us: "We are here, so do not take us for granted, and do not drag us in."*

8. The Window

Thursday, 26 October 2045 — 08:15 EDT

On 26 October, General Luo Tiancheng, commander of the PLA Rocket Force, gave a speech to officers at a Rocket Force academy. State media published extracts that evening, Beijing time. The same day, the PLA's official newspaper ran a long doctrinal article by "a commentator." Our translators had both on my desk by eight.

Luo's speech was built around one word: *window*. He argued that the United States, with Golden Dome — our partially fielded missile defense, including a few hundred experimental space-based interceptors — and with its surveillance of Chinese forces, was building toward the ability to destroy most of China's nuclear forces in a first strike and intercept the rest. "Today the window is still open," he said. "Our deterrent is still real. But every month the Americans place more eyes and more shields in the sky, the window narrows. History does not forgive a nation that waits until the window closes."

The newspaper article went further. China's long-standing policy of "no first use" of nuclear weapons, it said, "has always presupposed the survival of the nation's core strategic deterrent." If that deterrent were "threatened with destruction or neutralization," the article suggested, the policy would require "reinterpretation."

A successor should know that the phrase "window of vulnerability" has an American history. In

the late 1970s and early 1980s, American critics of arms control argued that Soviet missiles were becoming accurate enough to destroy our land-based missiles in their silos, opening a "window" in which Moscow might be tempted to strike. Reading Luo, I saw our own fears in a Chinese mirror. The tragedy of the Mirror Model is that it is almost always symmetrical: both sides believe the other is building toward a first strike, and both point to the same evidence.

At nine, Abigail Fenn gave WILLOW's context: Luo had given the speech without clearing the text with the General Secretary's office, and President Wen, whose heart condition we had learned of in July, had spent two days of the previous week out of public view. Premier Qiao's people were said to be furious. "This isn't policy yet," Abigail said. "It's a bid for policy."

"It's a bid aimed at us as much as at Wen," Helena said. "He wants us to think Golden Dome will get us killed."

Hale said the thing that I remember. "Mr. President, he isn't entirely wrong about what he sees. He's wrong about what we intend. But a man who believes his window is closing does not wait to be persuaded."

That night I wrote in my diary: *Luo has told us his fear in public. That is a gift and a warning. A frightened man's fear can be addressed; it cannot be ignored. What can we give that narrows his argument without opening ours?* The answer — a pause in launching new space interceptors, a promise not to strike China's nuclear forces or its early warning — would take another ten days and the most dangerous week of my life to deliver.

Tuesday, 31 October 2045 — 23:50

The last diary entry of October is short. Claire had gone to bed. Maya's ship was in Yokosuka for repairs and resupply; she had called her mother that afternoon, and they had talked about nothing for forty minutes. The forty-one letters were finished. The aspens on the Front Range had turned and fallen without me.

I wrote: *Phase three is over, if phases are real. We fought a battle at sea and a battle in orbit, and we lost men in both and in a third place nobody will ever see. We did not lose the bridge. Tomorrow is November. Hale says the satellites over Hami are watching. LANTERN says the work is not finished. I believe him.*

At 05:40 the next morning, the satellites over Hami saw the launchers move.

Chapter 5: The Nine Days, I: The Gamble

"The machine can tell me where every launcher is. It cannot tell me who is holding the key, or whether his hand is shaking." — Diary of Thomas Ellery Kane, 1 November 2045

A Note Before the Nine Days

A successor will want to skip to the next chapter, where the eleven minutes are. I understand. But the eleven minutes were not the decisive part of the crisis. They were the moment at which the decisions of the five days before them were tested. A control system is not judged by the instant the disturbance arrives. It is judged by how it was tuned before.

The first five days of November 2045 were the days in which I tuned it. They were also the days in which I was most alone, most tired, and most often wrong in small ways that I corrected in time only because people around me refused to let me be wrong in large ones.

I have reconstructed these days from my diary, which I have kept every night since the summer of 1997 and did not stop keeping in the Presidential Emergency Operations Center (PEOC), though the handwriting of that week is nothing to be proud of; from the Situation Room log; and from the memoranda of my National Security Advisor, Dr. Samuel Oyelaran, who wrote down what people said, not what they later wished they had said.

Times in this chapter are Eastern. A successor should note a detail that seemed trivial then and was not: daylight saving time ended in the United States at two in the morning on 5 November 2045. Until that moment Washington ran on Eastern Daylight Time (EDT), twelve hours behind Beijing. After it, on Eastern Standard Time (EST), we were thirteen hours behind. In a crisis in which a message delivered at the wrong hour of someone's night can be read by the wrong man, I had the Situation Room put two clocks for Beijing on the wall for the whole week and label them in red: *BEIJING (IF EDT)* and *BEIJING (IF EST)*. Someone thought it was overcautious. Nobody laughed by Wednesday.

1 November 2045 — 04:12 EDT (16:12 Beijing)

I was not asleep when the phone rang. I had not slept well since the night of the Bashi Channel, when for forty-seven minutes I did not know whether my daughter was alive. I had taken to lying still at about four in the morning and listening to the house: the ventilation, a floorboard in the hallway answering the weight of the Secret Service agent, and far below a sound like distant surf, which I was told is the building's water system and which I chose to believe was the Potomac.

“Mr. President, it’s Sam.”

Oyelaran does not apologize for waking me. We agreed long before that he would never begin a call with an apology, because the apology costs two seconds and tells me nothing.

“Go ahead.”

“ARGUS is showing dispersal from garrison at two locations in western and northern China. Hami and the Ordos field. Road-mobile launchers leaving their bases, and what the analysts are calling unusual activity at the silo fields: maintenance crews at a number of silo headworks, at the lids. It started about ninety minutes ago, their afternoon. DNI is on her way in. I don’t think you need to come down yet. I think you’ll want to.”

“How many launchers?”

“Seventeen in custody outside garrison at Hami and nine near Ordos as of four minutes ago. The number is moving.”

“Weather?”

He understood why I asked. “Clear over Hami. Overcast over Ordos. They’re working Ordos with radar.”

“I’m coming down.”

Claire was awake. She had been a pediatric oncologist for twenty-six years before she became the First Lady, and she has the doctor’s habit of waking fully and instantly, with no transition. She did not ask me what it was. She asked whether I wanted coffee or tea, and I said tea, because I knew my stomach, and she said she would have it sent down and that I should put on the grey sweater because the Situation Room was always cold. I put on the grey sweater.

In the corridor outside the bedroom, Lieutenant Colonel Jamal Brooks of the United States Marine Corps was already standing, dressed, with the black leather satchel that the newspapers call the nuclear football at his feet. He had been on the overnight rotation. He said, “Good morning, Mr. President,” as though it were nine o’clock and we were going to a ribbon cutting.

I have written in Chapter 1 about the football and the aides who carry it. A successor should know only this: the aide is not there for the satchel. He is there because the President must never be more than a few steps from a human being whose whole purpose is to make the worst decision of his life possible to execute correctly. Brooks was thirty-nine, from Columbus, Georgia, the son of a postal carrier and a schoolteacher. He never once, in all the months of the crisis, said anything to me that was not either necessary or kind.

We went down together.

What ARGUS Saw

I want to explain, for a successor who has not lived with it, what the screen in the Situation Room showed at 04:31 that morning, and what it did not show.

ARGUS is a program, not a satellite; I described it in Chapter 1. It fuses hundreds of sensors in orbit, some on National Reconnaissance Office (NRO) satellites and many hosted on the Halcyon Orbital constellation. Synthetic aperture radar (SAR), which sees through cloud and darkness by reading radio echoes off the ground, revisits any point in China roughly every ninety seconds; electro-optical satellites take pictures when the weather allows; others locate radio transmitters or read chemical

signatures. Software ties these together and maintains what the analysts call *custody*: an unbroken track on one object, so that the launcher that left its garrison at 15:40 Beijing time is known to be the one on a highway at 16:05.

Before September, the program office had claimed custody of 94 percent of Chinese road-mobile launchers outside garrison. After the counter-space attacks of 13 September, coverage over western China fell to about 70 percent for nine days. By November, with Halcyon's replacements in orbit, the claim was 88 to 91 percent, depending on the weather and on which analyst you asked. I had learned in September to ask more than one.

So the screen showed small icons, each one a launcher, each one with a thin trailing line behind it that recorded its path since it had left its base. At Hami, in the far west, in Xinjiang, the icons were moving out along roads into the desert in a pattern that a younger analyst, a woman from the National Geospatial-Intelligence Agency (NGA) who had been on the overnight watch, described with real precision:

"Sir, they're not going to one place. They're spreading. That's what dispersal is. In garrison they're vulnerable, because we know exactly where the garrisons are. Out in the field, each one is a separate problem for an attacker. This is what you do if you think someone may strike you. It's also what you do before you launch, because some of the pre-surveyed launch points are out there."

"Which is it?"

"I can't tell you that from the imagery, Mr. President."

I liked her very much for that sentence.

At the silo fields, the picture was stranger. China's three fields, Yumen and Hami in the west and Ordos in Inner Mongolia, hold roughly three hundred and fifty silos for the DF-41 and the heavier DF-5C, not all loaded; some are decoys. At perhaps two dozen silos at Hami and fewer at Ordos there were vehicles at the headworks and people on foot, and the sensors had picked up a signature consistent with work on the heavy lid that has to open before a missile can leave.

"Is that lid maintenance or lid preparation?" I asked.

The answer, which came from three people at once, was that there is not a meaningful difference visible from orbit. A crew that tests whether a lid will open is doing the same thing whether the test is routine or the prelude to opening it for real. We had seen lid maintenance before. We had never seen it at this many silos at once, in two fields, coinciding with dispersal of road-mobile forces.

Director of National Intelligence Abigail Fenn arrived at 04:46 in a tracksuit with her hair wet. She has a habit, when the news is very bad, of speaking more slowly. She spoke very slowly.

"Mr. President, my assessment, which is preliminary and which I'll revise in two hours, is that this is either a large-scale readiness exercise that they haven't announced, or a real increase in the alert status of Rocket Force units in the west and north. I lean toward the second. I want to wait for one more source before I tell you that."

"What source?"

She looked at the CIA Director, Gregory Laszlo, who had come in behind her. He shook his head slightly. Not here. The room had too many people in it, and some of them did not need to know that we had a man inside Base 64.

"Within a few hours, I hope," Fenn said.

The Questions I Wrote on the Pad

I keep two yellow legal pads in the Situation Room. On the first I write what I know; on the second, what I need to know. At 05:10 on 1 November the first had six lines and the second eleven. I reproduce the second as a template for the first hour of an adversary's nuclear alert change:

1. Who ordered it? (Not which unit. Which man.)
2. Does the man at the top know? Did he approve it?
3. Is it a response to something we did? What did we do in the last 72 hours that they could read as a threat?
4. What does it look like from their side? (Mirror.)
5. Is it the same in all three fields? Why Hami and Ordos and not Yumen?
6. Where are their submarines? (THRUSH.)
7. What are our forces doing right now that they can see?
8. Which of our channels to Beijing are open tonight? Who is awake there?
9. What do our allies see? Who in Tokyo and Canberra already knows?
10. When does this hit the open internet? (Commercial imagery companies can see dispersal too.)
11. What is the smallest thing I can do that is also correct?

The eleventh line was underlined twice. It was underlined twice on every such list I wrote that year.

To the third question the answer was uncomfortable. On 30 October a second carrier strike group had entered the Philippine Sea. On 29 and 30 October two B-21 Raiders had flown a long-range mission from Guam that I had approved as a signal. On the 28th a scheduled launch from Vandenberg had put twelve more space-based interceptor prototypes in orbit, bringing the number to about two hundred. Each I could defend. Together, seen from Beijing after Luo Tiancheng's speech of 26 October about "the window," they made a picture.

Tom, I wrote in the margin, *what do they see when they look at us?*

I did not yet know the answer. I knew I would need it within hours.

1 November 2045 — 10:07 EDT (22:07 Beijing)

The burst arrived at 09:58. It reached me nine minutes later, which I record because a successor will be told that human intelligence is slow, and it often is, but on that day it was not.

LANTERN, as I have described in earlier chapters, was a senior officer on the staff of Base 64 of the People's Liberation Army Rocket Force (PLARF) at Lanzhou, whose brigades included those at the Hami field. He had been ours since 2033, recruited by the CIA's China Mission Center chief, whom I will call Ruth Calder, after his daughter was treated abroad for a disease that would have killed her at home. He did not work for money. He communicated by dead drop, which was slow, and by a burst device concealed in a commercial vehicle navigation unit, designed to be used rarely. Since the Ministry of State Security (MSS) had taken KESTREL on 2 October and executed him on the 19th, every use was a much greater risk.

Calder came to the Oval Office herself with the text. Laszlo was with her. They came in through the door from the secretaries' office rather than the main door, a small thing that the press office noticed later in the visitors' log and never asked about.

The message, translated, was this:

BASE 64 ORDERED READINESS LEVEL 2 AT 1420 LOCAL. WARHEADS MATED TWO BRIGADES HAMI. OTHER BRIGADES PREPARING. ORDER CAME ON ROCKET FORCE COMMAND CHANNEL FROM LUO HEADQUARTERS. NOT SEEN CHAIRMAN AUTHORITY. STAFF DO NOT KNOW IF WEN APPROVED. SOME OFFICERS AFRAID. I WILL TRY AGAIN WHEN SAFE.

I read it three times. Then I asked Calder to read it aloud in Chinese, which she did, because I had learned in thirty years of engineering that a translation is a model of the original and every model has error, and I wanted to hear the shape of the sentences. She has a low voice and she read without expression. When she came to the phrase that we had rendered as *some officers afraid*, she paused.

"The word he used is stronger than 'afraid,' " she said. "It's closer to *frightened in the way you are frightened of something you can't stop*. There isn't one English word."

I have thought about that phrase often since.

What Readiness Level 2 Means

A successor needs to understand the scale, because it was the spine of everything that followed.

According to our intelligence and LANTERN's reporting over many years, the Rocket Force used a four-step readiness scale for its nuclear brigades; the names are our translation. Level 4 is peacetime: warheads held separately in storage, crews on routine duty. Level 3 is heightened readiness: crews recalled, leave cancelled, more communication checks. Base 64 had been at Level 3 since 12 March, the first time since 2034. Level 2 is pre-launch preparation: warheads mated to missiles, launchers dispersed to field positions, silo crews at their stations, the force able to launch within a short time of an order. Level 1 is launch.

The phrase *warheads mated* deserves a sentence of its own. For most of its history China kept warheads separate from missiles in peacetime, a posture meant to be secure and slow: a warhead in a tunnel far from its missile cannot be launched by accident, by a rogue officer, or in panic. In the 2030s, as I explained in Chapter 1, part of the Rocket Force moved to what its writers called "early-warning counterstrike," which we would call launch on warning: launching after detecting an incoming attack but before the warheads arrive. That posture requires mated warheads. It requires speed.

So LANTERN was telling me that two brigades at Hami, a number of missiles larger than the entire arsenal of some nuclear-armed states, could now be launched within a short time, and that the officers who received the order had not seen the authority of the Chairman of the Central Military Commission, Wen Zhaolin.

In the PLA's system as we understood it, a change in nuclear alert status of this kind requires the authority of the Central Military Commission Chairman. We knew the internal name of that authorization only as a translation, which in our files was rendered as the *Chairman's Seal Directive*. LANTERN had seen many such directives in his career. He had not seen one this time.

"Does that mean Wen didn't approve it?" I asked.

"No," Calder said. "It means LANTERN didn't see the approval. It could have come through a channel he doesn't see. It could be coming. It could have been given verbally and not yet documented. It could not exist."

"What would you bet?"

“I don’t bet, Mr. President.”

“Then what would LANTERN bet?”

She looked at the message again. “He wrote *staff do not know*. If he believed Wen had approved it, he would have said so, because it would make him less afraid. I think he is telling us he suspects it wasn’t approved, and he is too disciplined to say what he can’t prove.”

The Two-Source Rule, and Its Limits

I described in Chapter 1 the Two-Source Rule for Armageddon, built for the worst case, a warning that missiles are in flight. I found on 1 November that it applies, in a gentler form, to everything.

Here I had two independent phenomenologies telling me the same thing. ARGUS saw dispersal and lid activity. LANTERN reported the order. They agreed on *what*. Neither could tell me *why*, or *who*, with certainty. ARGUS could not see intent at all. LANTERN could see part of a chain of command, and he was telling me honestly that he could not see the top of it. Satellites see objects; agents see paper and faces; neither sees a mind.

By noon Fenn assessed that elements of the Rocket Force had moved to a pre-launch posture at Hami and probably at Ordos, where we had no human source; that the order appeared to originate with the Rocket Force commander; and, with *low confidence*, that it had been issued without the General Secretary’s explicit approval. Low confidence means fragmented or single-source information. I asked her to keep it there. I did not want to fall in love with a story.

THRUSH, our source at Yulin, reported that evening through a slower channel that no additional Type 096 submarines had sailed, but that the three that had left at the end of August had not returned, and that the support base was preparing to receive or to dispatch a submarine within days. That was ambiguous too. Everything was ambiguous.

1 November 2045 — 21:30 EDT

Claire found me in the small study off the Oval Office, the one with the window onto the colonnade. I had not eaten since breakfast. She had a plate of something from the Navy Mess, soup and a grilled cheese sandwich, which is what she gives me when she has decided I am a patient.

“You don’t have to tell me,” she said.

“I can’t tell you.”

“I know. Eat the sandwich.”

I ate the sandwich. She sat in the other chair and read the medical journal she always carries, and did not look at me, which was the kindest thing she could have done. After a while she said, without looking up, “Maya called. Not about anything. She couldn’t say where she is. She said to tell you she’s eating.”

My daughter was a lieutenant aboard the destroyer USS *Daniel Inouye* in the Seventh Fleet. She had been within sight of the USS *Thomas Hudner* when it was hit on 11 September and forty-one sailors died. She had told her mother since then that she slept fine. Claire, who has sat with dying children for a quarter of a century, did not believe her and did not say so.

“Did she sound all right?”

"She sounded like you sound," Claire said. "Like someone who has decided not to be afraid in front of people."

I wrote in my diary that night, in handwriting that I can barely read now:

1 Nov. Two brigades mated at Hami. L says order from Luo, not seen W's seal. ARGUS: TELs out, lids worked. I keep asking what we did. I approved all of it. Every piece sensible. The sum is a threat. That is the Mirror Model and I designed it and I still walked into it. — C. brought soup. M. is "eating." — Tomorrow the generals will ask to strike first. They will have good arguments. I must have a better one, or be honest that I don't.

2 November 2045 — 06:40 EDT: Going Under the East Wing

I decided on the night of 1 November that the National Security Council would meet the next morning in the PEOC rather than in the Situation Room. Megan Pruitt, my Chief of Staff, asked me whether that was a signal, and to whom.

"To us," I said.

The PEOC is a bunker beneath the East Wing, which I described in Chapter 1. The ceilings are low, the air has a filtered, metallic taste, and the carpet is the color of wet cement. There is a conference table for sixteen, and behind a glass partition a watch floor connected to the National Military Command Center (NMCC) at the Pentagon, to North American Aerospace Defense Command (NORAD), and to United States Strategic Command (STRATCOM) at Offutt Air Force Base in Nebraska.

I wanted the principals there for two reasons. If the military clock accelerated, I did not want to spend minutes moving people downstairs. And I wanted every person who advised me that day to feel, physically, where the conversation might end. It is easy to recommend a preemptive strike in a room with a portrait of Lincoln in it. It is harder under eighteen feet of concrete, breathing canned air, with a Marine against the wall holding the satchel.

Brooks sat against the wall. He always did. Most people who attended meetings in the PEOC forgot within ten minutes that he was there. I never forgot, and I think it made me a better President to have him at the edge of my vision.

Present at 06:40: the Vice President, Rosalind Achterberg; Oyelaran; Secretary of State Helena Marsh; Secretary of Defense Ray Castillo; the Chairman of the Joint Chiefs, General Patricia Dunmore; Fenn; Laszlo; Treasury Secretary Daniel Iwu; Pruitt. On the screen from Offutt, Admiral Victor Hale of STRATCOM, and from Camp H. M. Smith in Hawaii, Admiral Kenji Moriyama of Indo-Pacific Command (INDOPACOM). Calder sat behind Laszlo. Two note-takers from the NSC staff, cleared for the compartment.

The Overnight Picture

Fenn briefed first. By 06:00 ARGUS held custody of thirty-one road-mobile launchers dispersed from Hami and twenty-two near Ordos, some stopped at what the analysts believed were pre-surveyed launch positions. Lid activity had been observed at forty-four silos at Hami and twenty-six at Ordos. Yumen was quiet. Rocket Force command traffic had surged, and China's large early-warning radars had changed their operating patterns in a way consistent with heightened alert.

"Why not Yumen?" I asked.

"We don't know, sir. One hypothesis is that Yumen's brigades are subordinate to a different base whose commander is less closely aligned with General Luo. Another is that it's a reserve. Another is that we're missing it."

"Which do you believe?"

"I believe it's the first, with low confidence, because it fits with what WILLOW has told us about factions in the Commission. But I'd hate to be wrong."

General Dunmore's Brief

Then the Chairman briefed the military options. I want to record her case fairly and in full, because she was not wrong to make it, and because a successor will one day hear it again from a Chairman who is also not wrong to make it.

Patricia Dunmore is a career Air Force officer, a former bomber pilot, and one of the clearest thinkers I have known in uniform. She had commanded Air Force Global Strike Command before becoming Chairman. She did not raise her voice.

"Mr. President, we do not recommend any nuclear option, and we do not recommend striking silos or launchers. What we are presenting are left-of-launch options. Picture a timeline. At the center is the moment of launch. To the right is everything after a missile is in flight: missile defense, Golden Dome. Left-of-launch means acting before, to prevent the launch or make it less likely to succeed."

She put up a slide. I will describe it rather than reproduce it, and at a general level only.

"There are three families. One: cyber and electronic operations against the Rocket Force's command and control networks, the systems by which orders reach the brigades, to degrade or delay their ability to receive a launch order. Two: conventional strikes, with long-range precision weapons, against a small number of large early-warning radar sites on China's eastern and northern periphery, to blind them. Three: combinations of both, plus actions against specific command posts."

"What's the theory?" I asked. I knew the theory. I wanted the room to hear it.

"The theory, sir, is that their posture depends on early-warning counterstrike. The most dangerous thing in the world tonight is a Rocket Force at Level 2, controlled by a man who believes in a closing window, connected to a warning system that could tell him at any moment he is being attacked. Degrade his ability to get orders to the brigades and you buy time. Blind his radars and he cannot launch on warning, because he has no warning."

"And then?"

"And then, sir, we hope he understands why, and that the political leadership in Beijing takes control back."

The Vice President spoke. Rosalind Achterberg had sat on the Senate Armed Services Committee for two terms, and in that room she was the most consistent voice for strength. I valued her for it. "Mr. President, I'll say what Pat is too disciplined to say. If Luo launches, we lose cities. Many. We have a source telling us he's acting outside his chain of command, and we have the means to jam the gun in his hand. I don't know how you explain to the American people afterward that we saw this coming and did nothing."

Castillo, the Secretary of Defense, a retired Marine general who had commanded in the Middle East and had watched men die in ways that made him, as he once told me, "allergic to theories," said,

"I'm not sold on the radars. I'm more open on the cyber. It's quieter. It might not be visible at all."

Admiral Hale, from Offutt, said nothing yet. He was waiting, I think, to be asked.

I Asked Hale

"Victor, you've spent your life thinking about how the other side's nuclear commanders think. If you were the officer in charge of Chinese early warning tonight, and your radars on the coast went dark, one after another, struck by American precision weapons, what would you conclude?"

There was a delay on the line, the half-second that satellite links impose. Then Hale, who was a submariner and who has the submariner's way of answering a question with exactly the words it needs and no more, said:

"That the attack had begun, sir."

"Why?"

"Because that's what we would do first. In every serious first-strike scenario anyone has modeled, the attacker blinds the defender's warning and disrupts his command before the main blow. Any Chinese officer who has read our literature, and they have read all of it, knows that. If his doctrine is to launch on warning, and he loses his warning, his doctrine may tell him to launch on the loss of warning."

"Is that their doctrine?"

"We don't know, sir. That's the problem. Some of their writing suggests it could be."

"And the cyber option?"

"Better and worse. Less visible, but if it's detected, on their most heavily monitored networks, during a Level 2 alert, it looks the same. And we can't be certain of the effect. We might cause a brigade commander to believe his link to Beijing has been cut." He stopped. "I'll just say that degrading a nuclear command system can make it more dangerous, not less, depending on how it was designed to behave when degraded."

I want a successor to read that last sentence twice.

The Mirror

I stood up. I do that sometimes when I am about to say something I have thought about for a long time, because I think better on my feet. Pruitt says it makes me look like a professor. I was one, briefly, and I do not consider it an insult.

There was a whiteboard on the wall of the PEOC, and I drew on it, as I had drawn in Chapter 1 for a different audience, the Mirror Model. Two boxes. On the left, *US*. On the right, *PRC*. Between them, a mirror, drawn as a vertical line.

"On our side of the mirror," I said, "we see a man, Luo, who may be acting outside his chain of command, with mated warheads, and we're afraid he'll launch. That's real. Now step through." I tapped the right side. "On their side, here is what they see, and I want everyone in this room to see it with their eyes, not ours.

"They see Golden Dome, which their strategists have told them for twenty years is designed to catch whatever survives an American first strike. They see two hundred space-based interceptor prototypes

in orbit, twelve put up five days ago. They see ARGUS, which we have advertised for a year in order to deter them. They see a second carrier group, two B-21s from Guam, and a strike on Fujian on 12 September that we called 'complete,' with no way to know whether the next 'complete' response will be larger.

"Now add one more thing to that picture. Their coastal early-warning radars go dark. Their command networks start behaving strangely. What does a reasonable Chinese officer, not Luo, a reasonable one, conclude?"

Nobody answered. I answered for them.

"He concludes that the Americans have seen everything, can hit anything, can catch whatever is left, and have just begun to close his eyes. He concludes that this is the disarming first strike that his entire career has trained him to fear. And if his posture is launch on warning, and he no longer has warning, then the only thing in the world that can save his country's deterrent is to use it, now, before the next minute.

"We would not be taking the gun out of Luo's hand. We would be handing his argument to every officer in the Rocket Force who is currently frightened and unsure. LANTERN told us some officers are afraid. Right now, their fear is of Luo. A left-of-launch strike would turn their fear toward us. It would make Luo right."

I wrote on the whiteboard, under the right box: *THE MORE PERFECTLY WE SEE, THE MORE FRIGHTENED THEY ARE.* And under that: *DO NOT CLOSE THEIR EYES.*

Dunmore said, quietly, "Sir, I understand the model. I'd ask you to consider the alternative risk. If you're wrong about how they'd read it, and we do nothing, and he launches—"

"Then I'll have been wrong in a way that killed millions of Americans," I said. "And if I'm right and we strike, I'll have been right in a way that killed millions of Americans and Chinese. Pat, I'm not saying the risk of doing nothing is zero. I'm saying the risk of doing this particular thing is larger, and it's irreversible, and we have reversible paths we haven't tried yet."

Marsh

Helena Marsh, who had been our ambassador in Beijing for four years before she was Secretary of State and who speaks Mandarin with an accent that Chinese officials have told me sounds like a professor from Nanjing, had said nothing. Now she said:

"There's one more thing on their side of the mirror, Mr. President. Face. Wen Zhaolin, if he didn't approve this, is the man who'll be blamed for it either way. If we strike, he becomes the leader who let the Americans attack China's strategic forces. He can't survive that politically, and he won't try to. He'll do what any Chinese leader would do: he'll stand with Luo, in public, whatever he thinks in private. The one man in Beijing with the authority to order this reversed will become the man who can't. You'll unify them. Right now they're divided, and division is the only good news we have."

That was the argument I had not made, and it was better than mine.

The Decision

I asked for objections, a habit from my company days. Achterberg stated hers again, well. Castillo asked that the cyber options be kept "warm," prepared but not executed. Fenn said the intelligence did not support a judgment that a launch was imminent. Laszlo said LANTERN's message, read

carefully, was that of a man asking us not to make things worse. Iwu said the bond market would not survive a strike on China's strategic infrastructure; it was the only time in the crisis I heard him use the word *survive*.

At 08:17 I said:

"We will not conduct any strike, kinetic or cyber, against Chinese early warning, Chinese nuclear command and control, or Chinese nuclear forces. That is not a decision for today. It's a decision for this crisis, unless and until we have evidence that missiles are actually being launched, and even then, I want that decision to come back to me. Ray, keep the cyber options prepared but not executed, and nobody puts so much as a probe on those networks without my personal authorization, in writing. Pat, I want options for generating our own strategic forces, quietly, today. Helena, I want a message to Beijing through every channel we have, by tonight. Sam, write this down the way it happened."

He did. I reproduce here the record as it appears in the manual.

Decision	Options considered	Reversible?	Risk	Chosen	Outcome
Response to PLARF Readiness Level 2 at Hami and Ordos, 2 Nov 2045	(A) Conventional strikes on Chinese early-warning radars; (B) cyber/electronic operations against PLARF command and control; (C) combined A+B with strikes on command posts; (D) no strike on nuclear, early-warning, or command-and-control targets; generate US forces quietly; intensive diplomacy	A, B, C: **irreversible** (cannot un-blind a radar or un-detect an intrusion; effects on a nuclear command system unpredictable). D: **reversible** (generation can be stood down; diplomacy can be withdrawn)	A–C: Mirror Model predicts Chinese reading as opening phase of disarming first strike, possible launch on loss of warning; would unify Beijing behind Luo. D: Luo could launch anyway; US forces less prepared to limit damage	**D**, with cyber options held prepared but unexecuted; no probes without written presidential authorization	No US action against Chinese strategic systems occurred during the crisis. Held through the false warning of 6 November (Chapter 6). In 2047 a senior PLA officer told a US delegation that a strike on the coastal radars "would have been read as the first minute of the war"

OPERATING NOTE — The most seductive option in a nuclear crisis is the one that promises to make the danger go away by acting first. Before you take it, run the Mirror:

describe your proposed action as the adversary's most frightened competent officer would describe it to his commander. If your action, so described, is indistinguishable from the opening of a disarming strike, it will be treated as one. Do not try to close the other side's eyes. A blind nuclear power is not a safe one; it is a desperate one.

2 November 2045 — 10:30 EDT: Generating the Forces

Having decided what we would not do, I had to decide what to do with our own nuclear forces. If a Chinese commander might launch, I owed the American people a force that could survive his attack and answer it; that is deterrence. And I owed Beijing a posture that did not look like preparation for a first strike of our own. Most of 2 November was spent finding the narrow space between those two obligations.

What "Generating Forces" Means

A successor will hear the phrase *generate the force* and may think it means building weapons. It does not. It means taking forces that already exist and moving them from their day-to-day posture to a higher state of readiness, so that more of them are survivable and available on short notice. I will explain it as it was explained to me, at the level of public knowledge.

On an ordinary day only part of the American strategic force is ready to go. The Sentinel intercontinental ballistic missiles (ICBMs) in their four hundred silos are always on alert; they are also the most visible leg of the triad, and the leg an adversary most fears will be used first. Ballistic missile submarines, Columbia-class and the remaining Ohio-class boats, are the most survivable leg, because once submerged at sea they are extraordinarily hard to find; but on an ordinary day some are in port at Kings Bay, Georgia, or Bangor, Washington, where they are vulnerable. Generating the submarine force means sending every boat that can be made ready to sea. Bombers, the B-21 Raider and the B-52J, have not been on day-to-day nuclear alert since 1991. Generating them means putting aircraft and crews on alert and dispersing some, so that a single attack cannot destroy them on the ground.

Generation also includes less visible things: increasing the readiness of the airborne command posts, the E-4B and the E-6B aircraft I described in Chapter 1, which can keep the President and the commander of STRATCOM in contact with the forces if ground command centers are destroyed; testing communication links; recalling key personnel; moving senior officials so that the chain of succession is not concentrated in one city.

Some of this we had already done. On 20 September, after the Bashi Channel, I had moved the country to Defense Readiness Condition 3, DEFCON 3, a scale that runs from 5 (normal peacetime) to 1 (nuclear war imminent or under way). In the public record, American forces had gone to DEFCON 2 only once, Strategic Air Command during the Cuban Missile Crisis in October 1962. Going there in November 2045 would have been, by every precedent, a statement that we believed nuclear war was close.

The Question of DEFCON 2

Dunmore and Hale presented a plan. It had three parts, and they were candid that it was what STRATCOM would recommend if it were thinking only of survivability.

"Step one, sir: surge every available submarine from Kings Bay and Bangor. Step two: place a

further portion of the bomber force on alert, with dispersal. Step three: move to DEFCON 2, which formally raises the readiness of all forces and tells every commander in the world how serious this is."

"Who does step three tell, besides our commanders?" I asked.

"Everyone, sir," Hale said. "It's not a secret when it happens. It would leak within hours, and frankly, in the past, part of the purpose was that it be seen."

"Then step three is a signal," I said. "What does it say?"

Oyelaran answered. "It says, *we think you're about to attack, and we're ready to fight a nuclear war.* To Luo, it says *you were right about the window.* To Wen, if he's in control, it says *the Americans are escalating.* To our own public, it says *panic.* To the markets—"

"Don't," Iwu said. "Please."

I said, "I want every bit of survivability we can get that they can't easily read as preparation for a first strike. And I want nothing that looks like a declaration."

We worked it through, item by item, for nearly two hours. I will give a successor the logic rather than the specifics, because the specifics are still sensitive and, more importantly, because the logic is what will be useful.

First, we sorted every measure by two criteria: how much it improved the survivability of our deterrent, and how much it would look to Beijing like preparation for attack. Submarine surges scored high on the first and low on the second, because submarines are a second-strike weapon, the most stabilizing kind of nuclear force. Bombers are visible on commercial imagery, so we put more on alert at home bases and dispersed a smaller number quietly, away from anything that could be read as staging toward Asia. Changes to the ICBMs we ruled out entirely: they were already on alert, anything we did to them would be visible, and they were precisely what Chinese planners feared we would use against their silos.

Second, we chose not to go to DEFCON 2. At 11:48 I said: "The force will be generated to the level STRATCOM needs for survivability. The country will stay at DEFCON 3." Dunmore asked, properly, whether I understood that some commanders would be operating at a readiness higher than the DEFCON in force. I did. She had the orders written so that those commanders had no ambiguity about what was expected of them.

Third, at Hale's suggestion, we would tell Beijing the fact, not the numbers. A message on the Defense Telephone Link, the military hotline that has existed since 2008, would inform the PLA that we were taking defensive measures for the survivability of our forces, that our intercontinental ballistic missile force was unchanged, and that we had no intention of attacking Chinese strategic forces.

"They won't believe it," Castillo said.

"Some of them will," Hale said. "And the ones who don't will notice that we told them. A force that's preparing a surprise attack doesn't announce it's increasing readiness. The announcement is itself evidence."

> **OPERATING NOTE** — Generate for survivability, not for signal. The nuclear forces that are hardest to destroy are also the ones least able to start a war by surprise: a submarine at sea is reassurance to you and, properly explained, reassurance to your adversary. The forces that are easiest to see are the ones most likely to be read as a

> threat. When you raise readiness in a crisis, raise it where it makes you safer without making the other side feel more cornered, and think hard before you announce it in a way your adversary's hardliners can quote.

By 16:00 on 2 November, every submarine that could be made ready had orders to sail, [REDACTED] boats in all, additional bombers were on alert, and the airborne command posts had raised their readiness. I asked the Vice President to leave Washington for several days, which she did not like and did. The Speaker, Diane Okonkwo, and the Senate Majority Leader, Cal Whitford, were briefed at 17:30 with the rest of the so-called Gang of Eight. Whitford, who had opposed me on almost everything for four years, listened without interrupting and said only, "Mr. President, I'll be praying for you, and I'll be keeping my mouth shut. You tell me if you need the Senate." I have never forgotten it.

The Message

At 19:00, I sent three messages to Beijing.

The first, through the Defense Telephone Link, was the one I have described, from Castillo to his counterpart in Beijing, informing the PLA that we were taking defensive measures and had no intention of attacking Chinese strategic forces.

The second was a request, through the State Department's channel to the Chinese Foreign Ministry, for a call on the Jade Line, the secure leader-to-leader video link our two governments had agreed to build in 2043, upgrading the old hotline arrangements. We had spoken on it three times since April. In September he had let a message on it go unanswered live. I did not know whether he would come to the screen this time.

The third was through Singapore. Prime Minister Daniel Tan had hosted Marsh and Foreign Minister Shen Yilan on Sentosa in August, and since then his office had served as a quiet relay. Marsh called him at 19:20 our time, 07:20 on the morning of 3 November in Singapore, and asked him to convey a single sentence to Shen, in person if possible: *The United States sees the change in your strategic posture, has not changed its own intentions, will not strike, and asks to speak to President Wen directly.*

Tan said he would deliver it before noon. He did.

Before I lay down I wrote the diary entry for 2 November. It is one of the few entries from that week that I can read without wincing at the handwriting, because I wrote it slowly:

2 Nov. No strike on their eyes. Subs to sea. No DEFCON 2. Told them so. Whitford: "I'll be praying for you." — What frightens me tonight is not Luo. It is the shape of the thing. Two men who have never met, each with his hand near a switch, each told by machines that the other's hand is moving. Luo and I. He reads ARGUS in our submarines and Golden Dome in our orbits; I read his lids and his launchers. Neither of us can see the other's face. The whole art now is to put a face between the switches.

I slept from 01:00 to 04:30, on the couch in the small study, which is the only place in the White House where I could sleep that week. Brooks's relief, a Navy commander named Ellison, sat in the anteroom. When I woke, the first thing I did was ask whether Beijing had answered.

It had. The Jade Line call would take place at 21:00 Beijing time on 3 November, 09:00 our time.

3 November 2045 — 09:00 EDT (21:00 Beijing): Jade Line, Call Four

The Jade Line terminal is in a small room adjoining the Situation Room. It has a single screen, a camera, two chairs on my side for me and an interpreter, and a seat at the back for the National Security Advisor and the Secretary of State. The Chinese side of the link terminates, we believe, in a room in the Zhongnanhai leadership compound in Beijing, although on this night it did not.

A leader-to-leader call is not a conversation. It is a performance before note-takers on both sides, every sentence later read by people who wish you ill. In our three previous calls, beginning with the cold one on 2 April, there had been one or two sentences, never more, that each of us understood were meant for the other man alone.

The screen came on at 09:00:40. Wen was not in the room I knew. The wall behind him was plain and pale green, with no flag. He wore a dark jacket and an open white shirt, which I had never seen. Foreign Minister Shen sat slightly behind him, a personal secretary on his other side. On 2 April I had noted in my diary that he looked pale, and had put it down to the lighting. He did not look pale now. He looked, there is no other word for it, grey. When he lifted his hand to adjust his earpiece, the hand moved slowly.

Our interpreter was a State Department officer named Anna Liu, who had interpreted every one of the calls. She told me afterward that his voice had changed. “It wasn’t fear,” she said. “It was breath. He was stopping in the middle of phrases to breathe.”

I give the call as reconstructed from our transcript, abbreviated.

Wen: President Kane. Thank you for your message. I have received it.

Kane: Thank you for taking this call, Mr. President. I want to say one thing at the beginning, so that there is no misunderstanding. The United States has observed changes in the readiness of your strategic forces in the last forty-eight hours. We have taken measures to protect our own forces. Those measures are defensive. We have not changed our intercontinental missile force. We have no intention of attacking China’s strategic forces, its early warning, or its command systems. I am saying this to you personally.

Wen: (pause) Your measures. Your submarines. Your bombers. You call them defensive.

Kane: I do. And I told your military about them before you could discover them, which I would not do if they were anything else.

Wen: China also takes defensive measures. China’s posture is defensive. China has a policy of no first use. That has not changed.

Kane: I’m glad to hear it. Mr. President, I want to ask you something directly. Are the measures your forces have taken in the last two days measures you have personally directed?

There was a long silence. Liu said afterward that it was nine seconds. On the screen, Shen turned her head slightly toward Wen and then back.

Wen: The Central Military Commission directs the armed forces of China. I am the Chairman of the Central Military Commission.

Kane: I understand that. That is why I’m asking you.

Wen: (pause) President Kane, in China we have a saying. *When the tree wishes to be still, the wind does not stop.* (Liu rendered it later as: "The tree would like to be still, but the wind won't let it.") You have placed weapons in space that are designed to destroy our deterrent. You have ships at our door. You struck our soil in September. You speak of defense. Some of my generals believe you are preparing to make China defenseless. I must listen to my generals.

Kane: I understand why they believe it. I'm telling you that it isn't true. I'm prepared to show you that it isn't true. Not with words. With steps.

Wen: What steps?

I had prepared for this, with Marsh and Oyelaran, and I had prepared to say nothing specific. It was too early. We had not yet decided what we could offer, or through whom, and the people sitting beside Wen were not, I suspected, all people who wanted the same thing he did. I said:

Kane: Steps that cost us something, and that you would be able to verify. I will send them to you, privately, in writing, for you alone. I ask only one thing in return, now, tonight: that your forces take no further step up while we talk.

Wen: (long pause) I hear you.

Then he said something that was not in any prepared statement, and that Liu interpreted with a slight hesitation, because the grammar was unusual.

Wen: President Kane. You are an engineer. I was also trained as an engineer. Hydraulic. Dams. When the water rises behind a dam, the engineer must open the gates. But sometimes the gates are held by other men.

Kane: Then perhaps we should talk about who holds the gates.

Wen: (to someone off-screen, in Chinese, not interpreted at the time; later rendered as "That's enough.") Thank you, President Kane. We will consider your letter.

The screen went dark at 09:26.

What I Heard

I sat in the small room for some minutes after the screen went dark. Marsh sat with me. Oyelaran went out to begin writing his memorandum. Liu, who had been a professional for two decades and had heard every kind of conversation, sat with her hands flat on the table and did not move.

"He told you," Marsh said finally.

"He told me something. I'm not sure what."

"*The gates are held by other men.* Mr. President, that is as close as a General Secretary of the Chinese Communist Party will ever come to telling an American President that he doesn't fully control his own military. He said it in front of Shen and his secretary. He said it knowing it would be in our transcript. He wanted you to hear it."

"Or he wanted me to think he isn't responsible, so that I'll offer something to strengthen him, and the alert is a lever."

"That's possible," Marsh said. "I don't think it's likely. I've known him for eleven years. He doesn't bluff with metaphors. He bluffs with statistics."

I wrote on my pad:

W. — exhausted, ill? Breathing. Unusual room. "Gates held by other men." Did not deny. Did not affirm. Asked "what steps." Wants an offer. For himself or for the room?

And underneath: *If he is losing control, then the man I need to speak to is not the man I can speak to.*

> **OPERATING NOTE** — On a leader call in a crisis, listen hardest to the sentences that were not prepared. The prepared statement tells you the position of the government. The unprepared sentence, especially one spoken in front of witnesses, tells you the position of the man. But do not mistake the man for the government, and do not assume the man can deliver the government. Ask yourself, after every call: if what he hinted is true, who else must I reach?

3 November 2045 — 14:50 EDT (03:50, 4 November, Beijing)

WILLOW's message arrived at 14:50.

WILLOW, as I have described in earlier chapters, was a major and communications duty officer at the Central Military Commission's Joint Operations Command Center in the Western Hills outside Beijing. She saw not weapons but traffic: who sent orders to whom, under whose name, and, from the faces in the corridors, in what mood. Her channel, [REDACTED], was slower than LANTERN's and in some ways safer.

She was, at that moment, the bravest person I was responsible for. KESTREL was dead. The MSS, under its Minister Kong Dewei, was running a hunt for further American sources inside the strategic forces, and our counterintelligence analysts assessed that it had already narrowed its search to the Rocket Force and to the Joint Operations Command. Every message WILLOW sent increased the chance that she would be found.

Her message, translated:

> CHAIRMAN TAKEN TO 301 HOSPITAL 2200 LOCAL BY AMBULANCE AFTER EVENING CALL. HEART. NOT ANNOUNCED. OFFICE SAYS RESTING. GENERAL LUO AND ADMIRAL GAO NOW IN COMMAND CENTER PERMANENTLY. JOINT OPERATIONS COMMAND TRAFFIC UNDER LUO AND GAO NAMES ONLY SINCE 2300. VICE CHAIRMAN ZHANG NOT PRESENT. PREMIER QIAO OFFICE CALLED THREE TIMES, NOT CONNECTED. OFFICERS SAY LUO SAYS THE WINDOW IS SIX MONTHS AND IT IS CLOSING. GAO QUIET. I AM WELL.

The PLA General Hospital, which everyone in China calls 301, is where China's leaders are treated. It was the same hospital to which Wen had been taken, secretly, for four days in July, as WILLOW had reported then.

Wen had collapsed within hours of our call. I asked myself many times whether the call had caused it. Claire, when I told her only that a man I had spoken to that morning had been taken to hospital, asked about his breathing, listened, and said: "Tom, a man who breathes like that has been sick for weeks. You did not do that to him." It helped some.

What WILLOW's Message Meant

I convened the principals at 15:30 in the PEOC. The picture, as we assembled it over the next two hours, was the worst of the crisis to that point.

The picture was the worst of the crisis to that point. Two of China's three silo fields were at Level 2 on an order from a man who had spoken publicly of a closing window. The man whose authority over those forces was supreme was in a hospital bed. The Joint Operations Command was being run by Luo and by Admiral Gao Ruiming of the Eastern Theater, whose forces faced Taiwan and our fleet. The general most loyal to Wen, Vice Chairman Zhang Hesheng, was absent. And Premier Qiao Weiming, the leading voice for de-escalation, could not get through on the telephone.

"Is this a coup?" Achterberg asked, from the secure video link at her undisclosed location.

"No," Laszlo said. "Not yet. It's something more Chinese than a coup. It's a vacuum, and the people who are standing nearest to it are filling it. Nobody has declared anything. Nobody has to. Luo and Gao are simply in the room where the decisions are made, and the people who would stop them aren't."

"Do Qiao and Zhang know what Luo has ordered?"

That was the question. Qiao and Zhang surely knew the Rocket Force had raised its readiness. But if they did not know the order lacked the Chairman's Seal Directive, they would assume, as anyone would, that Wen had authorized it before he fell ill. And in the Chinese system, challenging it would then be challenging him. Neither would do that while he lived.

"So the one fact that would allow the people who want to stop this to stop it," Oyelaran said slowly, "is the fact that the order wasn't legitimate. And the only people who know that fact, outside of Luo's headquarters, are us."

The room was quiet for a long time.

"And we know it," Calder said, "because of one man in Base 64 and one woman in the Western Hills. If we tell Beijing what we know, the first question every counterintelligence officer in the MSS will ask is: *how do the Americans know?*"

3 November 2045 — 21:15 EDT

I went upstairs for an hour. I have told Claire that I would put this in the manual, and she said I should, because successors should know that a President is allowed to go upstairs.

She was in the Yellow Oval Room, reading. Our son Eli had sent her a long message from Austin about a project at his company, and she read it to me, not because either of us cared at that moment about distributed database architecture, but because it was Eli's voice. He had written, at the end, *Tell Dad I know he can't say anything. Tell him I'm proud of him and that I'm not watching the news.* I asked her to read that part twice.

Then she said: "You have to decide something."

"Yes."

"Is it the kind of decision you can take back?"

Claire had learned the Reversibility Matrix from me when we were first married, in an apartment in Cambridge, when I drew it on a napkin to explain why I was hesitating to leave MIT for a startup. She had used it since, I learned many years later, with the parents of children she was treating, to help them decide between treatments.

"No," I said. "It's not."

"Then do what you always told me. Take it slowly."

"I don't know that I have time to take it slowly."

"You have tonight," she said. "Use all of it."

I went back down at 22:20. I decided, on the stairs, that I would not decide anything until I had heard every argument at least twice, and that I would give myself until the following evening. I started a clock on my pad at 22:30. By the time I signed the authorization it had run for nineteen hours.

The Nineteen Hours: 3 November, 22:30 EDT, to 4 November, 17:30 EDT

I have kept the pad on which I timed those nineteen hours. It is in front of me as I write. Down the left margin are the hours. Next to each is a note of who I spoke to and what I learned. I reproduce the substance here, because a successor who faces an irreversible decision should know what it looks like, from the inside, to take one slowly when every instinct says to hurry.

22:30 — The Question, Stated

I asked Oyelaran to write the question on the whiteboard in the PEOC so that none of us could drift from it. He wrote:

Do we tell Premier Qiao and Vice Chairman Zhang that Luo's Level 2 order lacks the Chairman's authority, knowing that doing so may expose LANTERN and WILLOW?

Underneath, I added: *Is there any reversible way to achieve the same effect?*

We spent the first three hours on the second line, because the Reversibility Matrix obliges you to exhaust the reversible paths, not wave at them. A hint on the Jade Line would be heard, if at all, by people standing beside Luo. A public statement about "irregularities in the Chinese command chain" would tell Luo what we knew and tell the MSS everything a private message would, with the world watching. Waiting for Wen to recover meant waiting at Level 2, with no sign he would recover soon. Asking President Volensky to warn Beijing we rejected in four minutes; Russia would sell it.

By 01:30 the second line had an answer. No. There was no reversible way to put that fact in the hands of the two men who could use it.

01:30 — Calder

Ruth Calder asked to speak to me alone. We went into the small conference room at the back of the PEOC, the one with no screens.

She is in her late fifties, has spent most of her life being someone else in other countries, and recruited LANTERN in a hotel room twelve years before, when I was Governor of Colorado. She does not make speeches. She made one that night.

"Mr. President, I want you to understand what you're asking me to agree to. The fact we'd be passing is that the Level 2 order didn't come with the Chairman's Seal Directive. A satellite can't see that. It's a fact about paper, known to maybe a few hundred people: officers at Luo's headquarters, the base staffs who received the order, the duty officers at the Joint Operations Command. Kong Dewei's investigators already believe they have a penetration in the Rocket Force, because they found KESTREL. If Qiao and Zhang use this, Luo will know within a day where it came from, and the MSS will start with Base 64 and the night roster at the Western Hills. LANTERN is in Base 64. WILLOW is on that roster.

"I can't promise you either of them will be caught. I can promise you the chance goes up. I don't know by how much. Nobody does.

"I also want to say this. LANTERN came to us because he didn't want this war. Every message he's ever sent us has been, in one way or another, a request that we use what he gives us to stop it. I believe if you could ask him, he'd tell you to send it. But you can't ask him, and I'm not going to pretend that what he'd say makes it costless. I lost KESTREL six weeks ago. I'd like not to lose any more."

"Can we get them out first?"

"Not in time. An exfiltration takes weeks to prepare properly. A hurried one is more likely to get them killed than leaving them in place. If you send this, they will be in China when it lands."

I thanked her. I asked her one more question. "If I asked you for a recommendation, what would it be?"

"I'd ask you not to ask me that, sir," she said. "It's not my decision, and I'd be lying if I told you I could make it fairly."

02:00 to 06:00 — The Sanitization

While I waited, I asked the analysts to solve a different problem: suppose we did it. What is the least revealing way to say it?

Sanitizing intelligence means taking information from a sensitive source and rewriting it so that it conveys the essential fact without revealing how that fact was learned. It is an old craft and an imperfect one. Every sanitized report is a negotiation between what the reader must believe and what the source must not lose. If you strip out too much, the reader does not believe you. If you leave in too much, the reader's counterintelligence service can work backward to the source.

The analysts produced three versions over four hours. The first said too much: it described the routing of the order as only someone who had seen the documents could. The second said too little, and Marsh, reading it, said, "Qiao will put this in a drawer. It reads like we're trying to divide them. Which we are."

The third version, which I approved in substance at 05:50, had three parts. First, an ARGUS timeline, at a resolution that the Chinese would already assume we had, showing the dispersal at Hami and Ordos beginning at roughly 14:20 Beijing time on 1 November and showing that Yumen had not moved. Second, the observation that Yumen's silence was inconsistent with a nationwide

order from the Central Military Commission, which any PLA officer would understand. Third, a single sentence: *The United States has reason to believe, with high confidence, that the order raising these units to pre-launch readiness was issued through the Rocket Force command channel and was not accompanied by the authorization of the Chairman of the Central Military Commission.*

It did not say how we knew, and it did not use the phrase *Chairman's Seal Directive*, our own translation, which an investigator might trace. It invited the reader to suppose signals intelligence, which the Chinese already assumed, and it rested on the Yumen anomaly, which they could verify themselves. Calder said it was as good as it could be made. She did not say it was safe.

> **OPERATING NOTE** — When you must spend a source to save something larger, spend as little of it as the purpose requires, and design the message so that the reader can verify part of it independently. A fact the other side can check for itself is both more credible and less revealing. But do not let the craft of sanitization persuade you that you have made the decision safe. You have made it less dangerous. The person in the field still pays whatever it costs, and you will not be told the price for months.

06:00 — Claire, and the Window

At six I went up for a shower and a change of clothes. Claire was awake. She did not ask. She made toast in the small family kitchen on the second floor, which no President's wife is supposed to do and which she has done every morning since 2041. I stood at the window looking at the Washington Monument in the grey light and said, to no one in particular, "There are two people I've never met and never will, and I'm deciding whether to make them more likely to die."

"To make a lot of other people less likely to," Claire said.

"I don't know that."

"No," she said. "You don't. That's what makes it yours."

I have been asked since, by a historian I trusted, whether I prayed in those nineteen hours. I grew up a Methodist in Pueblo and drifted, as engineers do. I told the historian the truth: I did, once, standing at that window, and I did not ask for an answer. I asked to be able to live with the one I gave.

08:00 to 12:00 — The Channels

The second half of the decision was how. We had two channels to the two men.

ORCHID was an aide in Premier Qiao Weiming's office, a source of political intelligence whom, since the summer, the CIA had also used with my authorization as a deniable path to the Premier. ORCHID was never told about LANTERN or WILLOW, but carrying this would put ORCHID at risk too: if the Premier acted, Luo's allies would ask who in his office talked to Americans.

For General Zhang Hesheng, we had no source. We had Singapore. Prime Minister Tan's government had, over many years, built a quiet relationship of its own with senior PLA officers through defense dialogues, and Tan told Marsh in August that he could, "in a true emergency, put a sealed envelope on the desk of a senior member of the Commission within a day." Marsh called him at 08:40 our time, 20:40 in Singapore.

I listened to the call. Tan asked one question. "Helena, is what is in the envelope true?"

"Yes."

"Will it make things better or worse?"

"We believe better. We aren't certain."

"Then I will carry it," he said, "and I will tell no one in my own Cabinet what it contains, and if anyone asks me later, I will say Singapore carried a letter, as friends do." I have thanked him many times since, and he has always said the same thing: that small countries survive by being the place where large countries can stop.

Indonesia's President, Arya Wicaksono, was not asked to carry anything, but at 10:15 his foreign minister told Marsh that if the chair of the Association of Southeast Asian Nations (ASEAN) needed to call on "all nuclear-weapon states to exercise maximum restraint," Jakarta would sign within the hour. It mattered later.

12:00 to 16:00 — The Last Objections

At noon I convened the principals one last time and asked each to state the strongest argument against sending.

Fenn: "That we may be wrong. LANTERN didn't see the authority. That isn't the same as the authority not existing. If Wen did approve it, verbally, before he fell ill, then we will have told Qiao and Zhang something false, and when they check, our credibility with the two men we most need is gone, and we will have exposed our sources for nothing."

That was the argument that kept me awake. I asked Calder whether WILLOW's report helped. She said that WILLOW had seen traffic under Luo's and Gao's names only, and that in the Joint Operations Command a nuclear readiness change authorized by the Chairman would normally leave a trace in traffic WILLOW could see. Two sources, two different vantage points, both seeing an absence. It was not proof. It was, I judged, enough to justify the words *high confidence*, and not one word more.

Castillo: "That Luo, if he learns his own leadership has turned against him because the Americans told them, accelerates. Cornered men move."

Oyelaran answered that one, and I agreed with him. "Ray, he's already acting as though he's cornered. What he doesn't have yet is a leadership that knows enough to stop him. If he accelerates, he does it against Zhang's opposition instead of with Zhang's silence. That's a different country."

Achterberg, from her location: "That we're interfering in the internal politics of the Chinese Communist Party during a nuclear crisis, and if it ever comes out, they'll never trust a word we say."

"If it comes out," I said, "it will come out that we told them the truth. I can live with that."

Iwu said that he had no economic argument against. Pruitt said that she had no political argument she was willing to state in that room. Laszlo said he would carry out whatever I decided and would like the order in writing.

17:30 — The Decision

At 17:30 on 4 November, nineteen hours after I had drawn the line on the pad, I signed a memorandum authorizing the CIA to pass the sanitized package to Premier Qiao through ORCHID and the Department of State to pass an identical package to Vice Chairman Zhang through the Government

of Singapore, both to arrive before the start of the working day in Beijing on 5 November. I wrote by hand, at the bottom, under my signature: *I take responsibility for the consequences of this to our sources. T.E.K.*

Decision	Options considered	Reversible?	Risk	Chosen	Outcome
Whether to pass evidence that Luo's Level 2 order lacked the Chairman's authority to Premier Qiao and CMC Vice Chairman Zhang, 4 Nov 2045	(A) Say nothing; wait for Wen's recovery; (B) public statement on "irregularities"; (C) hint on Jade Line to whoever answered; (D) via Russia; (E) sanitized package to Qiao (via ORCHID) and Zhang (via Singapore)	A, B, C, D: reversible in form but each either ineffective or more revealing than E. E: **irreversible** (once known in Beijing, the fact and its implication, that the US has access, cannot be recalled)	E: could expose LANTERN, WILLOW, and ORCHID to the MSS mole hunt; could be false if Wen gave verbal authority; could provoke Luo to accelerate. A: Level 2 continues under Luo's control with no check	**E**, after 19 hours; sanitized to suggest signals intelligence and to rest on the independently checkable Yumen anomaly; presidential written responsibility for source risk	Delivered to Qiao about 07:30 and to Zhang about 09:00 Beijing, 5 Nov. Its effect is described in Chapter 6. All three sources survived the crisis; WILLOW's channel went silent for eleven weeks during the MSS investigation that followed

> **OPERATING NOTE** — The Reversibility Matrix does not tell you what to decide. It tells you how fast. An irreversible decision taken in nineteen hours, with every objection heard twice, is not a slow decision by the standards of history; it is a slow decision by the standards of the military clock, and that is the only standard that matters. Use the time to exhaust the reversible paths honestly, to shrink the irreversible step to its smallest effective form, and to put your own name, in your own hand, on the cost.

5 November 2045 — 02:00 EDT Becomes 01:00 EST

At two in the morning the clocks went back an hour. The watch officer reset them by hand and took down the red card that said *BEIJING (IF EDT)*. Beijing was now thirteen hours ahead. I had been given an extra hour of the night, and I spent it writing a letter.

I had promised Wen steps that cost us something and that he could verify. The package told Qiao and Zhang the truth about Luo; it gave them nothing to hold up in the Politburo Standing Committee as a reason to stand down without looking as though they had surrendered. A golden bridge must be wide enough to cross in public. The letter was the bridge.

I wrote the first draft by hand between 01:10 and 03:40 EST. Marsh removed two sentences she said "a Chinese reader would hear as a lecture." Oyelaran checked every commitment against what Castillo and Hale had confirmed we could deliver; Hale's view, which I accepted, was that pausing

new interceptor launches cost us schedule, not defense. The convoy pause I put to Lin Shu-wen by telephone at 07:15. She was a law professor from Tainan, fifty-four, a wartime president against her expectation, and after a long silence she said, “Seventy-two hours. Not seventy-three. And you will tell me before it ends whether it is extended.” I promised I would.

The letter was translated by Liu, sealed, and handed at 11:00 EST to Ambassador Tang Shiyu at the State Department for delivery to President Wen personally, wherever he was, with a copy conveyed through Singapore to Vice Chairman Zhang, so that if Wen could not read it a man loyal to him could. Tang read the address, looked at Marsh, and said, “I understand where it must go.” He did not say *hospital*.

I give the letter in full. It is also reproduced in Annex E.

The Letter

THE WHITE HOUSE WASHINGTON

5 November 2045

His Excellency Wen Zhaolin General Secretary of the Communist Party of China President of the People’s Republic of China Chairman of the Central Military Commission

Mr. President,

I am writing to you myself, and not through ministries, because on 3 November you told me that you were an engineer, and so am I, and engineers speak to each other most clearly in plain terms.

I will say first what I said to you then. The United States does not intend to attack China. We do not intend to attack China’s strategic forces, its early warning, or the systems by which its leaders command those forces. In the last four days some of my advisors have presented me with plans to do exactly that, on the grounds that your forces had moved to a higher state of readiness. I have refused them. I refused them because I understand what such an attack would look like from Beijing. It would look like the beginning of the war that your strategists have warned of for twenty years. I will not begin that war, and I will not give anyone in China a reason to believe I am beginning it.

You told me that the water was rising behind the dam, and that sometimes the gates are held by other men. I have thought about that sentence every hour since. I cannot open your gates. I can lower the water on my side of the dam. I am therefore prepared to take the following steps, and I want you to know that I offer them to you, and to those who serve you faithfully, and not to anyone else:

First, the United States will conduct no strike, by any means, kinetic, cyber, or electronic, against China’s nuclear forces, its early-warning systems, or its nuclear command and control, for as long as China’s nuclear forces do not launch. I have given this order to my own commanders, and it will remain in force.

Second, the United States will pause all new launches of space-based interceptors for the Golden Dome architecture, beginning today, for the duration of the talks between our governments that I propose below. No interceptor will be added to those now in orbit while this pause holds.

Third, the United States, with the agreement of the authorities in Taipei, will pause all escorted convoys across the waters around Taiwan for seventy-two hours, beginning at a time we will agree, so that no ship of ours is in a place where an accident could occur while our two governments step back.

In exchange, I ask one thing. I ask that the units of the People's Liberation Army Rocket Force that have been raised in recent days to pre-launch readiness be returned to the readiness level at which they stood before 1 November, which our services understand as Level 3, and that this be done within the same seventy-two hours. My own forces will then return in step with yours, and I will tell you, through the channel of your choosing, when they have done so.

We will both be able to see whether the other has kept his word. Your satellites watch our bases and our seas, and ours watch yours. In a crisis, that has made each of us more afraid. In this letter I am asking that it make each of us more certain.

I propose that, as soon as these steps are under way, our Foreign Ministers meet in Singapore, with the help of Prime Minister Tan, to begin talks without preconditions on all the matters between us, including those that concern Taiwan, and including the fears on your side about the systems I have described. I know that no letter can settle those things. I ask only that we give ourselves the time to try.

Mr. President, I hope that you are well, and that you will forgive me for saying that I know you have not been. I wish you health, as one man to another. Whatever happens between our countries, there are a great many people on both sides, of whom we are only two, who would like to grow old.

With respect,

Thomas Ellery Kane President of the United States of America

Marsh read it one last time before it was sealed, and said, "The last paragraph is the part he'll read twice." I said I hoped so. I did not tell her that I had written it with my own daughter in mind, standing a watch on a destroyer somewhere east of the Philippines.

OPERATING NOTE — A letter to an adversary leader in a crisis must do three things: state what you will not do, so that his fears have a ceiling; offer something concrete, costly, and verifiable, so that his moderates have something to hold up; and ask for one thing only, clearly, with a time attached. Write it to the man, but write it knowing his worst rival will read it too. Every sentence should be one you would be willing to see published.

5 November 2045 — 14:00 to 23:59 EST

Then there was nothing to do but wait, which is the hardest work a President does.

The packages had landed overnight. At 19:40 EDT on the 4th, Laszlo had reported that ORCHID signaled, by a very limited pre-arranged method, that the Premier had read his. At 21:15 Tan's office told Marsh that the envelope had been placed in General Zhang's hands at an early meeting in Beijing, and that Zhang had read it before the Singaporean officer, put it in his breast pocket, and said, "Thank your Prime Minister." Then nothing. At 18:40 EST on the 5th, Tang informed the

State Department that my letter had been delivered "to its addressee." He did not say whether the addressee was able to read it.

LANTERN was silent. Calder reminded me, at 19:00, that LANTERN's silence meant nothing, that he could transmit only when it was safe and that silence was, statistically, the normal condition of a source. I said I knew. She said she knew I knew; she was reminding herself.

WILLOW was silent too.

ARGUS showed, by 21:00, no change. The launchers at Hami and Ordos remained in their field positions. The lid crews were still at the silos. Over Ordos the cloud had thickened, and the analysts were working the field entirely with radar. A Space Force briefer told us at 21:30 that one tranche of the infrared early-warning satellites in the tracking layer, partly degraded since the September attacks, was running a software update to its data fusion that night, and that its confidence scores would be noisier than usual for some hours. I wrote it on the pad. I did not underline it. I have asked myself many times since whether I should have.

At 22:15 Claire came down to the PEOC, which she had never done before. She brought a thermos of tea and a sweater I had left upstairs, and she sat with me for twenty minutes in the small conference room without screens. She asked whether I wanted to come up to bed. I said I would come up at midnight. She said, "You won't, but come up when you can." Then she put her hand flat on my chest, the way she does to a frightened child to slow its breathing, and left.

At 23:00 Pruitt sent the staff home who could go home. Oyelaran stayed. Dunmore went to the Pentagon to sleep for three hours in her office near the National Military Command Center. Hale was at Offutt. Moriyama was at Camp Smith, where it was six in the evening, and the fleet was holding position.

At 23:40 I wrote in my diary:

5 Nov. Letter delivered. Package delivered, both. Z. put it in his pocket. Q. read it. W. — "delivered to addressee." No answer. L. silent. W. silent. ARGUS: no change. Clouds over Ordos. One IR tranche updating tonight, noisy. — Clocks went back this morning and I wrote a letter with the hour. I have offered what I can offer. Now the other side of the mirror must move, and I cannot see it move. — C. put her hand on my chest. Two people I have never met. A billion I have never met. Maya. — Go to bed, Tom.

At 23:52 Brooks's relief came on for the overnight rotation, and Brooks, who had been on duty for most of the previous seventy-two hours although he was not required to be, stood up from his chair against the wall, picked up the satchel, and handed it across in the brief, formal way they always did. Then he did something he had never done before. He turned to me and said, "Mr. President. Get some sleep, sir. We've got the watch."

I said I would. I went up to the residence a few minutes before midnight, still on the fifth of November. Claire was asleep. I lay down beside her in the grey sweater and listened to the house: the hum of the air, the floorboard under the agent's weight, the distant sound of water that I chose to believe was the Potomac.

In Beijing it was the middle of the afternoon. In the desert outside Hami the launchers were standing in the sun. Somewhere above them, a tranche of satellites that I had been told were noisy that night was staring down at the heat of the earth, and its software was deciding, as software does, what it was seeing.

I fell asleep soon after, and the last thing I remember thinking was that I had not underlined the line about the satellites.

Chapter 6: The Nine Days, II: The Eleven Minutes

"The machine said ninety-three percent. I said no. I have never been less certain of anything I turned out to be right about." — Diary of Thomas Ellery Kane, 6 November 2045, 04:10

A Note Before the Minutes

I have rewritten this chapter three times. Written as a story, it read like a thriller, which tells the reader that the ending was never in doubt. Written as an after-action report, it read like a furnace maintenance log. What follows is organized by minute, because that is how I lived it, and because a successor needs to feel how short a minute is when there are only eleven of them. Between the minutes I have put what I learned later: from the logs of the North American Aerospace Defense Command (NORAD), the National Military Command Center (NMCC), and Strategic Command (STRATCOM), from our human sources inside China, and from the investigation that followed.

A successor should know three things before reading on.

First, the timestamps. All American times in this chapter are Eastern Standard Time. Daylight time had ended the day before, on Sunday, which meant Beijing was thirteen hours ahead of Washington. When it was 02:31 in Washington it was 15:31 in Beijing, a working afternoon on a cold, clear Monday. That asymmetry mattered. I was asleep. The men on the other side were awake, at their desks, and some of them were looking for a reason.

Second, the context. By the night of 5 November we were six days into what my staff later called the Nine Days. On 1 November, the Rocket Force base our agent LANTERN served in had been ordered to Readiness Level 2, and warheads had been mated to missiles in two brigades of the Hami silo field. On 4 November I had taken the gamble described in the previous chapter: I had passed to Premier Qiao Weiming and General Zhang Hesheng, through the aide we called ORCHID and through Singapore, evidence that General Luo Tiancheng had raised the alert without the full authority of the General Secretary. On 5 November, at 19:12, my private letter to President Wen Zhaolin had been handed to the Chinese side in Singapore. It offered no American strikes on China's nuclear or early-warning assets, a pause on new space-based interceptor launches, and a seventy-two-hour pause in our convoys, in exchange for a return to Readiness Level 3. We had no answer. Wen was in a hospital bed. Luo and Admiral Gao Ruiming were running the Joint Operations Command Center in the Western Hills.

Third, the condition of our eyes. Since 13 September, when China blinded about sixty of our ARGUS and tracking-layer satellites with co-orbital weapons and ground-based lasers, our infrared

early-warning picture over western China had been rebuilt, but not fully. One tranche of the low-orbit tracking layer was still degraded: some satellites had damaged detector elements, others had drifted out of calibration after the dazzling, and the replacements Halcyon Orbital had launched in October were still being folded into the network. To compensate, in mid-October the Space Force had retrained the software that fuses all these sensors together. I had approved the funding for that retraining. I want that on the record before anything else, because a president who reads this chapter and concludes that the machines failed him has learned the wrong lesson. The machine was ours. I had bought it, praised it, and leaned on it. What failed was a system, and I was part of the system.

I went to bed at 01:20 on the morning of 6 November. My last diary line before sleep reads: *"Nothing from Beijing. Claire says I should sleep in the Lincoln Bedroom so the phone doesn't wake her. I refused. If the phone rings tonight I want her to be the first face I see."*

The phone rang seventy minutes later. But the minutes began before it rang, and they began a long way from Washington.

Minute 0 — 02:31:07

At 02:31:07 Eastern, an algorithm declared a launch.

I did not know it for nearly three minutes. A successor should understand why, because it is by design and the design is sound. The warning system is a chain, and the President is near the end of it. The beginning is in space.

The United States watches for missile launches with infrared sensors in orbit. The oldest layer, descended from the Defense Support Program of the Cold War and the Space-Based Infrared System that replaced it, sits in high orbits, geosynchronous and highly elliptical, staring down at the earth and looking for the hot plume of a rocket motor against the cooler background of land, sea, and cloud. In the 2030s that layer was supplemented by the Proliferated Warfighter Space Architecture, the tracking layer, hundreds of smaller satellites in low earth orbit that see launches from closer and, more importantly, keep watching the missile after its motors burn out, so that interceptors and warning crews know where it is going. By 2045 the hosted sensors of ARGUS on Halcyon Orbital's commercial buses added still more eyes. No single human being looks at all of this raw data. There is too much of it, arriving too fast. What the crews look at is the output of fusion: software that takes detections from every sensor, matches them in space and time, throws out the ones that are clearly not missiles (forest fires, sun glint, industrial flares, the exhaust of a jet on afterburner), and declares the ones that remain to be events.

The fusion engine in 2045 was part of the Golden Dome battle-management architecture. Its formal designation is long and dull. The crews called it PALISADE, and I will too. PALISADE did more than fuse. It assigned each event a classification, a launch point, an estimated missile type, a predicted trajectory, and a confidence score. It also, and this is the part that matters most, used context. It knew what ARGUS knew: which Chinese silos were believed to hold missiles with mated warheads, which brigades were at elevated readiness, what the political temperature was. Its designers had built it that way deliberately, the way a good doctor reads an X-ray differently for a patient who arrives coughing blood. In peacetime this made it smarter. In the first week of November 2045, as I would learn, it made it primed.

At 02:31:07 PALISADE reported four probable intercontinental ballistic missile launches from the

Hami silo field in Xinjiang. It placed all four launch points within the field, inside a cluster of silos belonging to one of the two brigades LANTERN had told us were mated. It classified the objects as probable DF-41. It projected trajectories that rose northeast, toward the pole. Its confidence score was 0.87.

The event appeared on consoles in three places at once: at the ground station at Buckley Space Force Base in Colorado, where the satellites' data is processed; at the Missile Warning Center inside Cheyenne Mountain, the granite bunker outside Colorado Springs that NORAD has used since the 1960s; and on the displays of the Command Center at NORAD and U.S. Northern Command headquarters at Peterson. In each place a human being in uniform looked at the same symbols. Their job, in the first minute, is not to decide whether the United States is under attack. Their job is to decide whether the sensor report is coherent enough to send up the chain.

The senior officer on the floor at the Missile Warning Center that night was a Space Force lieutenant colonel I will not name. She had been on shift for six hours. She had trained for this exact moment perhaps two hundred times. In her statement to the investigation she said: "The first thing I felt was that it was too clean. Four plumes, same field, same second. Real launches are messier. The second thing I felt was that my feeling was not data."

She did what the procedures require. She checked whether the event was an exercise inject. It was not. She checked whether the event came from one sensor or several. The fusion display showed contributions from two high-orbit infrared satellites and three low-orbit tracking satellites. Two of those three low-orbit satellites belonged to the degraded tranche. That fact was on her screen, in a small field of metadata most operators had stopped reading in October because the degraded satellites had been flagged on every event for weeks. She noticed it anyway. She typed a note into the crew log at 02:31:44: *"2 of 3 LEO contributors Tranche-degraded. Recommend caution."* Then she picked up the phone to the NORAD Command Director.

Thirty-seven seconds had passed.

> **OPERATING NOTE — The First Minute Belongs to People You Will Never Meet.** When you are told of a missile warning, it will already be two or three minutes old, and decisions will already have been made by officers you have never heard of: whether the event was real enough to report, whether it was an exercise, whether a sensor was suspect. Your system lives or dies on those people. Go to Colorado in your first year. Stand on the floor of the Missile Warning Center at three in the morning. Ask the crews what they distrust about their own machines. Then make sure the culture rewards the officer who writes "recommend caution" in a log, even when — especially when — she turns out to be wrong. On 6 November, she was right, and her note was the first true thing anyone said.

Minute 1 — 02:32:07

The NORAD Command Director that night was a one-star general, Air Force, a former F-35 pilot with the unhurried voice those people have. He received the report from the Missile Warning Center, looked at the same display, and did what his checklist told him to do: he called the National Military Command Center in the Pentagon and asked for a Missile Event Conference.

These conferences are a ladder, and each rung brings more senior people onto one line and starts more clocks. At the bottom is the Missile Event Conference, the routine conversation between

NORAD and the Pentagon that happens whenever any country launches anything. If the event looks threatening, it becomes a Threat Assessment Conference, adding the Commander of Strategic Command and the Chairman of the Joint Chiefs. If an attack on the United States cannot be ruled out, the President is brought on, and it becomes the conference whose only purpose is a decision on nuclear weapons. Its old name was the Missile Attack Conference; by 2045 it was called the National Decision Conference. I had joined it in exercises four times.

The Deputy Director for Operations in the NMCC that night, the officer who runs the floor and convenes these calls, was an Army brigadier general. He opened the Missile Event Conference at 02:32:01.

Twenty seconds later, PALISADE reported two more launches.

They were not from Hami. They were placed some two hundred kilometers to the southeast, outside the boundaries of the known silo field, on terrain where ARGUS had tracked road-mobile launcher activity during the previous week. PALISADE classified them as probable road-mobile DF-41. Their trajectories were ambiguous. The fusion engine could not yet tell whether they were rising north, toward the pole and the radars that guard it, or south.

South mattered. China had tested a fractional orbital bombardment system in 2021, a weapon that puts a warhead into a partial orbit so that it can approach the United States from the south, over Antarctica and the Southern Ocean, away from the great northern radars that were built during the Cold War to watch the polar route. By 2045 China had fielded a small number of them. The purpose of such a weapon is precisely to defeat the second half of our warning system. If those two objects were heading south, then the radars, our second source, would never see them until far too late.

The Command Director said, on the open line, “NORAD assesses six probable ICBM launches, four from Hami silo field, two from mobile launch area southeast. Two tracks trajectory undetermined, possible southern azimuth.”

The Deputy Director for Operations asked, “Confidence?”

“PALISADE confidence on the four is zero point eight nine and rising. On the two, zero point seven one.”

“Sensor status?”

There was a pause of about three seconds in the recording. Then: “Multiple contributors, high and low orbit. Some low-orbit contributors are from the degraded tranche.”

“Understood. I’m upgrading to Threat Assessment Conference.”

The time was 02:32:58. In the residence, Claire turned over in her sleep. Downstairs, a night usher was reading the menu for a lunch that would never happen.

I have played the recording of that minute many times. What strikes me is that nobody on the line raised his voice. The system was working as it was designed to work: it was climbing the ladder, rung by rung, with every officer doing precisely what he had been trained to do. That is both what I admire about it and what frightens me. A ladder carries you up whether the house is on fire or not.

Minute 2 — 02:33:07

At the Threat Assessment Conference two people joined who would matter for the next nine minutes.

The first was Admiral Victor Hale, Commander of Strategic Command. He was not asleep. He had been sleeping in his office at Offutt Air Force Base in Nebraska since 1 November, on a cot beside his desk, and he picked up the handset on the second ring. Victor was a submariner, and submariners have a particular temperament. They do not panic, because there is nowhere to run. They do not trust anything they cannot verify, because on a submarine the thing you did not verify is the thing that kills you.

The second was General Patricia Dunmore, Chairman of the Joint Chiefs, who was at her quarters at Joint Base Myer–Henderson Hall across the river in Virginia. Her line connected at 02:33:20. She was awake within a sentence. She told me later that her husband had heard her say, "Say again the number," and had got up and started making coffee without being asked, because in thirty years of marriage he had learned which phone calls do not end.

The NORAD Command Director repeated his assessment. PALISADE's confidence on the four Hami tracks was now 0.91. The two southeastern tracks remained undetermined.

Hale asked the first question that mattered. "What do the radars have?"

"Nothing yet, Admiral. On a polar trajectory from Hami, first radar acquisition expected between launch plus seven and launch plus ten minutes, depending on loft. We are at launch plus about two and a half."

A successor must understand this, because it is the hinge of the whole night. Infrared satellites see a missile while its engines burn: in the boost phase, the first three to five minutes of flight. That is their great advantage, because they see a launch almost at once. Their weakness is that they see heat, not objects, and many things are hot. Radars are the opposite. The great early-warning radars that ring the northern approaches, at Clear in Alaska, at Shemya at the end of the Aleutians, at Pituffik in Greenland, can measure an object's position and velocity directly. They can count warheads. They are very hard to fool with a fire on the ground. But radar beams travel in straight lines and the earth is round, so a radar cannot see a missile until it rises above the radar's horizon, and a missile launched from the middle of Asia does not rise above the horizon of an Alaskan radar for several minutes. The two kinds of sensor are not redundant. They are complementary. The infrared tells you early and uncertainly; the radar tells you late and surely. Between the two there is a gap, and in that gap sits the President.

Dunmore asked the second question. "Is anyone else seeing anything? Any launch notifications, any space launch announced?"

"Negative. No NOTAM for a space launch, no notification under the ballistic missile launch notification agreement, nothing on the Defense Telephone Link."

Hale said: "I recommend we bring the President onto the line."

The Deputy Director for Operations agreed. At 02:33:41 he directed the White House Situation Room to wake the President and to begin the procedures for moving the President to a protected location, and he directed the Secret Service, through the Situation Room, to begin moving the Vice President to a separate one. The National Airborne Operations Center, the E-4B aircraft we call the Nightwatch, was ordered to launch from Offutt. That is routine in this situation; its takeoff would be visible to a satellite, but it takes off often enough in peacetime that it carries little meaning on

its own. The airborne relay aircraft that talk to the submarines, the Navy's E-6B Mercury fleet, already had one aircraft airborne over the Pacific, as they always do.

It was 02:34:00 in Washington, 15:34 in Beijing, where, as I would learn the next evening, the duty officers in the Western Hills were watching something on their own screens.

Minute 3 — 02:34:07

The phone on my side of the bed is not loud. It is a soft double tone, unlike any other phone in the house. I had heard it twice before in my presidency, both times for things that turned out to be less than they seemed.

I picked it up on the second tone. I remember the room was cold. I remember that the clock read 2:34 in green numerals and that I thought, absurdly, *That's a prime number*, which is the kind of thing an engineer's brain does when it is buying time.

The voice was the Senior Duty Officer in the Situation Room, a Navy commander named Ortega. I had met her a dozen times. She said, in a voice I will never forget because it was so even: "Mr. President, this is the Situation Room. NORAD is reporting a possible missile attack on the United States. Six possible ICBM launches from China. The Secret Service is coming to take you to the Presidential Emergency Operations Center. Lieutenant Colonel Brooks is on his way. Dr. Oyelaran is being connected."

I said, "Understood." I said nothing else, because there was nothing else. I swung my legs out of bed. Claire was already sitting up. She did not ask what it was. She had heard the double tone before. She looked at my face and then she reached out and took my hand, and held it for perhaps two seconds, and let it go. Neither of us said anything. If you are married to a physician who has sat with dying children, you learn that some moments need no words and that words in them are an insult.

The door opened. Two agents of my protective detail came in. Neither was running, but neither was walking either. Behind them, already in the hall, was Lieutenant Colonel Jamal Brooks of the United States Marine Corps, in uniform trousers and a T-shirt, carrying the Presidential Emergency Satchel, the thing the public calls the football: a leather-covered aluminum case, unremarkable to look at, which contains the means for the President to choose among the prepared options for nuclear employment and to authenticate his identity to the military so that the order is obeyed. I wrote about it in Chapter 1. I will not describe its contents here or anywhere. A successor will be shown them on the first day. Brooks had been sleeping on the third floor of the residence, as the duty aide does, and he had been dressed and in the corridor in under ninety seconds. He told me later that he had put his boots on without socks and that it bothered him for the entire night.

We went down the stairs; the detail prefers stairs. I was in a T-shirt, sweatpants, and the slippers Eli had given me for my birthday, with the University of Texas seal on them. The card the public calls the biscuit, which carries the codes that authenticate the President, was in the pocket of the trousers I had worn that day; the agent behind me had taken them from the chair without being asked, because that too is in the procedures. On the stairs I put my hand in the pocket and felt the card's edge.

On the way down, the handset an agent passed me connected me to the conference. I heard the NMCC officer say: "The President is on the line." And I heard Victor Hale's voice, very calm, say: "Good morning, Mr. President. Admiral Hale. I will give you the picture."

He gave it in perhaps forty seconds. Six probable ICBM launches, four from silos in the Hami field, two from a mobile area to the southeast; trajectories on the four consistent with targets in the continental United States; the two undetermined, possibly southern; launch time approximately 02:30:50 Eastern; if real, earliest arrival over the United States in roughly twenty-six to twenty-eight minutes from now. PALISADE confidence 0.91 and rising. No radar confirmation, none yet expected. Two of the three low-orbit contributors on the Hami tracks flagged as degraded.

I heard myself ask: "How many?"

"Six, sir."

Six. I remember the number entering me the way cold water does. And I remember, already on the stairs, the first coherent thought I had that night. It was not about China. It was about a man named Stanislav Petrov.

What I thought about on the stairs

On 26 September 1983, a Soviet lieutenant colonel named Stanislav Petrov was the duty officer at a command center south of Moscow called Serpukhov-15, which processed data from the Soviet Union's new early-warning satellites, the system known as Oko. Shortly after midnight, Moscow time, the system reported that the United States had launched one Minuteman missile. Then it reported another, and another, until it had reported five. The computer displayed, in large red letters, the word for "launch." The relations between the superpowers were as bad as they had been in twenty years: the Soviets had shot down a Korean airliner three weeks before, killing everyone on board, and the leadership in Moscow believed, genuinely believed, that the Reagan administration might be preparing a surprise attack.

Petrov's duty was to report the warning up the chain. He did not. He reported a malfunction. His reasons, as he later explained them, were three. The system was new and he did not trust it. The ground-based radars had not confirmed anything. And — the reason that has always moved me most — a first strike by the United States would not consist of five missiles. It would consist of hundreds. Five made no sense. He was right. The satellites had mistaken sunlight reflecting off high-altitude clouds over North Dakota for the plumes of rockets.

I had told that story to every class of new staff in the White House Situation Room since 2041. I had told it to myself so often that it had become furniture. On the stairs, with Brooks's breath beside me and Victor Hale's voice in my ear, I discovered that it was not furniture at all. It was the only map I had.

Six missiles. Tom, what is six missiles? I asked myself.

The honest answer was: I did not know. Six is not a first strike. A first strike on the United States designed to disarm us would require many hundreds of warheads against our silos, our bomber bases, our submarine ports, our command centers. Six would do none of that. Six would leave us able to destroy China a hundred times over.

But I knew, too, that six could be something else. General Luo had spent October writing, in his own name and through the editorial pages of the state press, about the "window" that was closing as Golden Dome matured. There is a body of doctrine, in Russia and to a lesser degree in China, about the limited use of nuclear weapons to compel an adversary to stop — "escalate to de-escalate" is the Western shorthand for it. Six warheads on six empty places — or on six of our bases, or our missile-defense sites in Alaska and California — might be intended not to win a war but to end one,

on China's terms, by showing that China was prepared to go first. Six might also be the first wave of something larger, sent ahead to blind our radars or kill our missile defenses. I had sat in exercises where the red team did exactly that.

So the Petrov reasoning — six makes no sense — was not available to me in its pure form. Six made a terrible kind of sense. That was the first lesson of the night. History gives you analogies, not answers.

Minute 4 — 02:35:07

The Presidential Emergency Operations Center lies beneath the East Wing. I described it in Chapter 1 and I will not repeat the description, except to say that it smells of carpet glue, cold coffee, and the particular dry air of machines that have been running since the 1950s and have been replaced, part by part, so many times that nothing original remains but the purpose. When I came through the heavy door at 02:35:10 the room was already half lit. The watch officer who keeps the PEOC ready was at the console. Two screens showed the conference: on one, the NORAD display with its symbols over the map of Asia, six of them now, small and red; on the other, a grid of faces as video connected — Hale at Offutt in a blue sweater, Dunmore in her kitchen with a coat over her shoulders, the NMCC floor behind the Deputy Director for Operations.

Samuel Oyelaran came in eleven seconds after me. He lives fifteen minutes away and could not have been driven in that time; I learned later that he had been asleep in his office in the West Wing, as he had every night since 1 November, and had run. He was wearing a suit jacket over pajamas. He did not sit down. He stood behind my left shoulder, where he always stood, and said quietly, "I'm here." That was all. It was a great deal.

Brooks took his place at the end of the table and set the satchel down in front of him. He did not open it. He looked at me and waited. That is his whole job: to be ready and to wait.

The Secretary of Defense, Ray Castillo, was in an armored car on the George Washington Parkway; his voice came on at 02:35:30. The Vice President, Rosalind Achterberg, was being moved by her detail away from the city, to a location I will not name, so that if the White House were destroyed there would be a constitutional successor alive and connected. Her voice came on at 02:35:50, from a moving vehicle: "Mr. President, I'm on." I heard the tires and the wind. I said, "Thank you, Rosalind." I was more grateful than I could say that she was there and that she was elsewhere.

Hale spoke. "Mr. President, I am going to brief you on the situation and on your response options from the Gold Folder. Under the procedures you approved, I am required to do this whether or not we assess the attack to be real."

"Go ahead, Victor."

The Gold Folder is the successor to what earlier generations called the Black Book: a summary of the prepared options for nuclear employment in a form a President can absorb in the minutes available. The plans themselves are vast, written by thousands of people over years. The Gold Folder is what is left when you strip them to a menu — not a menu of cruelties so much as of logic, each option a bundle of targets and timings that serves a purpose. I will describe it only in the most general terms.

Hale described three families. The first was a limited and proportionate response: a small number of weapons against the specific launch sites and their supporting command elements, intended to

show resolve and to stop further launches, while holding everything else back. The second was a larger response against Chinese nuclear forces and the command structure that controls them. The third was the one every President is briefed on and every President prays never to hear described at three in the morning. He named each by its folder designation. I will not.

Then he said the sentence I had been waiting for, and dreading. "Sir, there is one more consideration I am required to put to you. If this attack is real and if it is followed by a larger attack, the window to launch our ICBM force before it could be destroyed in its silos closes at about launch plus eighteen to twenty minutes. You are not required to use that window. The submarine force is survivable regardless. But I am required to tell you that it exists."

"Understood," I said. "Victor, what is your assessment?"

There was a pause of two seconds. "Mr. President, my assessment is that we do not yet know. The infrared is coherent. The radar has not had its chance. The fusion engine is confident. I am not."

Thank God for submariners, I thought.

It was then, at 02:36:00 or a few seconds before, that I said what I had written into this manual's annexes years earlier, what I had rehearsed in four exercises, and what I had privately feared I would not have the nerve to say for real.

"I am invoking the Two-Source Rule. No decision on nuclear employment will be taken on the basis of one phenomenology. I want radar confirmation, and I want a human check from inside China if one is available in time. Director Laszlo, are you on?"

The CIA Director, Gregory Laszlo, was. He had been connected from his home in McLean at 02:35:40. "I'm on, Mr. President."

"Can you reach LANTERN?"

"Not on demand, sir. His device transmits on a schedule unless he triggers it. The next scheduled window is in about one minute. The China Mission Center watch is standing by. Ruth Calder is coming in."

"Tell them to put anything that comes in on this line the instant it comes in."

"Yes, sir."

The Two-Source Rule, and why it was written in advance

I described the Two-Source Rule for Armageddon in Chapter 1. The principle of dual phenomenology — confirmation by two physical methods that see different things and fail in different ways, infrared from space and radar from the ground — is old American doctrine; the false alarms of 1979 and 1980 were caught largely because the radars did not agree with the computers. What I added was the human check, and the commitment, written, signed, and briefed to every member of the conference, that the President would hold to it. A rule like this must be written in advance because in the moment it does not feel like a rule. It feels like cowardice. The system has been built for seventy years to guarantee that you will never act too late. Nothing in it guarantees that you will not act too early, except a promise you made when you were calm. I think I would have held without it. I am not sure. That is exactly why it exists.

> **OPERATING NOTE — Write Down What You Will Refuse to Do.** In the hour you most need judgment, you will have least of it. Decide in peacetime, in writing,

the conditions under which you will not act, and brief those conditions to the people who will be on the line with you. Then they are not asking you to be brave; they are asking you to keep a promise. Four of the officers on the conference on 6 November told the investigators afterward that the moment I invoked the Rule by name, the tone of the call changed. It stopped being a race to a decision and became a search for a second source. That is what a rule written in advance can do. It turns a panic into a procedure.

Minute 5 — 02:36:07

At 02:36:12 PALISADE issued a recommendation.

I had known, in the abstract, that it could. The Golden Dome battle-management system was designed to coordinate missile defense, and missile defense, unlike nuclear retaliation, is something the law and the doctrine of 2045 allowed to happen at machine speed. If a missile is coming, you do not wait for a President to authorize the interceptor. The system therefore had, built into it, a threat-evaluation function, and that function produced, as one of its outputs, an overall assessment for human commanders. On the screen it appeared as a single line in a box with an amber border:

ASSESSMENT: STRATEGIC ATTACK — HIGH CONFIDENCE (0.93). RECOMMEND: DEFENSIVE ENGAGEMENT AUTHORITY RELEASED; NATIONAL DECISION REQUIRED.

The first half of the recommendation was about interceptors. The second was a notice, not a request: the system was telling us that what came next was beyond it. I want to be very clear about that, because in the years since there has been a public argument about whether an American artificial intelligence "recommended nuclear war" that night. It did not. It had no authority to recommend anything of the kind, and its designers had deliberately kept it from even modeling retaliation options. What it did was worse in a subtler way. It declared, with a number to two decimal places, that we were under attack. And numbers have a gravity that words do not.

I looked at the number for what felt like a long time and was perhaps four seconds. *Ninety-three percent*, I thought. *Ninety-three percent of what?* That is the question an engineer is trained to ask of any probability. A probability is a statement about a model. The number is only as good as the assumptions under it. The question was not whether PALISADE was confident. The question was what PALISADE believed before it looked.

I asked it aloud. "Victor, what was PALISADE's prior? What did it think the probability of a Chinese launch was before it saw these plumes?"

There was a silence on the line. Then a voice I did not know — a Space Force colonel on the NORAD side, the crew's battle-management specialist — said: "Mr. President, the prior is context-weighted. Since November 1 the strategic indicators have raised it substantially. I don't have the number."

"Substantially. So it walked in expecting this."

"Sir, it walked in weighting this as more likely than in peacetime, yes."

"And two of the satellites telling it what it expected are the damaged ones."

"Yes, sir."

There it was. I did not know the cause yet. I would not know it fully for two days. But in that exchange I understood the shape of the danger. We had built an estimator that read the news. The

news was frightening. So when it saw something ambiguous, it resolved the ambiguity toward the frightening. That is what human beings do too. We had built a machine that shared our worst habit and expressed it with a decimal point.

The bombers

At 02:36:40 Hale put to me the first real decision of the night. It was not a decision to use nuclear weapons. It was a decision about whether to protect them.

"Mr. President, I recommend that you authorize the launch of the alert bomber and tanker force for survivability. If this is a real attack, the bomber bases will be among the first targets. Aircraft that are airborne survive. They can be recalled at any time. They do not proceed toward any target without a further positive order from you."

This is a sound recommendation. It is in every exercise. It is what is called a reversible step, and my own Reversibility Matrix says: take reversible decisions fast. The bombers would not be going anywhere near China. They would be flying in circles over the middle of the United States, waiting.

But I had a second model running. *What does Beijing see when it looks at us?* I asked myself. At that moment, I knew, the Chinese early-warning system — the same system that gave General Luo's Rocket Force the ability to launch on warning — was watching our bases from orbit. B-21s and B-52s and their tankers taking off together, in the dead of night, all at once, from bases across the country, would look to a Chinese satellite exactly like the opening move of an American attack. On any other night that might be understood as a precaution. On the sixth night of the crisis, with a general in the Western Hills who had spent October arguing that the United States was preparing to strike first, and with two brigades of Chinese missiles mated and at Readiness Level 2, it might be the thing that pulled the trigger we were trying to keep from being pulled.

I said: "Victor, crews to aircraft. Engines started if you judge it necessary. No takeoff without my order. I want the force able to launch in the time it takes to roll, and I do not want one airplane in the sky that a Chinese satellite can see."

He did not argue. He said, "Crews to aircraft, engines at commander's discretion, hold on the ground. Aye, sir." He told me later that he had been about to recommend the same compromise if I approved the launch, and that he considered my order the right one. He also told me that it cost us about four minutes of survivability margin, which on a real night might have meant several aircraft. I accepted that cost. I would accept it again. But I record it, because a successor should know that restraint is never free.

What I did not know, and could not know, was that even engines starting on the ground at a handful of bases would be seen.

The Vice President

At 02:37:00 Rosalind Achterberg spoke from her car. "Mr. President, for the record, I concur with holding. I'd like to understand what we do if the radar confirms."

It was a fair question and a brave one, because it was the question everyone else was avoiding. I said: "If the radar confirms the four from Hami on a track toward the United States, and if we have no human source contradicting it, we go to the options. Not before."

"Understood."

I have thought often about that answer. It was true. It was also the moment I realized that I was not certain what I would have chosen, among the options, if the radar had confirmed. I have spent years since trying to answer that question. I have not answered it. I suspect no honest President ever does.

Minute 6 — 02:37:07

At 02:37:19 Laszlo's voice came back on the line, faster than before. "Mr. President, LANTERN has transmitted. Ruth is reading it now. Stand by."

At the level that may be written: LANTERN communicated through a device disguised as commercial navigation equipment, which sent short encrypted bursts to a satellite during brief windows. In the summer, at Ruth Calder's request, it had been given a second mode: he could enter a short status line at any time, and the device would send his latest entry automatically at its next scheduled window, whether or not he could touch it again — so that if he were arrested, or if the war came, the last thing he knew would still reach us. The window that afternoon was 15:37 in Lanzhou: 02:37 in Washington. That was not only luck. It was a decision made in July by a case officer who understood that the most valuable intelligence in a crisis is the intelligence that arrives on time.

Ruth Calder's voice came on. She was in her car in the CIA's parking lot at Langley, connected through the agency's operations center. She read it exactly, without inflection, the way a good case officer reads a source's words, so that the listener hears the source and not the officer.

"Message composed fifteen thirty-four local. Text follows. 'No launch. Lids closed. RL2 holding. Duty net confused. Exercise fault?' End of text. Authentication checks are valid."

I asked her to read it again. She did.

Composed at 15:34 Beijing time. That was three minutes after PALISADE said the missiles had left their silos. LANTERN was on the staff of the very base whose brigades the sensors had placed the launches in. If four missiles had left the Hami field at 15:30, the duty net at Base 64 would not be "confused." It would be the loudest place on earth.

But I made myself ask the next question, because the system around me would ask it, and because a President who only asks the questions that lead where he wants to go is a danger to his country. "Ruth. Can it be coerced? The Ministry of State Security has been hunting for him for a month. If they have him, could they have written that?"

She did not hesitate, which is how I knew she had already asked herself. "It's possible in principle, sir. There are measures in the device and in his habits that would tell us if it were written under duress, and none of them are present. I assess with high confidence that this is from him and that he is free. I cannot assess what he can see from where he's standing. 'Lids closed' may mean the lids he can see, or the lids he's been told about."

"And 'exercise fault'?"

"I read that as his guess about what's going on, not something he knows. The question mark is his."

A question mark, I thought. *The machine gives me ninety-three percent and the man gives me a question mark.* And it was the question mark I believed, because it was the more honest of the two.

Hale came back on. "Mr. President, that is significant but it is one human source, and he may not have full visibility. I don't recommend we treat it as confirmation of no attack. I recommend we

treat it as strong reason to wait for the radars, which we are doing anyway."

"Agreed."

The time was 02:37:58. The football remained closed.

Minute 7 — 02:38:07

The radar window opened. On the NORAD display, a shaded wedge appeared over the Arctic Ocean north of Alaska and the Chukchi Sea: the volume of sky through which, if the four Hami missiles were real and bound for the United States on the shortest path, they would have to climb into view of the radars at Clear, Shemya, and Pituffik within the next three minutes. The Long Range Discrimination Radar at Clear had been fielded in the early 2020s precisely to look into that wedge with enough precision to tell warheads from decoys. The crew reported its status: operational, full power, searching.

The wedge stayed empty.

"No objects in the expected volume at launch plus seven," the Command Director said. "That is consistent with a low-probability early window. The main window is launch plus eight to ten."

Dunmore said: "Understood. Two southern tracks?"

"PALISADE now has a southerly component on one of the two. Still unresolved. If either is a fractional orbital system on a southern route, it will not appear in the northern volume at all."

That was the cold arithmetic underneath everything. Even if the northern radars stayed clear, they could not rule out the two southern tracks. Southern coverage existed — sea-based radars, the space tracking layer itself, other sensors I will not name — but it was thinner, and it was slower.

I had a minute in which there was nothing to decide, and so my mind went where minds go. It went to 1979.

What happened in 1979 and 1980

On the morning of 9 November 1979, the NORAD computers showed a massive Soviet attack on the United States: hundreds of missiles, from land and sea. Interceptor aircraft were launched. Missile crews were alerted. The National Security Advisor of the day, Zbigniew Brzezinski, was telephoned at home at about three in the morning by his military assistant and told that two hundred and twenty Soviet missiles had been launched against the United States. He asked for confirmation. A minute later the assistant called back: the number was now two thousand two hundred. Brzezinski, the story goes, did not wake his wife, because he assumed that in half an hour they would both be dead. He was preparing to call President Carter when the third call came. The radars, and the satellites, had seen nothing. Someone had loaded a training tape — a simulation of an attack — into a live NORAD computer. The alert lasted six minutes.

Seven months later, on 3 June 1980, it happened again, and then twice more in the following days. This time the cause was a single computer chip in a communications device, a part that cost forty-six cents. It had failed in such a way that it began writing the number two into fields that should have contained zero, so that the screens showed two missiles, then two hundred, then two thousand. Again the radars saw nothing. Again the system was saved by the fact that it had two independent kinds of eyes and that the people on the floor insisted on asking both.

I thought about Brzezinski not waking his wife. I thought about Claire's hand on mine two minutes and a lifetime ago. I thought: *He chose to let her sleep through the end. I had her hold my hand.* I do not know which was the braver choice. I know which I am glad I made.

And I thought: *In 1979 the radars saved them. Tonight the radars cannot see the two southern tracks. If they are real, the radars will never save me.* I understood then that the Two-Source Rule I had written was necessary but not, by itself, sufficient. For the southern tracks, I would need a third source. I had one, I hoped, coming in about forty seconds.

Minute 8 — 02:39:07

At 02:39:22 the director of the National Reconnaissance Office's operations center, who had been on the line without speaking since 02:36, said: "Mr. President, we have an ARGUS synthetic aperture radar pass over the Hami field. Imagery is being processed. Under a minute."

ARGUS was the system I had once, in an unguarded moment before the Senate Armed Services Committee, called "absolute." Its synthetic aperture radar satellites — radar imaging from orbit, which sees through cloud and darkness — revisited any point on earth roughly every ninety seconds, and its coverage over western China had been rebuilt since September. SAR shows not heat but shape: whether a silo door is open, whether ground is scorched. It was a third phenomenology, looking down at where the missiles came from rather than up at where they should be going.

Those forty seconds were, I think, the longest of my life. I spent them looking at the satchel. Brooks spent them looking at me. Neither of us said a word. Oyelaran's hand was on the back of my chair. Somewhere in the room the air handler cycled and I felt it on my neck. I remember thinking about Maya on the *Daniel Inouye*, somewhere east of Okinawa, asleep in a rack or standing a watch, and that she did not know any of this was happening, and that I hoped she would never have to know. I did not pray in words. I am a lapsed Methodist and I do not pray easily. But if wanting something with every part of you is a kind of prayer, I prayed.

At 02:40:02 the NRO officer said: "Imagery processed. All six silos at the coordinates PALISADE associated with the four Hami launches — and the two adjacent — show doors closed. No surface disturbance, no blast or thermal scarring consistent with launch. Reported launch points are at closed silos. Repeat, the silos are closed."

She added, a moment later: "Mr. President, there is a large thermal and smoke-related feature at an industrial site approximately thirty-eight kilometers northeast of the field. It does not correspond to a silo. And there is activity at a known engine test facility to the southeast, near the reported mobile launch area. We are working that now."

"Thank you," I said. My voice, on the recording, is steady. I did not feel steady. I felt as if I were standing on the rim of a very tall building and someone had just put a hand on my arm.

Minute 9 — 02:40:07

Twenty-nine seconds after the imagery report, the radars spoke.

At 02:40:31 the NORAD Command Director said: "Clear, Shemya, and Pituffik report no objects in the expected volume for the four Hami tracks. We are now at launch plus approximately nine minutes forty. On any credible polar trajectory from Hami to the continental United States, the objects would be in radar view. They are not there."

A beat. Then: “Tracking layer reports no midcourse tracks associated with any of the six events. PALISADE is re-evaluating.”

On the screen the amber box flickered and redrew itself.

ASSESSMENT: STRATEGIC ATTACK — LOW CONFIDENCE (0.41). SENSOR CONFLICT. HUMAN REVIEW.

The number dropped by more than half in under a minute, and the machine did exactly what it was designed to do: when its sensors disagreed, it said so. It is a good machine. That is the hardest thing to say about it, and the most important. A good machine fed by damaged eyes, in a world made to look like a war, with a prior that leaned toward war, will give you a bad number with great confidence. So will a good analyst. So will a good President.

Hale said: “Mr. President, on the four Hami tracks, I now assess no attack. On the two southeastern tracks, I assess probable misidentification of a ground thermal event, pending the southern sensors. Recommend we hold all postures and complete the conference.”

“Agreed.”

Minute 10 — 02:41:07

I did not want to end it yet. I wanted to end it more than I had wanted anything in my life, and that was precisely why I did not.

“I’m going to ask each of you in turn,” I said. “Is there anyone on this line who believes the United States is under attack? Victor?”

“No, sir.”

“General Dunmore?”

“No, Mr. President.”

“Ray?” Castillo had reached the Pentagon by then and was on the NMCC floor.

“No.”

“NORAD?”

The Command Director: “Sir, the southern sensors report no objects associated with either southeastern event. The thermal sources correlate with the test facility. NORAD assesses no attack.”

“Gregory? Ruth?”

“No attack, sir. And LANTERN is free and talking.”

“Rosalind?”

From her car: “No attack, Mr. President.”

Samuel Oyelaran had not been asked. He leaned down to my ear and said, so quietly that the recording does not catch it: “And if you’re wrong, there’s nothing anyone could do about it now anyway, Tom. But you’re not wrong.”

I turned to the watch officer. “Time?”

"Zero two forty-one forty, sir."

There was one more thing to do before the end. I said: "Victor, I want no posture change for twelve hours that a satellite can see. The bomber crews stay at their aircraft, engines off, for one hour, then return to alert facilities in the normal way. The Nightwatch completes a normal training profile and lands. I do not want Beijing watching us stand down in a hurry tonight, because a hurried stand-down will look like the end of a preparation. Everything should look like a dull Monday."

"Understood, sir. A dull Monday."

Minute 11 — 02:42:07

At 02:42:12, eleven minutes and five seconds after PALISADE first reported a launch, I said the sentence that ended it. I said it formally, because the procedures require the President's decision to be stated in words that cannot be misunderstood and recorded in the log.

"This is the President. I have reviewed the warning. The United States is not under attack. There will be no nuclear response. Terminate the conference. All forces maintain current posture. I want a full investigation of the cause, beginning now, and a first report to me by noon."

The Deputy Director for Operations repeated it back, word for word, and logged the time. One by one the faces on the screens went dark. Victor Hale's was the last. Before it went, he looked into his camera — not at me, I think, but into it, the way you look at a periscope eyepiece — and said: "Mr. President. That was correct."

Then the screen was blank and the room was quiet and I was a man in sweatpants and university slippers sitting in a basement under the White House at a quarter to three in the morning.

Brooks closed nothing, because he had opened nothing. He put his hand flat on top of the satchel for a moment, the way you might put a hand on the flank of a horse that has been spooked. Then he said: "Sir, permission to go find some socks?"

I laughed. It was the first sound I had made in eleven minutes that was not a question or an order, and it came out of me with a force that surprised everyone in the room, including me. Oyelaran laughed too. The watch officer looked away, smiling. Brooks did not smile. He was serious. His feet were cold.

"Granted, Colonel," I said. "And bring me one of those terrible muffins from the mess."

> **OPERATING NOTE — A Confidence Score Is a Statement About a Model, Not About the World.** When a machine tells you "ninety-three percent," your first question must be: ninety-three percent given what? Ask what the system believed before it looked — its prior — and what it was told about the world. A fusion engine that weighs political context will grow more alarmed as a crisis deepens, which means it will be most likely to see an attack exactly when an attack is most catastrophic to misjudge. That is not a reason to remove context from your machines. It is a reason never to let a number stand in for a second phenomenology. The same rule applies to people. Your most confident advisor in a crisis is often the one whose prior was set before the crisis began.

Decision Record 6-A: The Eleven Minutes, 6 November 2045, 02:31–02:42 EST

Decision	Options considered	Reversible?	Risk	Chosen	Outcome
How to treat the warning (minute 4)	(a) Proceed to response options on infrared and PALISADE assessment; (b) invoke the Two-Source Rule and wait for radar and a human check	(a) Irreversible once executed; (b) reversible until ICBM window closed at ~minute 18–20	(a) Nuclear war on a false alarm; (b) loss of part of the ICBM force if attack real	(b) Two-Source Rule	Radar, synthetic aperture radar imagery, and LANTERN all contradicted the warning by minute 9
Alert bomber and tanker force (minute 5)	(a) Launch for survivability; (b) crews to aircraft, hold on ground; (c) no change	(a) Reversible but highly visible to Chinese early warning; (b) reversible, less visible; (c) reversible	(a) Read in Beijing as the start of a US strike; (b) ~4 minutes' loss of survivability margin; (c) loss of aircraft if attack real	(b) Crews to aircraft, no takeoff	Engine starts at several bases were nonetheless detected by Chinese satellites (see below)
Trusting LANTERN's message (minute 6)	(a) Treat as confirmation of no attack; (b) treat as strong indicator, still wait for radar; (c) disregard as possibly coerced	All reversible	(a) Over-reliance on one human source under mole hunt; (c) discarding the only inside view	(b) Strong indicator	Corroborated by imagery at minute 8
Declaration (minute 11)	(a) Declare no attack at minute 9 on radar alone; (b) poll principals and southern sensors, declare at minute 11	Reversible either way	(a) Two southern tracks unresolved; (b) small delay	(b)	"No attack" stated 02:42:12; conference terminated

Decision	Options considered	Reversible?	Risk	Chosen	Outcome
Post-alarm posture	(a) Rapid visible stand-down; (b) no visible change for 12 hours	Reversible	(a) Beijing reads stand-down as end of preparation; (b) minor cost	(b) "A dull Monday"	No visible US change observed by Chinese side after 02:45 EST (per WILLOW)

After the Minutes: 02:43–04:10

Claire came down at 03:05, in my old Colorado School of Mines sweatshirt, carrying a pair of my socks, because she had seen Brooks in the corridor and drawn her own conclusions. She sat beside me. After a while she said, "Was it close?" and I said, "I don't know. I think it was eleven minutes long and I don't know how close it was." That is still the most accurate answer I can give.

At 03:20 Helena Marsh arrived from Foggy Bottom; the Secretary of State is not on the conference, which by design is a military chain with the President at its head. She looked at me and said, "Mr. President, someone in Beijing was watching us for those eleven minutes. We need to know what they saw."

She was right, and she was the first to say it. I asked Laszlo to task WILLOW, through whatever means were safe, for anything from the Joint Operations Command Center between 15:20 and 16:00 Beijing time.

I went back upstairs at 04:00. I did not sleep. I wrote in my diary. The entry for that night is the epigraph of this chapter, and then it continues: *"Six missiles that were not there. One man in Lanzhou with a question mark. One officer in Colorado who wrote 'recommend caution.' One submariner who said 'I am not.' A radar in Alaska that looked at an empty sky and told the truth. The system held because at four points in it there was a human being who refused to be rushed. I must find out whether the other side's system has as many."*

The Other Eleven Minutes: Beijing, 15:31–15:50

What follows I learned in three pieces: a WILLOW report that reached Langley at 19:40 Eastern on 6 November; a longer report from WILLOW on 8 November; and, many years later, testimony at General Luo's trial, portions of which were made available to the editors of this book. I have reconstructed it as best I can. Where the sources disagree I have followed WILLOW, who was in the room.

At 15:33 Beijing time — 02:33 in Washington, during our minute two — the Chinese early-warning center reported to the Joint Operations Command Center in the Western Hills an "abnormal concentration of American strategic indicators." It listed them. A ballistic missile submarine of the Ohio class, which Chinese intelligence had been tracking imperfectly in the Pacific, had made an unscheduled depth change and course change that their analysts associated with preparation to receive orders — in fact, as our Navy later confirmed, the submarine was evading a fishing fleet. The American airborne command aircraft, the Nightwatch, was being readied at Offutt. And at

15:37 Chinese satellites detected thermal signatures consistent with engine starts at three American bomber bases.

The last of those was my doing. I had kept the bombers on the ground precisely so that Beijing would not see them. Beijing saw their engines anyway.

General Luo Tiancheng was in the Joint Operations Command Center. Admiral Gao was with him. Premier Qiao was in a State Council meeting across the city. General Secretary Wen was in the PLA General Hospital, after his second cardiac episode in five days. At 15:39, according to WILLOW, Luo instructed the duty officer to prepare an order raising two brigades of Base 64 — the two whose warheads were mated — from Readiness Level 2 to Readiness Level 1, the level at which missiles are prepared for immediate launch on order. The Chinese term the order used, WILLOW reported, was "counter-strike readiness." He said, according to WILLOW, these words: "If they are coming, the window is minutes. We will not be the people who waited."

This is the mirror image of my minute five. Luo was looking at American indicators that were real but innocent, through a prior set by six days of crisis and ten years of conviction that the United States meant to disarm China. His estimator was primed exactly as PALISADE's was. He saw an attack because he expected one.

He did not, however, have the authority to send the order alone. Under the Chinese system as it stood in 2045, a change to Readiness Level 1 required the authorization of the Chairman of the Central Military Commission — Wen — or, in his incapacity, a procedure involving the Vice Chairman. The duty officer, a colonel whose name I do not know and whose courage I will always honor, asked under which authority the order was to be transmitted. Luo said, "Under mine, pending the Chairman." The colonel said that the procedure required the Vice Chairman to be informed. He telephoned General Zhang Hesheng's office. That telephone call, WILLOW reported, was made at 15:41.

It was also the source of the confusion LANTERN saw. When the duty net of Base 64 received a preparatory notice at 15:40 that an order changing readiness might follow, and then no order, its staff did what any staff does: they began asking one another what was going on. LANTERN, at his desk, wrote "Duty net confused. Exercise fault?" at 15:34 — before the preparatory notice — because the confusion had started even earlier: the base had been told at 15:31 to stand by for a high-level conference call that never came. I do not know who ordered that first stand-by. The trial record suggests it was linked to the deception operation I describe in the next section. [REDACTED]

General Zhang Hesheng reached the Western Hills by telephone at 15:44 and in person at 16:20. On the telephone, according to WILLOW, who heard the duty officer's side and later read the log, he said: "No order changes the readiness of any unit without the Chairman's word. The Chairman has not given it. I am countermanding. Hold at present level." Luo argued. Zhang said — and this line appears in the trial record, so I may quote it — "General, you are describing a war that you can see and I cannot. Until I can see it, there is no war."

At 15:50 the Chinese early-warning center reported that the American bomber engines had shut down and no aircraft had taken off. The order was never sent. Readiness Level 2 held. The two brigades never went to Level 1.

When I read the first WILLOW report at 20:05 on 6 November, sitting at the desk in the Treaty Room, I put it down and went to the window and stood there for a long time. On both sides of the Pacific, in the same nineteen minutes, a machine or a man had said "attack," and on both sides

someone else had said "until I can see it, there is no war." Zhang Hesheng and I had never met. We never would. I owe him as much as I owe anyone alive.

> **OPERATING NOTE — Your Restraint Must Be Visible to Be Useful, and Even Then It May Not Be Enough.** I kept the bombers on the ground so that Beijing would not see them rise. Beijing saw their engines start. In a crisis, every precaution you take is also a signal, and the adversary's sensors read your precautions through the same frightened prior that yours read theirs. Before you order any protective step, ask: what will this look like on their screen, to their most frightened officer, at the worst possible moment? Then choose the least legible precaution that still protects you. And recognize that the decisive restraint may not be yours at all — it may belong to someone in the other capital whom you will never meet. Do everything you can, in peacetime, to make sure those people exist and have the authority to say no.

The Cause

The investigation I ordered at 02:42 reported to me at noon on 6 November, again at 22:00, and finally in a seventy-page memorandum on 14 November, which became the basis of the classified report to Congress. I summarize its findings here at the level that may be written.

There were two causes, and either alone would probably not have produced the alarm.

The first was deliberate. In the weeks after the counter-space attacks of September, a unit under the Rocket Force's command — the investigation could not establish on whose specific authority — had prepared what the Chinese military literature calls a "strategic deception" operation against the American warning system. The intent, as best we could reconstruct it, was not to trigger a false alarm for its own sake. It was to probe: to learn how quickly, and in what form, the United States responded to an ambiguous warning, and so to measure how close to the edge the American system ran. On the afternoon of 6 November three things happened near Hami within a few minutes of one another. A large fire was started at an industrial facility northeast of the silo field — whether deliberately or not, the investigation could not determine with certainty, though its timing argues for deliberation. A scheduled static test of a large rocket engine took place at a test facility to the southeast. And a ground-based laser illuminated two of our low-orbit tracking satellites passing over the region — two of the satellites already damaged in September — at an intensity that did not blind them but distorted their readings. The effect, at the level of the sensor, was that bright, hot, plume-like signatures appeared at locations close enough to silos for the fusion software to associate them with launch points.

The second cause was ours. When PALISADE was retrained in October to compensate for the degraded tranche, the engineers had, sensibly, reduced the weight it gave to any single satellite's report and increased the weight it gave to agreement between satellites and to context. In normal conditions this made the system more robust. Under these conditions it made it fragile in a new way. The two damaged satellites agreed with each other, because they were being deceived in the same way. The high-orbit satellites, which saw the industrial fire and the engine test from much farther away, reported real heat at roughly the right places. And the context — mated warheads, Readiness Level 2, a general writing about windows — raised the prior. The fusion software combined three weak, partially true signals into one strong, wholly false one. The investigators had a phrase for it that I have since used in many briefings: *coherent error*. It is the most dangerous kind, because it looks like truth.

Neither cause alone would have been enough. Together they built a war out of a fire, a test, and a beam of light.

I asked the investigators one question at the end of their briefing on 14 November. “If the radars at Clear and Pituffik had been down for maintenance that night, what would have happened?” The lead investigator, a civilian scientist who had spent forty years on early-warning systems, said: “Then, Mr. President, you would have had LANTERN and one SAR pass. I hope you would have trusted them. I don’t know whether the system would have let you.”

I directed three changes before the end of November, which are described in Annex A: no fusion system’s confidence score shall be displayed to a President without the system’s prior displayed beside it; any warning in which more than half of the contributing satellites carry a degradation flag shall be marked as such in the top line, not the metadata; and the Two-Source Rule became a formal requirement of the procedures, not merely a President’s personal commitment. A fourth change — that there be a way to tell Beijing, in real time, when our machines see something we do not believe — took four more months and a summit in Singapore.

Tuesday, 7 November 2045, 10:38 Beijing — The Jade Line

In Washington it was still Monday, 21:38. In Beijing it was Tuesday morning, and Tuesday the seventh is the date the Chinese record gives to the call, so it is the date I use.

The decision to make the call was harder than the eleven minutes, because an afternoon is long enough for every good argument to be made.

We met in the Oval Office at 16:00. Castillo was against telling Wen anything: “You’d be telling them their deception worked, and roughly how.” Abigail Fenn, the Director of National Intelligence, feared that anything suggesting we knew what had happened in the Western Hills would tell the Ministry of State Security it had a source in the room. Dunmore said only, “Whatever you tell him, tell him nothing about the radars.” Marsh was for it: “His own Rocket Force commander tried to move to launch readiness without him. He doesn’t know whom to trust. If you tell him a truth that makes us look weak, he’ll have one person he knows isn’t lying to him.”

Oyelaran made the argument that decided me. He drew it on my notepad as a control loop: sensor, estimator, controller, actuator. “Tom, both our estimators failed last night, in the same way, at the same time. Each side’s sensors read the other side’s precautions as preparations. You can’t fix his estimator. But you can feed it a true measurement. Right now his inputs are Luo telling him we’re coming. Give him an input that contradicts that and comes from you. That’s the only control action available that reaches inside his loop.”

Decision Record 6-B: Disclosure to President Wen, 6–7 November 2045

Decision	Options considered	Reversible?	Risk	Chosen	Outcome
Whether and what to tell Wen about the false alarm	(a) Say nothing; (b) send a technical notice through the Defense Telephone Link; (c) tell Wen personally on the Jade Line that our system reported an attack and that I did not believe it, without sources or sensor detail	(a) Reversible; (b) partly; (c) irreversible — once said, cannot be unsaid	(a) Wen hears Luo's version only; (b) notice lost in military channels Luo controls; (c) reveals a vulnerability; may expose WILLOW if worded carelessly	(c), with wording cleared by the Director of National Intelligence	Wen convened the Politburo Standing Committee within 24 hours; Luo relieved 8 November

I made the call from the Oval Office at 21:38. Marsh sat beside me. Our interpreter, a State Department officer who had interpreted every Jade Line call since 2042, sat at the end of the desk. The video connected at 21:41. President Wen was propped up in a hospital bed, in a dark cardigan over hospital clothing, with a monitor lead visible at his collar. His own interpreter sat beside him. Behind him, at the edge of the frame, stood a man in uniform: General Zhang Hesheng. I had seen his photograph a hundred times. I had never seen him move. He did not speak during the call. I believe his presence was itself the message.

What follows is reconstructed from our transcript, compared afterward with the Chinese transcript provided under the Singapore Understanding.

WEN: President Kane. You asked for this call urgently. I am told it is late at night in Washington.

KANE: It is, Mr. President. Thank you for taking it, and I'm sorry you are unwell. I'll be brief. Mr. President, eleven minutes — no. Nineteen hours ago, my machines told me that you had attacked us.

(Pause. The interpreter rendered it. Wen did not move.)

KANE: At 15:31 yesterday, your time, our early-warning system reported six intercontinental missiles launched from western China toward the United States. Our battle-management computer assessed it as an attack with high confidence. I was woken and taken to the shelter beneath this house. My officers briefed me on my options. I did not believe the machines. I waited. Within eleven minutes, other means showed me that no missile had been launched. I declared that there was no attack and ordered no response.

WEN: Why do you tell me this?

KANE: For three reasons. First, because you may hear from your own system that American forces did something unusual during those minutes, and I want you to hear from me what it was: our aircraft crews went to their planes; I ordered them not to take off; they did not. Nothing more. Second, because I believe the false warning was caused in part by activity near Hami that afternoon, and in part by a fault in our own systems. I am not accusing you. I don't believe it came from you. But someone on your side was testing how close we would come, and I want you to know that the answer is: closer than either of us should ever allow.

WEN: And the third reason.

KANE: Because the next time the machine is wrong, it may be yours. When that happens, I need you to remember that I told you about mine.

(Wen was silent for nineteen seconds. I counted them.)

WEN: You could have hidden this. It makes you look weak. Your Congress would be angry if they knew you had told me.

KANE: Some of them will be. But a strong country can afford to tell the truth about its mistakes. A country that can't afford that isn't strong; it's only armed.

WEN: *(slowly)* In my country we have a saying: the one who points out your faults is your teacher. I did not expect an American President to be my teacher. *(He coughed, and someone off-camera handed him water.)* I was not told of any American activity yesterday. I was told nothing yesterday. That is a fact I find more interesting than your missiles.

KANE: Mr. President, I have not asked, and I won't ask, what happened on your side. That is yours. I'll only say that my letter of 5 November stands. Every word. No strikes on your nuclear forces or your early warning. A pause in new interceptor launches. Seventy-two hours with no convoys, beginning when you tell me. In return, I ask what I asked: Readiness Level 3, and a channel that is always open.

WEN: You have mentioned your letter. I have read it. I read it this morning. *(Pause.)* It had been in Beijing since the evening of the fifth. It reached my bed this morning. That, too, is interesting.

KANE: I understand.

WEN: I do not think you do, President Kane, but perhaps you understand enough. *(He looked to his left, toward Zhang, and back.)* Tomorrow I will convene the Standing Committee. I will be there in person, whatever my doctors say. I cannot promise you an answer tomorrow. I can promise you that the answer will be mine.

KANE: That's all I can ask.

WEN: One more thing. Your officers — the ones who told you not to believe your machines. Thank them. We have such officers too. Not everyone knows their names.

KANE: I will, Mr. President. And if you ever learn the name of yours, thank them from me.

WEN: *(a very faint smile, the only one I ever saw from him)* Perhaps someday we will compare lists.

The call ended at 22:04. Twenty-three minutes. Marsh sat back in her chair and closed her eyes. The interpreter, who had kept her voice perfectly level for twenty-three minutes, put her pen down

and I saw that her hand was shaking. Oyelaran, who had watched from the doorway, said nothing at all, which from him is the highest praise.

Why did that call work? I think because it gave Wen something he had lost: an accurate measurement. For a week his picture of the world had been assembled by people with reasons to frighten him and reasons to keep things from him. My letter had sat on someone's desk for a day and a half. The first true thing he learned about 6 November came from his adversary, and that showed him, as no evidence could, that the man running his Joint Operations Command was not telling him what the Americans were doing.

> **OPERATING NOTE — Tell Your Adversary When Your Machines Lie.** In a nuclear crisis, the most dangerous thing on the other side is not the adversary's intent. It is the adversary's estimate of your intent, assembled from sensors and subordinates you cannot see. You cannot control that estimate, but you can feed it a true input. When your own system has erred in a way that could have been seen as preparation, tell the other leader yourself, plainly and quickly, without sources and without boasting. It will cost you some secrecy. It may buy you their trust, and it will certainly cost the hawks on their side the monopoly on describing you. I disclosed a weakness to Wen Zhaolin and it was the strongest move I made in fourteen months.

Wednesday, 8 November — The Standing Committee

The Politburo Standing Committee of the Chinese Communist Party met in emergency session in Zhongnanhai at 09:00 Beijing time on 8 November — 20:00 on Tuesday in Washington. I learned what happened there from WILLOW, who saw the message traffic that followed, and from ORCHID, whose principal, Premier Qiao, was in the room.

Wen attended in person. ORCHID reported that he was brought in a wheelchair and walked the last ten meters to his seat on his own. He spoke, by ORCHID's account, for about forty minutes. He described the events of 6 November as they had been reported to him by Zhang, and then, without naming me, as they had been reported to him "by another source." He described the delay in delivering a letter from a foreign head of state. He described a preparatory order to change the readiness of strategic forces issued without his authorization while he was in hospital. Premier Qiao spoke in support. Zhang spoke in support. Two members who had been thought close to Luo's view, ORCHID said, did not speak at all.

At 13:10 Beijing time the Central Military Commission issued an order, which WILLOW saw transmitted, relieving General Luo Tiancheng of command of the Rocket Force "for reasons of health" and placing him at the disposal of the Central Military Discipline Inspection Commission. His deputy, a general whom LANTERN described as "a careful man who reads the regulations," was named acting commander. Admiral Gao Ruiming remained in his post as Eastern Theater Commander but was recalled from the Western Hills to his headquarters in Nanjing. Xinhua carried a two-line item that evening noting the "change of personnel." It made no mention of anything else.

I learned of the order at 00:40 on Wednesday morning, Washington time. I wrote one line in the diary: *"He took his country back. I hope it lets him keep it."*

Thursday, 9 November — Readiness Level 3

LANTERN's message came at 06:52 Eastern on Thursday, 9 November — 19:52 in Lanzhou. It was composed at leisure, and it was longer than any message he had ever sent. The operative portion read: *"Order received 17:00 local: Base 64 all brigades return Readiness Level 3. Warhead de-mating begins tomorrow under new acting commander. Duty net calm. Colleagues relieved. I am well."* Ruth Calder told me later that "I am well" was something he had never written before.

At 11:15 ARGUS imagery showed the first transporter-erector-launchers from the dispersal areas southeast of Hami moving back toward their garrisons. By 18:00 the National Geospatial-Intelligence Agency counted thirty-one returned. Silo maintenance crews at the Hami and Ordos fields had stood down. At 15:30 Ambassador Tang Shiyu delivered a two-sentence note to Secretary Marsh at the State Department: President Wen had received President Kane's letter of 5 November, and China would "welcome continued contacts through existing channels, including in Singapore." Marsh read it to me over the phone. "In diplomatic Chinese," she said, "that is a yes."

I addressed the nation that evening at 20:00 from the Oval Office. I had argued with my speechwriters, and with myself, about how much to say. I said nothing about the eleven minutes. That would come, in part, five months later, in a closed session of Congress, and in full only with the declassification of this book. A successor will face the same choice. My judgment was that the American people deserved to know that the danger had receded, but that telling them how close we had come, while the Chinese leadership was still reasserting control and while General Luo's supporters still held commands, would have given every hardliner on both sides a weapon. The excerpt below is from the delivered text.

> "My fellow Americans, for eight months you have lived with a crisis that most of you could see only in its shadows: in prices at the store, in the lights that went out in Texas, in the names of forty-one sailors we buried in September. Tonight I can tell you that in recent days the danger has receded. Over the past week, the leaders of the United States and the People's Republic of China have been in direct and continuous contact. Tonight Chinese forces that had been placed on a higher state of readiness are returning to a lower one. Ours will do the same. We have agreed to a pause.
>
> A pause is not a peace. I will not tell you tonight that our differences with Beijing are resolved. They are not. Taiwan's people still live under pressure, and our sailors still stand watch in those waters, among them my own daughter. But a pause is a door, and I intend to walk through it.
>
> I want to say a word tonight about the men and women you will never see: the officers who sit through the night watching screens in Colorado and Nebraska and in ships at sea; the people who carry burdens in silence so that the rest of us may sleep. In this crisis, at moments you will not read about for many years, some of them were asked to be certain, and they had the courage to say that they were not. Their doubt kept us safe. I have never been prouder to serve with anyone.
>
> President Kennedy said, in the summer after the Cuban crisis, that 'we all breathe the same air. We all cherish our children's future. And we are all mortal.' That was true in 1963. It is true tonight, in Washington and in Beijing. Good night, and God bless the United States of America."

After the cameras were off I sat at the Resolute Desk for some minutes. Claire came in and stood

behind me with her hands on my shoulders, as she had in the PEOC three nights before. Then the phone on the desk rang, an ordinary ring this time, and it was Maya, calling on a satellite line from the *Daniel Inouye*, where it was Monday morning. She had watched the address in the wardroom. "Dad," she said, "you said 'a pause' four times. Is that a real word, or a speechwriter word?" I said it was a real word. She said, "Okay. Then I'm going to go eat breakfast." And she did.

My diary entry for 9 November is the shortest of the crisis. It reads: *"Level 3. Thirty-one launchers home. Maya ate breakfast. We are not out. But we are on the way down."*

> **OPERATING NOTE — The Public Is Owed the Truth, but Not Necessarily Tonight.** There will be things you cannot tell your people while the danger is live, because telling them would arm the very people abroad you are trying to disarm. That is a legitimate reason for silence, and it is the only one. Write down, the same night, what you withheld and why, and set yourself a date by which you will tell the Congress and, eventually, the country. I told the Congress about the eleven minutes in closed session in April 2046. I should have told its leaders within days. Secrecy that outlives its reason stops protecting the country and starts protecting the President.

Chapter 7: The Long Way Down

"Every climber in Colorado knows the summit is only halfway. Most of the people who die on a mountain die on the way down." — Diary of Thomas Ellery Kane, 10 November 2045, 23:40

Why the Descent Gets Its Own Chapter

When I was fourteen my father took me up Mount Lindsey, in the Sangre de Cristo range, because he had decided that a boy who wanted to build rockets should first understand what it costs to go up. We left the truck at four in the morning. We stood on the summit a little before eleven, and I remember the wind, the light on the Blanca massif, and the feeling that we had done the hard part. My father let me enjoy it for exactly ten minutes. Then he tightened the straps of my pack and said, "Now pay attention. This is where people get hurt. They think they're finished. Their legs are tired and their heads are full of the view." On the way down I slipped twice on loose scree and once badly enough that he caught me by the collar. He did not say anything about it. He did not need to.

I thought of my father, who has been dead for many years now, on the night of 9 November 2045, after I had spoken to the country and told it, without saying how close we had come, that there was "a pause." The Rocket Force was going back to Readiness Level 3. General Luo Tiancheng was under investigation. President Wen Zhaolin, from a hospital bed, had taken back the reins. My staff were exhausted and, for the first time in eight months, a little giddy. Megan Pruitt had ordered pizza for the Situation Room watch. Someone had put a handwritten sign on the door of the Presidential Emergency Operations Center, the bunker under the East Wing that we call the PEOC, reading CLOSED FOR THE SEASON.

I let them enjoy it. Then I took the sign down myself, and in the morning I told the Principals that we were on the summit, and that the summit is only halfway.

This chapter covers the five months that followed, to the end of April 2046. No missiles were reported launched in them. No ships were hit. Yet a successor may learn more here than from the eleven minutes, because a crisis at its peak tests nerve and systems built in advance, while a crisis on its way down tests design. On the way down, each side must be given a path it can walk without falling, built while the people walking it are tired, grieving, and watched by their own hardliners and allies. A de-escalation is a structure. It carries load. It fails at the joints.

Monday, 13 November 2045 — 07:00 EST: The Descent Plan

The Principals met in the Situation Room, not the PEOC. That was itself a signal, and I wanted it to be. The people who work in the West Wing read the President's movements the way sailors read the sky.

Sam Oyelaran had prepared a paper overnight. Its title was "The Long Way Down," and I have taken it for the title of this chapter, because it was the right name. It was eleven pages long, and its first sentence was one I have since quoted to every new member of my National Security Council (NSC) staff: *"The danger has not passed; it has changed shape. We are no longer at risk of a war that begins in eleven minutes. We are at risk of a crisis that resumes in eleven weeks."*

Sam's argument, which I accepted, was that the pause of 9 November was the product of three things, all of them temporary. The first was fear: both leaderships had looked over the edge and recoiled, and fear is a powerful motive that fades fast. The second was Wen's restored authority, which rested on a sick man's stamina and on a Politburo Standing Committee that had backed him once and might not again. The third was American restraint, which my own Congress and some of my own generals regarded as a gift we had given too freely. None of those would last. The task, Sam wrote, was to convert a temporary condition into durable structure before the temporary condition expired. He estimated we had until the spring.

I put my own model on the whiteboard. In control-system terms, the crisis had been a feedback loop with positive gain: each side's sensors detected the other's action, its estimator read it as threatening, its controller responded with something stronger. Between 6 and 9 November the gain dropped, briefly, below one. Our task was to keep it there after the fear faded. In an engineered system you do that three ways: better sensors, better estimators, and damping, structures that absorb a shock instead of passing it on. I wrote three words on the board: EYES, MEANING, DAMPING.

Helena Marsh, who had been an ambassador in Beijing and had heard me draw diagrams for five years, said, "Mr. President, the Chinese will want a fourth word. They'll call it respect."

"Then we'll find a way to write it down," I said.

"It doesn't write down. It's how you treat them in the room."

She was right, and she would prove it over the following four months.

The four baskets

By the end of the meeting we had divided what we wanted, and what we believed Beijing wanted, into four baskets. I give them here because a successor negotiating his way out of a crisis will need some structure like it, and because the structure itself was a tool of diplomacy: it allowed both sides to concede in one basket while winning in another.

The first basket was *the sea*: the quarantine, the customs blockade, the no-sail zone east of Taiwan, our convoys, the rules for ships and aircraft operating near each other, and the prisoners and the dead from the summer and autumn.

The second was *the strategic*: the readiness of China's nuclear forces, the Rocket Force's launch-on-warning posture, our Golden Dome space-based interceptors, our early-warning satellites and theirs, and the machines that sat between sensors and leaders.

The third was *the political*: Taiwan, its referendum, our arms sales, our deployments in the Philippines, and the words each government would use about the island's status.

The fourth was *the economic*: sanctions, the chip ban, China's embargo on rare earths and gallium, the Treasury market, and trade lanes.

I set one rule for the American side. Nothing in the strategic basket would be traded for anything in the political basket in a way that could be described, truthfully, as selling Taiwan's security for our

own. I did not believe we could keep the baskets entirely separate. I did believe that if we let the Chinese side link a de-alerting of their missiles directly to a concession on Taiwan, we would have taught them that pointing nuclear weapons at us pays. That lesson would outlive me.

> **OPERATING NOTE — De-escalation Is a Separate Engineering Problem.** The skills that carry you through the peak of a crisis (speed, nerve, a small circle, secrecy) are not the skills that bring you down from it. Coming down requires patience, a wider circle, allies who feel consulted, legislators who will ratify what you build, and an adversary who can tell his own people he did not lose. Plan the descent as deliberately as you planned the response. Write it down. Give it a name. And do not let your team rest on the summit; the summit is where the next accident begins.

Choosing the channel

Beijing proposed Geneva. Moscow offered its good offices, which I declined with thanks; a negotiation brokered by Russia gives Russia a veto. Indonesia's President Arya Wicaksono generously offered Jakarta, and I asked Helena to thank him personally and keep him informed.

But the channel already existed. It ran through Singapore, through Prime Minister Daniel Tan Wei Ming, and it had carried my letter of 5 November to President Wen. It had been tested under the heaviest load a channel can carry, and it had held. Engineers do not swap out a component that has just passed its qualification test. I asked Helena to call Tan.

He answered on the second ring, at what was nearly midnight in Singapore. He listened, and then he said a thing that I wrote down that night and have thought about since. "Mr. President, I will host. But you must understand Singapore's interest. We do not want to be your friend against China or China's friend against you. We want both of you to be our customers for a very long time. If you use us to win, we will be finished as a meeting place. If you use us to stop, we will be grateful forever."

"We want to stop," I said.

"Then we will find you a quiet room."

Saturday, 25 November 2045 — Singapore, Ashore

The first round took place on 25 and 26 November in a colonial-era bungalow in the grounds of a government estate in central Singapore, with deep verandas, ceiling fans, and, Helena reported, the best kaya toast she had ever eaten. Tan cleared the estate of everyone but his own security service and two senior diplomats.

We sent two pairs. Helena Marsh would sit with Foreign Minister Shen Yilan. Sam Oyelaran would sit with Liang Qiwen, Director of the Office of the Central Foreign Affairs Commission, who reported, Sam believed, directly to Wen. Shen was the voice of the Chinese state; Liang was the Party's strategist, who could say things the state could not. A negotiation that talks only to a government's formal face produces only formal results.

I stayed in Washington. A President does not go to a first round, for the same reason a general does not lead a patrol. But I had told both of them that they could reach me at any hour, and on the night of 25 November, which was the morning of 26 November in Singapore, they did.

What Shen said first

I have Helena's notes, written in her small, neat hand on the flight home, and I reconstruct the opening from them.

Shen arrived at nine in the morning, Singapore time. She was polite and cold. She declined coffee, accepted tea, and opened with a statement that Helena believed had been written in Beijing and approved at the top.

"Madam Secretary, my government wishes me to say three things. First, the Chinese people will never forget that American missiles killed one hundred and twenty-three Chinese sailors and soldiers on 12 September. Second, the incident of 6 November, which your President described to mine, proves that the United States has built a machine that can mistake a fire for a war, and that such a machine, pointed at China, is intolerable. Third, Taiwan is China's internal affair. On that basis, I am prepared to listen."

Helena had prepared for this. She did not answer the three points. She said, in Mandarin, "Yilan, we have known each other fifteen years. I will answer your three statements for the record before we leave today, and you will not like my answers, and I will not like yours. But before that, I want to tell you something not for the record. My President asked me to tell you that on 6 November he did not know whether he would be alive in half an hour, and that he believes your President did not know either. He asked me to say that he does not want either of them ever to be in that room again. Everything else is detail."

There was a long silence, Helena wrote. Then Shen put down her tea and said, in English, "You saw everything and understood nothing."

"Tell me what we didn't understand."

What followed was, I believe, the most important hour of the negotiation. Shen described, without naming him, General Luo's argument as it had been put inside the Chinese system for a decade: Golden Dome's space interceptors would catch whatever survived; ARGUS, our fused surveillance system, watched every road-mobile launcher outside its garrison; our long-range strike forces could hit silos; our cyber operations, she said pointedly, had been found inside the Rocket Force's logistics network. China's leaders saw a country that could see everything, strike anything, and catch whatever was left. "And then you ask why our missiles are on early warning. If they wait, they die."

Helena wrote in her notes: *She is describing the Mirror Model back to us. She has been told about it, or she has built her own.*

I had built the Mirror Model, as I described in Chapter 1, to force myself to ask what Beijing saw when it looked at us. Here was the answer, from Beijing, in Singapore. The more perfectly we watched, the more frightened they became. The fear was not an argument. It was a fact about the world, as real as the silos, and any agreement that did not reduce it would not last.

What Liang said

In the other room, Sam's conversation with Liang Qiwen had gone differently. Liang was a quieter man, a former scholar of international relations who had studied for a year in Boston in the 2000s and who, Sam said, had read more American strategic literature than most American strategists. He did not make statements. He asked questions.

His first question was: “Dr. Oyelaran, when your machine said ninety-three percent, who in your system had the authority to say no?”

Sam told him: the President, and in practice, before the President, a chain of officers whose job was to doubt.

Liang’s second question was: “And if the President had been unreachable?”

Sam answered honestly that there were procedures for that, that he would not describe them, and that they did not include letting a machine decide.

Liang’s third question was the one that mattered. “In our system,” he said, “there is a man who would have been woken in the same way. On that afternoon he was in a hospital. And the men who were awake did not share your President’s doubts.” He paused. “We have both learned that the most dangerous moment is not when the leader decides. It is when the leader is not in the room.”

Sam called me that night from a secure phone in the residence of our ambassador. “Mr. President, I think Liang just told us that Wen wants the same thing we want, and for the same reason. He wants a rule that keeps the next Luo from acting while the next Wen is sick.”

“Then we offer him a rule that binds both of us,” I said. “A pledge that no machine has authority to initiate nuclear use. Human decision required, on both sides. It costs us nothing; we already work that way. It gives Wen something he can take to the Standing Committee that sounds like a restraint on us, and it is, in fact, a restraint on his own generals.”

That was the origin of what became the Human Control Pledge. It was born on the first night, from a question asked by a Chinese official about who could say no.

> **OPERATING NOTE — Find the Shared Fear.** Adversaries rarely share goals. They often share fears. Wen and I did not want the same Taiwan, the same Pacific, or the same world. But we had both learned, in the same week, that our own systems could carry us to war while we were asleep or ill or misinformed. A shared fear is the strongest foundation an agreement between enemies can have, because neither side has to pretend to trust the other. They only have to admit that they distrust the same thing. Look for it early. Name it privately. Build on it publicly.

The first round ends

The first round produced no text. It produced an agenda in four baskets, which the Chinese accepted after renaming two of them (the “political” basket became “core interests”; the “strategic” basket became “strategic stability,” which I preferred anyway). It produced an agreement to meet again in two weeks. And it produced a problem.

On 27 November an American newspaper reported that “senior American and Chinese officials” had met “in Southeast Asia,” quoting “a congressional aide familiar with the briefings.” Within six hours Beijing told Tan that if the next round leaked, there would be no round after it.

I did not order a leak investigation; a President who hunts leakers mid-negotiation spends his capital on the wrong war. I called the leaders of the intelligence and foreign affairs committees personally and asked them to hold what they knew. And Tan called Helena two days later with a solution.

“There is a place in Singapore,” he said, “where no one can follow you, there are no telephones, and every person present wears my country’s uniform. It is a ship.”

Friday, 8 December 2045 — Singapore Strait, Aboard RSS *Fortitude*

The Republic of Singapore Navy lent us a ship. She was one of its large multi-mission vessels, built in the 2030s to carry helicopters, landing craft, and a hospital, and to serve as a command ship for disaster relief across Southeast Asia. Her name, which I give because the Singaporeans have since made it public, was RSS *Fortitude*. For three days she steamed slowly in the waters east of Singapore, in the approaches to the South China Sea, within sight of the old Horsburgh Lighthouse on its rock, while the two delegations lived aboard.

The arrangements were Tan's. Each delegation was flown by helicopter to the ship at night and given one deck; between them, the hangar had been converted into a conference room with a long gray table and a Singaporean flag. Every telephone and computer was surrendered at the brow into a sealed locker. Each side kept one secure terminal to reach its capital. The crew had been told they were supporting a regional exercise. The commanding officer told Helena it was the most interesting week of his career and that he would never speak of it. As far as I know, he never has.

A ship matters for a reason beyond security: no one can leave early. On land, a delegation losing an argument goes to its embassy to "consult its capital," and the room's momentum is gone. At sea, both delegations eat in the same ship and walk the same flight deck at sunset. On the first evening Helena and Shen walked it together for forty minutes. On the third morning, Liang brought Sam a cup of coffee before the session, unasked. Small things. They are not small.

> **OPERATING NOTE — The Venue Is Part of the Agreement.** Where you negotiate shapes what you can agree. Choose a host both sides trust, a place that cannot leak, and conditions that make it hard to walk out. Let the host own the arrangements; it gives him a stake in success. And protect the people across the table from their own capitals as carefully as you protect yourself from yours: a negotiator who can be photographed is a negotiator who must posture.

The breakthrough on the eyes

The second round was meant to be about the sea. It became, on the second day, about the eyes.

The difficulty in the strategic basket was a simple one to state and a very hard one to solve. We wanted China to take its Rocket Force off launch-on-warning: to separate warheads from missiles, return its road-mobile launchers to garrison, and go back, in effect, to the posture it had kept before about 2030, in which its retaliation was assured but not instant. Beijing would not do that while it believed, as Shen had said, that waiting meant dying. And it would not believe otherwise while we kept building the things that frightened it.

On the morning of 9 December, Sam proposed, with my authorization, a principle I had carried in my briefcase since 5 November. We called it "the eyes stay open." Neither side would attack, jam, dazzle, or otherwise interfere with the other's early-warning satellites, its nuclear command-and-control satellites, or their ground stations. The Cold War treaties had protected the satellites that verified agreements; we proposed to protect the satellites that told each country whether it was under attack. If China's early-warning satellites were safe, its leaders need not fear being blinded before a strike, and the logic of launching on warning would weaken. If ours were safe, 13 September could not be repeated.

Liang listened and said, "You are asking us to give up the ability to blind you."

"We are asking both of us to give up the ability to blind each other. On 13 September you blinded

sixty of our satellites. On 6 November our partly blind machine told my President you had attacked. You blinded us, and it nearly killed you. That is the lesson. A frightened, half-blind nuclear power is the most dangerous thing on earth. We would prefer you be able to see. We would prefer that you know we prefer it."

Liang asked for a recess. He went to his deck and was gone for five hours. When he came back he said, "My government is prepared to discuss the principle, on two conditions. First, it must cover space-based interceptors. A satellite that can destroy a missile can destroy a satellite. Your Golden Dome interceptors are therefore counter-space weapons, whatever you call them. Second, there must be a way for each side to know the other is complying, without inspectors on our soil."

The first condition we had expected; my letter of 5 November had already offered a pause in new launches of space-based interceptors. The second was the opening I had been waiting for since 13 November, because it meant that Beijing would accept verification by our satellites. It meant, in other words, that the thing that had frightened them most, our ability to watch, could become the instrument that let them stand down.

The Jade Line, 9 December

That night, which was the morning of 9 December in Washington, I spoke to President Wen on the Jade Line, the video link between our two offices that had been upgraded in 2045 from the older hotline and that had carried, in November, the most important conversation of my life.

It was our first call since 7 November. He was out of the hospital, in his office in Zhongnanhai, the leadership compound beside the Forbidden City. He was thinner than I had ever seen him, and his voice was hoarse. Our intelligence had told me what his illness was; I will not record it here beyond what became public later, that it was his heart. He sat very still, the way men sit when movement costs them something.

I told him that our negotiators had reported progress, and that I wanted him to know personally that the United States would accept the principle Liang had put forward: protection of each other's early-warning and command satellites, linked to restraint on space-based interceptors.

He listened to the interpretation and then said, in Chinese, slowly: "President Kane, in our tradition there is a saying: *the one who tied the bell must untie it.* You tied the bell when you built your shield in the sky. I tied the bell when I allowed our missiles to wait on a hair trigger. I think we must each untie our own."

I said, "Then let us agree that neither of us will ask the other to untie his bell first."

He almost smiled. "Simultaneously," he said, in English. It was the only English word he spoke in the call.

I wrote in my diary that night: *He is tired and ill and I think he knows he has very little time. He wants this finished while he can still finish it. That is a gift to us and a danger. A man in a hurry signs things his successors repudiate. We must make it something his successors will want to keep.*

Tuesday, 19 December 2045 — 20:00 EST: The Hardest Call

There is a conversation in every presidency that the President would give anything not to have. Mine was with Lin Shu-wen.

Lin's Status Referendum Act, which would ask voters whether Taiwan should apply to United Nations bodies as "Republic of China (Taiwan)," had been the political spark Beijing used to justify the quarantine. The vote was non-binding and scheduled for 2046, and by December it was the largest obstacle to ending the crisis. "If we lift the blockade and the vote proceeds," Shen had said in the hangar of *Fortitude*, "we will have been defeated by a ballot. The hardliners will say Luo was right. The next Luo will not be in a hospital."

I believed her. That was the trouble.

What I owed Lin

A successor must understand what the United States owes Taiwan, legally and morally, because the two are not the same.

Legally, the relationship rests on the Taiwan Relations Act of 1979, passed when the United States switched recognition from Taipei to Beijing. It does not make Taiwan an ally. It commits us to provide arms "of a defensive character" and declares any effort to determine Taiwan's future by other than peaceful means a matter of "grave concern." Beside it stand the Six Assurances of 1982, including that the United States would not pressure Taiwan to negotiate with Beijing. The ambiguity has kept the peace for decades. In 2045 it almost failed.

Morally, I owed Lin more. Taiwan had held its nerve for nine months under blockade, cyberattack, and the deaths of two young sailors. And I was about to ask its elected president to postpone a vote her people had been promised, at the request of the government that had blockaded them. I asked myself how I would have felt as Governor of Colorado if a foreign president had asked me to cancel a ballot measure because a neighbor with nuclear weapons objected. I knew exactly how.

Preparing the call

I spent three days preparing for a twenty-five-minute conversation. That is the right ratio. Helena drafted, Sam redrafted, and I threw both drafts away and wrote my own on a legal pad, because this was a conversation in which the other person needed to hear my voice and not my staff's.

I set myself four rules.

First, I would not lie to her. I would not tell her that Beijing had offered things it had not offered, or that the United States was asking nothing.

Second, I would not ask her to cancel the referendum. I would ask her to postpone it, without a date, and I would not ask her to repudiate the idea behind it.

Third, I would offer her concrete things that strengthened Taiwan, not things that consoled it, and I would let her choose among them.

Fourth, I would promise that the decision would be hers in public. The United States would never describe it as something we had asked for. If she said no, I would accept no, and I would tell her so before she answered.

The fourth rule was the one my staff liked least. Ray Castillo said, "Mr. President, if she says no, the deal collapses." I said, "Then the deal deserves to collapse. A peace that requires us to coerce a democracy we are pledged to defend is not a peace I can defend to my successor." I believed that. I also believed, though I did not say so, that a person offered a real choice with real dignity will more often choose well than a person who is cornered.

The call

The call was placed at 20:00 Eastern on 19 December, which was 09:00 on 20 December in Taipei, on the secure line we maintained through the American Institute in Taiwan, which serves as our embassy in all but name. Lin was in her office in the Presidential Office Building. Her National Security Council secretary-general was with her. I had Sam.

She spoke first. “Mr. President, I know why you are calling. I have read the reports from Singapore that you have shared with us, and I have read the reports from Singapore that you have not shared with us.” She paused to let that land. Our intelligence relationships with Taipei were close, and theirs with Singapore were closer than we sometimes remembered. “So, please. Do me the courtesy of asking directly.”

I did. I told her what Beijing offered: a phased lifting of the blockade from January if the vote was postponed, a verifiable de-alerting of its nuclear forces, a formal expression of regret to the families of the two *Formosa Pearl* sailors, and release of the Taiwanese mariners detained from seized ships. I told her what we would give: a written reaffirmation that the United States does “not support” Taiwan independence, and a secret five-year commitment not to station intermediate-range missiles in the Batanes, the Philippine islands closest to Taiwan. I would not keep a secret from her that concerned her own island’s defense.

Then I asked her to postpone the referendum.

She did not answer at once. When she did, her voice was very level, and I could tell from Sam’s face across the table that he was hearing what I was hearing: a person working very hard not to say what she felt.

“Mr. President, my grandmother was born in Tainan under Japanese rule. She lived under martial law for most of her adult life. She voted for the first time in a presidential election in 1996, when she was sixty-one, while Chinese missiles were landing in the sea off our coast. She told me it was the proudest day of her life. Do you know why? Not because of who she voted for. Because China fired missiles to frighten her, and she went and voted anyway.”

“I know that story,” I said. “Not your grandmother’s. The story of 1996.”

“Then you know what you are asking. You are asking me to be the president who did not go and vote anyway.”

I had prepared for many things she might say. I had not prepared for that, and I did not answer with anything from my legal pad. I said, “Madam President, I’m asking you to be the president who kept the ballot box and the island both. I can’t promise you that the vote will happen in your term. I can promise that I will never ask you to give it up, and that no agreement I sign will say that the people of Taiwan may not someday decide their own future peacefully. That sentence is in our law. It will stay in our law. I’m not asking you to surrender the question. I’m asking you to choose the time.”

“And if I refuse?”

“Then I will tell Beijing that the President of Taiwan has refused, and that the United States respects the decision of a democratic government, and we will continue the convoys and the sanctions and the rest of it, and we will try to find another way down. It will be harder and it will be more dangerous. But I will not punish you for it, publicly or privately. I give you my word, and Dr. Oyelaran is witness to it.”

There was a silence of perhaps fifteen seconds, which on a call between heads of state is a long time.

"What does Taiwan receive?" she said. "Not what Beijing gives. What does the United States give Taiwan?"

I had five things, and I offered them as a menu, not a package, because a menu is a choice and a package is a price. Delivery within eighteen months, not five years, of the anti-ship missiles, air-defense batteries, and mines already approved, the weapons of the "porcupine" strategy. A strategic reserve of American liquefied natural gas stored on the island, so the next blockade would face months of fuel, not weeks. A trade agreement I would send to Congress in the spring. A permanent American-backed reinsurance arrangement for Strait shipping. And a public visit to Taipei by the Secretary of State within a year.

She said, "You are offering me a great deal to do something I do not want to do."

"I'm offering you what I think Taiwan needs to be stronger the next time, whatever you decide about the vote. I'd like you to have all five whether you say yes or no. I can't promise Congress on the trade agreement. The rest is within my power, and I'll do it either way."

"Either way?"

"Either way."

She was quiet again. Then she said, "Give me ten days. And Mr. President, if I do this, I will announce it in my New Year address, in my own words, as my own decision, and I will say that it was taken because Taiwan is strong enough to wait. I will not say that Washington asked. If anyone in your government ever says that Washington asked, I will say publicly that it is a lie."

"If anyone in my government ever says it, I'll fire them," I said. "And you'll be right."

On 1 January 2046, in her New Year address, President Lin Shu-wen announced that the referendum under the Status Referendum Act would be postponed "until the people of Taiwan can vote in a season of peace rather than under the guns of a blockade." She said the decision was hers. It was. She said Taiwan was strong enough to wait. It was. Her approval rating fell eleven points in a week and recovered most of it by spring.

I wrote in my diary that night: *She chose the time. I asked her to. Both of those sentences are true and I will carry the second one longer than she will carry the first.*

> **OPERATING NOTE — How to Ask an Ally for Something They Do Not Want to Give.** Ask directly; they already know. Tell them everything that concerns them, including the secrets. Offer things that make them stronger, not things that console them, and offer them whatever they answer. Let them say no, and mean it when you tell them they can. Let them own the decision in public and protect that ownership for the rest of your life. An ally will forgive you for asking. An ally will not forgive you for pretending you did not.

January 2046: Lifting the Blockade, One Valve at a Time

The third round in Singapore, again aboard *Fortitude*, from 4 to 6 January, turned the principles into a sequence. I want to spend some time on the sequence, because a successor will one day need to unwind a confrontation, and the sequence is where the unwinding succeeds or fails.

An engineer shutting down a high-pressure system does not open every valve at once. He opens them in a particular order, checks the pressure after each, and keeps the ability to close any valve again if the gauge moves the wrong way. A blockade is a high-pressure system. So were our convoys, our sanctions, and the alert postures of both militaries. The question was which valve to open first, who would open it, and what each side would watch to know that the other was keeping its word.

Each side wanted the other to move first. Sam and Liang solved it at the stern rail with a principle Sam called “interleaving”: each step by one side matched within days by the other, every step small enough to reverse without losing face.

The sequence that emerged was this.

Phase one, beginning 8 January. China Coast Guard vessels would cease inspecting ships carrying food, medicine, and fuel to Taiwan. The United States would suspend its escorted convoys, which had in any case been paused since 12 September, and would announce that the escorted humanitarian corridor would not resume “while commercial navigation proceeds unimpeded.” The detained Taiwanese mariners, forty-six men and women from four seized vessels, would be released to Taiwan’s custody through the International Committee of the Red Cross.

Phase two, beginning 19 January. China would rescind the requirement that goods bound for Taiwan clear customs in Fujian. The United States Treasury would suspend, but not revoke, the Phase A blocking sanctions on six of the fourteen firms designated in July, chosen because they had no direct connection to the Rocket Force. China would suspend its embargo on gallium and on two of the rare earths most needed by our defense and medical industries. The prisoner and remains exchange would take place.

Phase three, beginning 2 February. China would formally withdraw its declared no-fly and no-sail zone east of Taiwan. The Rocket Force would begin returning the two Hami brigades to their pre-crisis condition: warheads separated from missiles and returned to storage, road-mobile launchers returned to garrison. The United States would return its strategic forces to their normal day-to-day posture, which in practice meant fewer submarines at sea and the release of bombers from generated alert. The second carrier strike group would leave the Philippine Sea.

The advanced chip ban would stay. The remaining eight sanctioned firms would stay. Those were for the summit, and for later.

Decision	Options considered	Reversible?	Risk	Chosen	Outcome
Sequencing the lifting of the blockade and the US–China drawdown (January–February 2046)	(1) United States moves first as goodwill; (2) demand China move first; (3) simultaneous comprehensive lifting on a single date; (4) interleaved small steps, each matched by the other side within days, each individually reversible	(1) Partly: convoys can resume, but trust spent is hard to recover; (2) reversible but produces deadlock; (3) hard to reverse: if one side defaults, everything collapses at once; (4) each step reversible on its own	(1) Hardliners in Beijing read weakness; Congress revolts; (2) no movement, the window closes; (3) single point of failure; (4) slowness, complexity, many chances for an accident to break the chain	(4) Interleaving, three phases, with ARGUS, LANTERN, and WILLOW reporting at each gate before the next step	All three phases completed between 8 January and 16 February, one week behind schedule. One gate was held for six days when a China Coast Guard cutter inspected a Panamanian-flagged fuel tanker on 22 January; Beijing called it a local error, disciplined the commander, and the sequence resumed

The six-day hold deserves a word. Three members of my Cabinet recommended declaring the arrangement broken. I asked: was this a decision or a malfunction? Signals intelligence and a report from WILLOW, our source in the Joint Operations Command Center in Beijing, suggested the cutter's commander had simply not received the order. I treated it as a malfunction, warned Beijing privately that a second would be treated as a decision, and held the next gate until they showed us the corrective order. The instinct to treat every deviation as betrayal is the instinct that restarts wars.

Monday, 22 January 2046 — Changi: the exchange

The prisoner and remains exchange took place at a Singaporean military airfield on the morning of 22 January, local time, with the Red Cross presiding and Prime Minister Tan's foreign policy adviser as the only political official present. I watched it, again, on a screen in the Situation Room, at eight o'clock on a Sunday night Washington time, with Claire beside me. I did not have to watch. I wanted to.

From China came Lieutenant Commander Owen Delacroix, United States Navy, the pilot of the F-35C fighter shot down at the edge of the zone on 12 September. He had ejected over water and

been pulled out of the sea by a Chinese fishing militia boat, and he had spent four months in a military facility in Fujian. The Red Cross had visited him twice. He walked down the steps of a Chinese aircraft on his own, in a borrowed jacket, lost twenty pounds, and saluted the American officer who met him. Also from China came the formal letter, signed by the Chinese Minister of Transport, expressing "deep regret" for the deaths of the deckhand Lai and the engineer Hsu of the *Formosa Pearl*, and the bridge recordings of the cutter, which Taiwan's investigators had been asking for since March. It was not an apology. It was more than Beijing had ever given.

From the United States came fourteen Chinese sailors, survivors of the two ships sunk on 12 September, who had been pulled from the water by an American destroyer and a Japanese one and held, under the Third Geneva Convention, on Guam. And from the United States came twenty-nine coffins, each draped in a plain white cloth at China's request, holding the remains of Chinese sailors recovered from the sea in the days after the strike. I had approved the recovery in September, against the advice of some who said we had no obligation to retrieve the bodies of men who had fired on our ships. I had approved it because I had stood at Dover Air Force Base on 16 September and watched forty-one American coffins come off an aircraft, and I did not believe a Chinese mother wanted her son less than an American one did.

The Chinese officer receiving the remains, a senior captain of the PLA Navy, did a thing that was not in the protocol. As the last coffin was lifted into the Chinese aircraft, he turned to the American officer and saluted. The American officer, a Navy captain whom I later learned had served on *Thomas Hudner* years earlier, returned it.

I called Delacroix that night, on the aircraft to Hawaii. He kept apologizing for being shot down. I told him he had been flying cover for other people's lives, and that his mother, who had called the White House switchboard every week since September, could now stop. Then he went quiet, and I let him. The most useful thing a President can do on such a call is stop talking.

Claire said, when it was over, "That officer saluting the coffins. That's the whole agreement, isn't it? Everything else is paper."

She was right, and she was also wrong, because paper is what makes the salute repeatable. The rest of this chapter is about the paper.

February 2046: Writing the Singapore Understanding

By early February, the negotiators had a draft. It was called, at Tan's suggestion, the Singapore Understanding, because "understanding" was a word both governments could use without implying more than either was willing to promise. It had a public part and a secret part. The full public text appears in Annex E of this manual. Here I will describe its structure, and then I will do what I promised in the Preface I would do in every chapter: explain to a successor how the machinery works, in this case the machinery of American law that governs what a President may promise to a foreign government, in what form, and with whose consent.

What it said

The public part had five elements.

First, a new agreement on incidents at sea and in the air, which we called, without imagination, Incidents at Sea and in the Air Agreement 2.0. It was modeled on the American–Soviet agreement of 1972, which had for decades kept warships and aircraft of the two navies from colliding by accident or

provocation, and which had worked well enough that most Americans never heard of it. Ours covered coast guards and maritime militias as well as navies, because the *Formosa Pearl* had been rammed by a coast guard cutter, not a warship. It covered unmanned vessels and aircraft, because the Bashi Channel Incident had begun with a confused engagement between drones. And it established a standing military-to-military channel with duty officers on both sides around the clock.

Second, a joint statement, to be issued by the two Presidents in person, reaffirming that "a nuclear war cannot be won and must never be fought," the sentence first agreed by an American President and a Soviet General Secretary at Geneva in 1985, and adding a sentence of our own: "No artificial intelligence or automated system shall have authority to initiate the use of nuclear weapons; the decision shall be made by human beings, under the authority of each nation's highest leadership." We called it the Human Control Pledge.

Third, a Space Non-Interference Code: neither side would attack, jam, dazzle, or otherwise interfere with the other's early-warning satellites, its nuclear command, control, and communications satellites, or their ground stations. The negotiators' private shorthand, "the eyes stay open," leaked within a week and became the phrase by which the public knew it.

Fourth, a Joint Early Warning Data Exchange Center, to be built in Singapore and staffed by American and Chinese officers, where each side would share data on its own missile and space launches, and on launches it detected from third countries, so that neither would ever again have to guess at a plume. A successor may know that the United States and Russia agreed to build such a center in Moscow in 2000 and never did. I was determined that this one would be built, and I asked Tan, as a condition of hosting it, to make Singapore's own reputation depend on its completion. He agreed. He told me later it was the cleverest thing I did in the whole negotiation, because no one in either capital would want to explain to Singapore why it had failed.

Fifth, restored trade lanes and the partial relief from sanctions described above, with a joint commission to address the remainder.

The secret part had four elements. The United States reaffirmed that it "does not support" Taiwan independence, which American presidents had said publicly for decades but which Beijing wanted in writing; and committed not to station intermediate-range missiles in the Batanes islands for five years. Taiwan's referendum, already postponed by Lin's own decision, was recorded as shelved, without reference to any American request. China committed to lift the quarantine permanently and to return the Rocket Force to the posture it had kept before about 2030: warheads stored separately from missiles, road-mobile launchers in garrison, no forces on launch-on-warning; and accepted, implicitly, that the United States would verify this with its own satellites. The United States committed to pause deployment of space-based interceptors for Golden Dome at their current number, roughly two hundred, for five years. And both sides agreed to upgrade the leaders' hotline and to add a second, independent link through Singapore.

Treaties, executive agreements, and secret understandings

Here a successor needs a lesson that is usually taught in law school and rarely to engineers. I learned it in February 2046 from the Legal Adviser of the State Department and from the Counsel to the President, over four long evenings, and I wish I had learned it in 2041.

The Constitution lets the President make treaties "by and with the Advice and Consent of the Senate," with two-thirds of Senators present concurring. An Article II treaty is the strongest instrument we have, and in the divided Senate of 2046 nearly impossible to obtain.

Most international agreements are executive agreements. A *congressional-executive agreement* is approved by ordinary statute, by majorities of both houses; the 1972 interim agreement on offensive arms was approved this way, alongside the Anti-Ballistic Missile Treaty. A *sole executive agreement* rests on the President's own constitutional authority; the 1972 Incidents at Sea agreement and the 1963 hotline memorandum were such agreements. They bind the nation internationally but are fragile at home: a later President can withdraw, and Congress can override them.

Then there is a third category, which is not an agreement at all in the legal sense. A *political commitment* is a promise between governments that both sides intend not to be legally binding. It rests on good faith and on the reputations of the people who make it. Joint statements are political commitments. So, often, are the private assurances that leaders give each other in a crisis.

And there are secret understandings. In October 1962, to end the Cuban Missile Crisis, President Kennedy's brother told the Soviet ambassador that the American Jupiter missiles in Turkey would be removed, provided the Soviets never called it part of the deal. The secret held for a generation, and Americans learned the false lesson that firmness alone had won. The secrecy may have been necessary. The false lesson did harm. Both are true.

Our own law has since narrowed the room for secrecy, and properly so. Under the Case-Zablocki Act, first passed in 1972 and strengthened since, the executive branch must transmit to Congress the text of every international agreement it enters into, including classified ones, which go to the foreign affairs and intelligence committees under appropriate security. The strengthened version also reaches certain commitments that are not legally binding. And the Arms Control and Disarmament Act forbids the President to obligate the United States to limit its armed forces or armaments except by treaty or by act of Congress. That provision, more than any other, shaped what we did.

> **OPERATING NOTE — Choose the Form Before You Choose the Words.** The legal form of an agreement decides who can undo it. A treaty survives you but requires two-thirds of the Senate. A congressional-executive agreement requires majorities in both houses. A sole executive agreement can be signed by you alone and torn up by your successor alone. A political commitment binds only the honor of the people who make it. Decide early which you need and which you can get. Bring the Legal Adviser into the room on the first day, not the last. And never call a commitment secret when the law requires you to tell Congress; call it classified, and tell them.

What we chose

We chose a different form for each element. The incidents agreement and the Joint Early Warning Data Exchange Center became sole executive agreements, on the precedent of 1972 and 1963. The joint statement and the Human Control Pledge, which restated policies we already held, became political commitments. The pause on space-based interceptors was the hard case. A binding promise to stop deploying a weapon is exactly what the Arms Control and Disarmament Act reserves to Congress. So we did not make one. We made a unilateral statement of American policy, conveyed to Beijing as a political commitment, that I would not deploy additional interceptors for five years, and I said in writing, in the classified transmittal to Congress, that Congress could fund or forbid them as it chose. The same was true of the Batanes commitment. Every secret element went to the committees within the time the law allows, in full text.

Decision	Options considered	Reversible?	Risk	Chosen	Outcome
Legal form of the Singapore Understanding (February 2046)	(1) Single treaty submitted to the Senate; (2) congressional-executive agreement for the whole; (3) mixed forms: sole executive agreements, political commitments, unilateral policy statements; (4) entirely unwritten leader-to-leader understanding	(1) Most durable, but a failed vote is irreversible and would collapse the deal; (2) durable, but secret elements cannot pass a public vote; (3) each element reversible by a future President; (4) wholly reversible, wholly fragile	(1) No sixty-seven votes existed; (2) leaks and delay; (3) seen as evasion of Congress; weak against a hostile successor; (4) violates the reporting law and repeats the error of 1962	(3) Mixed forms, with full classified transmittal of every element to the committees and a public pledge to seek legislation for the Data Exchange Center	Elements held through the end of my term and, as of this writing, beyond. The charge that I evaded Congress was made, and was partly fair, and is answered below

Verification by national technical means

Every arms-control agreement depends on a question an engineer would ask first: how will we know? A successor should understand the answer, because in 2046 it was different from the answer of the twentieth century, and because it was the place where the frightening thing became the useful thing.

"National technical means" is the treaty phrase, from 1972 onward, for the ways each side watched the other without permission: imaging satellites, signals collection, radars. The treaties never described them; they forbade interfering with them or deliberately concealing from them. Later agreements added cooperative measures: under the first Strategic Arms Reduction Treaty, the Soviets agreed on request to open the roofs of their mobile-missile garages so satellites could count what was inside.

The Singapore Understanding adopted the same structure. China did not admit that ARGUS existed; it did not need to. It agreed not to impede "national technical means," and it agreed to cooperative measures modeled on those of the twentieth century. On notice, at agreed intervals, the garrisons of the Hami and Ordos brigades would open their launcher sheds and park their transporter-erector-launchers in the open during a satellite pass. The specialized vehicles that carry warheads between storage sites and missiles would move only under notification.

A successor must understand the limits. No satellite can see whether a warhead sits on a missile inside a closed silo. Satellites see indicators: warhead transport convoys, security posture at storage sites and silos, crews and vehicles at launch facilities, launchers in or out of garrison, signals between command posts. A de-mated force behaves differently from a mated one, and over weeks the difference shows. Verification is never perfect; it need only be adequate, good enough that cheating

large enough to matter is detected in time to respond. ARGUS made ours far better than adequate. The surveillance that had loaded Luo's gun was now the instrument that let Wen unload it.

There was a second layer that the agreement did not mention and never will. For the first months, the officer who told us most reliably whether the Hami brigades were de-mated was LANTERN. A successor should take from this a hard truth. An agreement verified by satellites alone is an agreement about what can be seen. A President who also has a human source will know more than the agreement says, and that knowledge will create obligations of its own, to the source, which I come to below.

Wednesday, 21 March 2046 — Singapore: The Summit

I landed at Paya Lebar Air Base on the evening of 20 March after twenty hours of flying, with a refueling stop in Guam that I spent, at my own request, on the pier beside the destroyer that had pulled fourteen Chinese sailors out of the sea in September. Claire came with me. Maya, whose ship was in Yokosuka for repairs, was not permitted to, and I did not ask; a President who rearranges the Navy to see his daughter has stopped being a President.

The first sight of Wen

The summit was held in the Istana, the official residence of Singapore's President, a white colonial building set in a park in the middle of the city. Tan met each of us at the portico. Wen arrived ten minutes after I did.

I had seen him on the Jade Line six times. I had not been prepared for how he looked in person. He came out of his car slowly, with an aide at his elbow whom he shook off at the foot of the steps, and climbed them alone, one at a time, with his hand on the rail. His suit hung on him. His face had the gray, waxen color I had seen in my father's in his last year, and which Claire, beside me, saw too; I felt her go still. When he reached the top he was breathing hard, and he waited, with great dignity, until he was not, before he offered his hand.

"President Kane," he said, in Chinese. "You are taller than your screen."

"And you are braver than yours," I said. I meant the steps. The interpreter rendered it. He understood what I meant and inclined his head.

Our intelligence had told me, through WILLOW, that his doctors had argued against the trip and that he had overruled them in a meeting of the Standing Committee by saying that he would rather die in Singapore than leave the agreement to a successor who might not finish it. I did not know whether that was true. I knew that he had come.

The garden

The afternoon plenary was theater: each of us read prepared statements for the note-takers behind us, so that no one at home could say we had not. A successor should not skip it; what follows depends on it. At 17:00, Tan led Wen and me, with two interpreters and no one else, along a path through the Istana's gardens to a pavilion beside a small pond full of carp. Then he left us. Two chairs had been set out, and a pot of tea, and a jug of water for Wen, who could not drink tea that spring.

We talked for an hour and forty minutes. I will not record all of it, because some of what he said was said to me alone and he is not here to release it. I will record what a successor needs.

He began by talking about his father, who had been a railway engineer in Sichuan and had been sent to the countryside during the Cultural Revolution. “He built bridges,” Wen said. “He used to tell me that a bridge is a promise to people you will never meet. I became an engineer because of him. So, I think, did you.” I told him about my father’s lathe. We talked about engineering for perhaps ten minutes, two old men comparing the tolerances of their youths, and I think it was the most honest part of the conversation, because neither of us was pretending.

Then he said, “On 6 November, when your machine told you that I had attacked you, what did you think about?”

I told him the truth. I said that I had thought about whether six missiles made sense as an attack, about the radars, about LANTERN’s scheduled transmission, though I did not name it and he did not ask, and that in the seconds between those thoughts I had thought about my daughter, who was on a ship in the Pacific.

He said, “I was in the hospital. They did not wake me. I learned two days later that my generals had been told your submarines were moving and your bombers were on the runways. I learned also that one of them wanted to fire.” He did not say Luo’s name. “I have asked myself every night since then what I would have done if they had woken me, and I was ill, and they showed me pictures, and they said ninety-three percent.”

“What is your answer?”

“I do not know. That is why I came.” He drank some water. “General Luo believed he was saving China. He believed your shield and your eyes meant that one day you would strike us and catch what remained, and that the only answer was to be ready to fire before your missiles arrived. I helped him believe that. For ten years I allowed it to be said, because it made our people proud and our generals loyal. When I was sick it was no longer my belief. It was the system’s belief. A belief that has become a system does not need its author.”

I wrote that last sentence in my diary that night and have read it many times. It is the best description I know of how great powers drift into war.

At the end, as the light went orange over the pond, he said, “President Kane, you told me the truth on 7 November when it would have been easier to say nothing. I have asked myself why. I think it was because you are an engineer, and an engineer reports a fault even when it is his own. So I will report one of mine. My heart will not last long. I want this agreement to be built so that the men after me cannot easily undo it, because I do not trust all of them. That is why I accepted your satellites. Let them watch. It is better that you see us standing down than that you imagine us standing up.”

I said, “Then let’s build it so neither of our successors can easily undo it.”

“Simultaneously,” he said, and this time he did smile.

22 March: the signing

The public elements were signed on the morning of 22 March in the Istana’s main hall, before the press of both countries. The incidents agreement was signed by our two defense ministers. The Space Non-Interference Code and the agreement on the Joint Early Warning Data Exchange Center

were signed by Helena and Shen. Wen and I then stood side by side at two lecterns and read the joint statement, alternating sentences, he in Chinese and I in English. When I read, “No artificial intelligence or automated system shall have authority to initiate the use of nuclear weapons,” a Chinese journalist in the second row put down her pen and simply listened. I noticed it because I had not expected anyone in that room to be moved by a sentence about software.

The secret elements were never signed by either of us. They were set out in a memorandum that Helena and Shen initialed in a small room off the hall, with Tan’s foreign policy adviser as witness, while the press photographed the leaders outside. That was deliberate. A secret commitment signed by a head of state becomes, when it leaks, a story about the head of state. A memorandum initialed by foreign ministers becomes a story about diplomacy. I would like to say that we chose this form only for the dignity of the arrangement. We chose it also because it was easier to defend.

At the end of the press conference, Wen turned to leave the stage and stumbled. His aide caught one arm. I caught the other. The photograph that ran around the world the next morning showed the President of the United States holding up the General Secretary of the Chinese Communist Party. In Washington, some of my critics said it was a picture of American weakness. In Beijing, I was told, some of his said it was a picture of Chinese humiliation. In Singapore, Tan told me, people said it was a picture of two old men who had not let each other fall.

Wen said to me, quietly, in English, as he steadied himself: “Thank you, Tom.” It was the only time he used my given name. I never saw him again.

April 2046: The Reckoning at Home

The secret elements did not stay secret from the public for long. On 30 March a news organization reported, accurately, that the United States had agreed “in a classified side arrangement” not to place missiles in the northern Philippines, and, inaccurately, that we had agreed to “freeze” Golden Dome. On 2 April, Senate Majority Leader Cal Whitford of Texas rose on the Senate floor.

I have described Whitford in earlier chapters: a former rancher and Air Force officer, my fiercest critic, and in the worst hours a man who kept his counsel. His speech was the best attack on my policy that anyone made, and I want a successor to read its central passage, because it was partly right.

“Mr. President,” he said, addressing the chair, “we are told that the crisis is over. I thank God for it. But I rise to ask what it cost, and who decided. We learn from the newspapers, not from the President, that the United States has made promises about the defense of an ally, the deployment of our forces, and the shield that protects every American city. These promises were not submitted to this body. They were not voted on by anyone. They were initialed in a back room by a Secretary of State while the cameras were outside. The Constitution does not have a back room.”

He also said, and this was unfair, that I had “traded Taiwan’s vote for Beijing’s smile.” That was untrue, and Lin Shu-wen said so publicly within a day, in the words she had promised me: that the decision had been hers and anyone who said otherwise was lying. Whitford, to his credit, withdrew the phrase in a statement that afternoon.

Closed session

I could have answered him with a press conference. Instead I asked to appear before the Senate in closed session. Presidents almost never testify to Congress; one did so publicly in 1974, about a

pardon, and my Counsel reminded me how that had gone. I went anyway. You do not answer a charge about consent by refusing to be questioned.

On 18 April I sat for three hours at a witness table in the secure hearing room of the Senate in the Capitol Visitor Center, before every Senator who chose to come, and eighty-one did. There were no cameras and no staff below the level of the committee directors. I brought Helena, Sam, and the Legal Adviser, and I answered every question myself.

I told them everything in the secret memorandum. I told them every element had been transmitted to the committees in full text within the time the law allows, which some had not known, because transmittal to a committee is not reading by a Senator. I told them the interceptor pause was my policy, not the nation's obligation, and that they could fund more if they chose. And I told them Whitford was right about one thing: I had relied on transmittal when I should have relied on conversation.

Whitford questioned me for forty minutes. His last question was this: "Mr. President, if your successor tears all this up on his first day, what was it worth?"

I said, "Senator, that's why I'm here. Help me put the parts that matter into law, so he can't."

He looked at me for a while. Then he said, "Send me the Data Exchange Center bill. I'll read it." He voted for it in 2047. He never stopped criticizing the secret parts, and he was right to keep criticizing them, and the Republic was better for his criticism than it would have been for his silence.

> **OPERATING NOTE — Secrecy Is a Loan, Not a Gift.** You will sometimes need secrecy to make peace. Every secret you keep from Congress and the public is borrowed against your credibility, and the loan comes due, usually at the worst moment. Keep the secret as small and as short as the purpose allows. Obey the reporting law to the letter, and then exceed it: tell the leaders of both parties in person before they read it. When the secret comes out, go to them and answer every question yourself. A President who will be questioned is believed far longer than one who will not.

Thursday, 26 April 2046 — 01:50 EDT: Operation NIGHT FERRY

LANTERN had been in danger since October, when the Ministry of State Security rolled up KESTREL at Yumen. I had kept him in place through the winter because his reports were our best evidence that the Rocket Force was truly standing down. By April that reason had expired: the Hami brigades had been de-mated for seven weeks, and ARGUS and the cooperative measures confirmed it.

And the danger had grown. The investigation of General Luo had widened into an inquiry into how the Party leadership learned of his unauthorized orders in November. Our gamble of 4 November had saved the peace, and it had also told a careful counterintelligence officer that someone inside the Rocket Force had talked. On 9 April Ruth Calder, head of the China Mission Center, came to the Oval Office with the Central Intelligence Agency Director, Gregory Laszlo. The mole hunt had narrowed to eleven officers at Base 64. LANTERN was on the list.

"How long?" I asked.

"Weeks," Calder said. "Maybe less. He's asked to come out. He's never asked before."

I approved the exfiltration that afternoon. I will not describe its methods; they will be needed

again. A President does not run such an operation. He approves it, accepts its risks, including the diplomatic cost if it fails publicly, and stays out of the way. Laszlo told me the route would run west through Xinjiang and across the border into Kazakhstan, and that I should expect silence for about seventy-two hours.

It began on 23 April. LANTERN's wife and his daughter, the young woman whose medical treatment abroad in 2033 had first brought him into our orbit, left first, separately, by an ordinary route that had been prepared over two years. LANTERN went on leave from Lanzhou on the evening of 24 April.

I did not sleep well for three nights. On the third, I gave up at about one in the morning and went down to the Oval Office and sat at the desk with my diary and a lamp, and Lieutenant Colonel Jamal Brooks, who was on duty, sat in the outer office with the satchel and did not ask me what I was waiting for. At 01:50 the secure phone rang. It was Calder.

"Mr. President, the ferry has docked."

"All three?"

"All three. He's tired. He asked me to tell you something. He said: tell him the lids are closed."

I sat for a long time after she hung up. Then I went out to the outer office and told Brooks, who did not know what had happened and never asked, that he could get some coffee. He said, "Good news, sir?" and I said, "The best I've had in a year." I went back in and wrote it down.

I met LANTERN once, in May, in a house in Virginia. What I said to him and he to me will appear in the Epilogue, if the editors judge it appropriate. Here I will say only that he was smaller than I had imagined, that he shook my hand with both of his, and that I thought of KESTREL, whose family we did not bring out, and whose name I wrote in my diary that night beside LANTERN's.

Decision	Options considered	Reversible?	Risk	Chosen	Outcome
Exfiltrate LANTERN (Operation NIGHT FERRY, approved 9 April 2046)	(1) Keep him in place to monitor compliance with the Singapore Understanding; (2) order him to go silent and remain; (3) exfiltrate him and his family now	(1) Irreversible if he is arrested; (2) partly reversible; (3) irreversible: the source is gone for good	(1) Arrest and execution, as with KESTREL; (2) mole hunt continues and may find him anyway; (3) failure would be public, could damage the Understanding, and ends our best human window into Base 64	(3) Exfiltration, family first, route through Xinjiang into Kazakhstan	Succeeded 26 April. Compliance thereafter verified by national technical means and other sources. The loss of LANTERN's reporting was felt, and was right

Saturday, 28 April 2046 — Wet Mountains, Colorado

My father built a cabin in the Wet Mountains southwest of Pueblo in the 1990s, on weekends, out of lumber from a mill that has since closed. It has two rooms, a woodstove, a porch that faces the Sangre de Cristos, and no telephone except the ones the Secret Service and the White House Communications Agency installed in 2041 over my objections. I had not been there since the summer before the crisis.

Claire and I flew to Colorado on the afternoon of 27 April. Maya arrived the next morning, on two weeks' leave, driving a rented car up the dirt road with the agents' trucks in front and behind her, and I stood on the porch and watched her get out and stretch the way she did as a child after long drives, and I found that I could not walk down the steps for a moment, and Claire went down them first.

We did not talk about the crisis that day. Maya talked about her ship, and about the boatswain's mate who had taught her to splice line, and about a lieutenant on *Thomas Hudner* she had known since the Naval Academy, who had been killed on 11 September and whose parents she had visited in Ohio the week before. Claire talked about her patients, and about the seven-year-old from Pueblo with neuroblastoma, who was in remission. I mostly listened. Brooks's relief that weekend, a Navy commander, sat at the far end of the porch with the satchel and read a paperback novel about fishing.

In the evening Maya and I walked up the ridge behind the cabin to where my father used to sit, and watched the light go off the snow on the peaks. She asked me, finally, whether it had been as close in November as people were saying.

I told her what I could. I told her it had been closer than people were saying, and not in the way they imagined, and that what had saved us was not courage but a set of rules written in advance, some machines that disagreed with each other, a man in China who sent a message on schedule, and a sick old man who decided to come to Singapore. She thought about it and said, "So it was a system."

"It was a system," I said. "And some people inside it who didn't do what the system expected."

"Like you."

"Like a lot of people. Me least."

She took my arm on the way down, which she has not done since she was fifteen, and I let her think it was to steady me on the scree.

That night I wrote in my diary, at the kitchen table, by the light of the woodstove: *Maya home. Claire laughing at something on the porch. LANTERN safe. Wen still alive. The lids are closed. I am not a believer the way my mother was, but tonight I said thank you to someone, and I meant it.*

Letter to My Successor

The White House, 4 July 2046

Dear Mr. or Madam President,

You are reading this because something has gone wrong, or because someone wise on your staff handed it to you before anything did. I hope it is the second. Either way, I want to tell you in a few

pages what took me eight months, forty-one sailors, one hundred and twenty-three Chinese dead, an agent executed in Yumen, and eleven minutes to learn. I write on the anniversary of the night I started keeping a diary, forty-nine years ago, because it seems the right night to hand something on.

First: you will not be ready, and that is not the point. No one is ready for the moment when the phone rings in the dark and a calm voice tells you that missiles are rising. What you can be is prepared, which is different. Readiness is a feeling. Preparation is a set of decisions made in daylight that you can lean on at night. Before your first year is out, sit down with the Commander of Strategic Command, the Chairman of the Joint Chiefs, and your National Security Advisor, and write down, in your own hand, what you will refuse to do and under what conditions. Mine was the Two-Source Rule: no nuclear use on warning without two independent kinds of detection and, where time allowed, a human check. Yours may differ. But write it down, and brief it to the people who will be on the line with you. In the moment, it will not feel like a rule. It will feel like cowardice. Keep it anyway.

Second: your machines will be confident; you must be curious. The systems you inherit will be extraordinary. They will fuse data from hundreds of satellites, assign probabilities to two decimal places, and present their conclusions with an authority no human briefer can match. On 6 November one of them told me, with ninety-three percent confidence, that China had attacked us. It was wrong because it had been trained to read the news, and the news was frightening. Ask every number where it came from. Ask what the machine believed before it looked. Keep human beings, and more than one kind of sensor, between the machine's conclusion and your decision. And never let anyone persuade you that speed is the same as safety. The system was built for seventy years to make sure you would never act too late. Nothing in it was built to make sure you would not act too early. That part is your job.

Third: what you see is not what they intend, and what you see frightens them. I came into office proud of our eyes. ARGUS could follow nearly every Chinese mobile missile outside its garrison. I believed that watching was the opposite of threatening. I was wrong. To Beijing, our perfect sight, joined to our missile defenses and our long-range strike forces, looked like the preparation for a first strike. The more completely we saw, the more frightened they became, and their fear put their missiles on a hair trigger, and the hair trigger nearly killed us all. Ask, every week, what your adversary sees when he looks at you. Not what you mean. What he sees. Then build something that lets him see you are not coming. In the end, the very satellites that frightened China became the instrument that allowed it to stand down. Surveillance can load an enemy's gun, and it can help him unload it. Which one it does depends on whether you are willing to let him know what you are watching for.

Fourth: run three clocks, and keep the fastest one slow. Every crisis runs on a military clock measured in minutes, a political clock measured in days and weeks, and an information clock measured in seconds. The danger comes when they fall out of step, when the generals can act faster than the diplomats can talk. Your task is to slow the military clock wherever you can, to speed the political clock by every channel available, and to keep control of what the public hears. In practice this means: prefer reversible steps; delay irreversible ones until no reversible path remains; and when you act, say plainly what you have done and what you have not, so that the other side's most frightened officer does not have to guess. The three words that saved more lives than any weapon in this crisis were "This response is complete."

Fifth: build the bridge before you need it. Every escalation should carry its own way back. Before you strike, decide what you will say to make it clear the strike is finished. Keep open channels

you do not trust, because the channel you need on the worst night is the one you maintained on an ordinary one. I owe the peace in part to a fifteen-year friendship between two diplomats and to a small country that had spent sixty years earning both sides' trust. Thank such people often.

Sixth: the person on the other side is a person, and his system is not. President Wen was not my friend, and I do not pretend otherwise. His government blockaded an island of twenty-two million people and killed our sailors. But in a garden in Singapore he told me the truest thing I heard in the whole crisis: that a belief which has become a system no longer needs its author. Your adversary's leader may want peace while his system drives toward war, and the same may be true of you. Learn to tell the leader from the system. Give the leader something he can carry home. And remember that the most dangerous moment is not when the leader decides. It is when the leader is not in the room.

Seventh: allies are not instruments. You will one day have to ask an ally to do something she does not want to do, for a peace she did not negotiate. Ask directly. Tell her everything, including the secrets that concern her. Offer things that make her stronger, whatever she answers. Let her say no. Let her own the decision in public and protect that ownership for the rest of your life. I asked a democratic president to postpone a vote. She chose to, and she was right, and I will carry the asking longer than she carries the choosing. That is as it should be.

Eighth: Congress is not your audience; it is your partner, and sometimes your brake. I kept secrets to make the peace, and I would keep most of them again. But secrecy is a loan, and mine came due on a Senate floor. Obey the reporting laws to the letter, and then go further: tell the leaders of both parties in person, before they read it in a newspaper. When they challenge you, answer them yourself. Put the parts that matter into law, so that your successor cannot undo them with a signature. The opposition leader who attacked me most fiercely in this crisis did more to make the peace durable than many who praised me. Your critics are part of the system. Treat them that way.

Ninth: the people who carry your secrets carry your burdens. Somewhere tonight there are men and women, in uniform and out of it, in our country and in others, who are risking their lives because you might need what they know. One of ours died at Yumen because I judged that his information, and that of a colleague, was worth the risk. I may have been right. He is dead either way. Know their code names. Ask about them. When their usefulness is over, bring them home before it is too late, even when it costs you your best window into an adversary. Write their names in your diary, so that someone remembers them as people.

Tenth: take care of the instrument. Your sleep, judgment, and temper are components with failure rates. Eat something at three in the morning. Keep one person near you who never tells you what you want to hear; mine was my wife. Take twenty minutes alone before an irreversible decision. Go to Dover when the dead come home. Count the enemy's dead as well as your own.

And finally: keep a log. My mother told me, the morning after I stayed up to watch a little spacecraft bounce across Mars, that engineers keep logs. Write down what you saw before you write down what you think. You will need to know the difference later. Write one true thing every night, even on the nights when you can manage only three lines. When the crisis comes, the log will tell you what you actually knew, rather than what you later wish you had known. And when it is over, it will let you hand on what you learned to the person who comes after you, as I am handing it to you now.

You will not get all of it right. I did not. The editors who will one day publish this document will, I

hope, say plainly what I got wrong, and I have asked them to. But you will get more of it right if you remember that the job is not to win. The job is to bring your country, and your adversary, and the whole frightened world, down the mountain alive.

The summit is only halfway. Pay attention on the way down.

With respect, and with more sympathy than you can yet imagine,

Thomas Ellery Kane 49th President of the United States

Epilogue

After the Minutes: Prepared by the Editorial Team, 2071

> "An officer should not report what he does not know." — attributed to the source code-named LANTERN, in conversation with the author, 2046

Editorial Preface to the Epilogue

President Kane finished this document on 30 June 2046 and left office in January 2049. The world he described went on without him, much of it in the direction he had hoped. This Epilogue, written in 2070 and 2071 by the editorial team, does three things. It records what became of the agreements, the machines, and the people that the President describes. It gives, for the first time and with the written consent of his family, the true name of the source the document calls LANTERN, and an account of his later life. And it offers an assessment of what the author got wrong. The author asked for that last section. In his letter to the review panel in the spring of 2070 he wrote: "Do not let the epilogue be a eulogy. I am not dead yet, and even when I am, I will not want one. Tell them where I failed, so the next person fails somewhere new."

We have tried to honor that request and to be fair: to judge each decision by the conditions in which it was made, the standard this team set for itself in the Prologue.

1. The Agreements

The Singapore Understanding

The Singapore Understanding, signed in public form by President Kane and President Wen Zhaolin on 22 March 2046, outlasted both of its signatories in office and one of them in life. Its public text is in Annex E. Its five public elements held, with strains, through the end of the 2060s. The Incidents at Sea and in the Air Agreement 2.0 was invoked more than three hundred times in its first two decades, most often by duty officers resolving close encounters between coast guard vessels and unmanned aircraft in the Taiwan Strait and the South China Sea. Twice, in 2052 and 2059, collisions occurred despite it. In both cases the standing military-to-military channel the agreement created was used within the hour, and neither collision became a crisis. Historians regard the channel as the agreement's real legacy.

The joint reaffirmation that "a nuclear war cannot be won and must never be fought," and the Human Control Pledge that accompanied it, were adopted in substantially the same words by all five nuclear-weapon states recognized under the Treaty on the Non-Proliferation of Nuclear Weapons in a joint statement of 2053, and later by two other nuclear-armed states. The pledge was a political

commitment, not a law, and it was never verified in any technical sense. Its value, one former Strategic Command officer told the editors, was that "it gave every officer in every crew the words to refuse."

The Space Non-Interference Code, "the eyes stay open," survived a test in 2057, when a Chinese accusation of American interference with an early-warning satellite was withdrawn within thirty hours after a joint review of the data in Singapore. In 2064 it was extended to the ground stations of commercial operators whose satellites carry military sensors, a legacy of the Halcyon Orbital episode.

The Joint Early Warning Data Exchange Center (JEDEC) in Singapore began limited operations in late 2047 and became fully operational in September 2048, the first such center ever built after a half-century of proposals. It fulfilled the fourth change the author sought after the Eleven Minutes, a way to tell Beijing in real time when American machines see something American officers do not believe, and it was used for exactly that purpose at least four times before 2071.

The economic element was the least durable. The joint commission on sanctions resolved perhaps half of its agenda in eleven years, and the advanced chip ban was never fully lifted.

The secret elements

The secret elements, declassified by agreement during the 2060s, are summarized in Annex E. China lifted the quarantine of Taiwan permanently. It did not return. The Rocket Force returned the two Hami brigades, and in time the rest of its force, to a posture close to the one it had kept before about 2030, with warheads stored separately from missiles and road-mobile launchers in garrison. American intelligence assessed with high confidence that the posture held through at least 2060; the cooperative display measures were carried out on schedule in all but three of the first sixty satellite inspections, and each exception was remedied within days.

The American pause on new space-based interceptors for Golden Dome was observed for its full five years. In 2051, after a long debate in Congress, it was replaced by a negotiated ceiling, set slightly above the 2046 number, in exchange for a Chinese commitment not to deploy fractional orbital bombardment systems beyond those already fielded. The commitment not to station intermediate-range missiles in the Batanes islands lapsed in 2051 and was not renewed in writing. No such missiles were stationed there before 2071.

Taiwan's referendum on the name to be used in applications to the United Nations was never held. President Lin Shu-wen completed her term in 2048. In her memoir (Taipei, 2061) she called the call of 19 December 2045 "the night I learned what an ally is, in both senses of the word," and concluded that Kane kept every promise he made to her and asked of her a sacrifice no democratic leader should have been asked to make alone.

2. The Machines

PALISADE was withdrawn from the warning chain in early 2047 and replaced by a system built under the three rules the author directed in November 2045 (Annex A, Section 2). The Two-Source Rule was written into the instructions for the National Decision Conference in 2047, made a condition of funding by Congress in 2050, and remains in force in modified form.

ARGUS was renamed twice and expanded continuously. Its architects observe that the Singapore Understanding could not have been verified without it. That is true. The author himself called

it, in Chapter 7, the surveillance that "had loaded Luo's gun" before it let Wen unload it. Both statements are true at once, which is the hardest kind of truth for a government to act on. Golden Dome grew within the limits negotiated in 2051; its effectiveness against a deliberate attack by a nuclear peer was never tested.

3. The People

In Beijing

Wen Zhaolin returned to his duties in late November 2045 and presided over the summit in Singapore in March 2046, visibly frail. He died of heart failure in Beijing in 2047, at sixty-nine. His funeral was attended by a delegation led by the American Vice President. The President did not attend; the record shows that he asked to, and that his advisors, citing the political risk in Beijing for Wen's successors of an American President at the graveside, persuaded him not to. His diary that night reads: *"I owed him the trip. I let them talk me out of it. He believed me once when it mattered. I should have gone."*

His successor, a provincial party secretary with no role in the crisis, governed cautiously for two decades and never repudiated the Understanding.

General Luo Tiancheng was relieved of command of the Rocket Force on 8 November 2045, officially for reasons of health, and placed under investigation by the Central Military Discipline Inspection Commission. In 2047 a military court convicted him of violating the command authority of the Central Military Commission and of "seriously endangering the strategic security of the state." He was sentenced to life imprisonment. Portions of the trial record, provided to the editors through channels that remain confidential, inform the account in Chapter 6 of the events in Beijing between 15:31 and 15:50 on 6 November. Luo died in custody in 2063, maintaining to the end that he had acted to protect China from an American first strike he believed was coming. The editors record his belief without endorsing it.

General Zhang Hesheng, whose words on the telephone that afternoon ("Until I can see it, there is no war") are now quoted in staff colleges on both sides of the Pacific, retired from the Central Military Commission in 2048 and died in 2059. The author wrote to him once, privately, in 2050, through the Singapore channel. The letter was delivered. No reply was received, and none was expected.

Premier Qiao Weiming retired in 2048; **Foreign Minister Shen Yilan** served until 2052; **Minister Kong Dewei** was moved to a lesser post in 2046 for reasons never made public.

The colonel on duty at the Joint Operations Command Center on the afternoon of 6 November, who asked under what authority the order was to be sent and then telephoned General Zhang's office, has never been identified by any American source. The author ends Chapter 6 by saying he will always honor that officer's courage. Twenty-five years later, he still does not know the colonel's name. Neither do we.

In Washington

President Kane served out his second term and left office on 20 January 2049. He returned to Colorado with Dr. Claire Kane, who retired from medicine in 2056 and has since led a foundation for children's cancer care in the Mountain West. At the time of this release both are living.

Lieutenant Maya Kane remained in the Navy, commanded a destroyer of the Seventh Fleet in the 2050s, and retired with the rank of captain. She asked only that the editors note that the crew of USS *Daniel Inouye* in September 2045 "did their jobs and were not thinking about the President." **Eli Kane** lives in Austin.

Dr. Samuel Oyelaran returned to Georgetown University, where he taught a seminar on crisis decision-making until 2066 without once mentioning, his students report, that he had lived through one. **Secretary Raymond Castillo** died in 2058 and was buried at Arlington National Cemetery; at his family's request, the folded paper with the names of the forty-one sailors of USS *Thomas Hudner*, which he had carried throughout the crisis, was placed in his casket. **Lieutenant Colonel Jamal Brooks** retired as a colonel in 2060. The review panel consulted him in 2070 on several passages of Chapter 6. He confirmed them all, including the socks.

Ruth Calder, the name under which the chief of the CIA's China Mission Center appears in these pages, retired in 2049. Her true name remains withheld at her own request. She was present at the burial of the man she recruited.

Marcus Thorne sold Halcyon Orbital in 2054.

4. LANTERN: Fu Jianguo

The source this document calls LANTERN was **Senior Colonel Fu Jianguo** of the People's Liberation Army Rocket Force. His name is published here for the first time, once, with the written consent of his widow and his daughter.

Fu was born in 1988 in Tianshui, in Gansu province, the son of a railway signalman. He joined the Second Artillery Corps, the predecessor of the Rocket Force, as a young officer, and spent most of his career in the missile bases of the northwest. By 2045 he was deputy chief of staff of Base 64, headquartered at Lanzhou, with responsibility for the staff work of the brigades that manned the Hami silo field. Colleagues remembered him as precise, unambitious, and the only officer on the base staff who read poetry.

The circumstances of his recruitment are described in outline in the Prologue and in Chapter 4: a daughter, eleven years old in 2031, diagnosed with a rare and aggressive childhood cancer; treatment abroad that saved her life; slow contact over two years; and, in 2033, a decision that was his own. He accepted help with medical debts. He never accepted payment for information. He told Ruth Calder at their third meeting, in words she recorded that night, that he had spent his career preparing weapons whose use he had never let himself imagine, and that sitting beside his daughter's hospital bed in a foreign city had made him imagine it. "I know now what a city is," he said. "It is a hospital full of children."

For twelve years he reported on the readiness of the Hami brigades: their alert levels, their exercises, and, in 2045, the mating of warheads to missiles. His automated burst at minute six of the Eleven Minutes ("No launch. Lids closed. RL2 holding. Exercise fault?") is described in Chapter 6. The editors note what the author also noted: the message ended with a question mark. Fu did not know what was happening at his own base. He reported what he saw and marked what he did not know.

After the arrest of KESTREL in October 2045, the counterintelligence investigation reached Base 64. The author's decision to keep LANTERN in place, while preparing his exfiltration and asking him, through a single brief exchange, whether he wished to come out, is recorded in Decision Record

form in Chapter 4. Fu's answer, delivered in late October, has been released. It read in full: *"Not yet. The work is not finished."*

He was brought out in April 2046 under Operation NIGHT FERRY. His wife and daughter had left China some weeks earlier on travel arranged under the cover of the daughter's medical follow-up abroad. Fu himself crossed into Kazakhstan on foot at night, after a journey from Lanzhou whose details remain classified. The operation took nine days. He was absent from his post for six of them, and the investigators of the Ministry of State Security were, the archival record suggests, perhaps two days behind him. In 2047 a Chinese military court tried him in absentia and sentenced him to death.

He lived for twenty-two years in Virginia under another name. He advised the intelligence community for some years, then stopped, saying he had told them everything he knew and did not wish to begin guessing. He kept a vegetable garden. He taught Chinese calligraphy at a community center on Saturday mornings, where none of his students knew who he was. His daughter became an engineer.

He met the author once, in May 2046, a few weeks after his arrival, at a farmhouse in Virginia, with Calder present. Both men later described the meeting to the editors, Kane in writing and Calder in an interview. They spoke for three hours, mostly in English, which Fu had learned from technical manuals. Kane asked him why he had put a question mark at the end of the message of 6 November. Fu answered: "Because I did not know. An officer should not report what he does not know." Kane told him that the question mark had been the most important punctuation of his life. Fu said, according to Calder, that he did not think a President should say such things about punctuation, and they both laughed.

Before they parted, Kane asked whether there was anything he could do. Fu asked for one thing: that the family of the man code-named KESTREL be found and helped, if it could ever be done safely. Kane promised to try.

Fu Jianguo died in Virginia in 2068, at the age of eighty, of a stroke. At his family's request his headstone bears his true name. His daughter's statement, provided to the panel with the family's consent to this release, reads in part: "My father did not think of himself as a spy. He thought of himself as an officer of the Rocket Force who had understood what the Rocket Force was for. He would want you to know that he loved his country, and that he did what he did because he loved it."

KESTREL's family, and the others

The promise Kane made to Fu was not kept, though not for lack of attempts. Inquiries through Singapore in 2047 and 2048, and through a third government in 2052, about the widow and two children of the source code-named KESTREL received no answer. A fund established for them in 2046 has never been claimed. Their fate remains unknown. KESTREL's true name remains withheld.

The later lives of the sources code-named WILLOW, THRUSH, and ORCHID are among the matters the review panel withheld. The editors may say only that none of them was arrested during the crisis or in the year that followed.

5. What the Author Got Wrong

The author asked us to tell readers where he failed. The editorial team did not agree on every point below, and where we disagreed we say so. We list six failures, and then one matter that we leave

open.

He trusted the eyes too much, and built more of them

The most consequential error of the Kane presidency was not made in 2045. It was made in the years before, when the President championed ARGUS and the fusion of commercial and national sensors into a system that claimed "custody" of 94 percent of China's road-mobile launchers. Kane understood the Mirror Model better than anyone in his government: he wrote in Chapter 1 that absolute surveillance "loads the enemy's gun." But he built the surveillance anyway, and he continued to expand it after the crisis. The Eleven Minutes were, in part, a product of the system he championed. PALISADE was retrained in October 2045 to compensate for satellites lost in September, and the retraining that made it more robust in ordinary conditions made it fragile to coordinated deception. The coherent error that nearly started a war was an error of his architecture. The author knew this and said it. Knowing a thing and acting on it are different.

He kept KESTREL's information flowing at KESTREL's expense

The arrest of KESTREL in October 2045 is usually described as the result of the counterintelligence hunt that followed the escalation of the autumn. The archival record suggests a further cause. During the Pressure Season of the summer, the National Security Council staff raised the requirement for alert-status reporting from the Yumen field from weekly to near-daily. The CIA warned, in a memorandum of late August 2045, that the increased frequency of KESTREL's transmissions raised his risk of detection. The President approved the requirement without asking what it cost the source. He did not see the memorandum until after KESTREL's arrest.

The author confesses his decision to keep LANTERN in place. He does not, in this document, confess the earlier and smaller decision that may have cost KESTREL his life. When the panel showed him the August memorandum in 2070, he wrote in the margin of the review copy: *"I did not know. I should have asked. That is the same thing."*

He kept too much from Congress for too long

The author's legal handling of the Singapore Understanding, described in Chapter 7, was careful, and every secret element was transmitted to the committees as the law required. But the law sets a minimum, not a standard. For the nine days of November 2045, and for most of the negotiation that followed, the congressional leadership of the opposition party knew almost nothing of the gamble of 4 November, of the Eleven Minutes, or of the concessions under discussion. Senate Majority Leader Cal Whitford learned of the Golden Dome pause from a classified transmittal, not from the President. His public charge of "secret concessions" was overstated; his private complaint, that he had been treated as an obstacle rather than a partner, was justified. The Nuclear Crisis Consultation Act of 2050, which requires the President to brief the bipartisan congressional leadership within forty-eight hours of any agreement affecting the posture of strategic forces, was passed with the votes of Kane's own party as a direct response.

He gambled with sources he had promised to protect

The decision of 4 November 2045 to pass evidence of Luo's unauthorized orders to Premier Qiao and General Zhang, through ORCHID and through Singapore, worked. A minority of the editorial team, and a majority of the intelligence representatives on the review panel, regard it nonetheless as the most dangerous decision of the crisis, more dangerous than the Eleven Minutes. He took it slowly,

as his own rules required. But it put the lives of at least two sources in the hands of officials whose loyalties he could not know. It succeeded because Qiao and Zhang behaved as Kane hoped. The editors are not certain that this was skill.

He asked Taiwan to pay a price that he did not

The author writes honestly about the call of 19 December 2045. But its outcome, a referendum shelved at American urging and an American commitment that Taiwan independence was not supported, was a cost borne by a democracy that had not been at the table in Singapore. The author argues that Taiwan's security was improved, and twenty-five years support him. Taiwanese historians reply that the "golden bridge" he built for Beijing was paved partly with Taiwan's choices. Both judgments deserve to stand.

He never patched his own operating system

The author wrote that an operating system never patched becomes dangerous. His was never patched. The three presidents who held office between 2049 and 2061 signed for it, and two left notes in the margins. After 2061 the register is blank. No Edition 2 was written. The Two-Source Rule was absorbed into procedure; other rules were forgotten. The author, after 2049, declined every invitation to revise the document, on the ground that it belonged to the office and not to him. That was a principled position. It may also have been a mistake.

A matter we leave open

On 12 September 2045, the day after the Bashi Channel Incident, the President ordered a proportionate conventional strike that killed approximately one hundred and twenty Chinese military personnel. For forty-seven minutes the night before, he had not known whether his daughter was alive. The author describes those minutes in Chapter 4. He does not say whether they shaped the decision of the next day, and when the panel asked him in 2070, he answered: "I have asked myself that every year since. I do not know. I would distrust any President who said he did." The editors do not know either.

6. Coda

In the spring of 2070, the review panel sent the author a list of questions. The last was whether, knowing what he knew now, he would write the document differently. His answer, in a letter written by hand from his home in Colorado, is the last word in the archival register, and we give it the last word here.

"I would write it more slowly. I would put fewer rules in it and more doubts. I would say more about the colonel in the Western Hills whose name I will never know, and about the officer in Colorado who wrote 'recommend caution,' and about the man in Lanzhou who put a question mark at the end of the most important message of my life. I would tell my successors that the machines will keep getting better, and that this will make the problem harder, not easier, because the better the machine, the less anyone will want to say no to it. And I would tell them that every time they are tempted to believe the screen, they should picture a hospital full of children, because that is what a city is. A friend taught me that. He is dead now. I am not, yet. I am still keeping the log."

Annex A: The Operating System

The author's own text, typed June 2046. Editorial notes are in italics and marked as such.

Editorial note: The author's cover sheet for this Annex, and the Table of Modules that follows it, describe "forty rules and six protocols." A seventh protocol, on intelligence assets, was typed on separate sheets dated 29 June 2046 and inserted by the author between the original fifth and sixth sections. He renumbered "Managing Yourself" as Section 7 by hand. We reproduce the Annex as he left it.

How to Use This Annex

This Annex is the only part of the document I ask you to memorize. Everything in it has a story behind it in Chapters 1 to 7. Know the story before you rely on the rule. The rules are grouped by subject. The protocols are checklists, written to be used by a tired person in a bad hour. Keep a copy in the PEOC (the Presidential Emergency Operations Center), one in the Situation Room, and one with the military aide. Write in the margins.

Part One: The Forty Rules

I. Time and Speed

1. **Take reversible decisions fast and irreversible decisions slowly.** Take an irreversible step only when no reversible path remains.
2. **Slow the military clock. Speed the political clock. Control the information clock.** Danger lives where the three clocks drift apart.
3. **When you are pressed to act irreversibly and fast, treat the pressure itself as a warning.** That corner of the Reversibility Matrix (Quadrant III) is the most dangerous place a President can stand.
4. **The first report is wrong.** Not always. Often enough to wait for the second.
5. **You may decide not to decide.** Ride out and retain. Confirm before you act.
6. **Put a clock on your deliberation, and then use all of it.** "You have tonight. Use all of it."

II. Seeing and Knowing

7. **Never act on a nuclear warning from one kind of evidence.** Two sources before Armageddon; a third where time permits. This is the Two-Source Rule.
8. **A confidence score is a statement about a model, not about the world.** Ask what the machine believed before it looked.

9. **Seeing is not knowing.** Beware coherent error: several weak, partly true signals fused into one strong, false one.
10. **Ask every briefer three questions:** How do you know? How confident are you, as a number? What would change your mind?
11. **Write down what you saw before you write down what you think.**
12. **Know which of your eyes can be put out, and plan for the night they are.**
13. **A human source is a sensor with a conscience.** Weigh what he says, protect who he is, and never treat him as a switch.
14. **Distrust your most confident advisor.** His prior was set before the crisis began. So was the machine's.

III. The Adversary

15. **Model the other side's fears, not only its intentions.** Ask: what do they see when they look at us?
16. **The more perfectly you see, the more frightened they become.** Surveillance is a weapon that also loads the enemy's gun.
17. **Every move is a message. Make it legible.** If you cannot say in one sentence what a move means, do not make it.
18. **Build the golden bridge before you need it.** An adversary with no way down will come up.
19. **Bound every strike in target, time, and word.** Declare it complete, publicly, the moment it is.
20. **Do not strike the other side's eyes or its command.** Against a force on launch-on-warning, an attack on early warning or command reads as the first act of a disarming strike.
21. **Your restraint must be visible to be useful.** Before any precaution, ask what it will look like on their screen to their most frightened officer, and choose the least legible step that still protects you.
22. **Tell your adversary when your machines lie.** Honesty about your own false alarm is a strategic act.

IV. Channels and Allies

23. **Open the back channel before you need it.**
24. **Keep more than one channel, and never let one carry everything.**
25. **Consult before you decide, or do not call it consultation.**
26. **An ally's dignity is a strategic asset.** Spend it only in private, and repay it in public.
27. **Find the shared fear.** It is a firmer foundation than shared interest.
28. **De-escalation is a separate engineering problem.** Design the way down while you are still going up.

V. Your Own Government and Your People

29. **Decide the principle before the policy.**
30. **Write down what you will refuse to do, and tell your commanders before the crisis.**
31. **In a long crisis, the President's voice is a stabilizer, not an accelerator.**
32. **The public is owed the truth, but not necessarily tonight.** Set the date on which you will tell it, write that date down, and keep it.

33. **Every secret agreement is a loan against the trust of Congress.** Record the terms and plan the repayment.
34. **Economic weapons have recoil, and your debt is part of your defense posture.**

VI. The People Who Serve

35. **Do not drive the ship.** Give commanders intent and limits, then let them fight.
36. **The first minute belongs to people you will never meet.** Train them, trust them, and protect their right to write "recommend caution."
37. **You owe your sources a promise and an accounting.** Make decisions about their lives slowly, and write down why.

VII. Yourself

38. **Sleep is a weapons system.** A President who has not slept is a degraded sensor, and should be treated as one.
39. **When the crisis is personal, say so aloud to someone who can overrule you.**
40. **Keep a log. Every night, by hand.**

Section 1: The First Hour of a Crisis

Use this when you are woken, or interrupted, by news that may become a crisis. Times are from the moment you are told.

Minutes 0–5: Orient

- Look at the clock. Write the time down. Write down who told you and exactly what they said.
- Ask: Is anyone in immediate danger whom I can help in the next hour? If yes, that comes first.
- Ask: Is this a nuclear warning? If yes, go to Section 2 now and stop reading this one.
- Ask the three briefer questions (Rule 10). Accept "we don't know yet" as an answer. Do not accept a guess dressed as a fact.

Minutes 5–15: Stabilize

- Name a single point of contact for the next hour, normally the National Security Advisor. Every report comes through that person.
- Ask the Chairman of the Joint Chiefs: What are our forces doing right now, on their own authority, under standing orders? Is anything happening that I have not ordered and would not want?
- Direct that no new public statement be made by anyone in the government until you have approved its words. Silence for one hour is reversible. A wrong statement is not.
- Ask the Director of National Intelligence for the other side's likely view of the same event. What do they think just happened? (Rule 15.)

Minutes 15–30: Map the Clocks

- Draw the three clocks on paper. Where is each one? Which is running fastest?
- Military clock: what decisions will be forced on you, and by when? Get the deadlines in writing.
- Political clock: whom must you call, and in what order? (Allies before adversary, unless lives depend on the adversary.)

- Information clock: what is already public? What false version is already spreading?

Minutes 30–45: Sort the Decisions

- List every decision in front of you. Place each in the Reversibility Matrix.
- Make the reversible, urgent ones now. Defer every irreversible one you can.
- For each deferred decision, write down what information would let you make it, and who will bring it.

Minutes 45–60: Signal

- Decide the one sentence you want the other side to hear, and through which channel.
- Decide what you will say to the public, if anything, and when. "We are aware and assessing" is a complete statement.
- Schedule the next meeting. Then eat something.

Do not, in the first hour:

- Order any irreversible military action except to defend forces under attack.
- Speak to the press.
- Accept the first casualty figure, the first attribution, or the first assessment of intent.

Section 2: Nuclear Warning Protocol

Read this until you can recite it. The protocol below describes my own standard and the procedures I directed in 2045 and 2046. It gives no codes, authentication methods, or system details, and none should ever be written into a document like this. The minute marks are illustrative. A ballistic missile launched from the far side of the world takes roughly half an hour to arrive; one from a submarine may take much less. You will have minutes, not hours.

Before any warning ever comes

- In your first month, decide your personal standard for responding to a warning. Write it down. Tell the Chairman and the Commander of Strategic Command (STRATCOM) what it is.
- Read the first page of the Gold Folder until you know it: *You may decide not to decide. Ride out and retain. Confirm before you act.*
- Take part in at least two exercises a year in which the warning turns out to be false, and at least one in which you are not told in advance.
- Learn the names of the duty officers at the National Military Command Center (NMCC) and the North American Aerospace Defense Command (NORAD). They will be on the line before you are.

The Two-Source Rule

No report of a nuclear attack on the United States is to be treated as confirmed unless:

1. **Source A:** space-based infrared sensors report the heat of a missile launch; **and**
2. **Source B:** ground-based radars independently report physical objects in flight on a consistent path. Two different physical effects, seen by two different kinds of sensor, must agree. This is called dual phenomenology.
3. **Where time permits, a third check:** a human source, an independent overhead image, or a direct leader channel.

A fusion system's recommendation is not a source. It is a combination of sources and must be broken back into them.

The three standing requirements of November 2045

- **The prior beside the score.** No fusion system's confidence score is to be displayed to the President without the system's prior (what it believed before the event) displayed beside it.
- **Degradation in the top line.** Any warning in which more than half of the contributing satellites carry a degradation flag is to be marked as such in its first line, not in the metadata.
- **The rule is the procedure.** The Two-Source Rule is a requirement of the conference, not a preference of the President.

During a warning: the checklist

When you are woken (about minute 2–3)

- Get up and move. Talk on the way. Do not wait to reach the PEOC before you start asking questions.
- Confirm that the military aide is with you. Do not open anything yet.

Joining the conference (about minute 3–5)

- Ask the Commander of NORAD: Which phenomenology do we have? Infrared only, or infrared and radar?
- Ask: How many contributing sensors, and how many carry degradation flags?
- Ask: What is the prior?
- Ask: When will radar have its chance? Write that minute down.

The decision point (about minute 4–6)

- If you have one phenomenology: state aloud, for the record, that you are invoking the Two-Source Rule, and that no response options will be executed until a second source reports.
- Ask the Commander of STRATCOM for his own assessment, not the machine's.
- Ask the CIA Director whether any human source or leader channel can be reached in the time available.

Protective steps (about minute 5–8)

- For every protective measure proposed (bombers, airborne command posts, submarines), ask: Is it reversible? What will it look like on their screens? (Rule 21.)
- Prefer crews to aircraft over aircraft in the air when time allows.
- Begin the movement of the Vice President to a separate location. That is routine and reversible.

Waiting for the second source (about minute 6–10)

- Keep the line quiet. Ask for reports, not opinions.
- Ask for any independent imagery of the reported launch points.
- Ask what the other side's warning systems are likely to be seeing of our own actions at this moment.

Declaring (when the evidence allows)

- Poll the principals. Ask each: Does anyone have a reason to believe this is real?

- State your decision in words that cannot be misunderstood: "There is no attack" or its opposite. Have it logged with the time.
- Order no visible stand-down for twelve hours. A hurried stand-down looks like the end of preparation.

After a false warning

- Within the hour: order an investigation, to report to you by noon.
- Within the day: tell the other side's leader, through the leader channel, that your machines reported an attack that you did not believe. Tell the truth. (Rule 22.)
- Within the week: brief the bipartisan congressional leadership in closed session.
- Within the month: fix the fault, then fix the culture that allowed it.

What never to do

- Never execute a response on a single phenomenology.
- Never let a confidence score stand in for a source.
- Never strike an adversary's early-warning or command systems in the hope of preventing a launch you have not confirmed.
- Never decide your standard during the alarm. The alarm will decide it for you.

Section 3: Managing Diplomatic Channels

Keep a channel register. One page, updated daily by the NSC Executive Secretary. For each channel: who speaks on each end, what it is for, how fast it works, how secure it is, and when it was last used. In 2045 mine read:

Channel	Ends	Best for	Speed
Jade Line (leader video hotline, agreed 2043)	President to General Secretary	Leader-level commitments and honesty	Minutes to arrange
Defense Telephone Link (since 2008)	Secretary of Defense to Chinese defense leadership	Military incidents	Hours, often unanswered
Ambassadors	State Department to Foreign Ministry	Formal positions, démarches	Hours to days
Singapore (Prime Minister Tan)	Secretary of State to Foreign Minister; NSA to Liang Qiwen	Deniable proposals, documents	A day
ORCHID	CIA to the Premier's office	Sanitized intelligence; one use only	Unpredictable

Rules for every channel

- One voice per channel. If two Americans are saying different things on the same line, you have no channel.
- Know whose night it is. Put two clocks for the other capital on the wall when daylight time changes on one side and not the other.

- Use your own interpreter and keep your own notes. Follow every leader call with a short written summary through a second channel within six hours, so that what was said cannot be rewritten by either side's staff.
- Confirm receipt. A message not acknowledged has not been delivered.
- Never lie on the leader channel. You may decline to answer. You may not deceive. That channel's whole value is that both men believe it.
- Never use a channel in a way that burns a source unless you have decided, slowly and in writing, that the source is the price (Rule 37).
- When a channel goes silent, do not fill the silence with threats. Ask a neutral party what the silence means.

Section 4: Managing Allies

- **Call allies before you call the adversary**, unless lives depend on the order being reversed. Japan, Australia, the Philippines, and Taiwan each learned of my convoy decision before Beijing did.
- **Tell them what you have decided only after you have heard them.** If you have already decided, say "informed," not "consulted" (Rule 25).
- **Give each ally a role it can describe at home.** Escorts, basing, surveillance, humanitarian cargo, diplomatic sponsorship. An ally without a role becomes a critic.
- **Never ask an ally in public to bear a cost you will not bear.** Ask in private, explain why, and say what you will give in return.
- **Protect the ally's dignity when you ask it to step back.** Let it announce its own decision, in its own words, on its own day.
- **Keep the neutrals neutral.** A mediator is valuable only while both sides trust it. Do not use Singapore, or any Singapore, to win. Use it to stop.
- **Keep the large neutrals informed even when they are not in the room.** India and Indonesia were not parties to the talks, and both were told what they needed to know before the press was.
- **Watch the adversary's partners.** Russia's restraint in 2045 was worth more than any alliance statement. Speak to its leader directly, and do not ask him for anything he cannot be seen to give.

Section 5: Managing Congress and the Public

Congress

- Brief the bipartisan leadership of both houses, and the chairs and ranking members of the intelligence and armed services committees, within forty-eight hours of any major decision, and before any public announcement where possible.
- Comply with the War Powers Resolution's reporting requirement in letter and in spirit. File the report within forty-eight hours of introducing forces into hostilities, and say what you are doing in plain words.
- Transmit every international agreement, classified or not, to the committees as the law requires. Do not call a commitment secret when the law requires you to tell Congress. Call it classified, and tell them.
- Treat the opposition leader as a partner in the national interest, not as an obstacle. He will remember how he learned things longer than what he learned.

The public

- Speak rarely, calmly, and at a fixed time. A President's voice should lower the temperature (Rule 31).
- Say what you know, what you do not know, and when you will say more. Then say more on that day.
- Do not reveal sources, methods, or how close you came while the danger lasts. Do decide when you will reveal it, and write the date down (Rule 32).
- Work with the Treasury on every statement that could move markets. In 2045 the information clock ran through the bond market.
- Answer a fabricated video or false report quickly with evidence, not adjectives. Speed matters more than eloquence on the information clock.

Section 6: Managing Intelligence Assets

Added 29 June 2046.

- **Every requirement has a price.** Before you ask a human source for more frequent or more sensitive reporting, ask the case officer what it costs the source in risk. Get the answer in writing. Read it yourself.
- **Keep an exfiltration plan ready for every source you rely on,** current and tested, from the first day of a crisis, not the day you need it.
- **Ask the source, if you can, before you decide for him.** If you cannot ask, record why not, and record what you believe he would say.
- **Know who his family is and what will happen to them.** A plan that brings out the source and leaves his family has not finished.
- **Sanitize before you share.** Intelligence passed to anyone outside the United States Government, including a foreign official you hope will act on it, must be stripped of everything that points to its origin. Assume the first question asked will be: how do the Americans know?
- **Never let a source become your only check.** A source under a mole hunt can be caught, turned, or coerced. Weigh him alongside your satellites, never instead of them.
- **Never mention a source, even obliquely, in a public statement,** a leak to the press, or a conversation with a foreign leader.
- **Count the cost afterward.** When a source is lost, ask what you asked of him, and when, and whether you would ask it again. Write the answer in the margin of this page.

Section 7: Managing Yourself

Sleep

- Sleep in blocks of at least four hours whenever the military clock allows. Name a principal, normally the National Security Advisor or the Vice President, to hold the watch while you sleep, with written authority over everything except what only you can decide.
- Before you go to sleep, ask: What will they wake me for? Make the list short, and make sure the watch officer has it.
- After two nights with less than four hours, assume your judgment is degraded. Tell your National Security Advisor so, and ask him to tell you when he sees it.

Body

- Eat at regular times, even in the PEOC. Walk outside once a day, even for five minutes.
- See the White House physician weekly during a crisis, and tell the truth.

Family

- Decide early what you can tell your spouse and children, and tell them that much. Secrecy without explanation is worse than secrecy.
- If a member of your family is in danger because of your orders, say so to your National Security Advisor and your Chief of Staff, and ask them to tell you if they think it is affecting your decisions (Rule 39).
- Go upstairs. A President is allowed to go upstairs.

Faith

- Whatever you believe, keep a practice that slows you down: prayer, silence, a walk, the log. I am a lapsed Methodist. I prayed more in 2045 than in the twenty years before. I am not sure to whom. I am sure it slowed me down.
- Do not mistake your conviction for certainty, or your certainty for permission.

The log

- Every night, by hand, at least ten minutes. What happened, when, what you did, and what you were afraid of. Do not dictate it. Do not let anyone read it (Rule 40).

Annex B: Chronology

Prepared by the Editorial Team. United States times are Eastern (EST or EDT as in force); Beijing times are given where known. Daylight time began in the United States on 12 March 2045 and ended at 02:00 on 5 November 2045. Beijing was 13 hours ahead of Eastern Standard Time and 12 hours ahead of Eastern Daylight Time.

Date	Time	Event	See
5 Feb 2026		New START expires; no successor treaty	Prologue
20 Jan 2041		Kane inaugurated; inauguration-day nuclear briefing at Blair House	Ch. 1
2043		Jade Line leader video hotline agreed	Prologue
Jan 2044		Lin Shu-wen elected President of Taiwan (inaugurated May 2044)	Prologue
5 Nov 2044		Kane re-elected by a narrow margin	Prologue
20 Jan 2045		Kane's second inauguration	Prologue
Feb 2045		Status Referendum Act debated in the Legislative Yuan	Prologue
3 Mar 2045	19:00 Beijing	PRC announces "law-enforcement quarantine" of Taiwan and exercise Strait Thunder-2045	Ch. 2
4 Mar 2045	03:10 EST	Kane woken; first NSC meeting of the crisis	Ch. 2
4 Mar 2045	16:00 Beijing	Quarantine takes effect	Ch. 2
9 Mar 2045		CCG cutter rams *Formosa Pearl*; two Taiwanese crew killed	Ch. 2

Date	Time	Event	See
12 Mar 2045		LANTERN reports Base 64 at Readiness Level 3, first time since 2034; US daylight time begins	Ch. 2
30 Mar 2045		Kane decides on the escorted humanitarian corridor	Ch. 2
2 Apr 2045		Jade Line call 1; Kane tells Wen of the convoy privately	Ch. 2
6 Apr 2045		Convoy publicly announced	Ch. 2
20 Apr 2045		First convoy (Operation OPEN DOOR) departs Subic Bay	Ch. 2
22 Apr 2045	16:40 local	Convoy arrives Kaohsiung after PLA close pass	Ch. 2
May 2045		Quarantine partly eased; warhead-mating drills at Hami	Ch. 2
18 May 2045		Luo Tiancheng visits Lanzhou	Ch. 2–3
6 Jun 2045		PRC “customs blockade”: goods for Taiwan must clear in Fujian	Ch. 3
14–17 Jun 2045		Cyber attacks on West Coast ports and Texas grid; rolling blackouts	Ch. 3
Jun–Jul 2045		Treasury yields reach 7.1%; China sells about $200 billion in Treasuries; Phase A sanctions; total chip export ban; PRC rare-earth and gallium embargo	Ch. 3
4 Jul 2045		Kane’s address, “The Patience of Free People”	Ch. 3

Date	Time	Event	See
Jul 2045		Defense Production Act order to Halcyon Orbital; Oval Office meeting with Marcus Thorne	Ch. 3
28 Jul 2045		WILLOW reports Wen's secret four-day hospitalization	Ch. 3
19 Aug 2045		Secret Marsh–Shen meeting at Sentosa, Singapore	Ch. 3
30 Aug 2045		THRUSH reports three Type 096 submarines sailed from Yulin in 48 hours	Ch. 3
8 Sep 2045		PLA declares no-fly, no-sail zone east of Taiwan	Ch. 4
11 Sep 2045	21:40 EDT	Bashi Channel Incident; USS *Thomas Hudner* hit, 41 killed; 47 minutes before Kane learns Maya is safe	Ch. 4
12 Sep 2045		Proportionate US strike on three PLAN vessels and a Fujian missile battery; about 120 PLA killed; F-35C pilot Lt. Cmdr. Owen Delacroix captured	Ch. 4
13 Sep 2045		Chinese counter-space attack blinds about 60 ARGUS and tracking satellites	Ch. 4
14–22 Sep 2045		Halcyon and Space Force launch 34 replacement satellites; coverage back to about 90%	Ch. 4
16 Sep 2045		Kane at Dover Air Force Base for the return of the *Hudner* dead	Ch. 4, 7

Date	Time	Event	See
20 Sep 2045		US at DEFCON 3; bomber dispersal; submarine surge	Ch. 4
2 Oct 2045		MSS arrests KESTREL near Yumen	Ch. 4
4 Oct 2045		Kane decides to keep LANTERN in place	Ch. 4
19 Oct 2045		KESTREL executed (reported by WILLOW on 20 Oct)	Ch. 4
Oct 2045		Kane–Volensky call; Russian bombers to the Far East	Ch. 4
26 Oct 2045		Luo's "window" speech; doctrinal article in state media	Ch. 4
1 Nov 2045	04:12 EDT	ARGUS detects dispersal at Hami and Ordos; LANTERN reports Readiness Level 2 and warheads mated in two brigades	Ch. 5
2 Nov 2045	06:40 EDT	NSC in the PEOC; "left of launch" strike rejected; quiet generation of strategic forces; no public DEFCON 2	Ch. 5
3 Nov 2045	09:00 EDT	Jade Line call 4	Ch. 5
3 Nov 2045	14:50 EDT	WILLOW reports Wen hospitalized again; Luo and Gao running Joint Operations Command	Ch. 5
3–4 Nov 2045	from 22:30 EDT	Nineteen-hour deliberation; evidence on Luo's orders passed to Qiao and Zhang via ORCHID and Singapore	Ch. 5
5 Nov 2045	02:00	US daylight time ends	Ch. 5
5 Nov 2045	19:12 EST	Kane's letter to Wen handed to the Chinese side in Singapore	Ch. 5

Date	Time	Event	See
6 Nov 2045	02:31:07–02:42:12 EST	The Eleven Minutes: false warning of six ICBM launches; “no attack” declared	Ch. 6
6 Nov 2045	15:31–15:50 Beijing	Luo orders preparation for Readiness Level 1; Zhang countermands	Ch. 6
7 Nov 2045		Kane tells Wen of the false alarm on the Jade Line	Ch. 6
8 Nov 2045		Politburo Standing Committee emergency session; Luo relieved	Ch. 6
9 Nov 2045		Readiness Level 3 confirmed; Kane addresses the nation and announces “a pause”	Ch. 6
13 Nov 2045	07:00 EST	“The Long Way Down” descent plan adopted	Ch. 7
14 Nov 2045		Seventy-page investigation of the false alarm delivered	Ch. 6
25 Nov 2045		First Singapore round, ashore	Ch. 7
8 Dec 2045		Second round aboard RSS *Fortitude*; agreement in principle on “the eyes”	Ch. 7
19 Dec 2045	20:00 EST	Kane–Lin call; referendum postponed	Ch. 7
4–6 Jan 2046		Third round sets the three-phase sequence	Ch. 7
8 Jan–16 Feb 2046		Blockade lifted in three phases (one six-day hold after 22 Jan tanker inspection)	Ch. 7
22 Jan 2046	morning, Singapore	Prisoner and remains exchange at Changi; Delacroix released	Ch. 7
Feb 2046		Singapore Understanding drafted; legal forms chosen	Ch. 7
21–22 Mar 2046		Kane–Wen summit in Singapore; public text issued 22 March	Ch. 7, Annex E

Date	Time	Event	See
Apr 2046		Senate debate; Kane's closed testimony; Operation NIGHT FERRY brings LANTERN out	Ch. 7, Epilogue
30 Jun 2046		PCOS Edition 1 completed	Front matter
2047		Wen dies; Luo convicted; PALISADE withdrawn; Two-Source Rule written into procedure	Epilogue
Sep 2048		JEDEC fully operational in Singapore	Epilogue
20 Jan 2049		Kane leaves office	Epilogue
2049–2061		Three Presidents sign for PCOS-1	Front matter
2050		Nuclear Crisis Consultation Act	Epilogue
2051		Space-based interceptor pause replaced by negotiated ceiling	Epilogue
2068		Fu Jianguo (LANTERN) dies in Virginia	Epilogue
Oct 2069–Feb 2071		Interagency declassification review	Front matter
4 Mar 2071		Archivist accepts the panel's findings; release authorized	Front matter

Annex C: Dramatis Personae

All persons listed are fictional. Positions are as of 2045–2046.

United States

Name	Position	Role in the crisis
Thomas Ellery Kane	49th President	Author; engineer; keeper of the log
Dr. Claire Kane	Pediatric oncologist; the President's wife	His conscience; taught the Reversibility Matrix on a napkin
Lt. Maya Kane, USN	Officer, USS *Daniel Inouye* (DDG-118), Seventh Fleet	The President's daughter; near the Bashi Channel Incident
Eli Kane	Software engineer, Austin	The President's son; lost power in the Texas blackout
Rosalind Achterberg	Vice President	Steady, hawkish-leaning; concurred in holding on 6 November
Dr. Samuel Oyelaran	National Security Advisor	Closest strategist; author of "The Long Way Down"
Helena Marsh	Secretary of State	Former ambassador in Beijing; negotiator in Singapore
Gen. (ret.) Raymond Castillo	Secretary of Defense	Carried the 41 names of the *Hudner* dead
Gen. Patricia Dunmore, USAF	Chairman of the Joint Chiefs of Staff	Advocate of "left of launch" options
Adm. Victor Hale, USN	Commander, US Strategic Command	Submariner; "The fusion engine is confident. I am not."
Adm. Kenji Moriyama, USN	Commander, US Indo-Pacific Command	Commanded convoys and the 12 September strike
Abigail Fenn	Director of National Intelligence	Intelligence assessments
Gregory Laszlo	Director, Central Intelligence Agency	Ran the human sources
"Ruth Calder" (pseudonym)	Chief, CIA China Mission Center	Recruited LANTERN
Daniel Iwu	Secretary of the Treasury	Sanctions and the bond market

Name	Position	Role in the crisis
Megan Pruitt	White House Chief of Staff	Disliked the document's title
Lt. Col. Jamal Brooks, USMC	Military aide	Carried the emergency satchel; asked for socks
Cal Whitford	Senate Majority Leader (R-Texas)	Critic of "secret concessions"
Diane Okonkwo	Speaker of the House (D)	Congressional leadership
Lt. Cmdr. Owen Delacroix, USN	F-35C pilot	Captured 12 September; released 22 January
Marcus Thorne	Chief executive, Halcyon Orbital	Resisted, then supported, military use of his satellites

People's Republic of China

Name	Position	Role in the crisis
Wen Zhaolin	General Secretary, President, Chairman of the Central Military Commission	Cautious, ill; regained control 7–8 November
Qiao Weiming	Premier	Voice of de-escalation; received the 4 November evidence
Gen. Zhang Hesheng	Vice Chairman, Central Military Commission	Countermanded Luo on 6 November
Gen. Luo Tiancheng	Commander, PLA Rocket Force	Believed the "window" was closing; relieved 8 November
Adm. Gao Ruiming	Commander, Eastern Theater	With Luo in the Joint Operations Command Center
Shen Yilan	Foreign Minister	Marsh's counterpart
Kong Dewei	Minister of State Security	Ran the mole hunt
Liang Qiwen	Director, Office of the Central Foreign Affairs Commission	Oyelaran's counterpart in Singapore
Tang Shiyu	Ambassador to the United States	Formal channel in Washington
Unnamed colonel	Duty officer, Joint Operations Command Center	Asked "under what authority?" on 6 November

Intelligence Sources (CIA)

Code name	Identity	Reporting
LANTERN	Senior Col. Fu Jianguo, deputy chief of staff, PLARF Base 64, Lanzhou (named in the Epilogue only)	Alert levels and warhead status, Hami field

Code name	Identity	Reporting
WILLOW	Maj. Song Lihua, duty officer, Joint Operations Command Center, Western Hills	Message traffic, leadership mood, Wen's health
THRUSH	Mei Zhen, civilian maintenance engineer, Yulin naval base	Submarine sailings
KESTREL	Technician, Yumen field power and communications (name withheld)	Field readiness; arrested 2 October, executed 19 October 2045
ORCHID	Aide in the Premier's office (name withheld)	Political intelligence; back channel on 4 November

Taiwan

Name	Position	Role
Lin Shu-wen	President (Democratic Progressive Party)	Former law professor from Tainan; postponed the referendum
Lai and Hsu	Deckhand and engineer, *Formosa Pearl*	Killed 9 March 2045

Other Countries and Organizations

Name	Position	Role
Daniel Tan Wei Ming	Prime Minister of Singapore	Mediator and host
Arkady Volensky	President of Russia	"Neutrality with teeth"
Hiroshi Saeki	Prime Minister of Japan	Convoy partner
Grace Holloway	Prime Minister of Australia	Convoy partner
Andrés Villanueva	President of the Philippines	Host of Subic Bay convoys
Ananya Rao	Prime Minister of India	Kept informed as a major neutral
Arya Wicaksono	President of Indonesia	Supported an ASEAN restraint statement; offered Jakarta as a venue
Astrid Lindqvist	Secretary-General of the United Nations	Offered New York as a venue

Annex D: Glossary of Terms and Acronyms

Real terms are explained at the level of public sources. Terms marked (fictional) were invented for this book.

Term	Spelled out	Explanation
AI	Artificial intelligence	Software that learns patterns from data; used in 2045 to fuse sensor reports and recommend assessments
ANMCC	Alternate National Military Command Center	Backup to the NMCC at Raven Rock Mountain Complex ("Site R"), Pennsylvania
ARGUS	Not an acronym (fictional)	Classified US surveillance program fusing commercial and national satellites; revisits any point about every 90 seconds
ASAT	Anti-satellite weapon	A weapon that destroys, disables, or blinds satellites: missiles, co-orbital craft, lasers, jammers
ASEAN	Association of Southeast Asian Nations	Regional grouping of ten Southeast Asian states, including Singapore and Indonesia
B-21, B-52J	Bomber designations	The B-21 Raider stealth bomber and the modernized B-52J; the air leg of the US triad
Biscuit	(informal)	Card carried by the President with codes used to identify himself to the NMCC
CCG	China Coast Guard	Chinese maritime law-enforcement service; enforced the 2045 quarantine
CCP	Chinese Communist Party	The ruling party of the PRC

Term	Spelled out	Explanation
CIA	Central Intelligence Agency	US agency responsible for human intelligence abroad
CMC	Central Military Commission	The Party and state body that commands China's armed forces; chaired by the General Secretary
COG	Continuity of government	Plans to keep the government functioning if leaders or the capital are lost
CUES	Code for Unplanned Encounters at Sea	Voluntary naval safety code adopted in 2014 at the Western Pacific Naval Symposium
DDG	Guided-missile destroyer	Hull designation, as in USS *Daniel Inouye* (DDG-118)
DEFCON	Defense Readiness Condition	US military alert scale from 5 (normal) to 1 (maximum readiness)
DF-41, DF-5C, DF-31AG, DF-26	Dongfeng ("East Wind") missiles	Chinese ballistic missiles: solid-fuel ICBM, liquid-fuel silo ICBM, road-mobile ICBM, intermediate-range missile
DNI	Director of National Intelligence	Head of the US intelligence community
DPA	Defense Production Act	1950 US law letting the President prioritize contracts for national defense
DPP	Democratic Progressive Party	Taiwanese party of President Lin
DTL	Defense Telephone Link	US–China military hotline established in 2008
E-4B	Aircraft designation	National Airborne Operations Center ("Nightwatch"), a flying command post
E-6B	Aircraft designation	Navy aircraft for communicating with submarines (TACAMO) and serving as an airborne command post
EDT / EST	Eastern Daylight / Standard Time	US East Coast time; 12 or 13 hours behind Beijing
EO	Electro-optical	Imaging by visible or infrared light; limited by cloud and darkness
E.O.	Executive Order	A presidential directive with legal force

Term	Spelled out	Explanation
ERCOT	Electric Reliability Council of Texas	Operator of the Texas power grid
FOBS	Fractional orbital bombardment system	A warhead placed into partial orbit to approach from an unexpected direction
Gold Folder	(fictional)	Book of pre-planned response options carried in the emergency satchel
Golden Dome	Not an acronym	US missile-defense architecture announced in 2025
HUMINT	Human intelligence	Information from human sources
IAS 2.0	Incidents at Sea and in the Air Agreement 2.0 (fictional)	2046 US–China safety agreement modeled on the US–Soviet agreement of 1972
ICBM	Intercontinental ballistic missile	Land-based missile with a range over 5,500 kilometers
INDOPACOM	US Indo-Pacific Command	US military command for the Pacific and Indian Ocean regions
IR	Infrared	Heat radiation; early-warning satellites detect the infrared plume of a rising missile
Jade Line	(fictional)	US–China leaders' video hotline agreed in 2043
JEDEC	Joint Early Warning Data Exchange Center (fictional)	US–China center in Singapore for sharing launch data
JL-4	Julang ("Giant Wave") 4	Chinese submarine-launched ballistic missile
JOCC	Joint Operations Command Center	China's military command center in the Western Hills, Beijing
KMT	Kuomintang	Taiwanese opposition party
Launch on warning		Launching missiles after detecting an incoming attack, before it arrives
LEO / GEO	Low Earth orbit / geosynchronous orbit	About 300–2,000 km altitude / about 36,000 km
LGM-35A	Missile designation	Sentinel ICBM, which replaced Minuteman III
LRASM	Long Range Anti-Ship Missile	US anti-ship cruise missile
MSS	Ministry of State Security	China's civilian intelligence and counterintelligence ministry
NC2 / NC3	Nuclear command and control / nuclear command, control, and communications	The authority over nuclear forces, and the sensors, links, and centers that support it

Term	Spelled out	Explanation
NGA	National Geospatial-Intelligence Agency	US agency that analyzes imagery
NMCC	National Military Command Center	Pentagon center that convenes warning conferences and transmits orders
NORAD	North American Aerospace Defense Command	US–Canadian command that detects and assesses air and missile threats
NRO	National Reconnaissance Office	US agency that builds and operates reconnaissance satellites
NSA	National Security Agency / National Security Advisor	Signals-intelligence agency; also the President's chief national security aide (context decides)
NSC	National Security Council	The President's forum for national security decisions
NSPM	National Security Presidential Memorandum	Formal presidential national security directive
PALISADE	(fictional)	Crew name for the Golden Dome sensor-fusion engine
PBSC	Politburo Standing Committee	China's top leadership body
PCOS	Presidential Crisis Operating System (fictional)	This document
PDB	President's Daily Brief	The daily intelligence briefing to the President
PEOC	Presidential Emergency Operations Center	Protected command center beneath the East Wing
Phenomenology		The physical effect a sensor detects (heat, radar reflection, signals)
PLA / PLAN / PLARF	People's Liberation Army / its Navy / its Rocket Force	China's armed forces; the Rocket Force controls land-based missiles
PRC	People's Republic of China	
PWSA	Proliferated Warfighter Space Architecture	US Space Development Agency constellation in low orbit, including a missile-tracking layer
RL	Readiness Level (fictional scale)	PLA alert levels: 4 peacetime, 3 heightened, 2 pre-launch preparation, 1 launch
ROE	Rules of engagement	Orders defining when forces may use weapons
SAR	Synthetic aperture radar	Radar imaging that works through cloud and at night

Term	Spelled out	Explanation
SIGINT	Signals intelligence	Intelligence from intercepted communications and electronic signals
SLBM	Submarine-launched ballistic missile	
SSBN	Ballistic-missile submarine, nuclear-powered	Columbia-, Ohio-, and Type 096-class submarines
STRATCOM	US Strategic Command	Command responsible for US nuclear forces
TACAMO	"Take Charge and Move Out"	Mission of relaying orders to submarines from aircraft
TEL	Transporter-erector-launcher	Vehicle that carries, raises, and fires a road-mobile missile
Two-Source Rule	(fictional)	Kane's rule: no nuclear warning confirmed without two phenomenologies, plus a third check where time permits
UEWR	Upgraded Early Warning Radar	Large ground radars, such as those at Clear (Alaska) and Pituffik (Greenland)
USMC / USN / USAF	US Marine Corps / Navy / Air Force	
YJ-21	Yingji ("Eagle Strike") 21	Chinese anti-ship ballistic missile

Annex E: Documents

E-1. The Singapore Understanding: Public Text

Joint Statement of the President of the United States of America and the President of the People's Republic of China Singapore, 22 March 2046

The President of the United States of America and the President of the People's Republic of China met in Singapore on 21 and 22 March 2046 at the invitation of the Prime Minister of the Republic of Singapore.

Conscious of the dangers of the past year, of the lives lost at sea, and of their special responsibility as nuclear-weapon states for the peace of the Asia-Pacific region and the world, the two Presidents have agreed as follows.

I. On nuclear war and human control

1. The two sides reaffirm that a nuclear war cannot be won and must never be fought. Recognizing that any conflict between them could have catastrophic consequences, they emphasize the importance of preventing any war between them, whether nuclear or conventional.
2. No artificial intelligence or automated system shall have authority to initiate the use of nuclear weapons; the decision shall be made by human beings, under the authority of each nation's highest leadership.
3. Each side will maintain and strengthen its national measures to prevent unauthorized, accidental, or mistaken use of nuclear weapons.

II. On safety at sea and in the air

4. The two sides have today concluded the Agreement on the Prevention of Incidents at Sea and in the Air (Incidents at Sea and in the Air Agreement 2.0).
5. The Agreement applies to naval vessels, military aircraft, coast guard and other government vessels, maritime militia, and unmanned vessels and aircraft operated by or for either government.
6. The Agreement establishes a standing military-to-military communications channel, staffed at all hours by duty officers of both sides, for the immediate resolution of encounters and incidents.

III. On the security of early-warning and command systems in space

7. Neither side will attack, jam, dazzle, or otherwise interfere with the satellites of the other that provide early warning of missile launches, or with the satellites and ground stations that support the command, control, and communications of its nuclear forces.

8. The two sides will consult promptly through the channels established by this Statement on any event that either side believes may be inconsistent with paragraph 7.

IV. On a Joint Early Warning Data Exchange Center

9. The two sides will establish, in the Republic of Singapore, a Joint Early Warning Data Exchange Center, staffed by officers of both sides.
10. Through the Center each side will provide advance notification of its own ballistic missile and space launches, and will share information on launches it detects by third parties, in order to reduce the risk of a launch being mistaken for an attack.
11. The two sides express their gratitude to the Government of the Republic of Singapore for agreeing to host the Center and ask it to report publicly each year on the Center's progress until it is fully operational.

V. On commerce and navigation

12. The two sides affirm the importance of the freedom and safety of commercial navigation, including in the waters around Taiwan, and welcome the restoration of normal commercial shipping completed on 16 February 2046.
13. The two sides will take further steps to ease restrictions on trade adopted during 2045, and establish a Joint Commission on Trade Measures to address the remaining measures, to meet at least twice a year.

VI. Final provisions

14. The two sides will review the implementation of this Statement at the level of foreign ministers each year.
15. The two Presidents express their gratitude to the Prime Minister and the people of Singapore for their hospitality and their service to peace.

Done at Singapore, on the twenty-second day of March, 2046, in the English and Chinese languages.

Editorial note on legal form: As the author explains in Chapter 7, the elements of this Statement were given different legal forms. The Incidents at Sea and in the Air Agreement 2.0 and the arrangement for the Joint Early Warning Data Exchange Center were sole executive agreements; paragraphs 1 to 3 were political commitments. The full texts of the Agreement and its annexes are not reproduced here.

E-2. The Secret Elements (Summary)

The secret elements of the Understanding were declassified by agreement of the governments concerned during the 2060s. In summary:

- The United States reaffirmed in writing that it "does not support" Taiwan independence, and stated that it would not station intermediate-range missiles in the Batanes islands for five years. Taiwan's referendum was recorded as shelved, without reference to any American request.
- China undertook to lift the quarantine permanently and to return the Rocket Force to the posture maintained before about 2030, with warheads stored separately from missiles, road-mobile launchers in garrison, and no forces on launch-on-warning; and agreed not to impede verification by national technical means, with cooperative display measures on notice.
- The United States stated as a matter of national policy, conveyed as a political commitment, that it would not deploy additional space-based interceptors for five years beyond the number

then deployed, about two hundred.

- Both sides agreed to upgrade the leaders' hotline and to add a second, independent link through Singapore.

E-3. The President's Letter of 5 November 2045

The text of President Kane's private letter of 5 November 2045 to General Secretary Wen Zhaolin, handed to the Chinese side in Singapore at 19:12 Eastern Standard Time that day, appears in Chapter 5 and is not repeated here. Its offer was: no American strikes on China's nuclear or early-warning forces; a pause in launches of new space-based interceptors; and a seventy-two-hour pause in American convoys, in exchange for a return of the Rocket Force to Readiness Level 3.

Annex F: Command Structures

Prepared by the Editorial Team from public sources current in 2045, simplified. Fictional elements are marked. Nothing here describes authentication, codes, or communications methods.

F-1. United States: Nuclear Command, Control, and Communications

The President alone has the authority to order the use of United States nuclear weapons. The Secretary of Defense is in the chain of command for executing the order. The Chairman of the Joint Chiefs of Staff is the President's principal military advisor and is in the chain of communication, but not of command.

The decision chain

- **The President** (with the military aide, the emergency satchel, the Gold Folder (fictional name), and the biscuit)
  - **Secretary of Defense** (chain of command)
  - **Chairman of the Joint Chiefs of Staff** (advice; chain of communication)
  - **National Military Command Center**, Pentagon (Alternate: Raven Rock Mountain Complex)
    - **Commander, US Strategic Command**, Offutt Air Force Base, Nebraska
      - ICBM wings: F.E. Warren (Wyoming), Malmstrom (Montana), Minot (North Dakota), LGM-35A Sentinel
      - Ballistic-missile submarines: Columbia and Ohio classes, Pacific and Atlantic
      - Bomber wings: B-21 Raider and B-52J
      - Airborne command and relay: E-4B National Airborne Operations Center; E-6B (TACAMO and airborne command post)

The warning chain

- **Sensors**
  - Space-based infrared satellites (geosynchronous and other orbits)
  - Proliferated Warfighter Space Architecture tracking layer (low orbit)
  - ARGUS overhead imagery and radar (fictional)
  - Ground radars: Clear (Alaska), Cobra Dane (Shemya, Alaska), Pituffik (Greenland), and others
- **Fusion**: Golden Dome battle management, including PALISADE (fictional, to 2047)
- **Assessment**: NORAD and US Strategic Command
- **Conferences convened by the NMCC**

1. Threat assessment conference (NMCC, NORAD, STRATCOM)
2. Missile event conference (adds the President, Secretary of Defense, Chairman)
3. National Decision Conference (the President decides)

Succession (Presidential Succession Act): Vice President; Speaker of the House; President pro tempore of the Senate; then Cabinet secretaries in order of their departments' creation.

F-2. People's Republic of China: Nuclear Command (as reconstructed for 2045)

- **Politburo Standing Committee** (political authority)
  - **Central Military Commission**, Chairman Wen Zhaolin
    * Authorization of changes in nuclear readiness by the Chairman ("Chairman's Seal Directive," a translated internal term; fictional)
    * Vice Chairman Zhang Hesheng (role in the Chairman's incapacity)
    * **Joint Staff Department / Joint Operations Command Center**, Western Hills, Beijing
      · **PLA Rocket Force Headquarters**, Qinghe, Beijing (Commander Luo Tiancheng to 8 November 2045)
        - Missile bases (Bases 61–66), each commanding several brigades
          - **Base 64**, Lanzhou: brigades including those of the Hami silo field (LANTERN on the base staff)
          - Bases responsible for the Yumen and Ordos (Hanggin Banner) fields and road-mobile DF-41 and DF-31AG brigades
        - Warhead storage and handling base (Base 67 in public literature)
        - Early-warning center (space and radar)
      · **PLA Navy**: Type 096 ballistic-missile submarines, Yulin, Hainan (JL-4)
      · **PLA Air Force**: H-20 bombers
      · **Eastern Theater Command**, Adm. Gao Ruiming (conventional forces facing Taiwan)

Readiness Levels (fictional scale): 4 peacetime; 3 heightened; 2 pre-launch preparation, warheads mated; 1 launch on order.

Separately: the Ministry of State Security (Minister Kong Dewei), China's counterintelligence service, reported to the Party leadership outside the military chain.

Annex G: Note on Real-World Sources

This note is addressed to the reader of the book, not to the fictional reader of 2071.

Everything technical in *The Deciding Minutes* is built from public, unclassified sources, and kept at their level of detail or below. ARGUS, PALISADE, the Jade Line, JEDEC, the Gold Folder, the Two-Source Rule, the PLA "Readiness Levels," the Chairman's Seal Directive, and every person and operation in the story are invented. The real works below informed the background. All were checked to exist; they are listed with author or issuing body and year.

On nuclear command, control, and communications

- Congressional Research Service, *Defense Primer: Command and Control of Nuclear Forces* (In Focus IF10521), first issued 2016 by Amy F. Woolf and updated regularly since (for example, 2024). The source for the President's sole authority, the warning conference, and the Chairman's role in the chain of communication.
- Anya L. Fink, Congressional Research Service, *Defense Primer: Nuclear Command, Control, and Communications (NC3)* (In Focus IF11697), 2020, updated through 2026.
- Bruce G. Blair, *Strategic Command and Control: Redefining the Nuclear Threat* (Brookings Institution, 1985).
- Eric Schlosser, *Command and Control: Nuclear Weapons, the Damascus Accident, and the Illusion of Safety* (Penguin Press, 2013).
- Scott D. Sagan, *The Limits of Safety: Organizations, Accidents, and Nuclear Weapons* (Princeton University Press, 1993). On the warning false alarms of 1979 and 1980.
- William J. Perry and Tom Z. Collina, *The Button: The New Nuclear Arms Race and Presidential Power from Truman to Trump* (BenBella Books, 2020).
- Annie Jacobsen, *Nuclear War: A Scenario* (Dutton, 2024). On the minute-by-minute shape of a warning and decision.

On China's forces and posture

- US Department of Defense, *Military and Security Developments Involving the People's Republic of China* (annual report to Congress), especially the 2024 and 2025 editions. On the silo fields, the warhead projections, and early-warning counterstrike.
- Hans M. Kristensen, Matt Korda, Eliana Johns, and Mackenzie Knight, "Chinese Nuclear Weapons, 2024" and "Chinese Nuclear Weapons, 2025," Federation of American Scientists Nuclear Notebook, *Bulletin of the Atomic Scientists* (2024 and 2025). On bases, brigades, and missile types.
- Graham Allison, *Destined for War: Can America and China Escape Thucydides's Trap?* (Houghton Mifflin Harcourt, 2017).
- Paul Bracken, *The Second Nuclear Age: Strategy, Danger, and the New Power Politics* (Times

Books, 2012).

On missile defense and space sensors

- Hannah D. Dennis, Congressional Research Service, *Defense Primer: The Golden Dome for America* (In Focus IF13115), 2025.
- Anya L. Fink, Congressional Research Service, *Golden Dome: Potential Strategic Stability Considerations for Congress* (Insight IN12568), 2025. On why missile defense alarms other nuclear powers, the root of the Mirror Model.
- Defense Intelligence Agency, *Golden Dome for America: Current and Future Missile Threats to the U.S. Homeland* (2025). Includes the fractional orbital bombardment system.
- Space Development Agency, *Proliferated Warfighter Space Architecture: Tracking Layer* fact sheet (2024).
- US Government Accountability Office, *Missile Warning Satellites: Space Development Agency Should Be More Realistic and Transparent About Risks to Capability Delivery* (GAO-26-107085, 2026).

On the end of the Cold War and its near misses

- David E. Hoffman, *The Dead Hand: The Untold Story of the Cold War Arms Race and Its Dangerous Legacy* (Doubleday, 2009). Includes the 1983 Petrov incident.

Documents quoted or echoed

- John F. Kennedy, Commencement Address at American University, Washington, D.C., 10 June 1963 (John F. Kennedy Presidential Library).
- Joint Soviet–United States Statement on the Summit Meeting in Geneva, 21 November 1985 (Ronald Reagan Presidential Library), source of "a nuclear war cannot be won and must never be fought."
- Joint Statement of the Leaders of the Five Nuclear-Weapon States on Preventing Nuclear War and Avoiding Arms Races, 3 January 2022.
- The White House, readout of the US–PRC leaders' meeting in Lima, Peru, 16 November 2024, in which both sides affirmed "the need to maintain human control over the decision to use nuclear weapons," the real-world seed of the fictional Human Control Pledge.

On form

- B. J. Habibie, *Detik-Detik yang Menentukan: Jalan Panjang Indonesia Menuju Demokrasi* (Jakarta: THC Mandiri, 2006). The model for this book's diary-based first-person narrative, its decision models, and its Prologue and Epilogue written by a team in the third person.